

AD and DA Conversion

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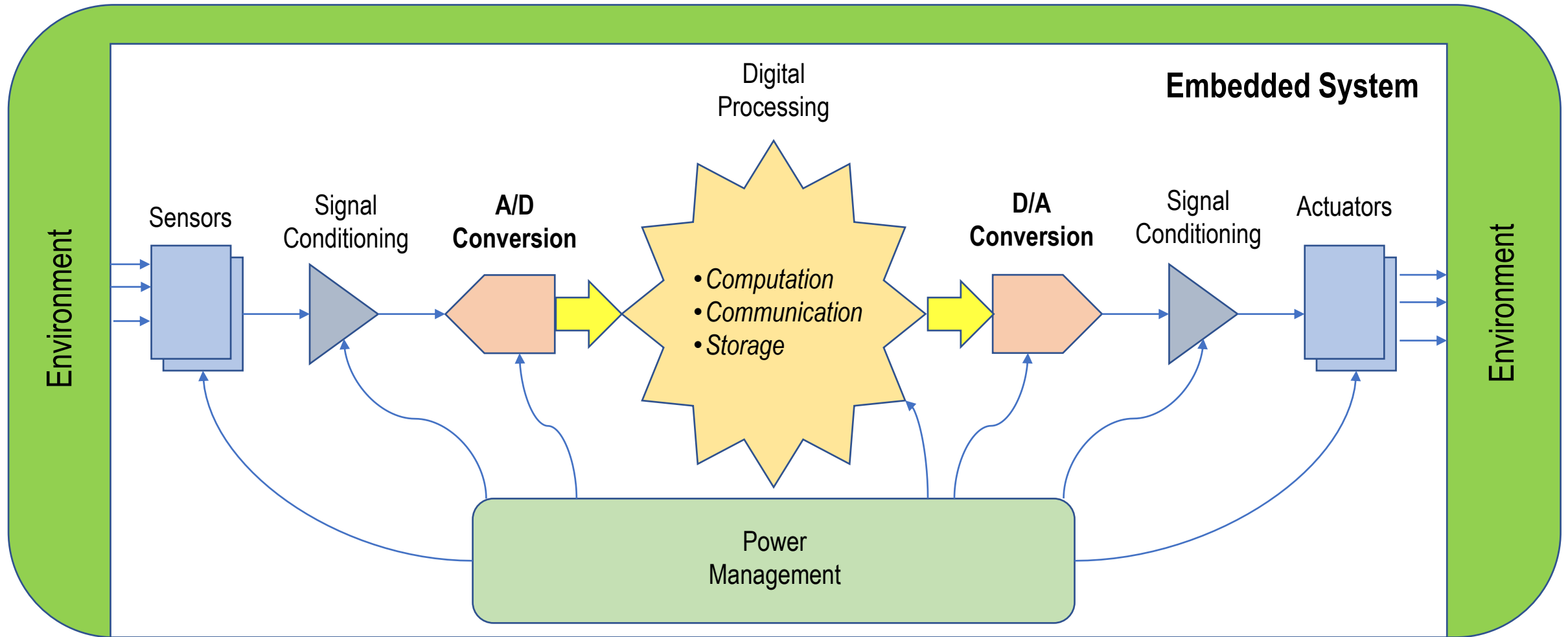
(Electronics for Embedded Systems)

Outline

- Introduction
 - Sampling
 - Quantization
- Digital to Analog Conversion
 - Errors
 - Circuits
- Analog to Digital Conversion
 - Errors
 - Circuits
- Acquisition systems
 - Sample & Hold Circuits

Introduction

- Typical architecture of an Embedded System

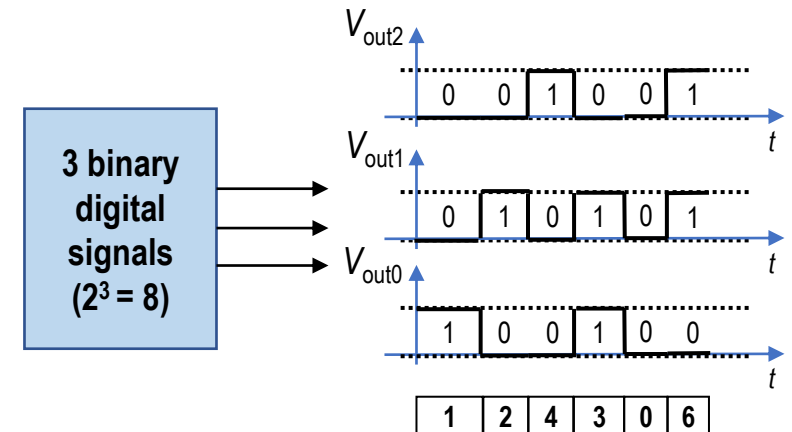
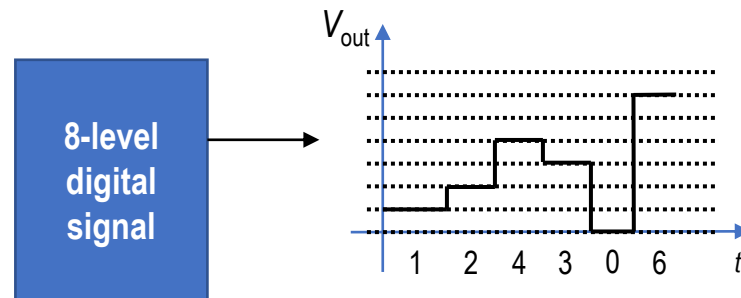


Introduction

- Analog signal
 - Continuous in time
 - It takes on a value for each time instant
 - Continuous in values
 - It may take on any value within a given dynamic range
- Digital signal
 - Discrete in time
 - It is defined only for certain time instants
 - Discrete in values
 - It can only take on values in a finite set of allowed values
 - If the set of values includes only two allowed states, then the signal is known as *binary*

Introduction

- Multiple binary signals, considered together and with proper weights, may represent a digital signal on more than two values
 - With n signals there are 2^n different values
 - Advantage
 - Better noise margin on individual wires compared to a single wire with 2^n discrete values
 - Disadvantage
 - More wires, hence more expensive
 - Examples: unsigned integer, two complements



Introduction

- Why digital?
 - Resilience to noise
 - A value affected by noise is interpreted as the closest allowed discrete value
 - High noise can introduce errors in the digital signal
 - Costs
 - Digital circuits are less expensive than analog circuits
 - Digital circuits are easier to design
 - Many more advance design tools are available
 - Flexibility
 - Digital circuits are easier to modify and adapt to new scenarios
 - Digital circuits can often be updated in the field (FPGAs or SW)

Introduction

- Analog-to-Digital (AD) Conversion
 - Sampling
 - Conversion from continuous time to discrete time
 - Quantization
 - Conversion from continuous values to discrete values
 - Converting from analog to digital implies some loss of information
 - Cardinality of continuum (uncountable) is strictly greater than the cardinality of an infinite discrete set (countable)
- Digital-to-Analog (DA) Conversion
 - Dequantization and reconstruction
 - Opposite conversions
 - Cannot restore information lost during the initial AD conversion

Introduction

- Goal
 - We want to convert a signal from analog to digital or viceversa
 - We want to limit the loss of information as much as possible
- When discretizing some noise is introduced
 - Noise corresponds to the loss of information
 - Evaluate the noise
 - Find techniques to keep the noise within the desired limits
- Physical quantities involved
 - Voltage, current
 - Frequency, phase, duty cycle

Introduction

• Sampling

- Analog signal in the time domain $x(t)$

- Spectrum of analog signal is $X(f) = \mathfrak{F}(x(t))$

- Sampled using an ideal train of δ functions $\psi_{T_S}(t) = \sum_{m=-\infty}^{m=+\infty} \delta(t - mT_S)$, $m \in \mathbb{Z}$

- Dirac comb with period T_S

- Sampling frequency $f_S = \frac{1}{T_S}$

- Spectrum of Dirac comb is $\Psi_{T_S}(f) = \mathfrak{F}(\psi_{T_S}(t)) = \frac{1}{T_S} \sum_{m=-\infty}^{m=+\infty} \delta\left(f - m\frac{1}{T_S}\right) = \frac{1}{T_S} \sum_{m=-\infty}^{m=+\infty} \delta(f - mf_S)$

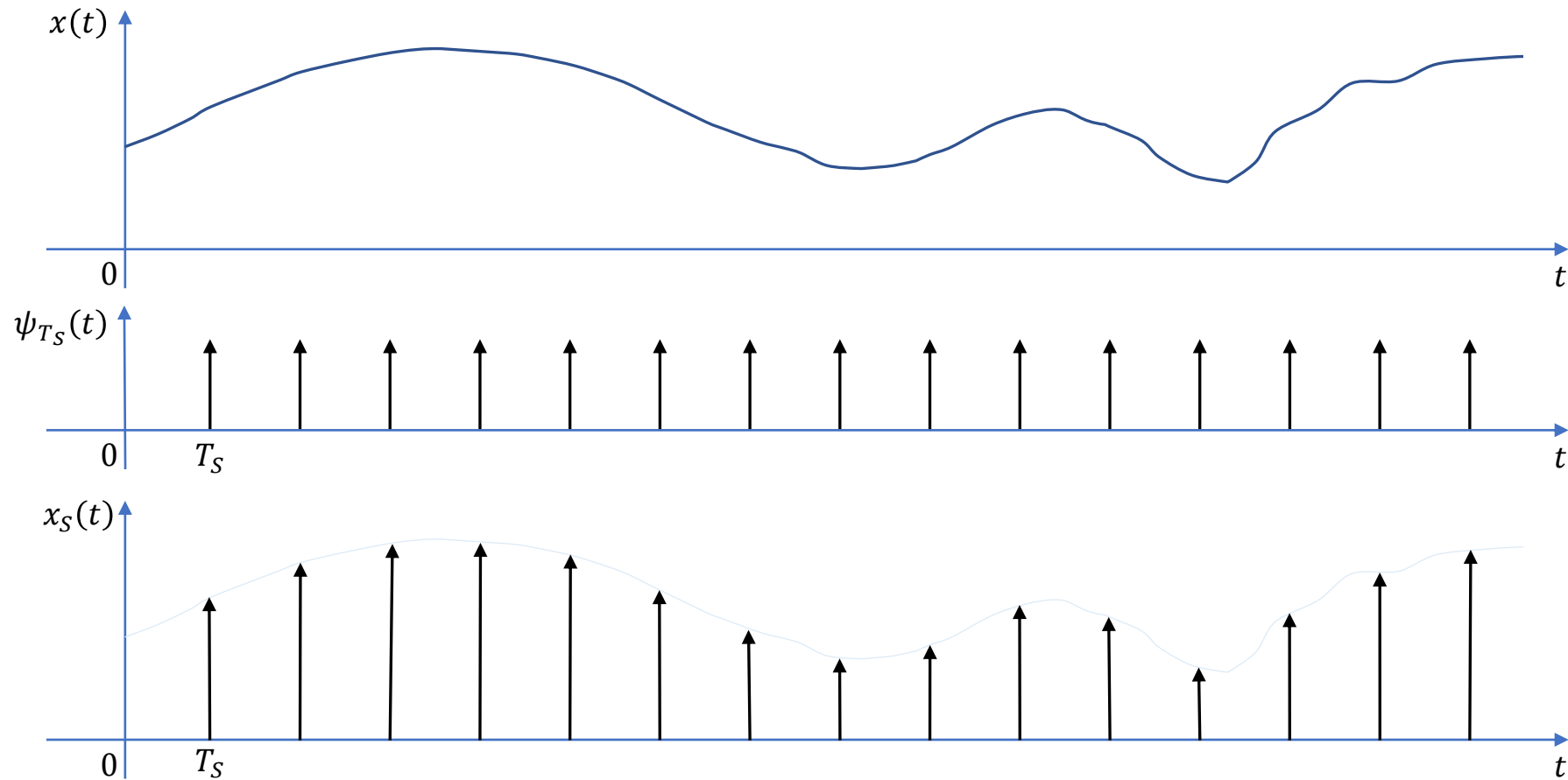
- Sampled signal is $x_S(t) = x(t) \cdot \psi_{T_S}(t)$

- Spectrum of sampled signal is

$$X_S(f) = \mathfrak{F}(x_S(t)) = \mathfrak{F}(x(t) \cdot \psi_{T_S}(t)) = \mathfrak{F}(x(t)) * \mathfrak{F}(\psi_{T_S}(t)) = X(f) * \frac{1}{T_S} \sum_{m=-\infty}^{m=+\infty} \delta(f - mf_S) = \frac{1}{T_S} \sum_{m=-\infty}^{m=+\infty} X(f - mf_S)$$

Introduction

- Sampling (continued)



Block diagram illustrating the sampling process:

The input signal $x(t)$ is multiplied by the sampling train $\psi_{T_S}(t)$ to produce the sampled signal $x_S(t)$.

$$\psi_{T_S}(t) = \sum_{m=-\infty}^{m=+\infty} \delta(t - mT_S)$$

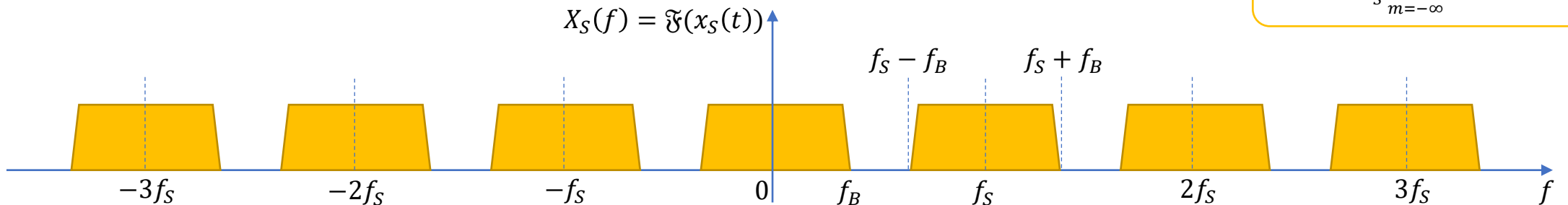
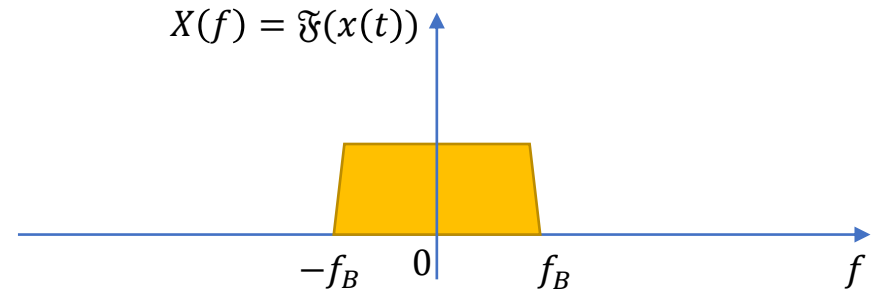
$$x_S(t) = x(t) \cdot \sum_{m=-\infty}^{m=+\infty} \delta(t - mT_S)$$

Introduction

- Sampling (continued)

- Initial hypothesis

- The analog signal is limited in bandwidth
 - f_B = maximum frequency with non null power spectrum
 - Sampling performed at regular intervals T_S
 - $f_S = 1/T_S$ is the sampling frequency
 - Assume that $f_S > 2 \cdot f_B$
 - The sampled signal power spectrum has infinite non overlapping replicas of the original signal power spectrum
 - Replicas are f_S apart in the spectrum
 - An ideal low pass filter can recover the original signal from the sampled signal
 - In these conditions, sampling does not introduce any loss of information or noise



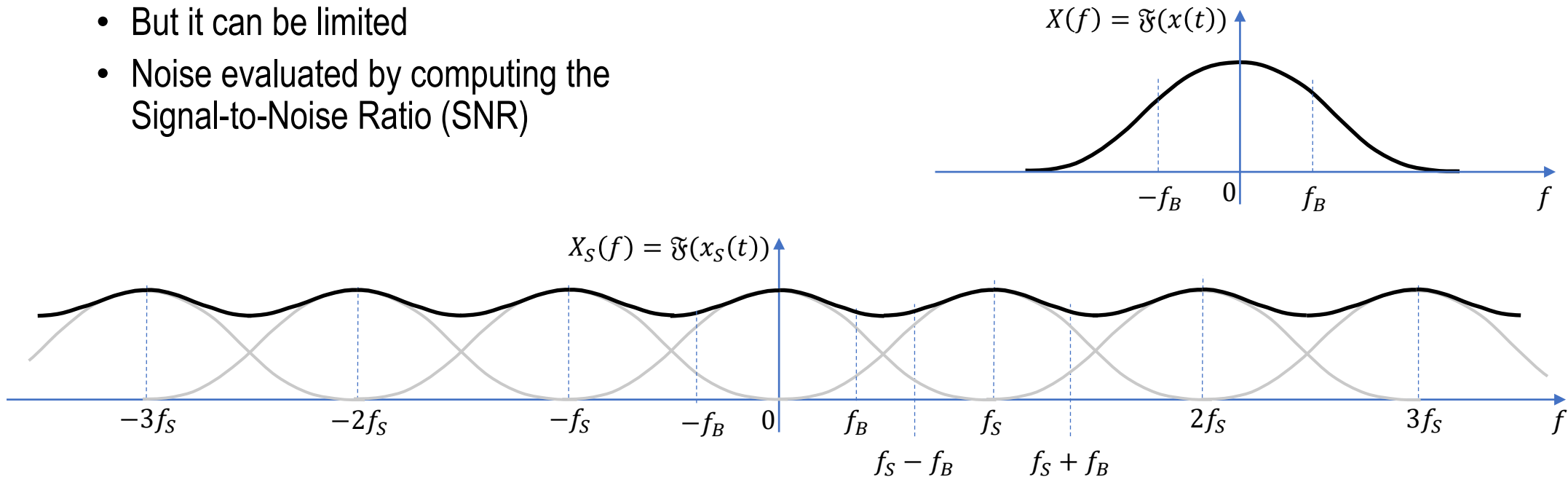
$$X_S(f) = \frac{1}{T_S} \sum_{m=-\infty}^{m=+\infty} X(f - mf_S)$$

Introduction

- Sampling (continued)
 - Condition for non overlapping replicas
 - $f_S - f_B > f_B \implies f_S > 2 \cdot f_B \implies f_{Smin} = 2 \cdot f_B$
 - Nyquist frequency $f_{Ny} = 2 \cdot f_B$
 - First problem
 - A signal limited in bandwidth is infinite in time
 - Not feasible in practice
 - All real signals have infinite bandwidth ($f_B = \infty$)
 - Although some have very small content above a certain frequency
 - It is then impossible to have $f_S > 2 \cdot f_B$
 - All sampled signal show some degree of *aliasing*
 - Replicas overlap, cannot isolate the original signal spectrum to recover it perfectly
 - Noise is introduced, loss of information
 - Second problem
 - An ideal low pass filter for reconstruction doesn't exist

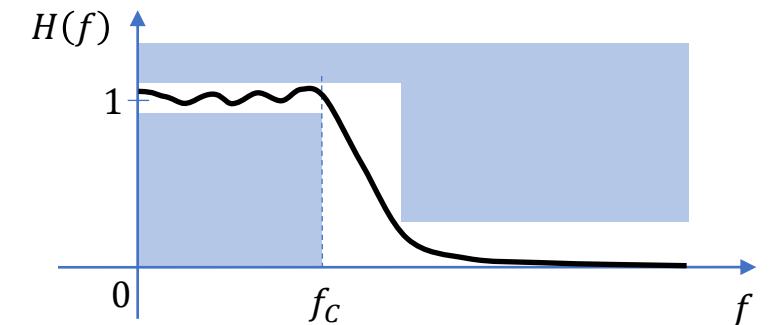
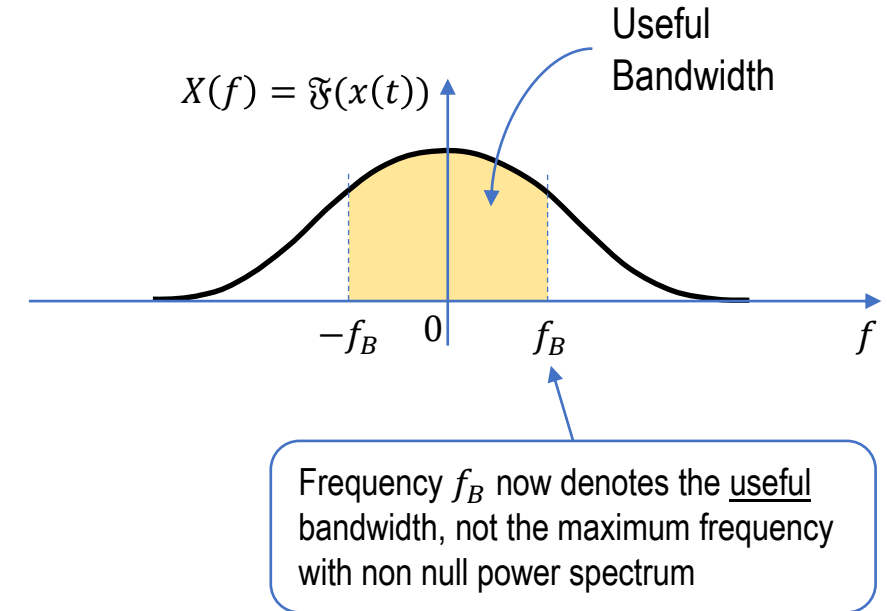
Introduction

- Sampling (continued)
 - Aliasing makes the spectrum of the original signal unrecoverable from the sampled signal
 - Even using an ideal reconstruction filter
 - Loss of information $\rightarrow$ noise
 - Aliasing cannot be avoided
 - But it can be limited
 - Noise evaluated by computing the Signal-to-Noise Ratio (SNR)



Introduction

- Sampling (continued)
 - Useful signal bandwidth
 - Portion of the power spectrum of interest for the application
 - From frequency 0 to frequency f_B
 - The input signal power spectrum above f_B is not necessarily zero
 - It might also be a band not starting from 0
 - The power spectrum outside this band is not considered useful
 - Anti-aliasing filter
 - A filter that attenuates the original signal outside the useful bandwidth
 - Characterized by:
 - Cut-off frequency f_c
 - Number of poles n_{poles}
 - Each pole introduces an attenuation of 20 dB/dec after f_c
 - Applied before sampling



Introduction

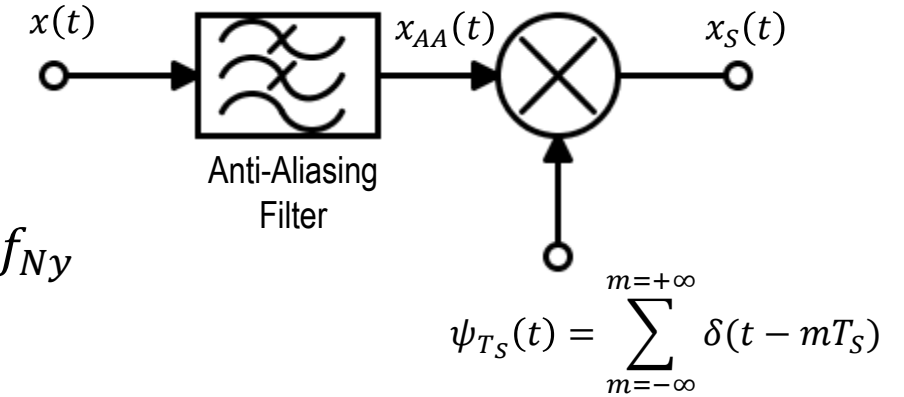
- Sampling (continued)

- Sampling frequency for real signals

- $f_S > 2 \cdot f_B$
 - Usually at least 2 to 5 times greater than the Nyquist frequency f_{Ny}
 - $f_S = k \cdot f_{Ny} = 2k \cdot f_B$

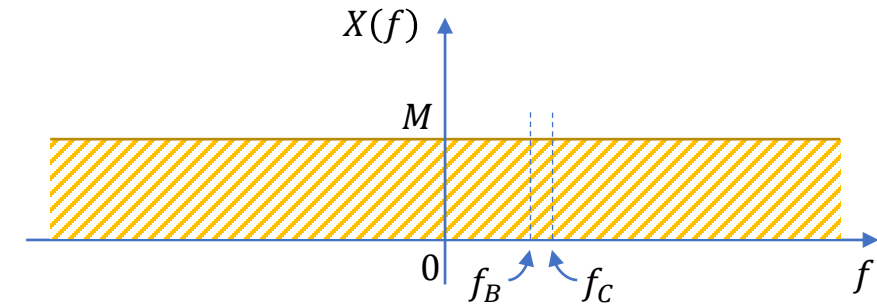
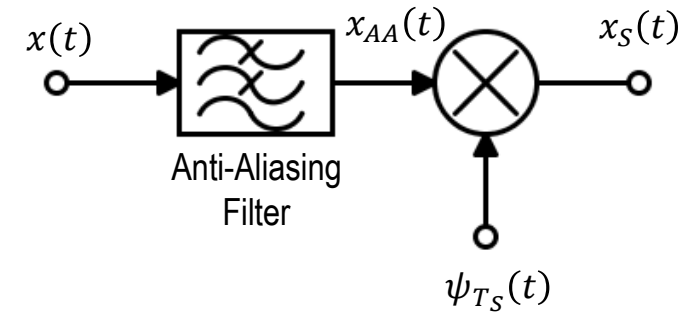
- Aliasing reduced by the filter

- It attenuates the tail of the original signal spectrum that is turned back into base band
 - The higher the attenuation, the lower the noise
 - Increase the number n_{poles} of poles of the filter
 - Increase the sampling frequency f_S to bring replicas further apart
 - The higher the number of poles, the higher the costs
 - The higher the sampling frequency, the higher the costs
 - Need to find a tradeoff



Introduction

- Estimate the amount of aliasing noise
 - Assume the input signal power spectrum $X(f)$ is flat
 - Even above f_B
 - The power spectrum magnitude is M
 - Sampling the input signal at frequencies $f_S > 2f_B$ gives a strong aliasing noise
 - Replicas contribution in base band is equal to the wanted input signal contribution
 - It is impossible to separate the contributions
 - An anti-aliasing filter is required



All graphs in this and next slides have linear scales both horizontally and vertically. Hence they are not Bode diagrams.

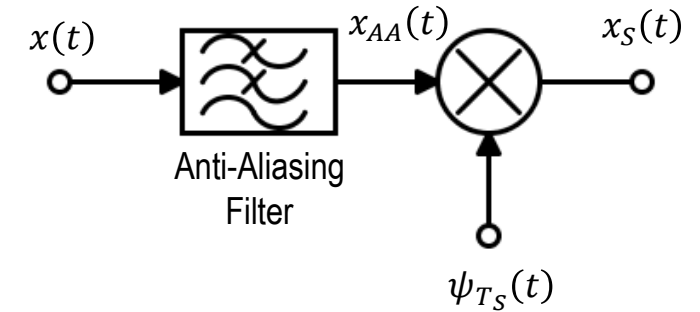
Introduction

- Estimate the amount of aliasing noise
 - The anti-aliasing filter is a low pass filter with n_{poles} poles at the cut-off frequency f_C

$$AA(s) = \frac{1}{\left(\frac{s}{\omega_c} + 1\right)^{n_{poles}}}$$

$$\omega_c = 2\pi f_C$$

Laplace transform of the anti-aliasing filter



- The magnitude of the anti-aliasing filter frequency response is

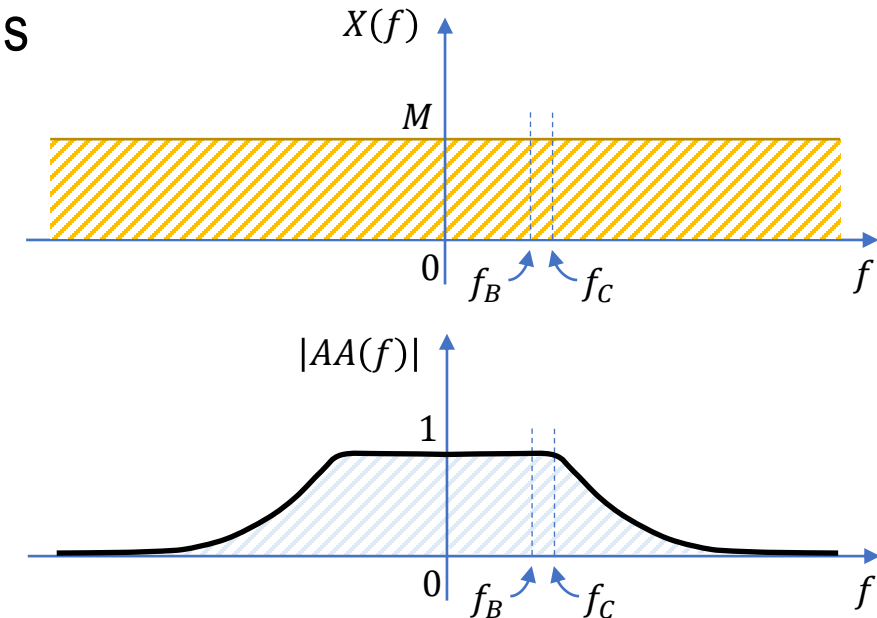
$$|AA(j\omega)| = \frac{1}{\sqrt{\left(\frac{\omega^2}{\omega_c^2} + 1\right)^{n_{poles}}}} = \frac{1}{\sqrt{\left(\frac{f^2}{f_C^2} + 1\right)^{n_{poles}}}} = \left(\frac{f^2}{f_C^2} + 1\right)^{-\frac{n_{poles}}{2}}$$

Response in linear scale at the generic frequency f

- At frequencies $f \gg f_C$, neglecting the term $+1$, the filter response is

$$|AA(j\omega)| \simeq \left(\frac{\omega}{\omega_c}\right)^{-n_{poles}} = \left(\frac{f}{f_C}\right)^{-n_{poles}}$$

Approximated response in linear scale at the generic frequency f



Introduction

- Estimate the amount of aliasing noise
 - The input signal power spectrum is attenuated by the anti-aliasing filter
 - The spectrum $X_{AA}(f)$ is flat with value M up to f_C , then it is attenuated
 - Attenuation is $n_{poles} \cdot 20$ dB/dec starting at f_C
 - With $n_{poles} = 3$ at $f = 10f_C$ the spectrum $X_{AA}(f)$ is attenuated by 60 dB
 - At frequencies $f \gg f_C$ the magnitude of $X_{AA}(f)$ is approximately

$$|X_{AA}(j\omega)| = |X(j\omega)| \cdot |AA(j\omega)| \simeq M \left(\frac{\omega}{\omega_C} \right)^{-n_{poles}} = M \left(\frac{f}{f_C} \right)^{-n_{poles}}$$

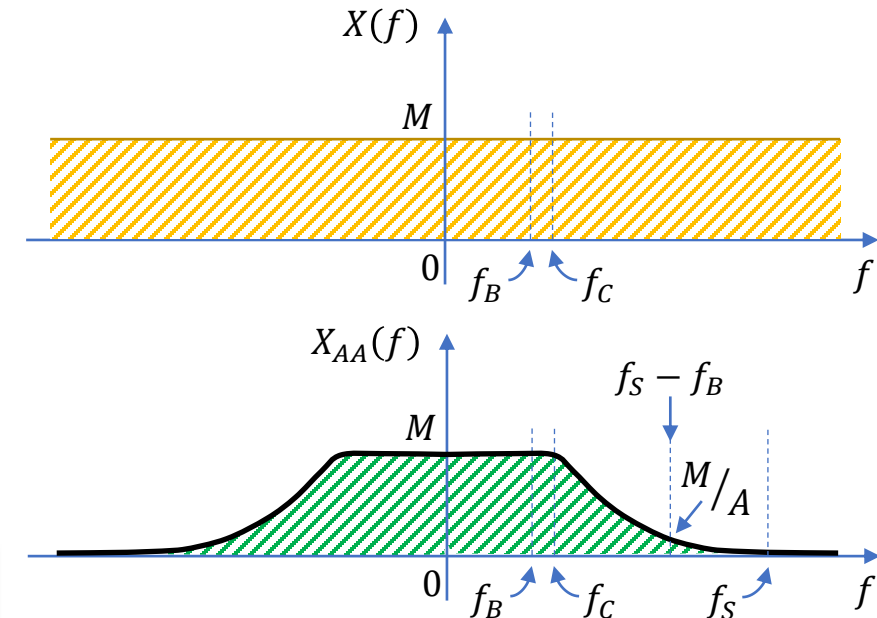
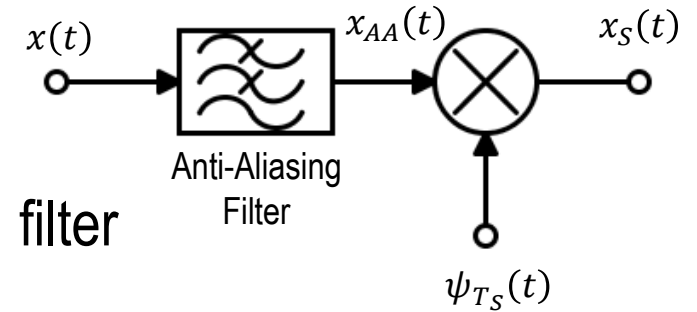
- In the next step we need the magnitude of the spectrum $X_{AA}(f)$ at frequency $f = f_S - f_B$

$$|X_{AA}(f_S - f_B)| = M \left(\frac{f_S - f_B}{f_C} \right)^{-n_{poles}} = \frac{M}{A}$$

- The attenuation due to the filter at frequency $f = f_S - f_B$ is

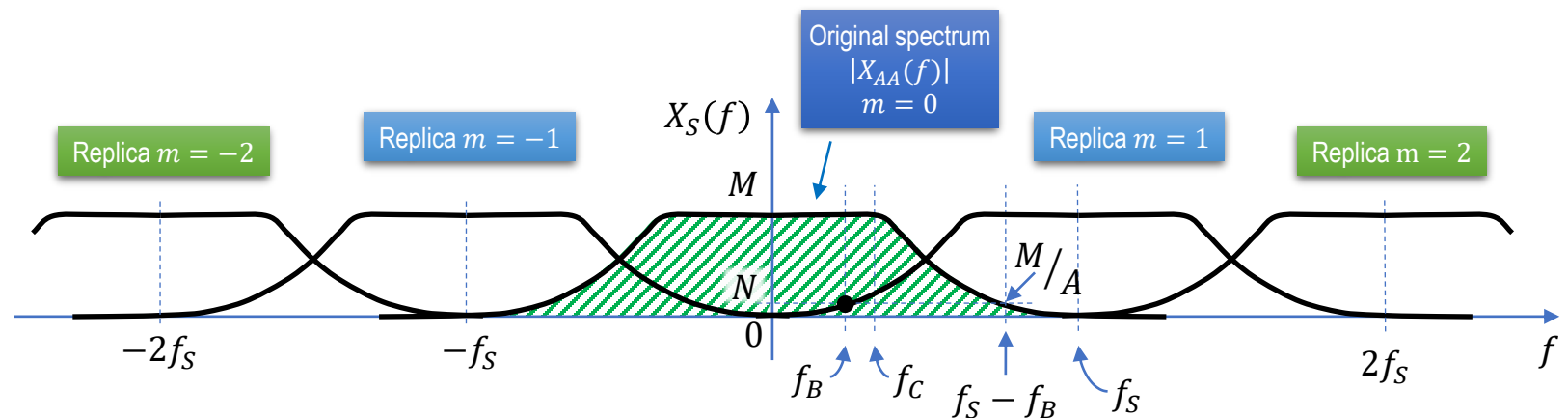
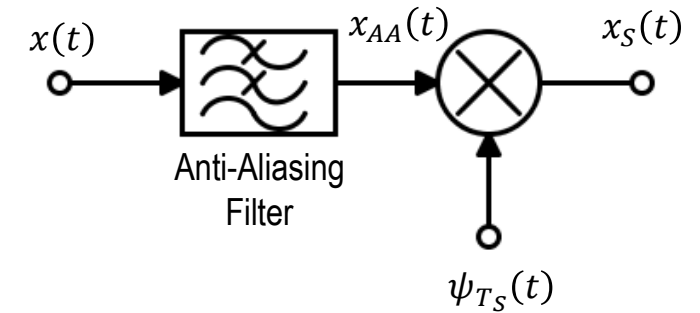
$$A = \left(\frac{f_S - f_B}{f_C} \right)^{n_{poles}}$$

Attenuation due to the anti-aliasing filter in linear scale at the frequency $f = f_S - f_B$ (the reciprocal of the approximated response)



Introduction

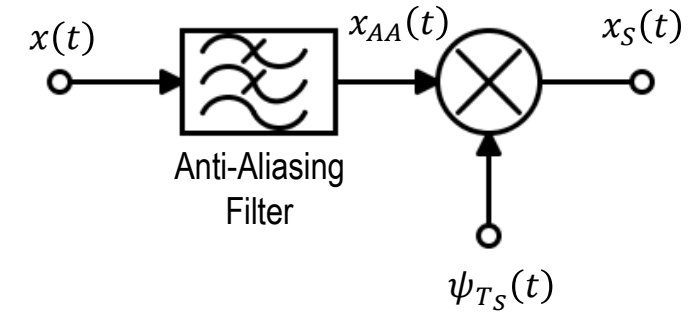
- Estimate the amount of aliasing noise
 - The spectrum $X_S(f)$ of the sampled signal shows the replicas
 - At integer multiples m of the sampling frequency f_S
 - To recover the original signal, a reconstruction low pass filter with cut-off frequency f_B is used
 - It isolates the base band
 - The tails of the spectra of the replicas contribute to the base band spectrum
 - They cause aliasing noise
 - The greater contribution is due to the first replica ($m = 1$) at frequency f_B
 - Let this noise contribution be of magnitude N
 - What is the value of N ?



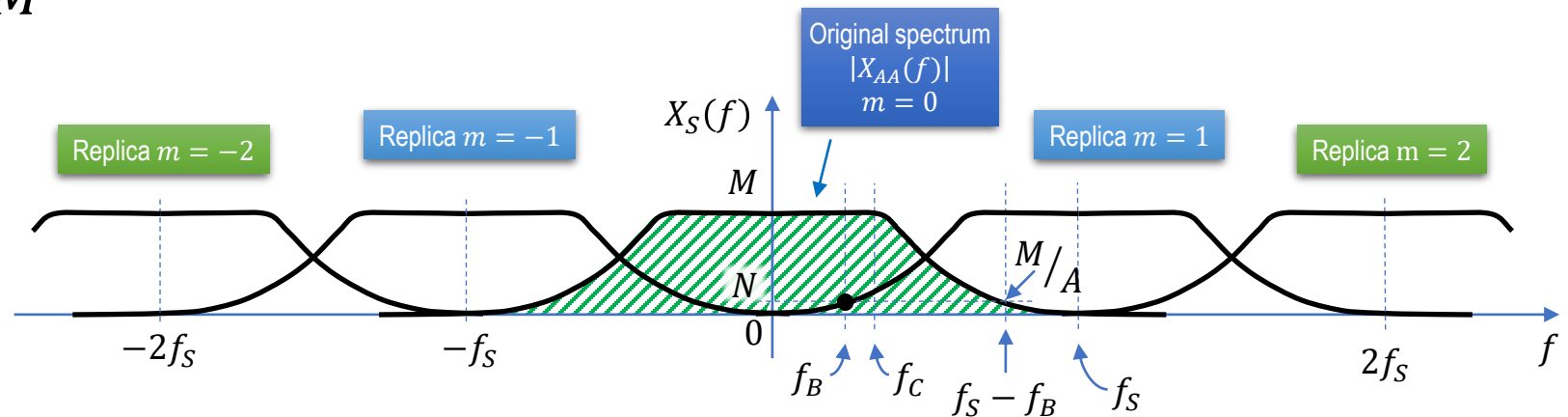
Introduction

- Estimate the amount of aliasing noise
 - The greater contribution is due to the first replica ($m = 1$) at frequency f_B
 - The magnitude N of the first replica at frequency f_B is equal to the magnitude of the spectrum $X_{AA}(f)$ of the filtered signal $x_{AA}(t)$ at frequency $f_S - f_B$

$$N = |X_{AA}(f_S - f_B)| = M \left(\frac{f_S - f_B}{f_c} \right)^{-n_{poles}} = \frac{M}{A}$$



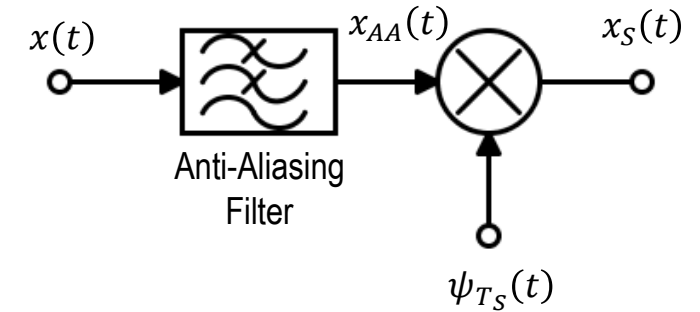
- Let's assume the magnitude of the spectrum of the filtered signal $x_{AA}(t)$ at frequency f_B is exactly M
 - It may be different, depending on the kind of anti-aliasing filter and its cut-off frequency f_c



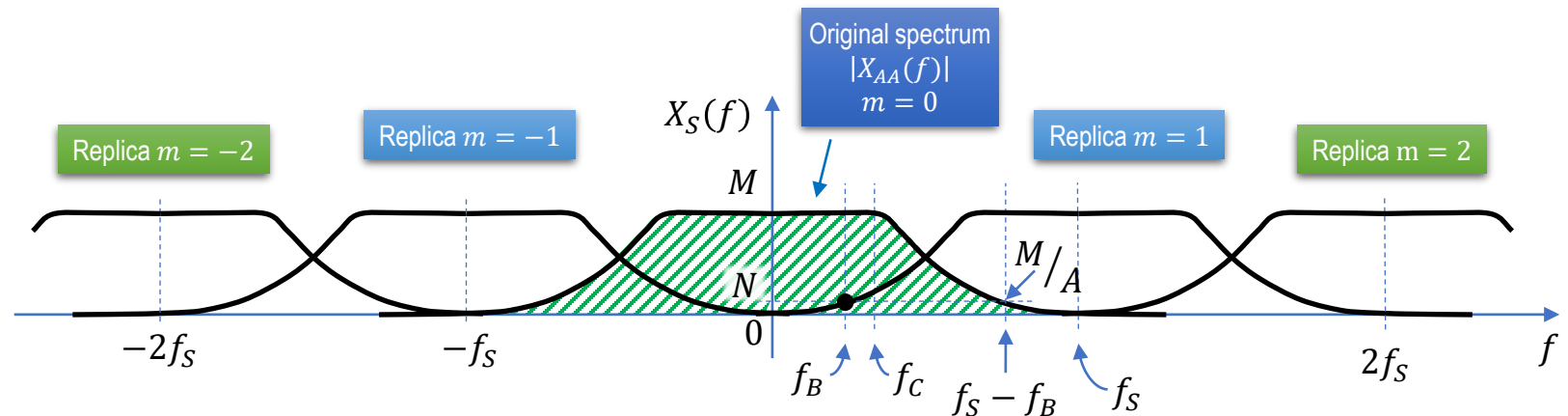
Introduction

- Estimate the amount of aliasing noise
 - Let's define the Signal-to-Noise Ratio SNR_a due to aliasing as the ratio between the magnitude M of the spectrum of the original signal and the magnitude N of the noise contribution at frequency f_B

$$SNR_a = \frac{M}{N} = \frac{M}{M \left(\frac{f_S - f_B}{f_C} \right)^{-n_{poles}}} = \left(\frac{f_S - f_B}{f_C} \right)^{n_{poles}}$$



- The value is in linear scale
- It is an approximation
 - We neglected several contributions
 - Some expressions were approximated

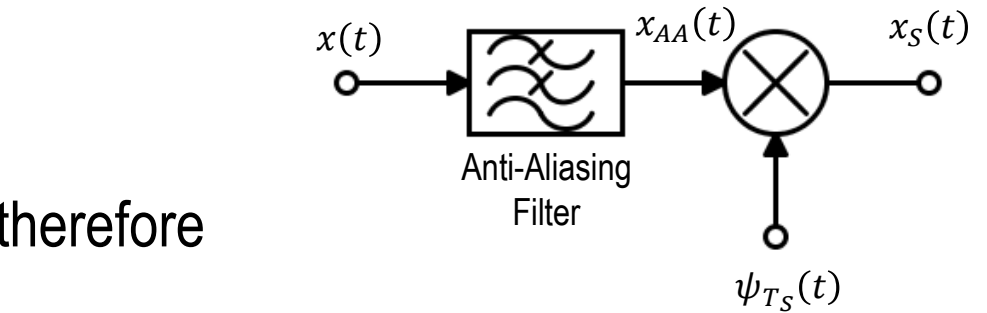


Introduction

- Estimate the amount of aliasing noise
 - The Signal-to-Noise Ratio SNR_a expressed in dB is therefore

$$SNR_a|_{dB} = 20 \log_{10} SNR_a = 20 \log_{10} \left(\frac{f_s - f_B}{f_c} \right)^{n_{poles}}$$

$$SNR_a|_{dB} = 20 \cdot n_{poles} \cdot \log_{10} \left(\frac{f_s - f_B}{f_c} \right)$$

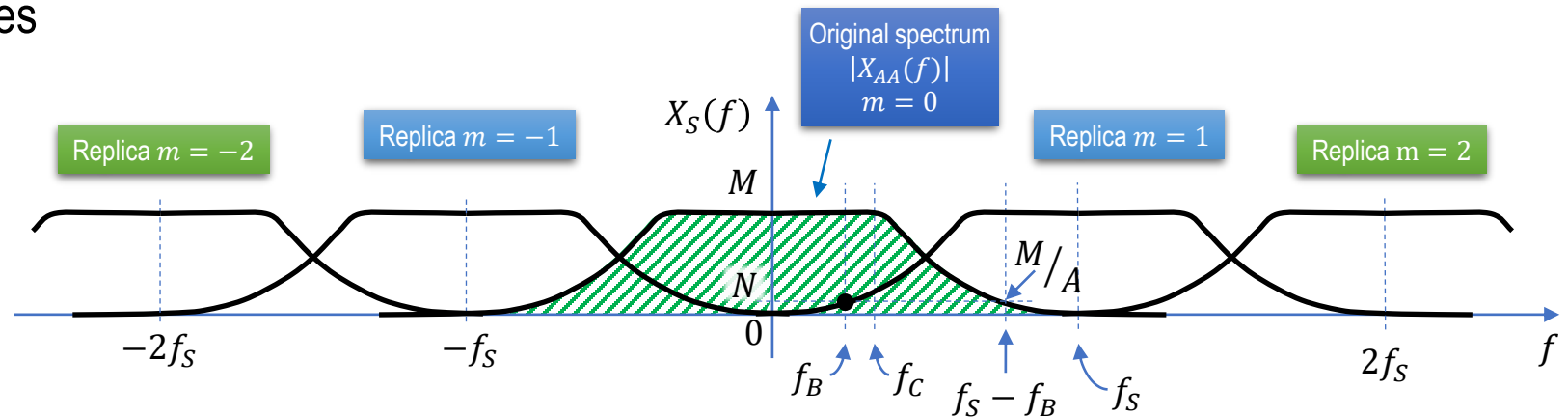


$$n_{dec} = \log_{10} \frac{f_s - f_B}{f_c}$$

Number of decades
between f_c and $f_s - f_B$

$$SNR_a|_{dB} = 20 \cdot n_{poles} \cdot n_{dec}$$

- To increase SNR_a
 - Increase the number of poles
 - Anti-aliasing filter more expensive
 - Increase the sampling frequency
 - Sampler more expensive



Introduction

- Estimate the amount of aliasing noise
 - If the original signal power spectrum is not flat, find the highest contribution to aliasing noise
 - Check how and where replicas overlap in the sampled signal
 - Estimate its power compared to the useful bandwidth power
 - This is the SNR_a without an anti-aliasing filter
 - If it is higher than desired
 - Insert an anti-aliasing filter with a proper cut-off frequency and number of poles
 - Estimate the new SNR_a with the anti-aliasing filter

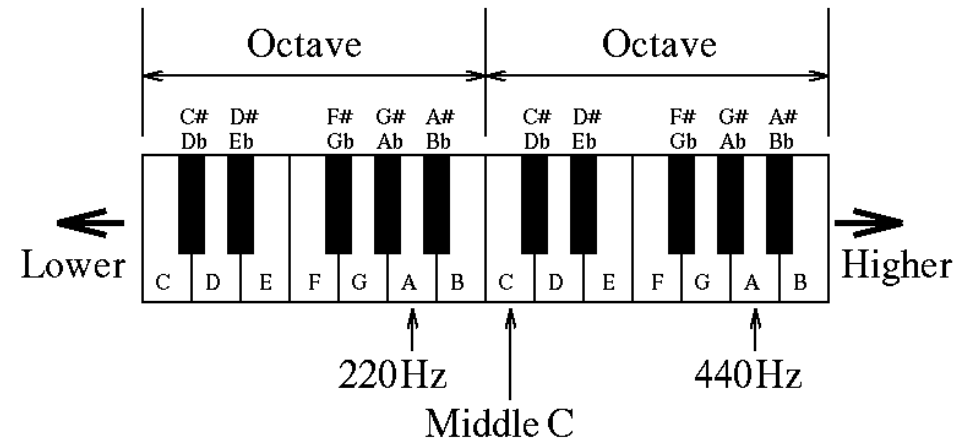
Introduction

- Sampling example 1

- Audio signal with useful bandwidth $f_B = 8$ kHz
 - Microphone has a flat response up to $f_{mic} = 20$ kHz, and then decreases by 12 dB/oct
 - Equivalent to 40 dB/dec

- Requirement

- $SNR_a > 60$ dB
- Nyquist frequency is $f_{Ny} = 16$ kHz
- Decide to sample at 3 times f_{Ny}
 - $f_S = 3f_{Ny} = 3 \cdot 16 = 48$ kHz
- How many poles should the anti-aliasing filter have?



Introduction

- Sampling example 1 (continued)
 - How many poles should the anti-aliasing filter have?
 - Assume the cut-off frequency is $f_C = 10$ kHz
 - Note: it is different from f_B
 - Arbitrary choice
 - Critical frequency is $f_S - f_B = 48 - 8 = 40$ kHz
 - Input signal naturally attenuated by the microphone response
 - $A_{mic}|_{db} = 12 \cdot \log_2 \frac{f_S - f_B}{f_{mic}} = 12 \cdot \log_2 \frac{48-8}{20} = 12$ dB
 - It is 12 dB/oct, and the number of octaves is the $\log_2$ of the ratio of the frequencies
 - The anti-aliasing filter should introduce an additional attenuation of 48 dB
 - $A_{AA} = 20n_p \log_{10} \frac{f_S - f_B}{f_C} > 48 \implies n_p > \frac{48}{20 \log_{10} \frac{48-8}{10}} = 3.98$
 - Now we are considering decades, so the logarithm is in base 10
 - Note also the different value at the denominator of the argument of the logarithm
 - We need 4 poles
 - If the microphone response is not considered, then 5 poles are needed

Introduction

- Sampling example 2

- $y(t) = \frac{S}{2} \sin(\omega_A t) + \frac{S}{5} \sin(3.5\omega_A t + 1.5)$

- Useful signal is only the first sinusoidal wave

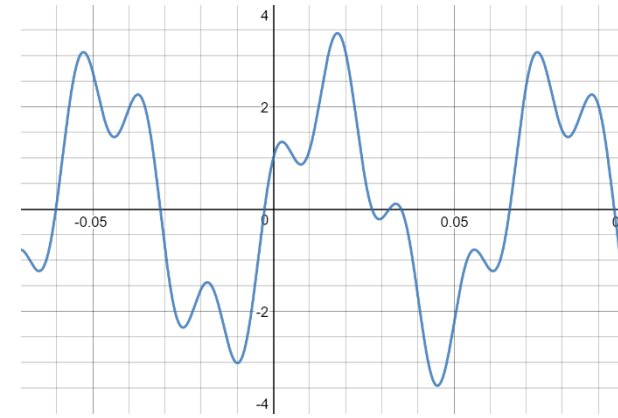
- Sample at $\omega_S = 3\omega_A$

- Anti-aliasing filter has $\omega_C = \omega_A$ and $n_p = 5$

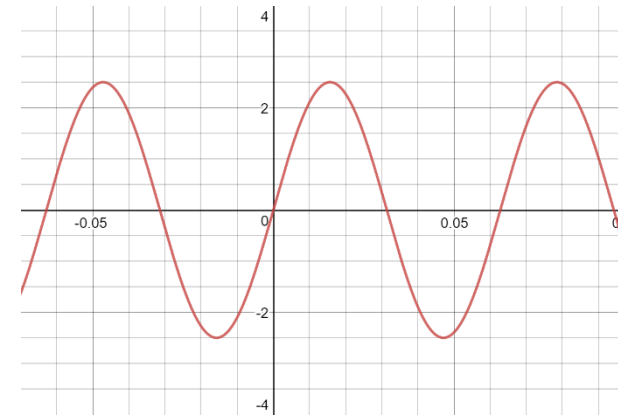
- What is the SNR_a ?

- Spectrum $X(\omega)$ of the input signal

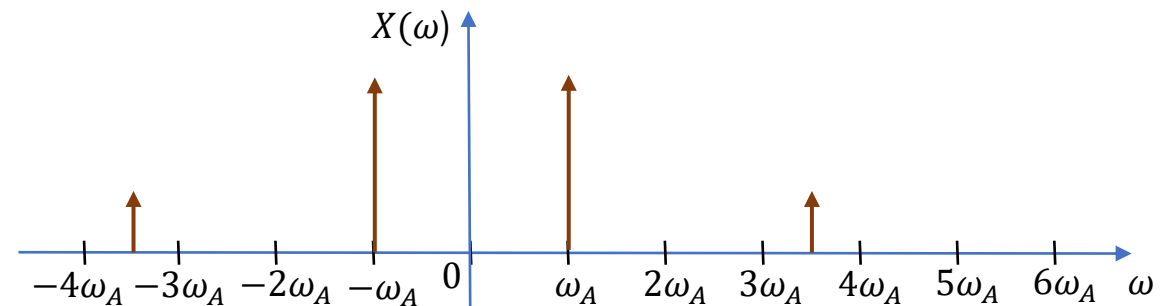
- One line at $\omega = \omega_A$
 - Useful signal
 - One line at $\omega = 3.5\omega_A$
 - It is 2/5 of the previous one
 - Not useful



Input signal

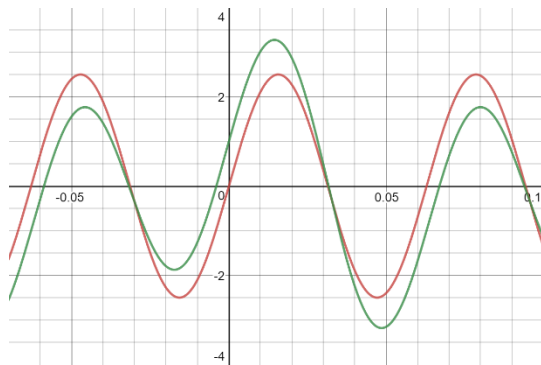
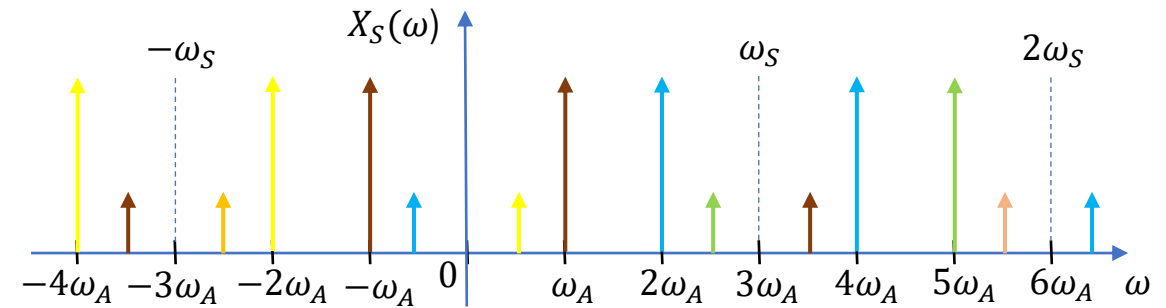


Wanted signal

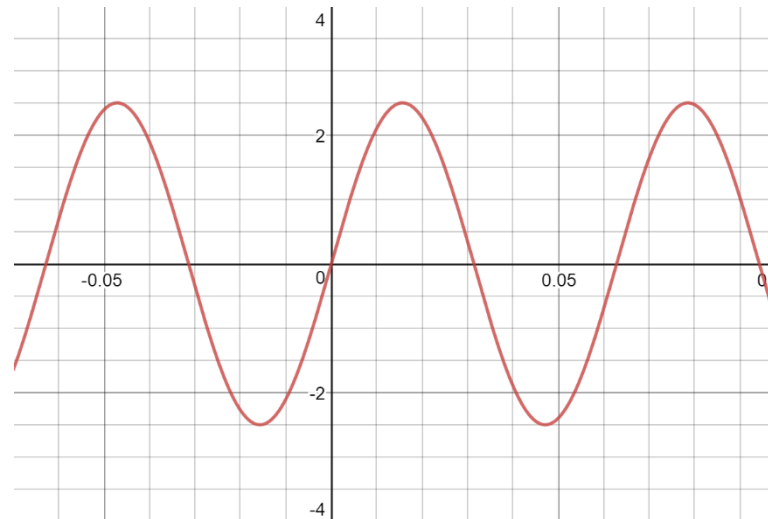


Introduction

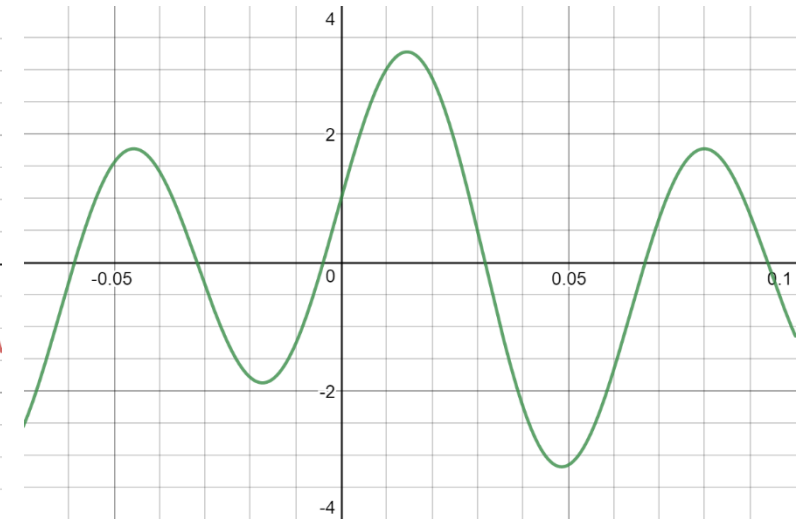
- Sampling example 2 (continued)
 - Without the anti-aliasing filter
 - Line at $\omega = 3.5\omega_A$ turned into baseband
 - At $\omega = \frac{1}{2}\omega_A$
 - When extracting baseband the obtained signal is not the wanted one
 - $\text{SNR}_a = 5/2 = 2.5$
 - $\text{SNR}_a|_{dB} = 8 \text{ dB}$



Comparison



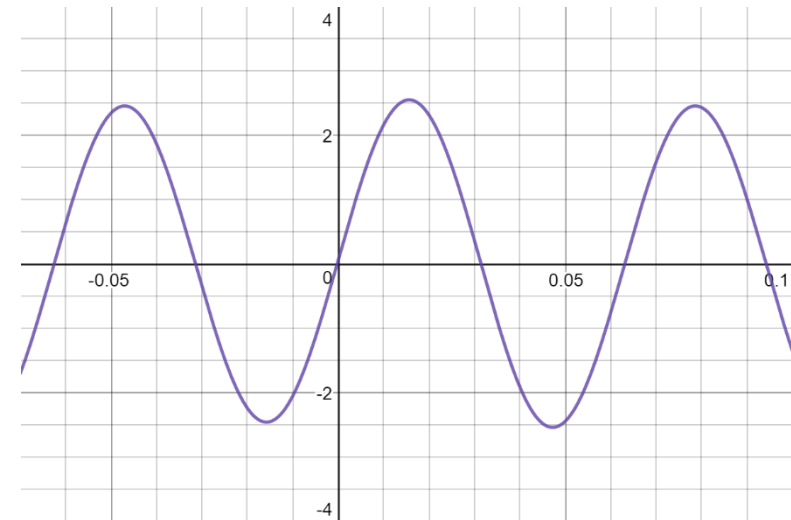
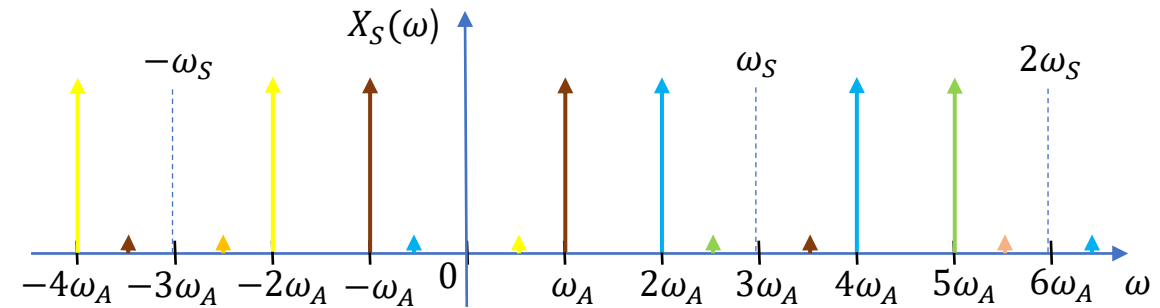
Wanted signal



Obtained signal

Introduction

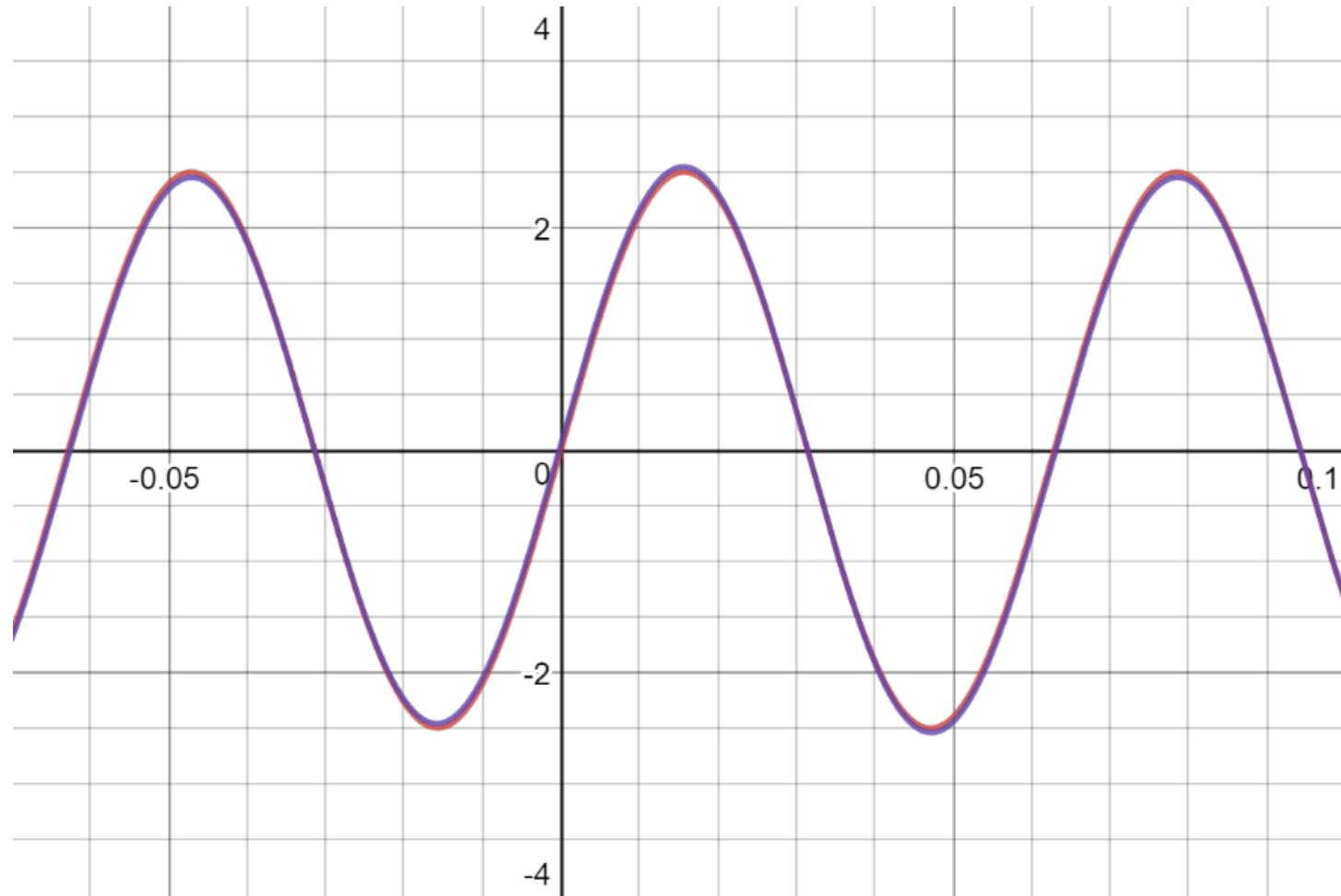
- Sampling example 2 (continued)
 - With the anti-aliasing filter
 - Line at $\omega = 3.5\omega_A$ attenuated
 - General attenuation (at $\omega_S - \omega_A$)
 - $A = 20n_p \log_{10} \frac{\omega_S - \omega_A}{\omega_A} = 30 \text{ dB}$
 - Attenuated by 32 times
 - $\text{SNR}_a = (5 \cdot 32)/2 = 80$
 - $\text{SNR}_a|_{dB} = 38 \text{ dB}$
 - Exact attenuation (at $3.5\omega_A$)
 - $A = 20n_p \log_{10} \frac{3.5\omega_A}{\omega_C} = 54,4 \text{ dB}$
 - Attenuated by 536 times
 - $\text{SNR}_a = (5 \cdot 536)/2 = 1340$
 - $\text{SNR}_a|_{dB} = 62,4 \text{ dB}$



Obtained signal

Introduction

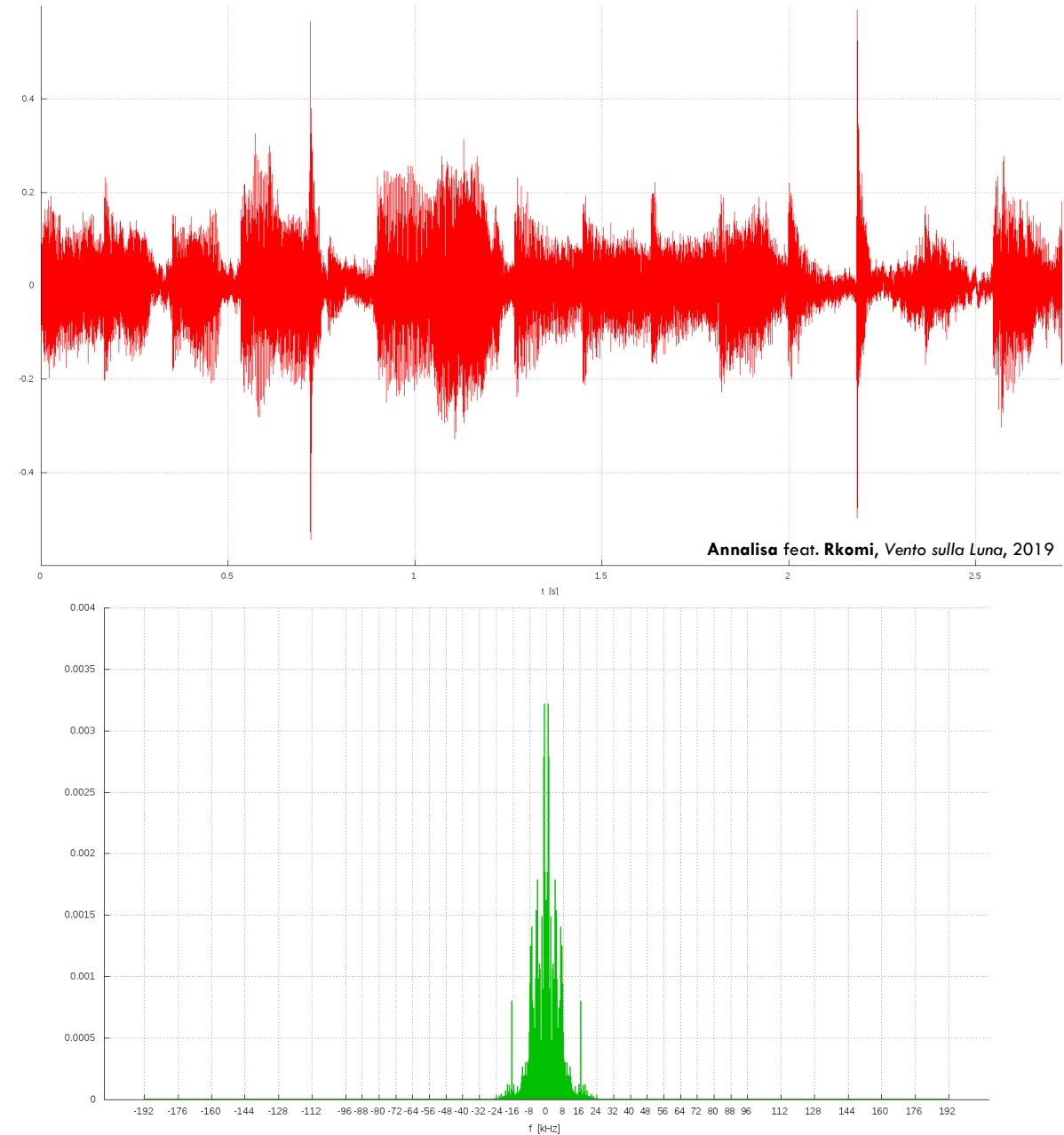
- Sampling example 2 (continued)
 - With the anti-aliasing filter
 - Line at $\omega = 3.5\omega_A$ attenuated
 - General attenuation (at $\omega_S - \omega_A$)
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 - Attenuated by 32 times
 - $\text{SNR}_a = (5 \cdot 32)/2 = 80$
 - $\text{SNR}_a|_{dB} = 38 \text{ dB}$
 - Exact attenuation (at $3.5\omega_A$)
 - $A = 20n_p \log_{10} \frac{3.5\omega_A}{\omega_C} = 54.4 \text{ dB}$
 - Attenuated by 536 times
 - $\text{SNR}_a = (5 \cdot 536)/2 = 1340$
 - $\text{SNR}_a|_{dB} = 62.4 \text{ dB}$



Comparison with $\text{SNR}_a = 38 \text{ dB}$

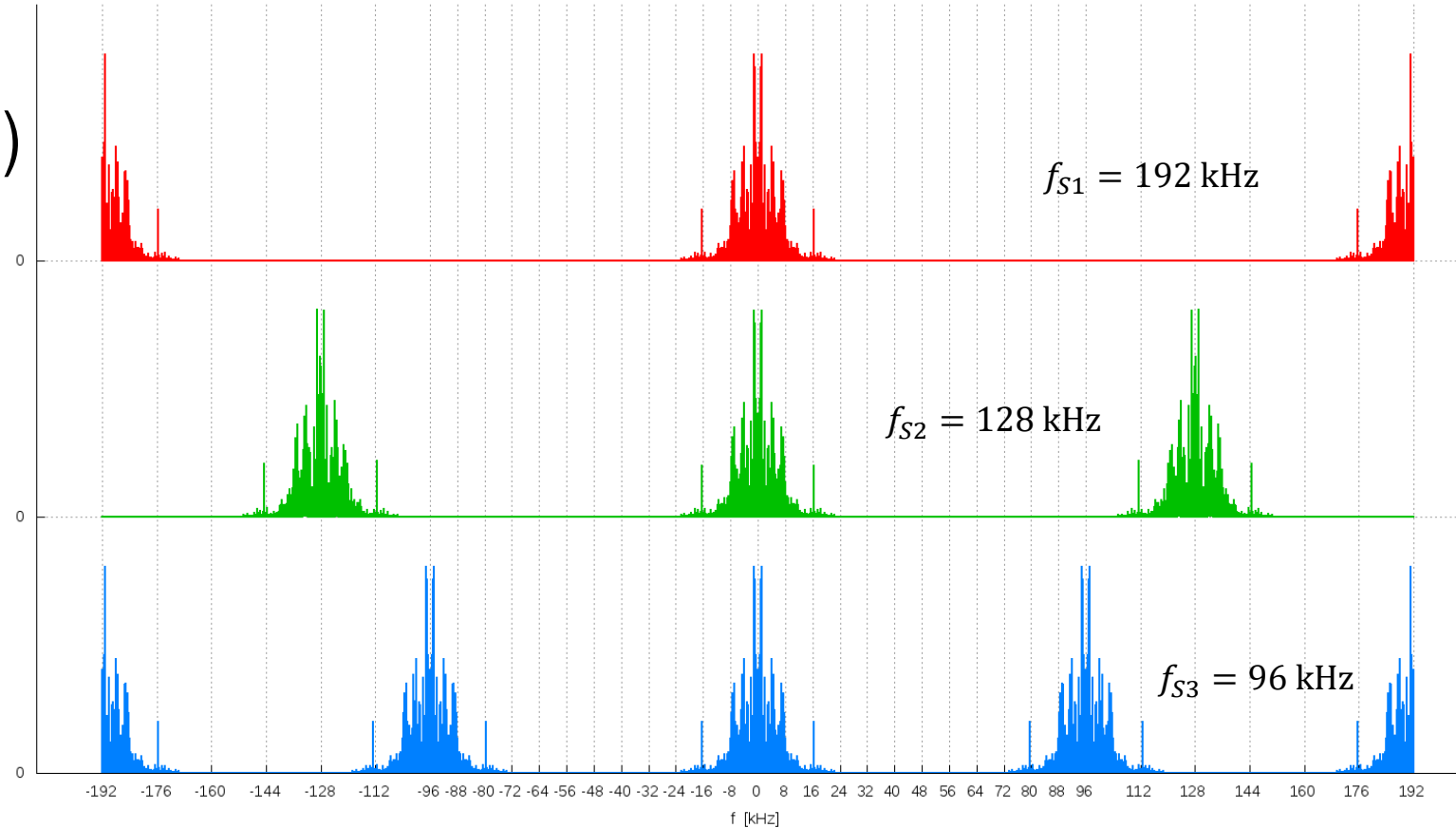
Introduction

- Sampling example 3
 - Audio signal (music)
 - Several seconds of signal
 - No anti-aliasing filter
 - Spectrum
 - Main contributions in the band [0 – 10 kHz]
 - But not negligible up to 20 kHz
 - Noise at around 16 kHz introduced by the acquisition system (bad microphone)
 - Sampled at different frequencies
 - Nyquist frequency around $f_{Ny} = 40$ kHz



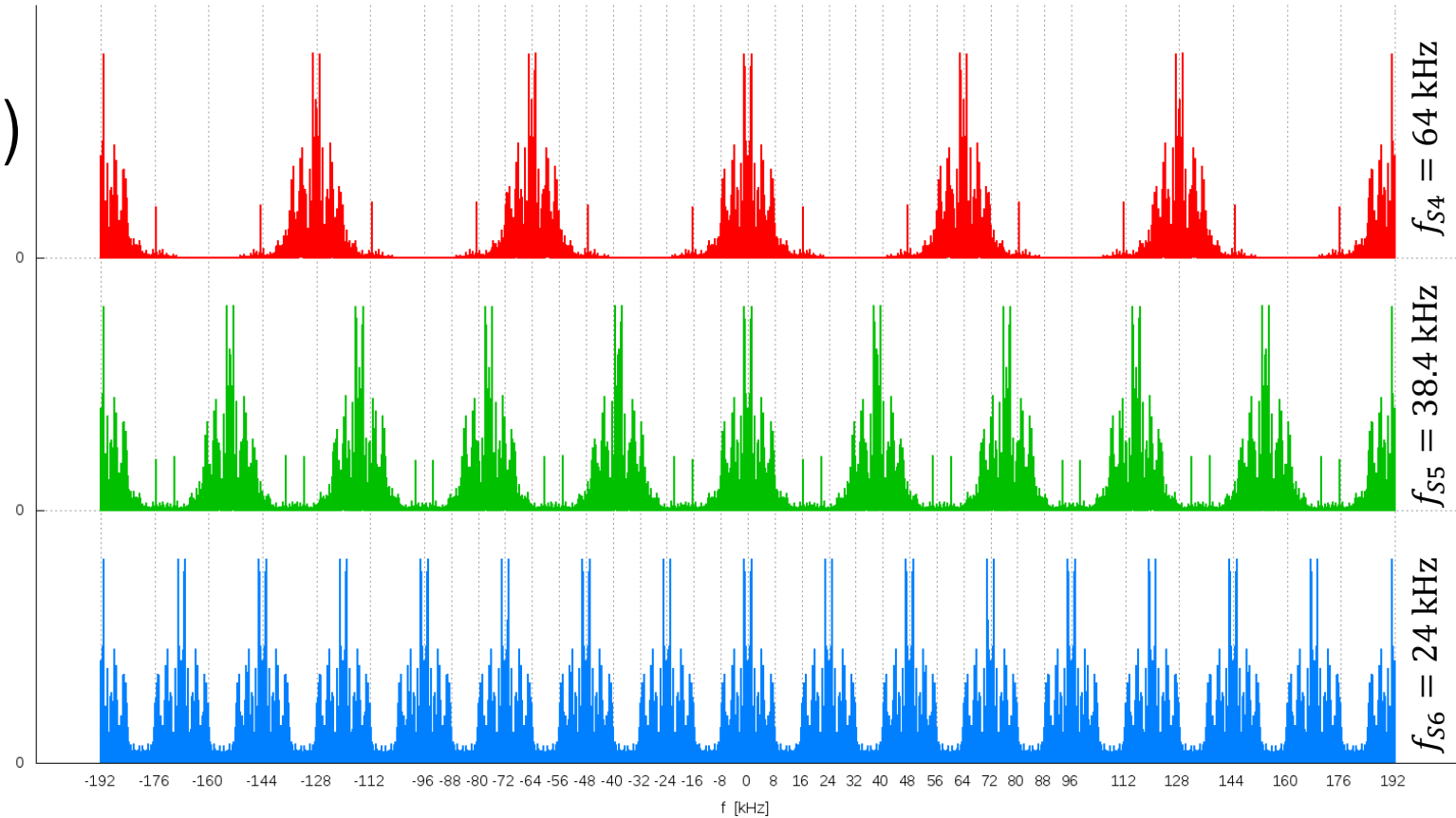
Introduction

- Sampling example 3 (continued)
 - High sampling frequencies
 - $f_{S1} = 192$ kHz
 - $f_{S2} = 128$ kHz
 - $f_{S3} = 96$ kHz
 - No aliasing
 - Replicas are sufficiently far apart
 - The original spectrum is almost negligible outside the useful bandwidth
 - An antialiasing filter is not even needed, or it is very simple and cheap



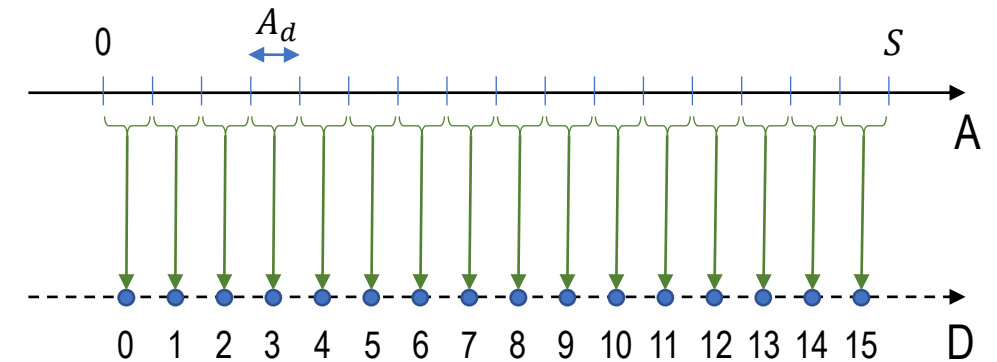
Introduction

- Sampling example 3 (continued)
 - Low sampling frequencies
 - $f_{S4} = 64$ kHz
 - $f_{S5} = 38.4$ kHz
 - $f_{S6} = 24$ kHz
 - Aliasing
 - Already slightly present at f_{S4}
 - Can be easily noticed at f_{S5}
 - Sampling is around the Nyquist frequency
 - It strongly affects the spectrum at f_{S6}



Introduction

- Quantization
 - Divide the analog dynamic range into N identical ordered intervals
 - N is the number of allowed discrete levels
 - $k = \log_2 N$ is the minimum necessary number of binary signals to represent the discrete signal
 - All analog values within an interval correspond to the same discrete level
 - Uniform quantization
 - Non-uniform quantization is also possible, and often desirable
 - Recovering the original analog value from the corresponding discrete value is impossible
 - Can only identify the original interval
 - Loss of information, introduction of noise

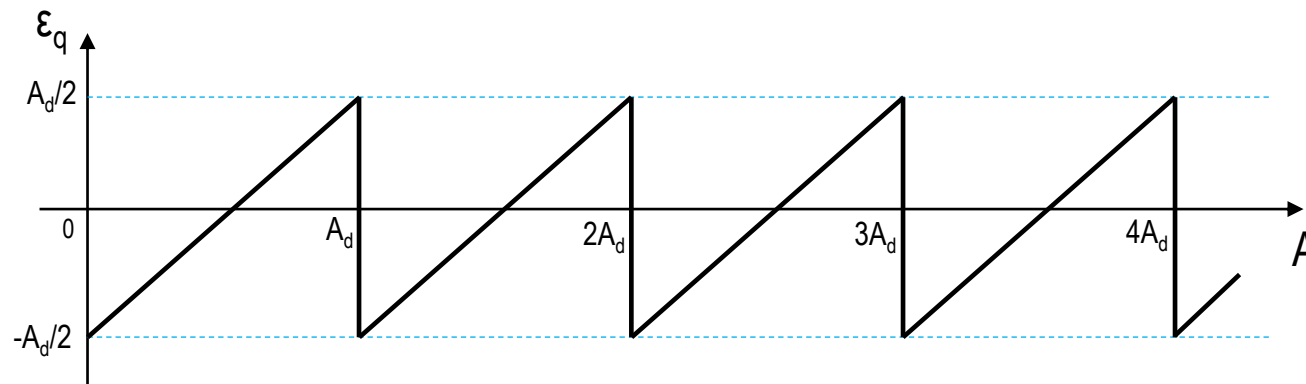


Introduction

- Quantization

- Let S be the analog dynamic range

- Each interval is $A_d = \frac{S}{N}$
- When recovering analog values from digital, use the center (middle value) of the corresponding interval
- The maximum error introduced by quantization is $|\varepsilon_{qmax}| = \frac{A_d}{2}$
- The error ε_q varies linearly within an interval $\left[-\frac{A_d}{2}, \frac{A_d}{2}\right]$



Introduction

- Quantization

- Example

- Random signal between 0 and 10 with normal distribution

- $S = 10$

- Quantized at 8 bits

- $k = 8, N = 2^8 = 256$

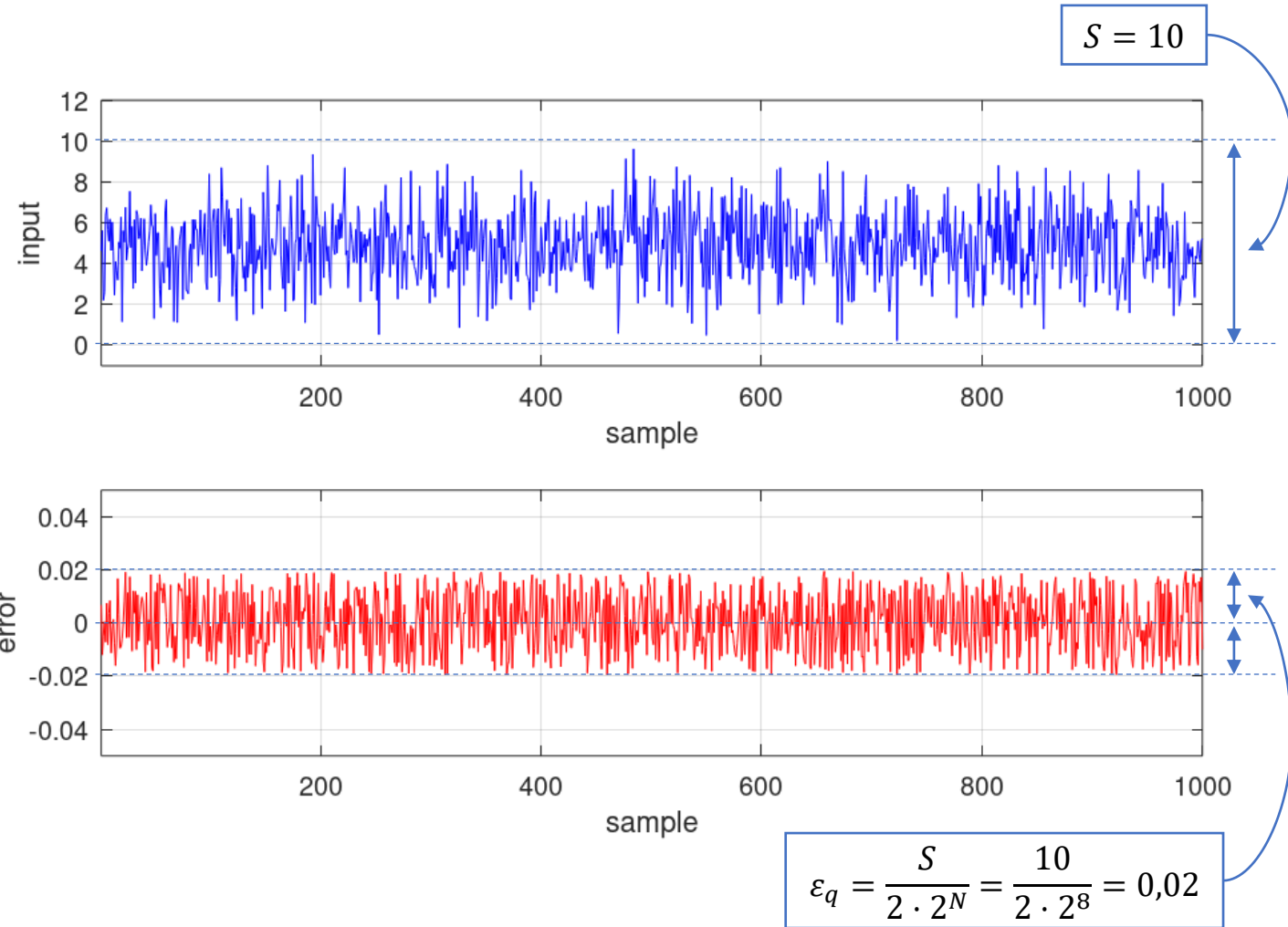
- $A_d = \frac{S}{N} = \frac{10}{256} = 0,04$

- Quantization error

- $|\varepsilon_{qmax}| = \frac{A_d}{2} = \frac{0,04}{2} = 0,02$

Matlab/Octave code

```
D = 1.67*randn(1000,1)+5;  
D(D<0 | D>=10) = [];  
B8 = floor(D * 25.6);  
E8 = (B8+0.5)/25.6 - D;  
subplot (2,1,1)  
plot (D, "b")  
grid on  
axis ([-inf, inf, -1, 12])  
ylabel ("input")  
xlabel ("sample")  
subplot (2,1,2)  
plot (E8, "r")  
grid on  
axis ([-inf, inf, -0.05, 0.05])  
ylabel ("error")  
xlabel ("sample")
```



Introduction

- Quantization

- Example

- Sinusoidal signal between 0 and 10

- $S = 10$

- Quantized at 4 bits

- $k = 4, N = 2^4 = 16$

- $A_d = \frac{S}{N} = \frac{10}{16} = 0,625$

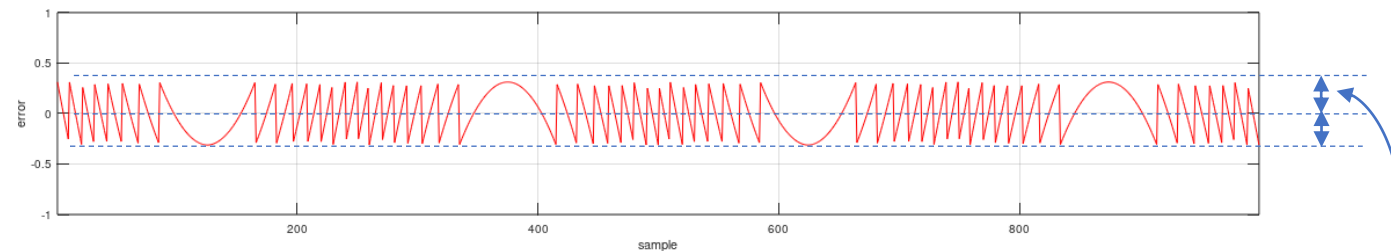
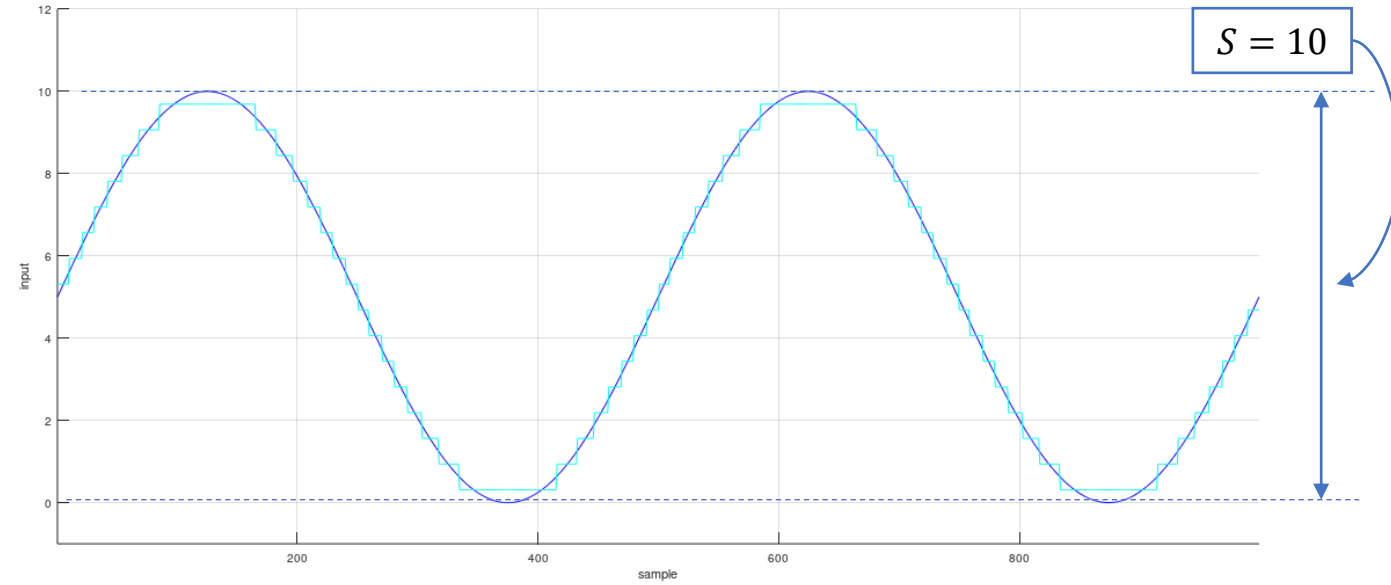
- Quantization error

- $|\varepsilon_{qmax}| = \frac{A_d}{2} = \frac{0,625}{2} = 0,3125$

Matlab/Octave code

```
D = 5*(sin(2*pi*[0:0.002:2]))+1;  
D(D<0 | D>=10) = [];  
B4 = floor(D * 1.6);  
E4 = (B4+0.5)/1.6 - D;  
subplot (3,1,1:2)  
hold on  
plot (D, "b")  
plot ((B4+0.5)/1.6, "c")  
grid on
```

```
axis ([-inf, inf, -1, 12])  
ylabel ("input")  
xlabel ("sample")  
subplot (3,1,3)  
plot (E4, "r")  
grid on  
axis ([-inf, inf, -1, 1])  
ylabel ("error")  
xlabel ("sample")
```



$$\varepsilon_q = \frac{S}{2 \cdot 2^N} = \frac{10}{2 \cdot 2^4} = 0,3125$$

Introduction

- Quantization
 - Estimate the amount of quantization noise
 - Use the noise power and the signal power
 - Power proportional to the variance of the analog signal
 - Variance is the square of the standard deviation σ
 - The variance depends on the amplitude distribution ρ
 - To compute the variance, determine the amplitude distribution ρ
 - Of the analog signal
 - Of the noise
 - Signal-to-Noise Ratio
 - $\text{SNR}_q = \frac{P_S}{P_{\varepsilon q}} = \frac{\sigma_S^2}{\sigma_{\varepsilon q}^2}$
 - $\sigma_y^2 = \int_{-\infty}^{+\infty} y^2 \rho(y) dy$

Introduction

- Quantization
 - Estimate the amount of quantization noise
 - Signal amplitude distribution ρ
 - It is the relative probability of a signal to assume a certain value
 - Also known as Probability Density Function, PDF
 - Note that for a continuous function the absolute probability to assume a certain value is always zero
 - Because there are infinite possible outcomes
 - To compute it, use the Cumulative Density Function (CDF)
 - Probability of a signal to assume a value less or equal to a certain one
 - $\rho(y) = \text{PDF}_y = \frac{d\text{CDF}_y}{dy}$
 - The CDF can be obtained by inverting a function and imposing
 - $\lim_{y \rightarrow -\infty} \text{CDF}(y) = 0$
 - $\lim_{y \rightarrow +\infty} \text{CDF}(y) = 1$
 - CDF should always be positive and between 0 and 1

Introduction

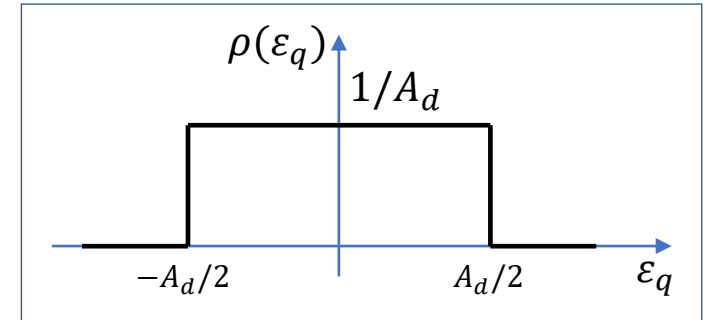
- Quantization

- Distribution of quantization noise:

- A_d is small, same probability to get any value within A_d , for any input waveform
 - We don't need the CDF in this case

- $\rho(\varepsilon_q)$ is constant

- $\rho(\varepsilon_q) = \frac{1}{A_d}$ in $\left[-\frac{A_d}{2}, \frac{A_d}{2}\right]$, zero otherwise



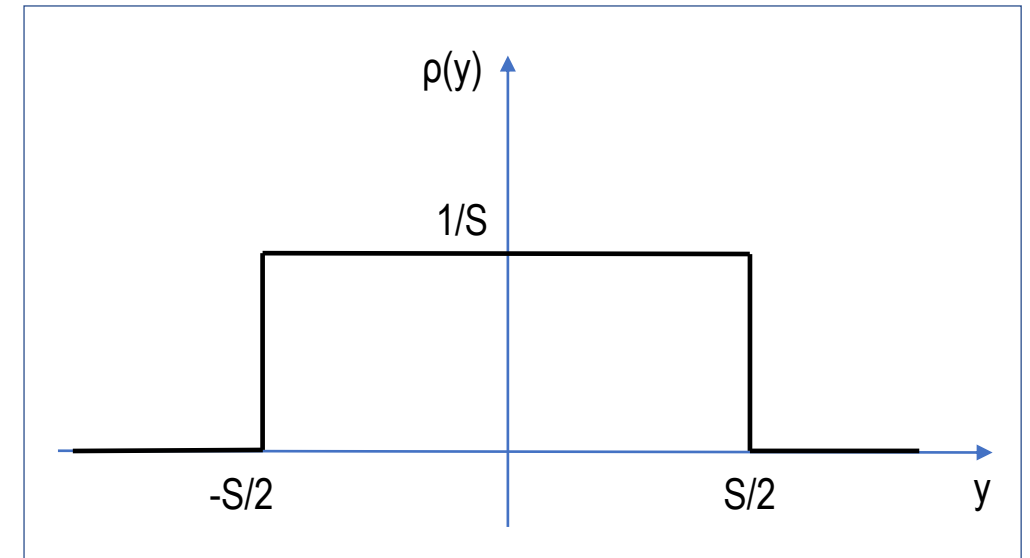
- $$\sigma_{\varepsilon_q}^2 = \int_{-\infty}^{+\infty} \varepsilon_q^2 \rho(\varepsilon_q) d\varepsilon_q = \int_{-\frac{A_d}{2}}^{\frac{A_d}{2}} \varepsilon_q^2 \frac{1}{A_d} d\varepsilon_q = \frac{1}{A_d} \left[\frac{1}{3} \varepsilon_q^3 \right]_{-\frac{A_d}{2}}^{\frac{A_d}{2}} = \frac{1}{3A_d} \left[\frac{A_d^3}{8} - \left(-\frac{A_d^3}{8} \right) \right] = \frac{A_d^2}{12}$$

- With $n = \log_2 N$, $N = 2^n$, $A_d = S/N$

- $$\sigma_{\varepsilon_q}^2 = \frac{1}{12} \frac{S^2}{2^{2n}}$$

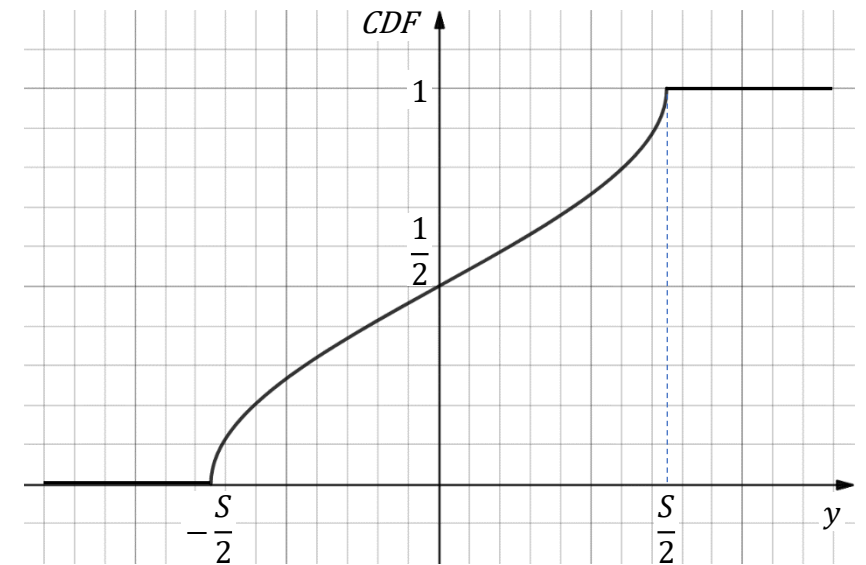
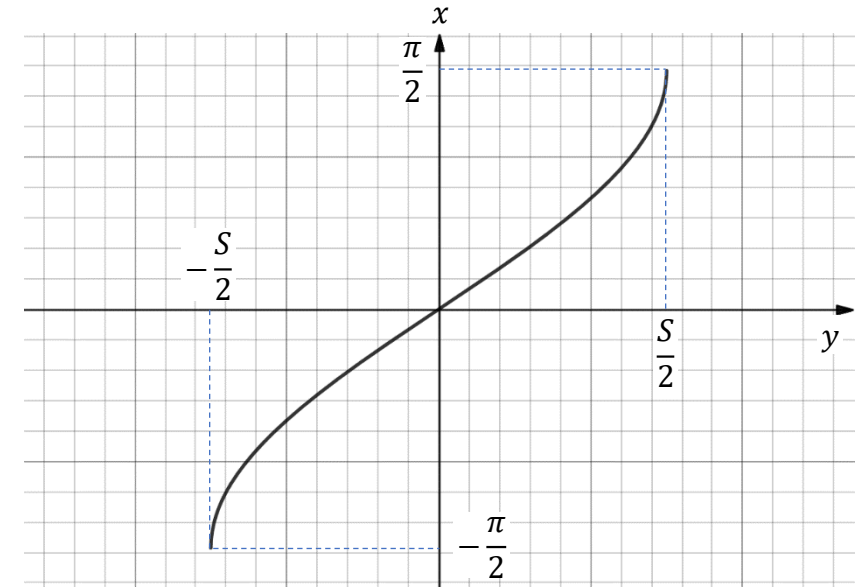
Introduction

- Quantization
 - Triangular signal with dynamic range S
 - Uniform distribution in the dynamic range
 - Regardless of the duty cycle of the triangular signal
 - $\rho(y) = \frac{1}{S}$ (zero outside the dynamic range)
 - Given the previous discussion for the noise, it follows:
 - $\sigma_y^2 = \frac{S^2}{12}$
 - Therefore:
 - $\text{SNR}_q = \frac{\sigma_y^2}{\sigma_{\varepsilon q}^2} = \frac{S^2}{12} \cdot \frac{12 \cdot 2^{2n}}{S^2} = 2^{2n}$
 - $\text{SNR}_q|_{dB} = 10 \log_{10} 2^{2n} = 6n$



Introduction

- Quantization
 - Sinusoidal signal with dynamic range S
 - Let's consider $y = \frac{S}{2} \sin x$
 - To find the Probability Density Function, first compute the Cumulative Density Function
 - Inverse is $\sin^{-1}\left(\frac{2}{S}y\right)$
 - But it is not a CDF because it varies between $-\frac{\pi}{2}$ and $+\frac{\pi}{2}$, while it should be between 0 and 1
 - First divide it by π
 - It is now between $-\frac{1}{2}$ and $\frac{1}{2}$, with a total range of 1
 - Then add $\frac{1}{2}$, so that it is between 0 and 1
 - $$\text{CDF} = \frac{\sin^{-1}\left(\frac{2}{S}y\right)}{\pi} + \frac{1}{2}$$
 - 0 for $y < -S/2$
 - 1 for $y > S/2$



Introduction

- Quantization

- To compute the PDF it is sufficient to derivate the CDF

- $\text{PDF} = \frac{d\text{CDF}}{dy} = \frac{1}{\pi S} \frac{2}{\sqrt{1 - \frac{4}{S^2} y^2}}$

- Zero outside $-S/2$ and $+S/2$

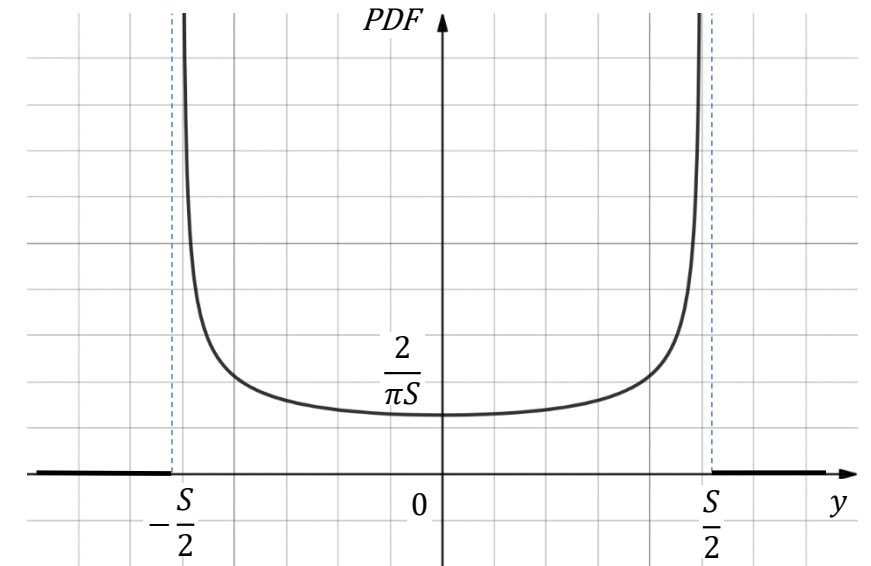
- Therefore

- $\sigma_S^2 = \int_{-\frac{S}{2}}^{\frac{S}{2}} \frac{1}{\pi S} \frac{2 y^2}{\sqrt{1 - \frac{4}{S^2} y^2}} dy = \frac{S^2}{8}$

- Signal to Noise Ratio

- $\text{SNR}_q = \frac{S^2}{8} \cdot \frac{12 \cdot 2^{2n}}{S^2} = \frac{3}{2} 2^{2n}$

- $\text{SNR}_q|_{dB} = 10 \cdot \log_{10} \left(\frac{3}{2} 2^{2n} \right) = 6n + 1.76 \text{ dB}$



Introduction

- Quantization
 - Derivation of integral

$$\sigma_S^2 = \frac{1}{\pi} \int_{-\frac{S}{2}}^{\frac{S}{2}} \frac{2}{S} \frac{y^2}{\sqrt{1 - \frac{4}{S^2} y^2}} dy = \frac{1}{\pi} \frac{S}{2} \int_{-\frac{S}{2}}^{\frac{S}{2}} \frac{\frac{4}{S^2} y^2}{\sqrt{1 - \frac{4}{S^2} y^2}} dy$$

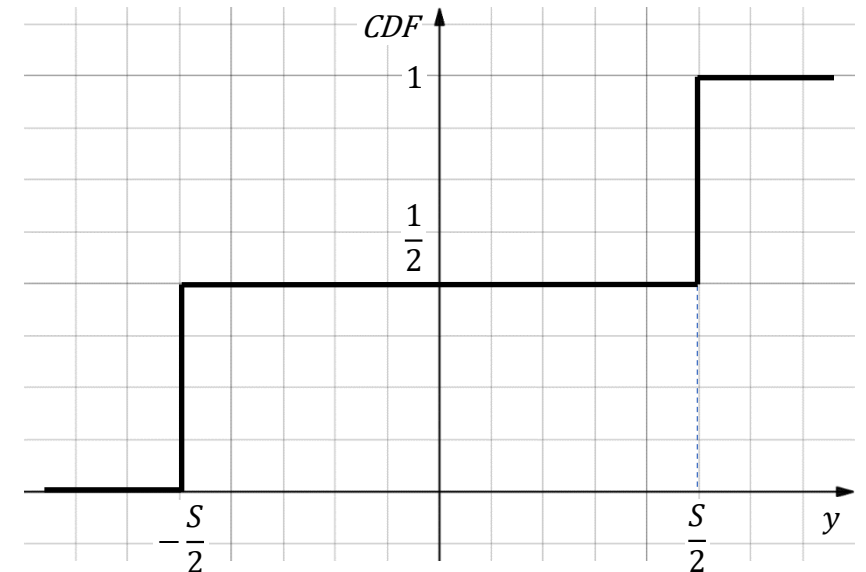
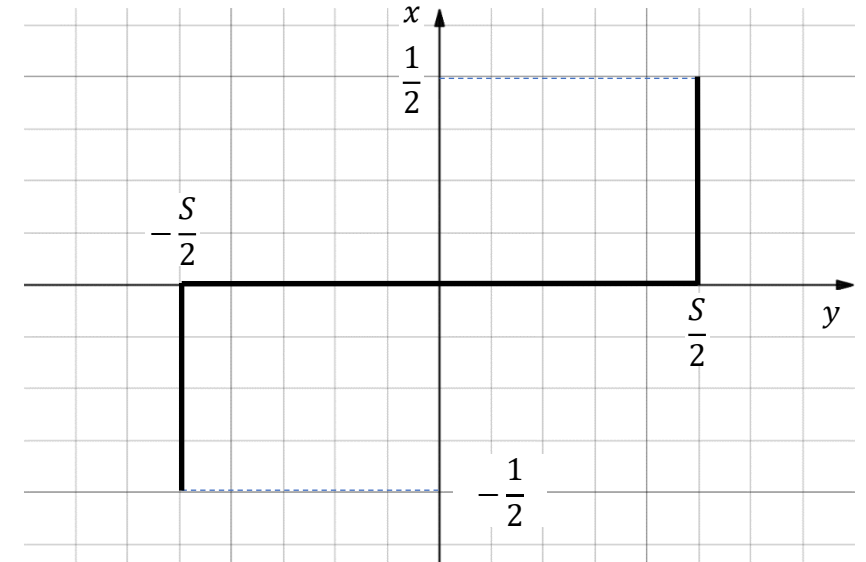
$$\text{Let } t = \frac{2}{S} y, \quad dt = \frac{2}{S} dy, \quad dy = \frac{S}{2} dt$$

$$\sigma_S^2 = \frac{1}{\pi} \frac{S}{2} \cdot \frac{S}{2} \int_{-1}^1 \frac{t^2}{\sqrt{1 - t^2}} dt = \frac{S^2}{4\pi} \left[-\frac{t\sqrt{1 - t^2}}{2} + \frac{1}{2} \sin^{-1} t \right]_{-1}^1 = \frac{S^2}{4\pi} \left[\frac{1}{2} \sin^{-1}(1) - \frac{1}{2} \sin^{-1}(-1) \right] =$$

$$= \frac{S^2}{4\pi} \left[\frac{1}{2} \frac{\pi}{2} - \frac{1}{2} \left(-\frac{\pi}{2} \right) \right] = \frac{S^2}{4\pi} \left[\frac{\pi}{4} + \frac{\pi}{4} \right] = \frac{S^2}{4\pi} \cdot \frac{\pi}{2} = \frac{S^2}{8}$$

Introduction

- Quantization
 - Square signal with dynamic range S
 - To find the Probability Density Function, first compute the Cumulative Density Function
 - Inverse is not really a function
 - And it is not a CDF because it varies between $-\frac{1}{2}$ and $+\frac{1}{2}$, while it should be between 0 and 1
 - Add $\frac{1}{2}$, so that it is between 0 and 1
 - 0 for $y < -S/2$
 - 1 for $y > S/2$



Introduction

- Quantization

- Take the derivative of the CDF to compute the PDF

- $\text{PDF} = \frac{1}{2} \delta\left(y + \frac{S}{2}\right) + \frac{1}{2} \delta\left(y - \frac{S}{2}\right)$

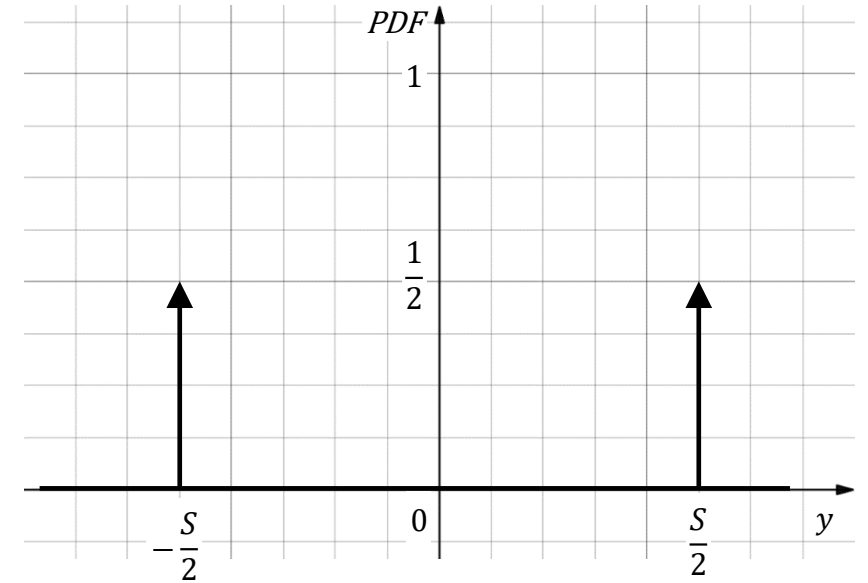
- Therefore

- $$\begin{aligned} \sigma_S^2 &= \int_{-\infty}^{+\infty} y^2 \left[\frac{1}{2} \delta\left(y + \frac{S}{2}\right) + \frac{1}{2} \delta\left(y - \frac{S}{2}\right) \right] dy = \\ &= \frac{1}{2} \frac{S^2}{4} + \frac{1}{2} \frac{S^2}{4} = \frac{S^2}{4} \end{aligned}$$

- Signal to Noise Ratio

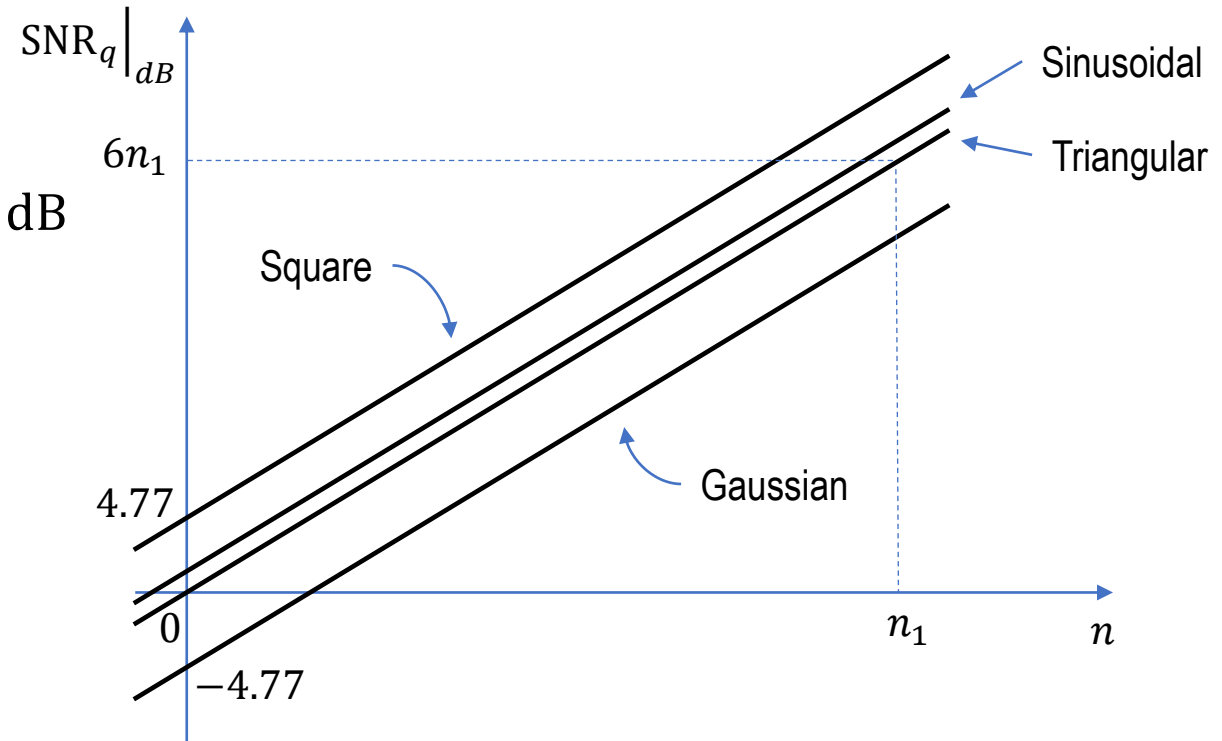
- $\text{SNR}_q = \frac{S^2}{4} \cdot \frac{12 \cdot 2^{2n}}{S^2} = 3 \cdot 2^{2n}$

- $\text{SNR}_q|_{dB} = 10 \cdot \log_{10}(3 \cdot 2^{2n}) = 6n + 4.77 \text{ dB}$



Introduction

- Quantization
 - Signal has a Gaussian distribution (i.e., voice)
 - $\sigma_S^2 = \frac{S^2}{36}$
 - Signal to Noise Ratio
 - $\text{SNR}_q = \frac{S^2}{36} \cdot \frac{12 \cdot 2^{2n}}{S^2} = \frac{1}{3} \cdot 2^{2n}$
 - $\text{SNR}_q|_{dB} = 10 \cdot \log_{10} \left(\frac{1}{3} \cdot 2^{2n} \right) = 6n - 4.77 \text{ dB}$
 - How to increase SNR_q
 - Increase the number of bits
 - More expensive, slower



Introduction

- Quantization
 - How to increase SNR_q
 - For each additional bit of quantization SNR_q increases of 6 dB

Octave example

Use the rand() function to create an input vector with dynamic range $S = 10$ to be quantized and to compute the variance

```
D = 10 * rand(10000, 1);  
D(D<0 | D>=10) = [];  
figure 1;  
hist(D, 100);  
varD = (std(D))^2  
...
```

varD = 8.3785

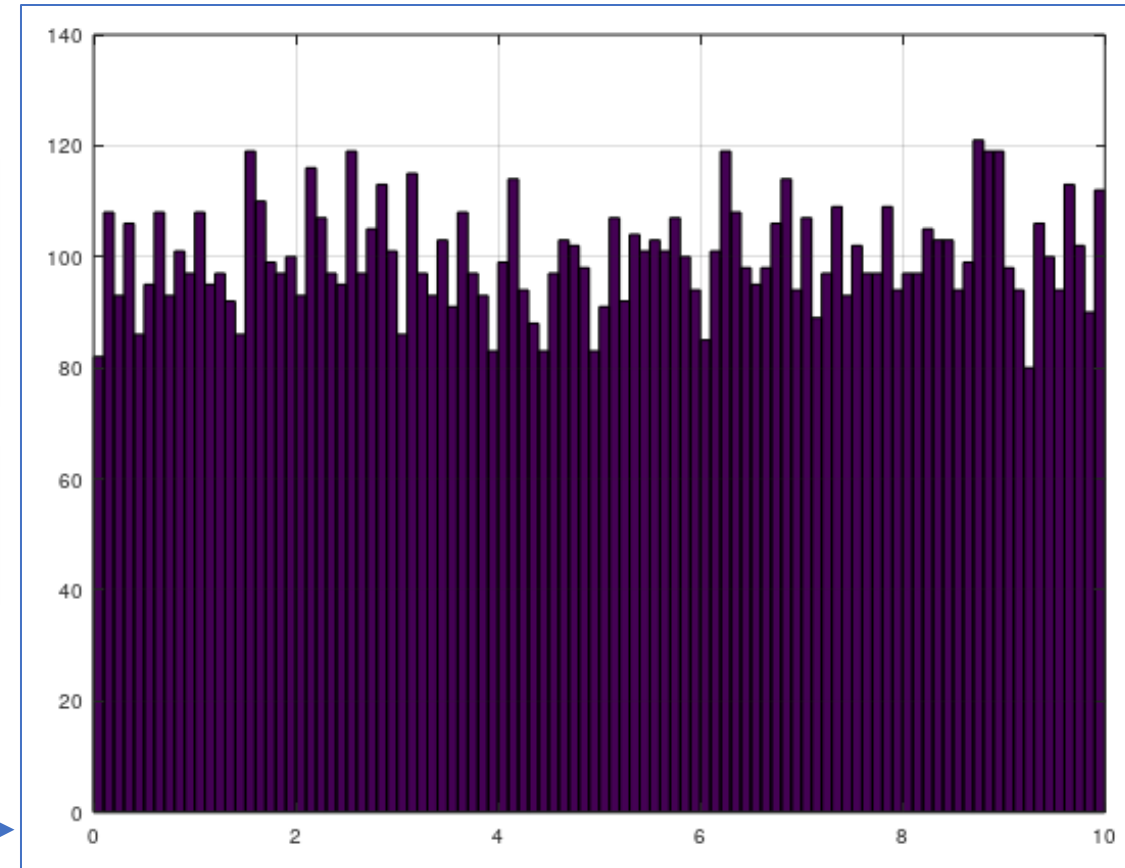
Create a column vector D of 10000 random elements with uniform distribution between 0 and 10

- Matlab uses a slightly different random number generator function

Eliminate possible elements outside the wanted dynamic range (not in this example)

Compute the D vector variance as the square of the standard deviation

Distribution of values in D



Introduction

- Quantization
 - How to increase SNR_q
 - For each additional bit of quantization SNR_q increases of 6 dB

Octave example

Convert to discrete values (quantization) on 5 or 6 bits (32 or 64 levels) using the floor function after scaling data

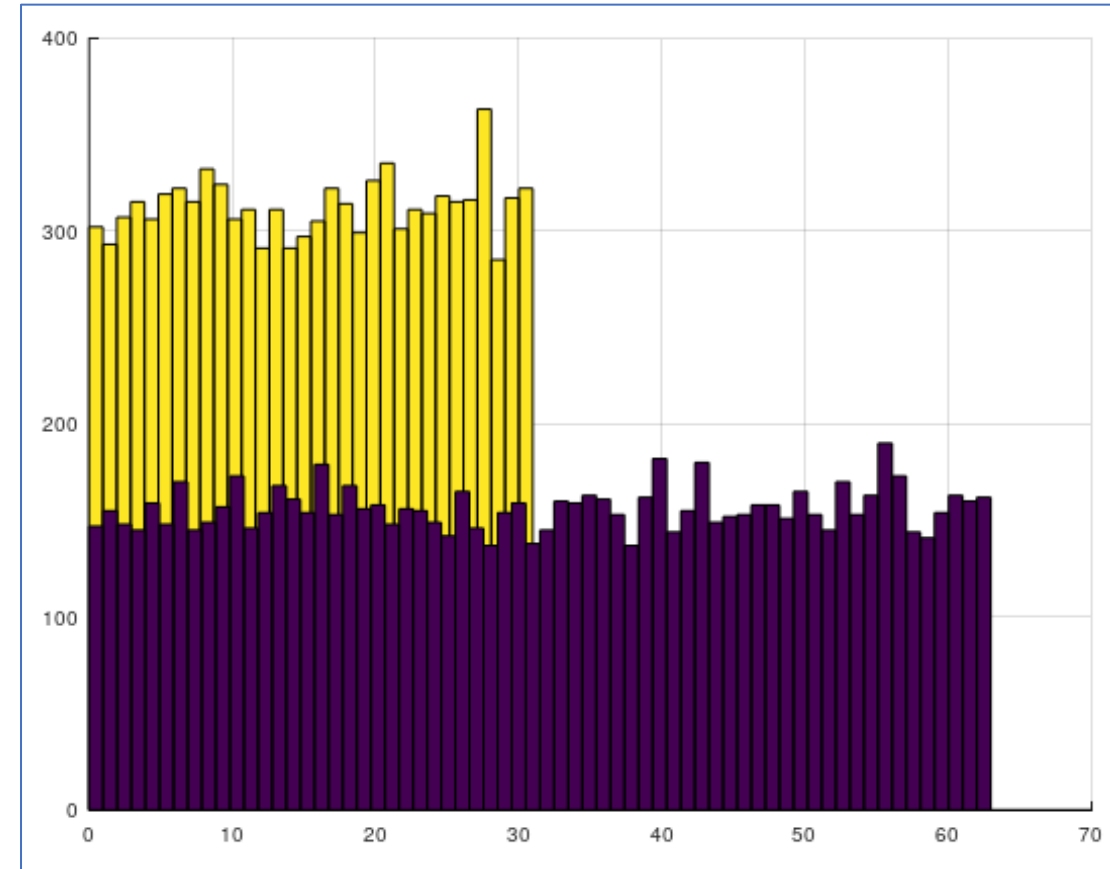
```
...  
B5 = floor(D * 3.2);  
B6 = floor(D * 6.4);  
figure 2;  
hold on;  
hist(B5, 32);  
hist(B6, 64);  
...
```

$n = 5$

$n = 6$

To make the conversion, I multiply all data by $2^n/S$ and then I take the integer part. It may not be the best way, but it works! The resulting vectors B5 e B6 only contain integer values between 0 and $2^n - 1$

Converted values distribution
(vectors B5 and B6)



Introduction

- Quantization
 - How to increase SNR_q
 - For each additional bit of quantization SNR_q increases of 6 dB

Octave example

Compute the quantization error by converting the discrete values to the middle of the corresponding continuous interval and then subtracting the input; then compute the variance

```
...  
E5 = (B5+0.5)/3.2 - D;  
E6 = (B6+0.5)/6.4 - D;  
figure 3;  
hold on;  
hist(E5, 100);  
hist(E6, 50);  
varE5 = (std(E5))^2  
varE6 = (std(E6))^2  
...
```

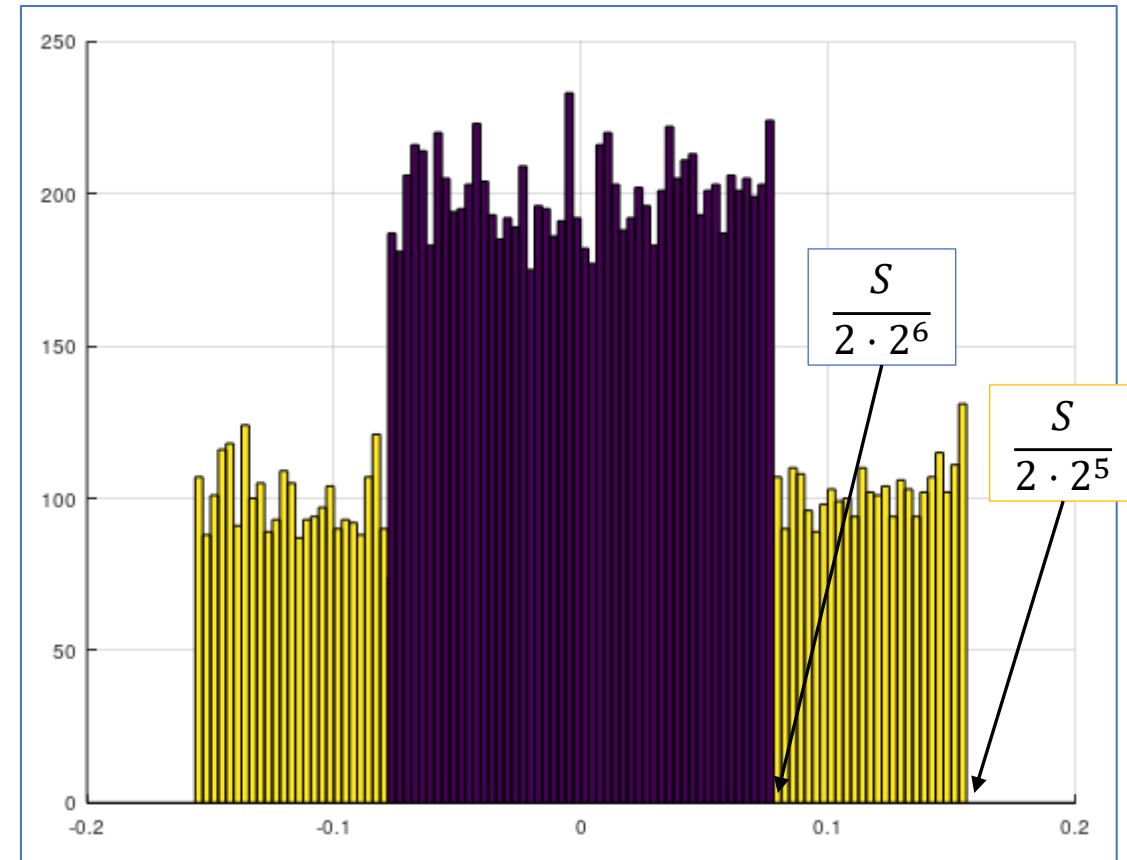
$n = 5$

$n = 6$

To convert back to continuous, I divide the discrete value by $2^n/S$ and center them in the middle of the corresponding interval in the S dynamic range, and then I subtract data from vector D. Vectors E5 and E6 have all values in the range $\pm A_d/2 = \pm S/(2 \cdot 2^n)$ and have a uniform distribution (regardless of the distribution of the input data)

varE5 = 8.1398e-03
varE6 = 2.0319e-03

Quantization error distribution
(vectors E5 and E6)



Introduction

- Quantization
 - How to increase SNR_q
 - For each additional bit of quantization SNR_q increases of 6 dB

Octave example

Compute the Quantization Signal-to-Noise Ratio (SNR_q) as the ratio between variances, both in linear and logarithmic scales; compute the ratio between SNRs in linear scale, and the difference between the SNRs in logarithmic scale

```
...
SNR5 = varD / varE5;
SNR6 = varD / varE6;
SNRratio = SNR6 / SNR5
SNR5dB = 10 * log10(SNR5)
SNR6dB = 10 * log10(SNR6)
SNRdBdiff = SNR6dB - SNR5dB
```

As predicted by the theory, the ratio of SNRs on a linear scale is about 4, while the difference between SNRs on a logarithmic scale is about 6 dB. Furthermore for $n = 5$ we have an SNR equal to $6n = 30$ dB, and for $n = 6$ we have an SNR equal to $6n = 36$ dB

SNRratio = 4.0061

SNR5dB = 30.035
SNR6dB = 36.062
SNRdBdiff = 6.0272

The uniform distribution corresponds to a signal with a triangular waveform. You can try with different distributions of the input signal, for instance sinusoidal or Gaussian. As predicted, the absolute values of the Signal-to-Noise Ratios change, because they now include the additional terms + 1,76 dB (sinusoidal) and -4,77 dB (Gaussian), but the ratio between SNRs and their difference do not change. Moreover, the quantization error distribution remains uniform, regardless of the distribution of the input signal.

```
D = 5*(sin(2*pi*rand(10000,1))+1);
D(D<0 | D>=10) = [];
```

Sinusoidal distribution

```
D = 1.67*randn(10000,1)+5;
D(D<0 | D>=10) = [];
```

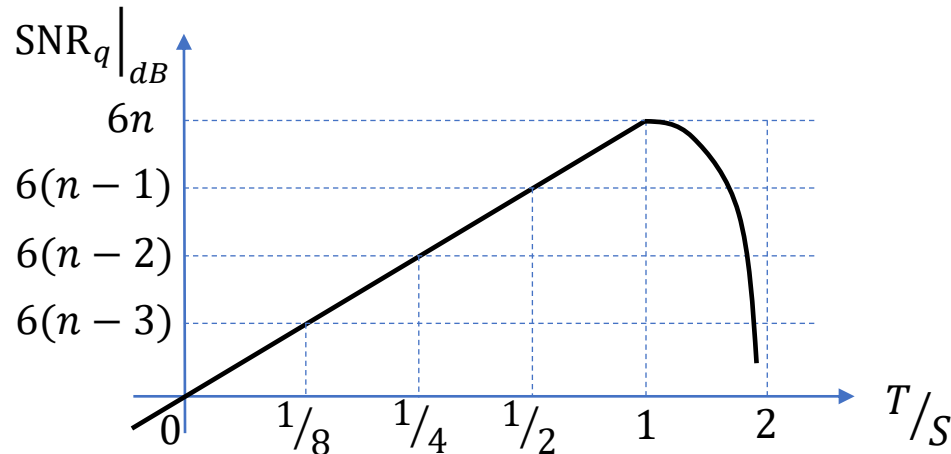
Normal or Gaussian distribution with standard deviation $\sigma = 1/6$

Introduction

- Quantization

- Input signal with a lower dynamic range
- Consider a triangular signal with dynamic range $\left[-\frac{T}{2}, +\frac{T}{2}\right]$, with $T < S$
- Compute SNR_q assuming quantization is over a range $\left[-\frac{S}{2}, +\frac{S}{2}\right]$
 - $\text{SNR}_q|_{dB} = 10 \log_{10} \left(\frac{T^2}{12} \cdot \frac{12 \cdot 2^{2n}}{S^2} \right) = 6n + 10 \log_{10} \frac{T^2}{S^2} = 6n - 20 \log_{10} \frac{S}{T}$
 - $\text{SNR}_q|_{dB} = 6n - 6 \log_2 \frac{S}{T} = 6n - 6t = 6(n - t)$, with $2^t = \frac{S}{T}$

$$\log_2 \frac{S}{T} = \frac{\log_{10} \frac{S}{T}}{\log_{10} 2}$$



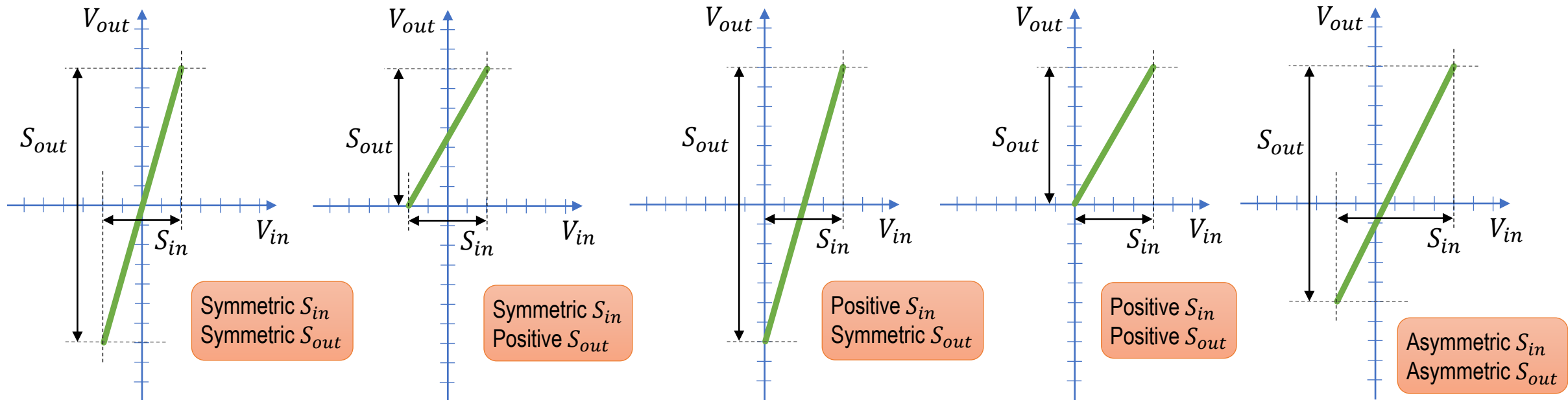
Introduction

- Quantization

- Input signal with a lower dynamic range
- Use an amplifier to adapt the input signal dynamic range to the quantization dynamic range

- $V_{out} = GV_{in} + V_{OFFSET}$

- $G = \frac{S_{out}}{S_{in}} = \frac{V_{outmax} - V_{outmin}}{V_{inmax} - V_{inmin}}, \quad V_{OFFSET} = \frac{V_{outmax} + V_{outmin}}{2} - G \frac{V_{inmax} + V_{inmin}}{2}$



Introduction

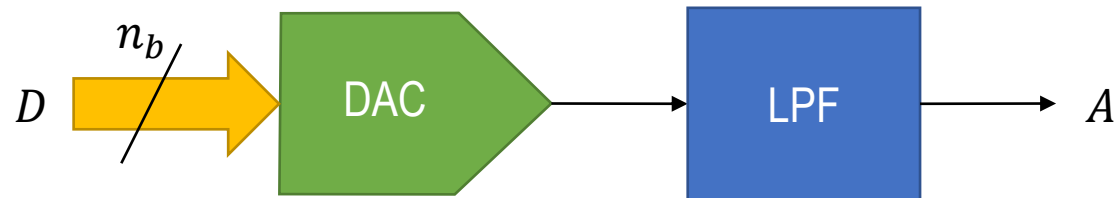
- Total Signal-to-Noise Ratio
 - Many noise sources with uncorrelated errors
 - Each noise source can be characterized by a SNR_i
 - Compute the total Signal-to-Noise Ratio
 - P_S = power of signal, P_{Ni} = power of noise for source i
 - $\text{SNR}_i = 10 \log_{10} \frac{P_S}{P_{Ni}} \Rightarrow \frac{P_{Ni}}{P_S} = 10^{\frac{-\text{SNR}_i}{10}}$
 - $P_{N_{tot}} = \sum P_{Ni} = P_S \sum 10^{\frac{-\text{SNR}_i}{10}}$
 - $\text{SNR}_t = 10 \log_{10} \frac{P_S}{P_{N_{tot}}} = -10 \log_{10} \sum 10^{\frac{-\text{SNR}_i}{10}}$
 - Example: two noise sources with $\text{SNR}_i = 60$ dB
 - $\text{SNR}_t = -10 \log_{10} \left(10^{\frac{-60}{10}} + 10^{\frac{-60}{10}} \right) = -10 \log_{10} (2 \cdot 10^{-6}) = -3 + 60 = 57\text{dB}$

Introduction

- Effective Number Of Bits (ENOB)
 - Alternative expression of the total Signal-to-Noise Ratio
 - Computed from SNR_t (measured or evaluated with full-scale sine input signal)
 - $\text{ENOB} = \frac{\text{SNR}_t - 1.76}{6}$
 - Includes all noise/error sources
 - Represents the number of actually useful bits of the A/D conversion system
 - Always less than the number of bits used for quantization
 - If the major source of error is not quantization
 - → Increasing the number of bits does not significantly increase the ENOB
 - SNR_t is dominated by other sources

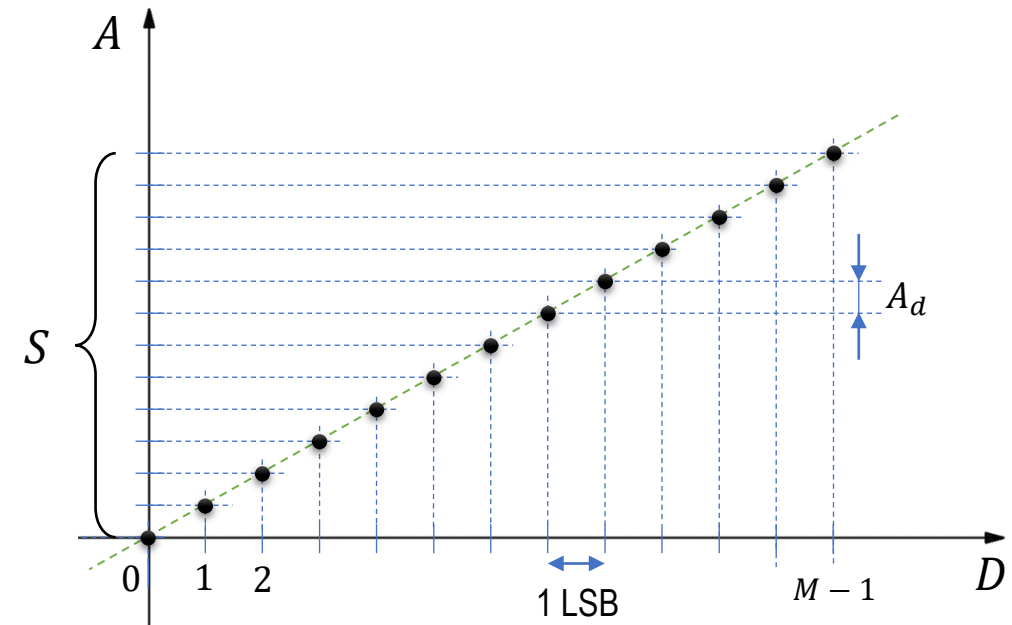
Digital to Analog Conversion

- Dequantization and reconstruction
 - From n_b binary wires to a single wire
 - From discrete values to continuous values
 - The single wire still shows $M = 2^{n_b}$ possible different levels
 - Digital to Analog Converter (DAC) circuit
 - From discrete time to continuous time
 - Reconstruction filter
 - Low pass filter to isolate the baseband spectrum



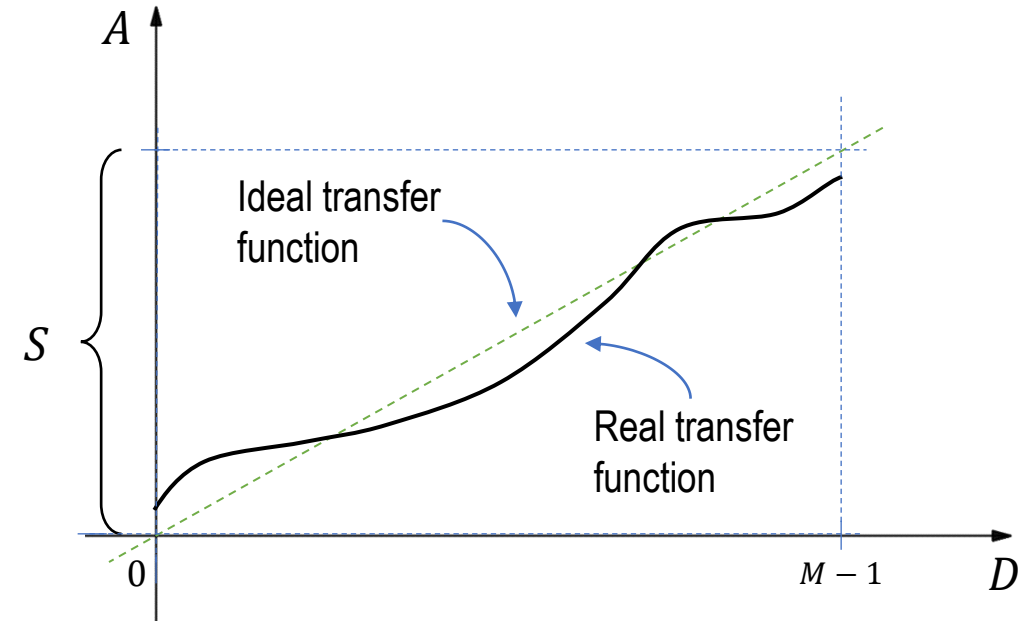
Digital to Analog Conversion

- DAC Transfer Function
 - Computed statically
 - Constant input
 - Steady state behavior
 - Input is discrete
 - M values
 - $M = 2^{n_b}$
 - Output is analog
 - Dynamic range S
 - Transfer function is a sequence of dots
 - With a constant A_d the dots are aligned
 - Uniform conversion
 - With a large M the dots sequence becomes a continuous straight line



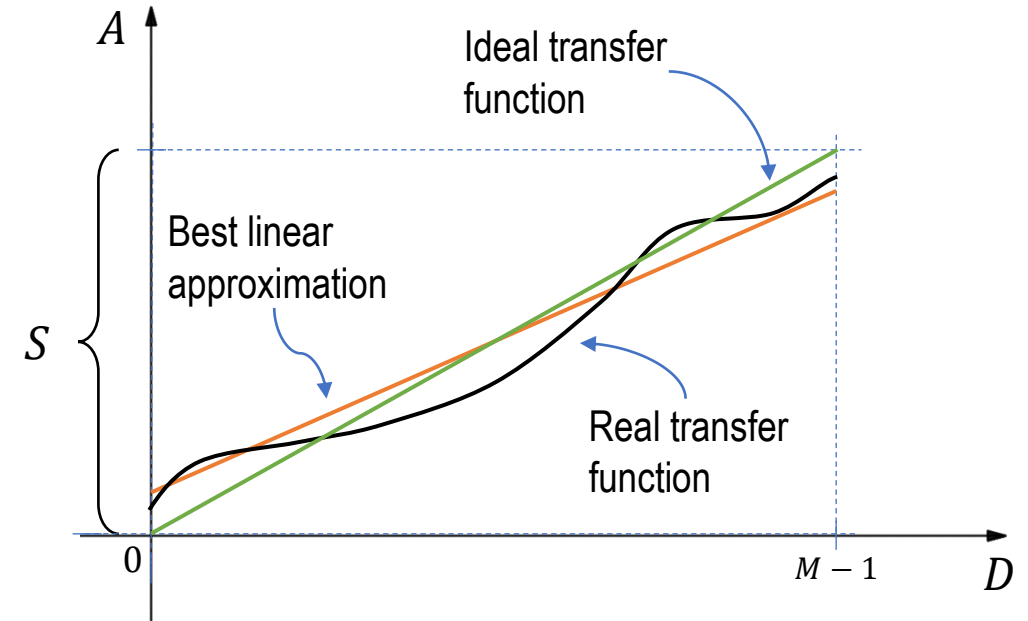
Digital to Analog Conversion

- DAC Transfer Function
 - A real DAC shows deviations from the ideal transfer function
 - Static errors
 - Constant input
 - Errors appear in the $A(D)$ transfer function
 - Dynamic errors
 - Variable input signal
 - Transient behavior
 - Errors appear in $A(t)$



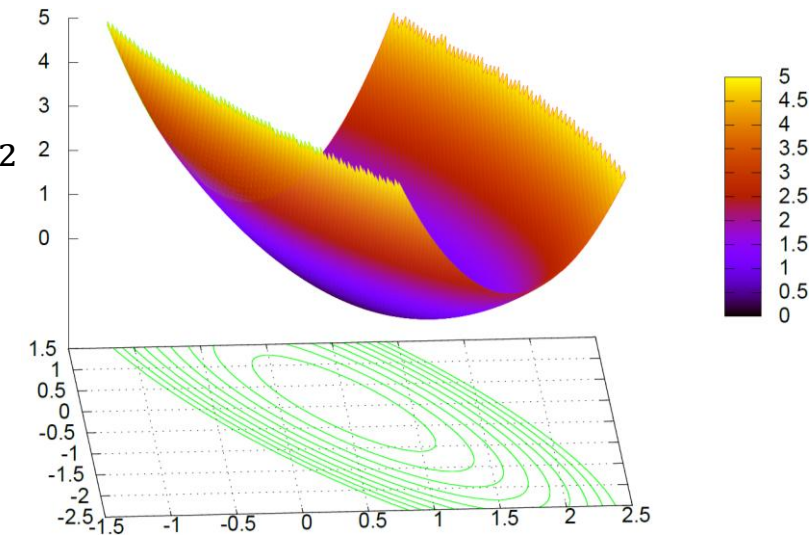
Digital to Analog Conversion

- DAC Transfer Function
 - Static errors
 - The real transfer function is not a perfect straight line
 - Find the best linear approximation of the real transfer function
 - Compare the best linear approximation with the ideal transfer function
 - → Linear errors
 - Compare the real transfer function with the best linear approximation
 - → Non linear errors



Digital to Analog Conversion

- DAC Transfer Function
 - Find the best linear approximation of the real transfer function
 - Minimum Mean Square Errors (MMSE) technique
 - The expression of a general straight line is
 - $A = m_{la}D + q_{la}$
 - For each one of the M points in the real transfer function we have a couple $P_i = (D_i, A_i)$
 - The error and the square error in each single point are
 - $\varepsilon_i = A_i - (m_{la}D_i + q_{la})$, $\varepsilon_i^2 = (A_i - (m_{la}D_i + q_{la}))^2$
 - The Mean Square Error is
 - $\varepsilon_{MSE}(m_{la}, q_{la}) = \frac{1}{M} \sum_{i=0}^{M-1} \varepsilon_i^2 = \frac{1}{M} \sum_{i=0}^{M-1} (A_i - (m_{la}D_i + q_{la}))^2$
 - Minimize the mean square error
 - It is a positive concave paraboloid (it holds water)
 - The coordinates (m_{la}, q_{la}) of the minimum give the parameter of the best linear approximation
 - The minimum is the vertex



Digital to Analog Conversion

- DAC Transfer Function Linear Errors

- Compare the best linear approximation to the ideal transfer function

- Offset error ε_o

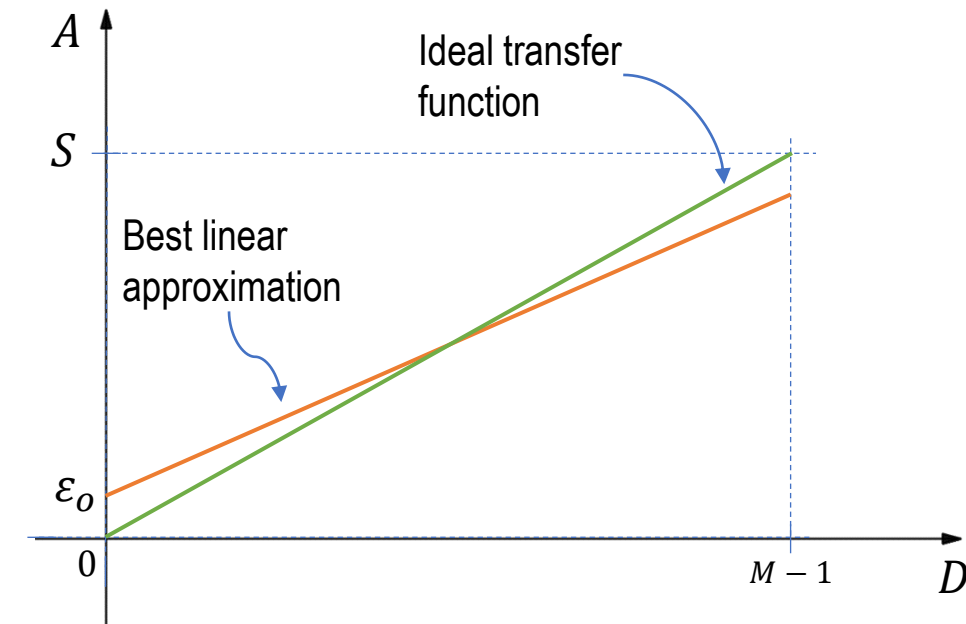
- Intercept of approximating line with A axis
- Can be compensated by adding a constant

- Gain error ε_g

- Difference among the ideal slope and the best linear approximation slope
- Can be compensated by gain correction
 - Add an amplifier with proper gain

- Compensation is expensive

- Need to characterize and compensate each single unit separately



Digital to Analog Conversion

- DAC Transfer Function Linear Errors

- Compare the best linear approximation to the ideal transfer function

- Offset error ε_o

- Intercept of approximating line with A axis
- Can be compensated by adding a constant

- Ideal transfer function

- $D = 0 \rightarrow A = 0$

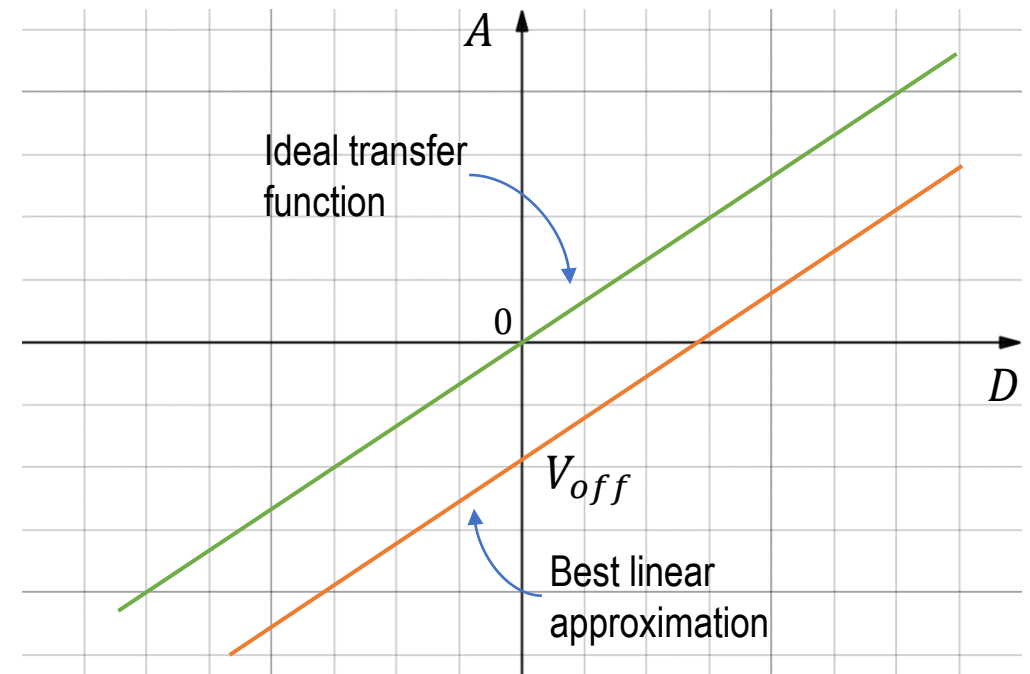
- Best linear approximation transfer function

- Not through 0,0
- $D = 0 \rightarrow A = V_{off}$

- Offset error

- $\varepsilon_o = V_{off}$

Absolute error, it has the same unit of measure of the analog input signal



Digital to Analog Conversion

- DAC Transfer Function Linear Errors

- Compare the best linear approximation to the ideal transfer function

- Gain error ε_g

- Difference among the ideal slope and the best linear approximation slope
- Can be compensated by gain correction

- Ideal transfer function

- $A = kD$

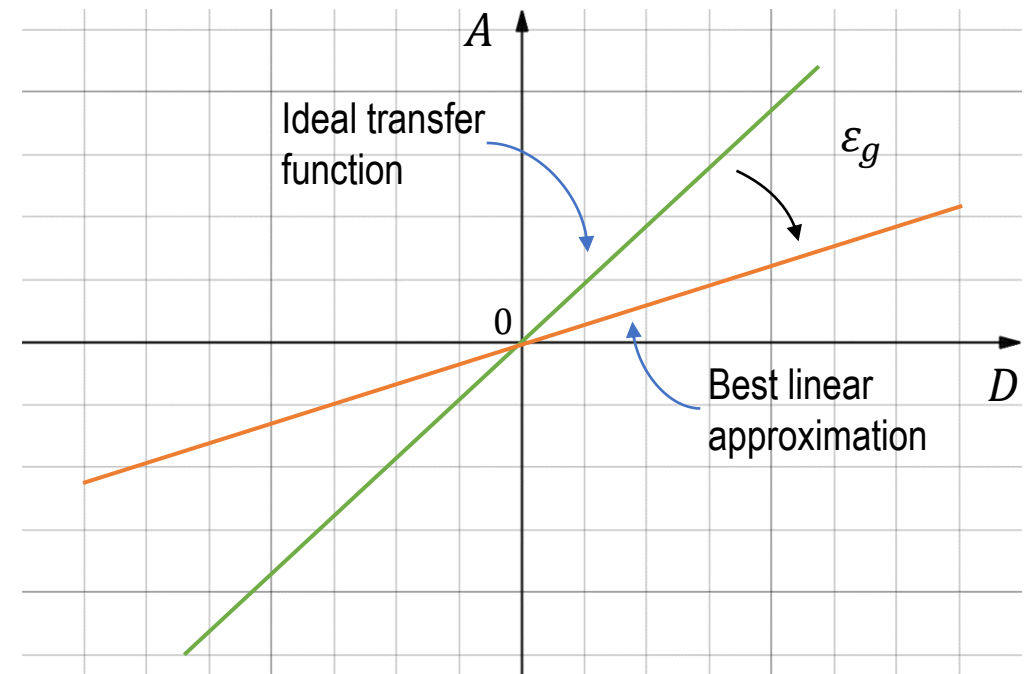
- Best linear approximation transfer function

- Different slope
- $A = k'D$
- $k' = k + \Delta k$

- Gain error

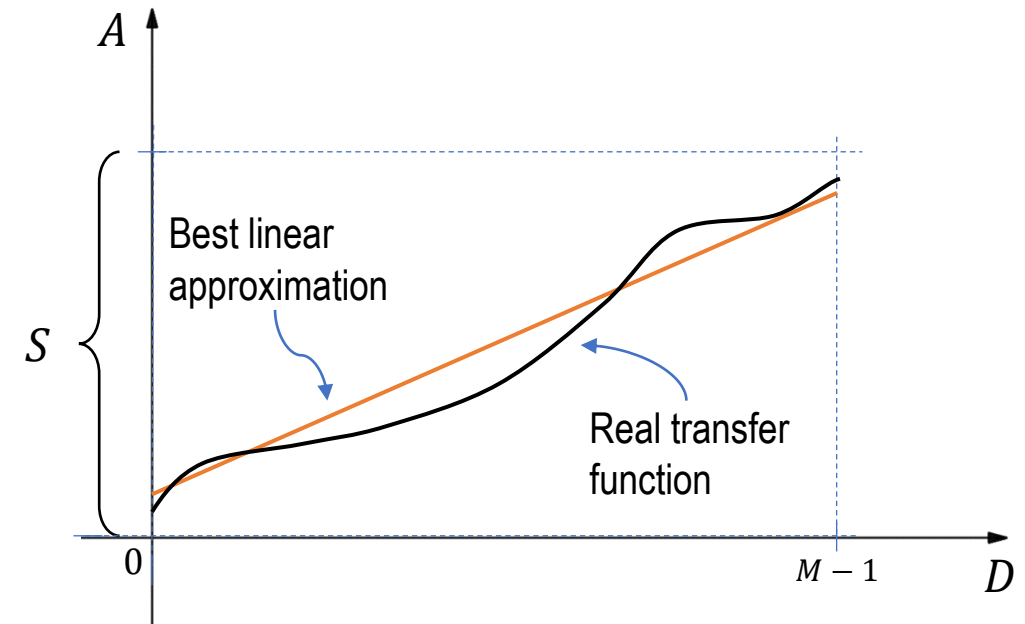
- $$\varepsilon_g = \frac{k' - k}{k} = \frac{\Delta k}{k}$$

Relative error, it is a pure number or it is expressed as a percentage



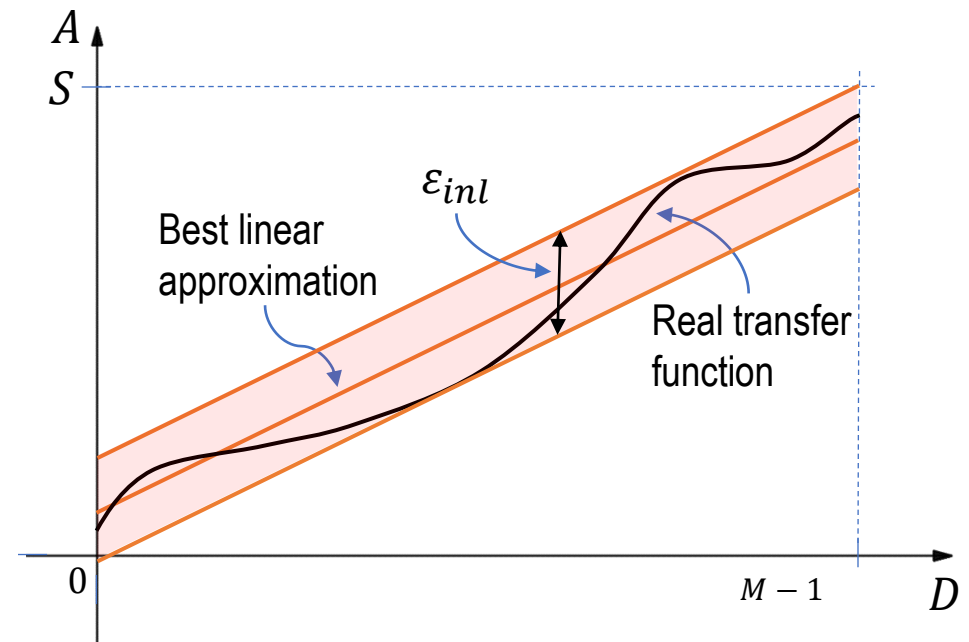
Digital to Analog Conversion

- DAC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Linear errors are factored out
 - Integral non linearity
 - Maximum deviation of the real transfer function from the best linear approximation
 - Global value
 - Differential non linearity
 - Difference between two consecutive analog values
 - Local value
 - Need to give 2^{n_b} different data
 - Often only an upper bound is given



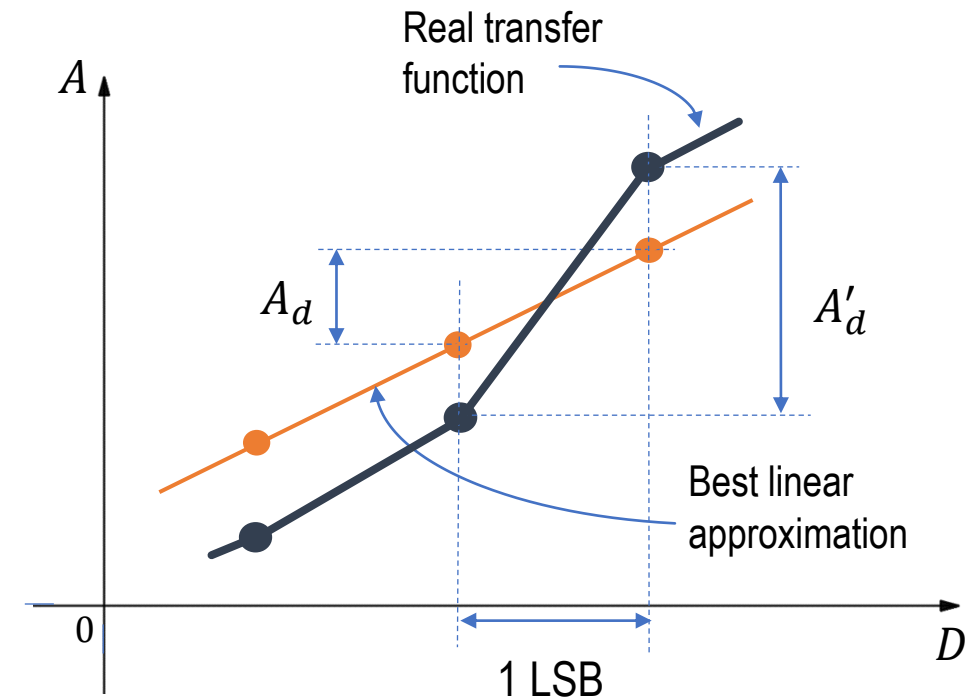
Digital to Analog Conversion

- DAC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Integral non linearity
 - Maximum deviation of the real transfer function from the best linear approximation
 - Global value
 - Measured along the vertical (analog) axis
 - As an absolute value with the unit of measure of the analog quantity
 - As a fraction of an LSB
 - Cannot be compensated



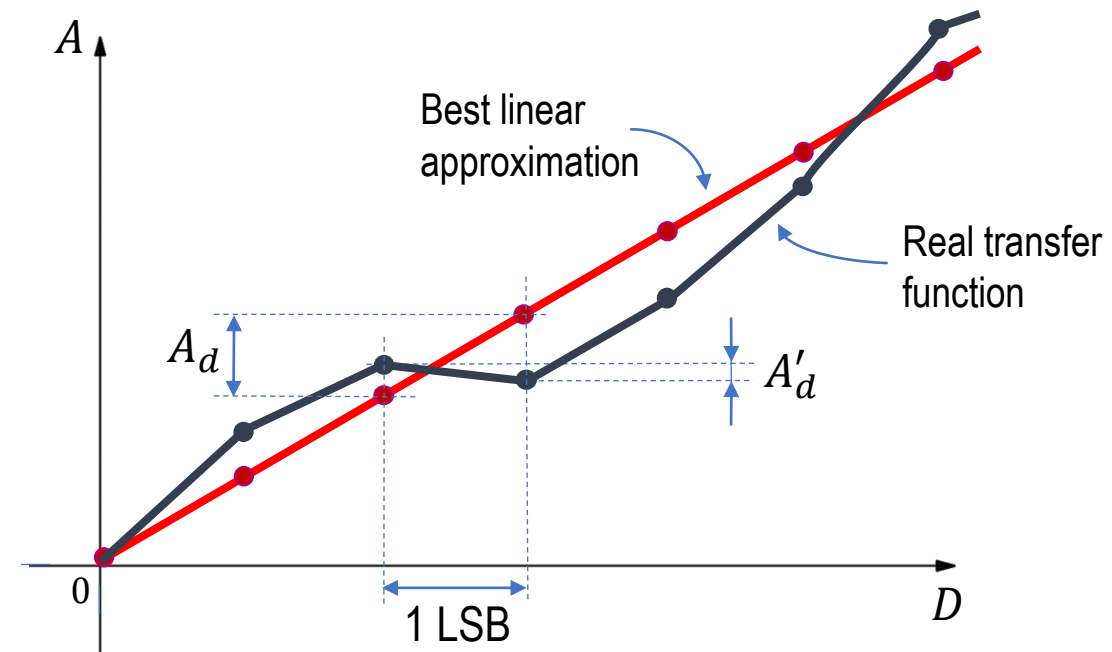
Digital to Analog Conversion

- DAC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Differential non linearity
 - Difference between two consecutive analog values
 - Local value
 - Need to give 2^{n_b} different data
 - Often only an upper bound is given
 - $A'_d \neq A_d$
 - $\varepsilon_{dnl} = A_d - A'_d$
 - Measured along the vertical (analog) axis
 - As an absolute value with the unit of measure of the analog quantity
 - As a fraction of an LSB
 - Cannot be compensated



Digital to Analog Conversion

- DAC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Differential non linearity
 - Difference between two consecutive analog values
 - Non Monotonicity Error
 - $A'_d < 0$
 - $\varepsilon_{dnl} = A_d - A'_d > 1 \text{ LSB}$
 - Slope inversion

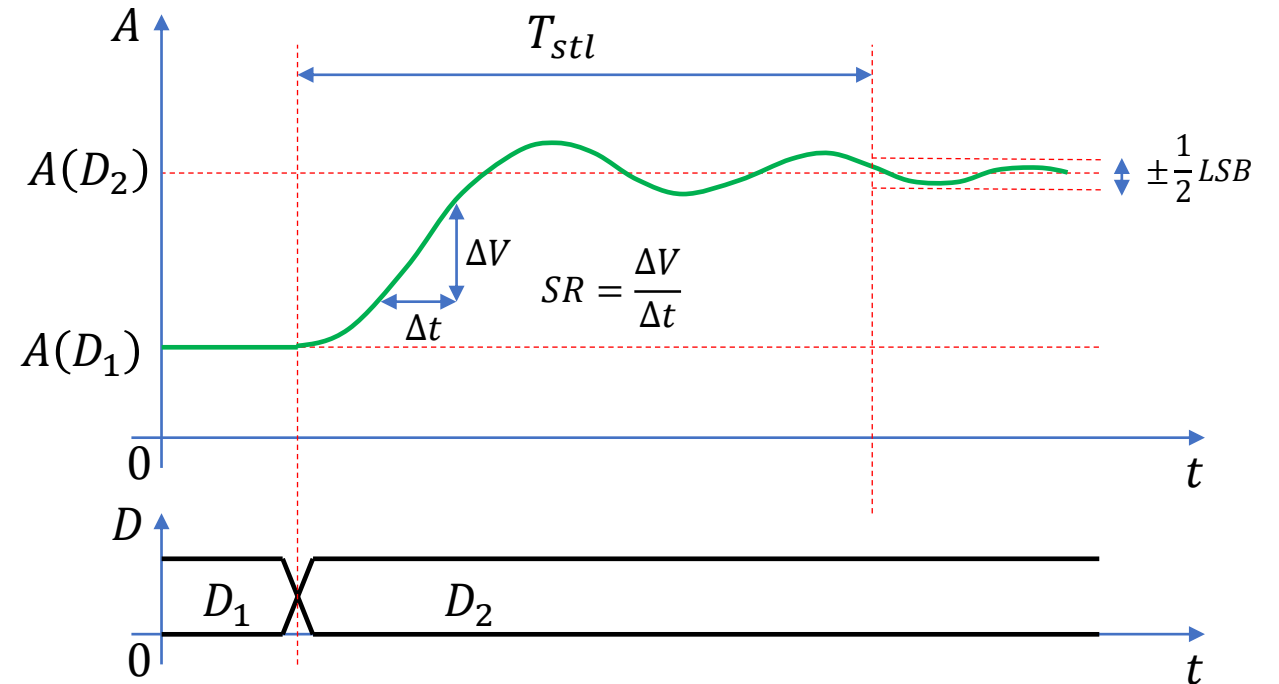


Digital to Analog Conversion

- DAC Transfer Function Non Linear Errors
 - Examples
 - Draw the conversion characteristic for a 3-bit DAC with:
 - $\varepsilon_{dnl} = +\frac{1}{4}$ LSB from 000 to 011 (MSB = 0)
 - $\varepsilon_{dnl} = -\frac{1}{4}$ LSB from 100 to 111 (MSB = 1)
 - Draw the conversion characteristic for a 3-bit DAC with:
 - $\varepsilon_{dnl} = +\frac{1}{4}$ LSB when LSB = 0
 - $\varepsilon_{dnl} = -\frac{1}{4}$ LSB when LSB = 1
 - Compare the two cases
 - Evaluate integral nonlinearity ε_{inl}
 - Evaluate offset and gain errors

Digital to Analog Conversion

- DAC Transfer Function Dynamic Errors
 - Settling Time T_{stl}
 - Time it takes to the DAC output to reach the new value
 - Within a specified error
 - Due to
 - Slew Rate SR of the circuit
 - Low pass effects of parasitic capacitances and resistances
 - Second order effects
 - Oscillations



Digital to Analog Conversion

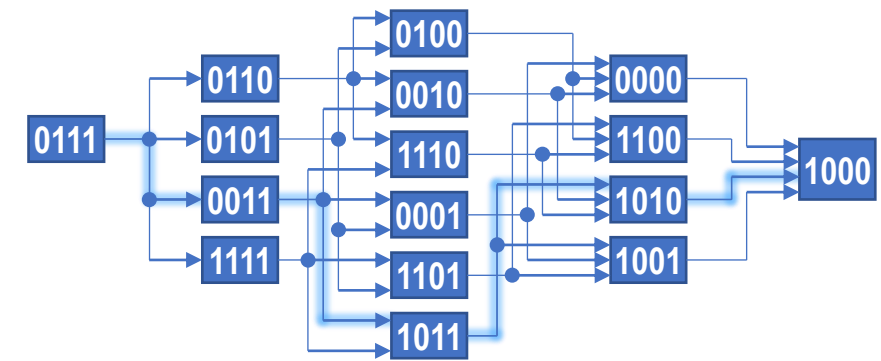
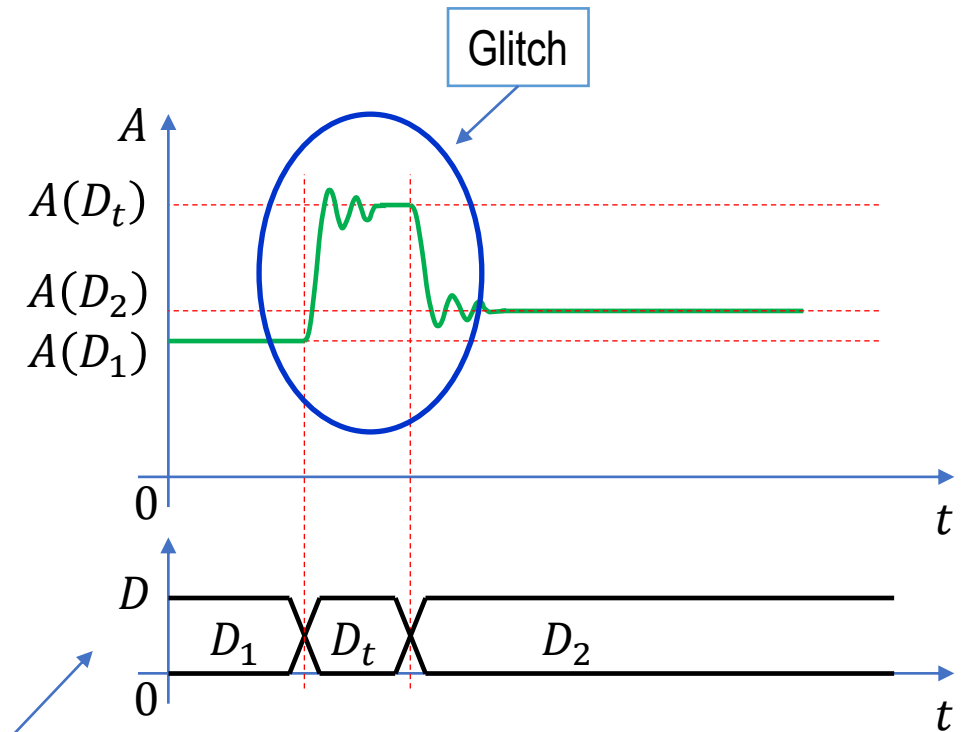
- DAC Transfer Function Dynamic Errors

- Glitch

- Transient when the output takes a widely wrong value
 - Caused by differences in switching delays of various bits

- Example

- Digital input changes from $D_1 = 0111$ to $D_2 = 1000$
 - Change is 1 LSB
 - Many possible transient behaviors
 - $D_t = 1111$ (change of d_3 , then change of $d_2 d_1 d_0$)
 - $D_t = 0000$ (change of $d_2 d_1 d_0$, then change of d_3)
 - $D'_t = 0011$ (change of d_2)
 - $D''_t = 1011$ (change of d_3)
 - $D'''_t = 1010$ (change of d_0)
 - $D_2 = 1000$ (change of d_1)



All possible path changing one bit at a time

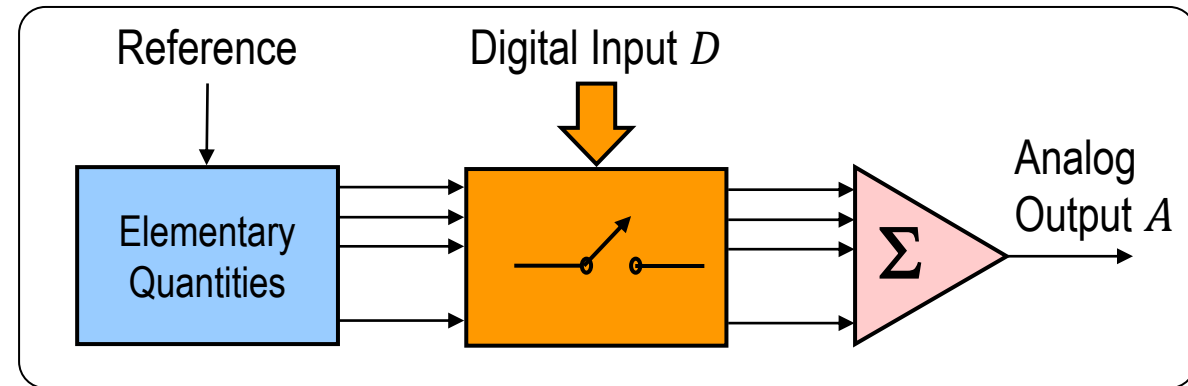
Digital to Analog Conversion

- DAC Transfer Function Errors
 - Linear errors
 - Gain ε_g
 - Offset ε_o
 - Non linear errors
 - Integral non linearity ε_{inl}
 - Differential non linearity ε_{dnl}
 - Dynamic errors
 - Settling time T_{stl}
 - Glitches
 - Measured as
 - % of full scale S
 - Absolute value (mV, μ V)
 - Fraction of LSB ($\frac{1}{2}$ LSB, $\frac{1}{4}$ LSB, ...)

Digital to Analog Conversion

• DAC Circuits

- Input is a digital signal
 - A binary value on n_b wires, with $N = 2^{n_b}$ possible different combinations
 - Expressed as an integer D from 0 to $N - 1$
- Output is an electrical quantity
 - A voltage
 - A current
- Circuit
 - Generate a number of elementary quantities
 - Uniform Elementary Quantities
 - Weighted Elementary Quantities
 - Sum elementary quantities based on the input digital signal



Digital to Analog Conversion

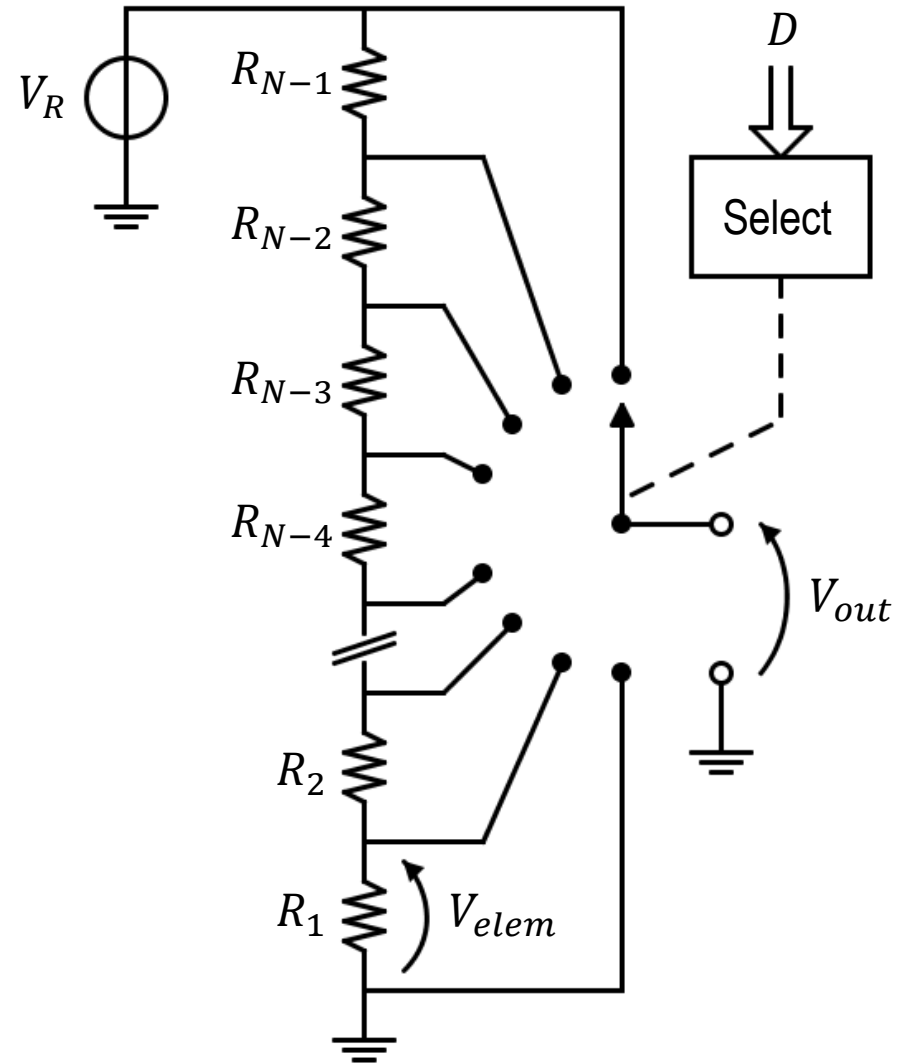
- DAC Circuits
 - Uniform Quantities
 - Sum elementary units with the same weight (1)
 - Each elementary unit corresponds to 1 LSB
 - The number of elementary units is controlled by the digital input D
 - $A = D \cdot \text{LSB}$
 - Example
 - $13_{10} = 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$
 - Advantages
 - All elementary units are identical
 - Disadvantages
 - Need an exponential number of elementary units on the number of bits

Digital to Analog Conversion

- DAC Circuits
 - Weighted Quantities
 - Sum elementary units with different weights
 - Weights correspond to powers of 2 (1, 2, 4, 8, 16, ...)
 - Each elementary unit is controlled by the corresponding bit with the same weight in D
 - $A = \sum(2^i d_i) \cdot \text{LSB}$
 - Example
 - $13_{10} = 1101_2 = 2^3 \cdot 1 + 2^2 \cdot 1 + 2^1 \cdot 0 + 2^0 \cdot 1 = 8 + 4 + 0 + 1$
 - Advantages
 - The number of elementary units is linear on the number of bits
 - Disadvantages
 - The elementary units are all different

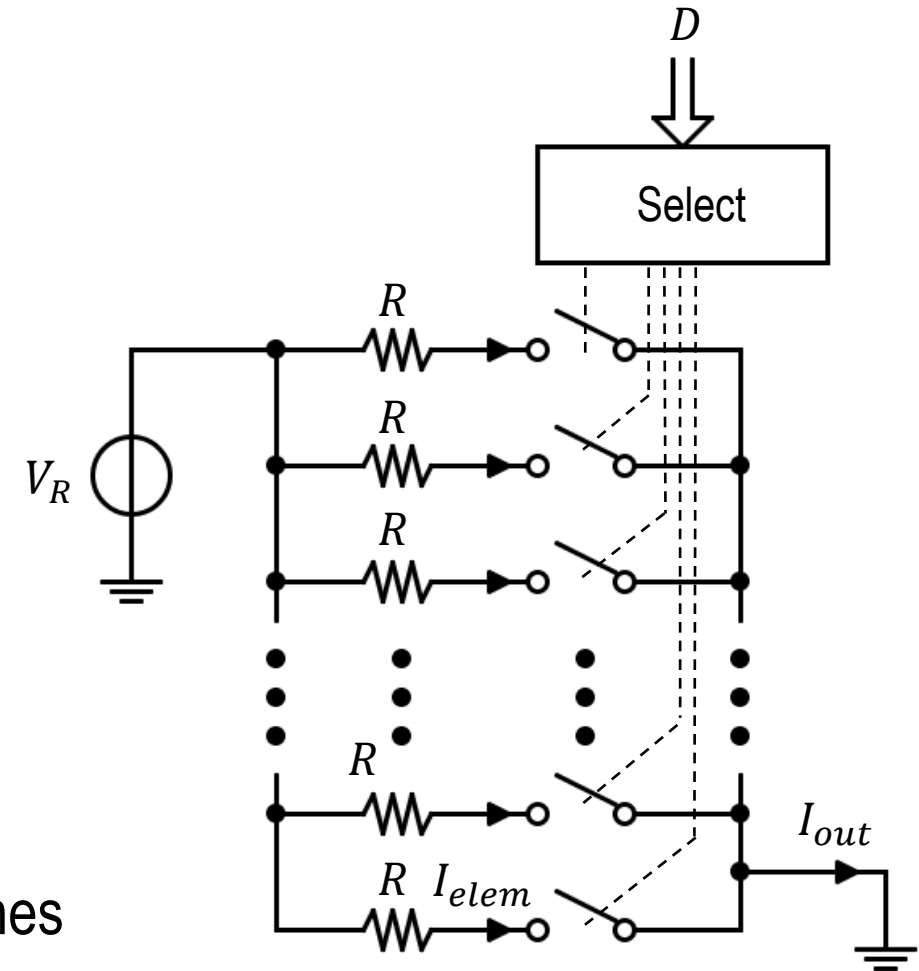
Digital to Analog Conversion

- DAC Circuit with Uniform Quantities
 - Reference voltage V_R
 - Elementary units are voltage drops across identical resistances
 - There are $2^{n_b} - 1 = N - 1$ resistances
 - All voltage drops are identical
 - $V_{elem} = \frac{R}{(N-1)R} V_R = \frac{V_R}{N-1}$
 - Exponential on the number of bits
 - Analog output is a voltage V_{out}
 - Dynamic range between 0 and V_R
 - Elementary units are summed by controlling a switch
 - Controlling signal is the digital input D
 - Potentiometric DAC



Digital to Analog Conversion

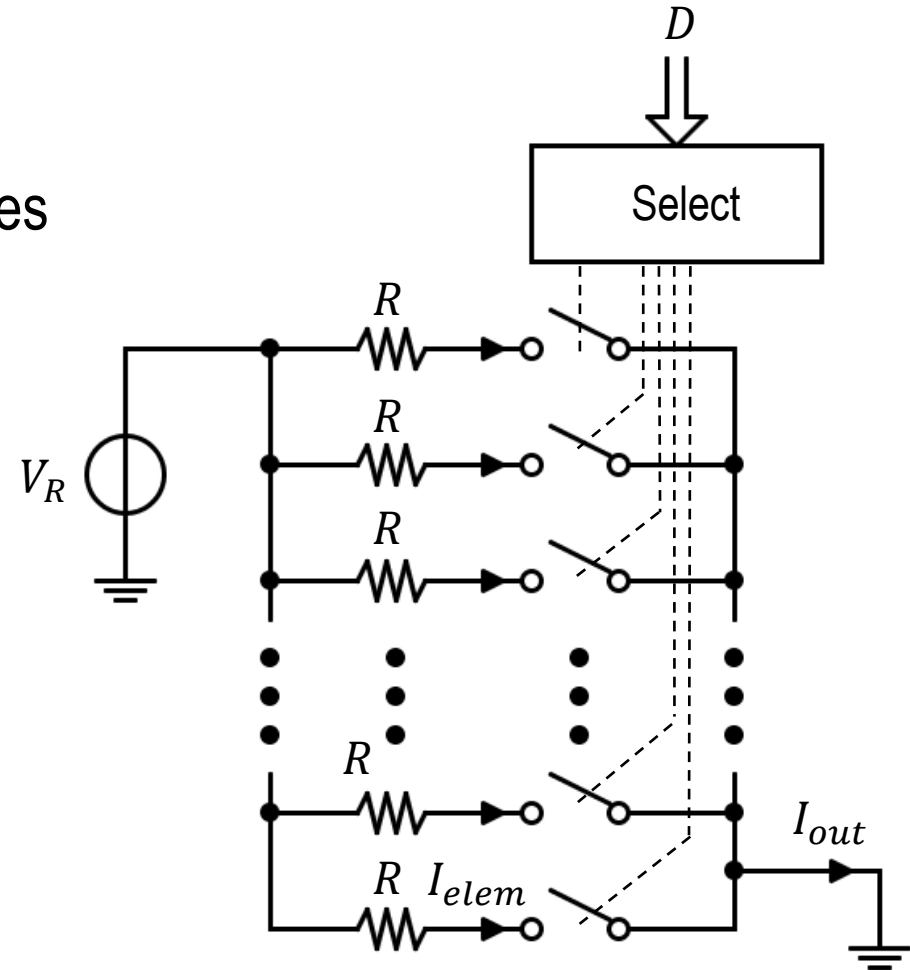
- DAC Circuit with Uniform Quantities
 - Reference voltage V_R
 - Elementary units are currents in identical resistances
 - There are $2^{n_b} - 1 = N - 1$ resistances
 - All currents are identical
 - $I_{elem} = \frac{V_R}{R}$
 - Exponential on the number of bits
 - Analog output is a current I_{out}
 - Dynamic range between 0 and $(N - 1) \frac{V_R}{R}$
 - Elementary units are summed by controlling $N - 1$ switches
 - Controlling signal is the digital input D on n_b bits
 - An encoder generates the $N - 1$ signals driving the switches
 - Sum obtained using the Kirchoff law



Digital to Analog Conversion

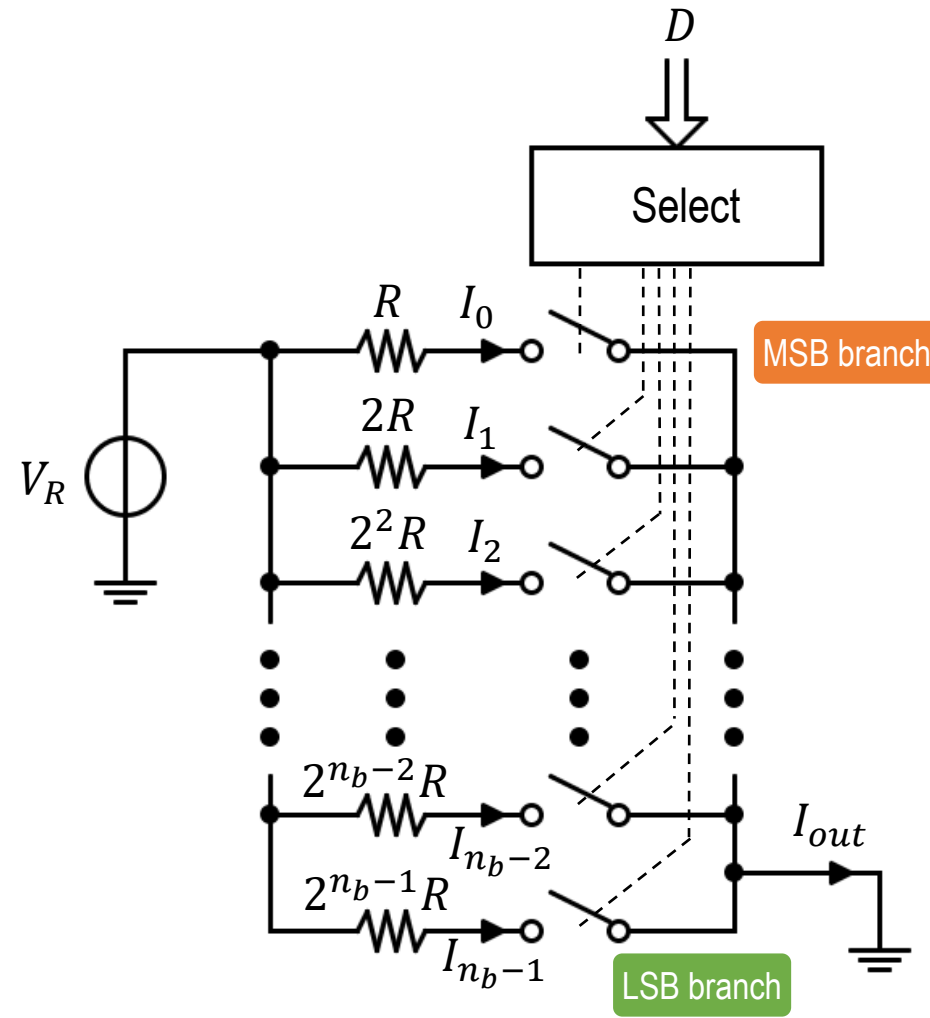
- DAC Circuit with Uniform Quantities
 - Elementary units are summed by controlling $N - 1$ switches
 - Controlling signal is the digital input D
 - The digital circuit controlling the switches has a thermometric output

d_3	d_2	d_1	d_0	s_{14}	s_{13}	s_{12}	s_{11}	s_{10}	s_9	s_8	s_7	s_6	s_5	s_4	s_3	s_2	s_1	s_0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1
0	1	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1
0	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1
1	0	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
1	0	1	1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1
1	1	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
1	1	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1



Digital to Analog Conversion

- DAC Circuit with Weighted Quantities
 - Reference voltage V_R
 - Elementary units are currents in different resistances
 - There are n_b resistances
 - All currents are different
 - $I_i = \frac{V_R}{2^i R}, \quad \forall i \in [0, n_b - 1]$
 - Linear on the number of bits
 - Analog output is a current I_{out}
 - Dynamic range between 0 and $\approx 2 \frac{V_R}{R}$
 - Elementary units are summed by controlling n_b switches
 - Controlling signal is the digital input D
 - Sum obtained using the Kirchhoff law



Digital to Analog Conversion

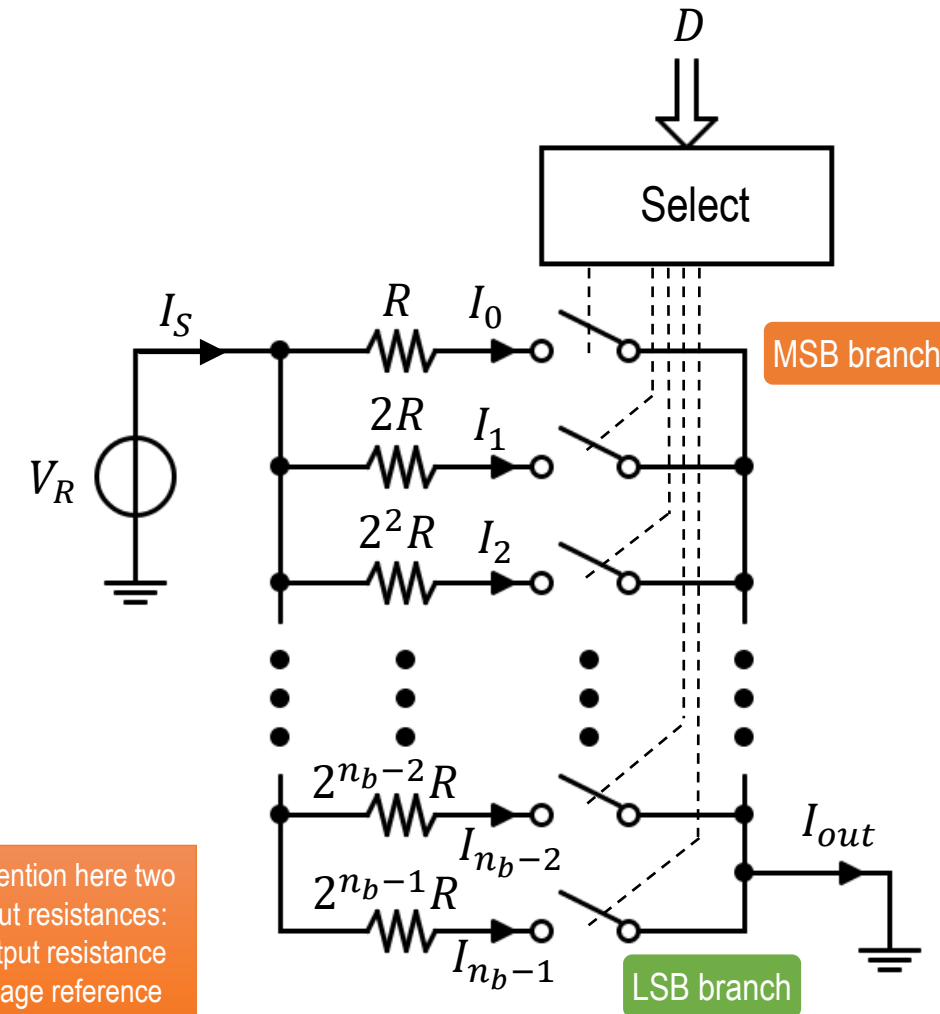
- DAC Circuit with Weighted Quantities

- Reference voltage V_R

- The output current I_S of the voltage reference is not constant
- It depends on the input digital value D
- Same dynamic range as the output current I_{out}
 - Minimum current is 0
 - Maximum current is $\approx 2 \frac{V_R}{R}$
- Voltage reference is hard to design
 - Should have a very small output resistance R_S
 - The voltage drop across the output resistance R_S of the voltage source causes errors in the output current

- Output resistance R_{out}

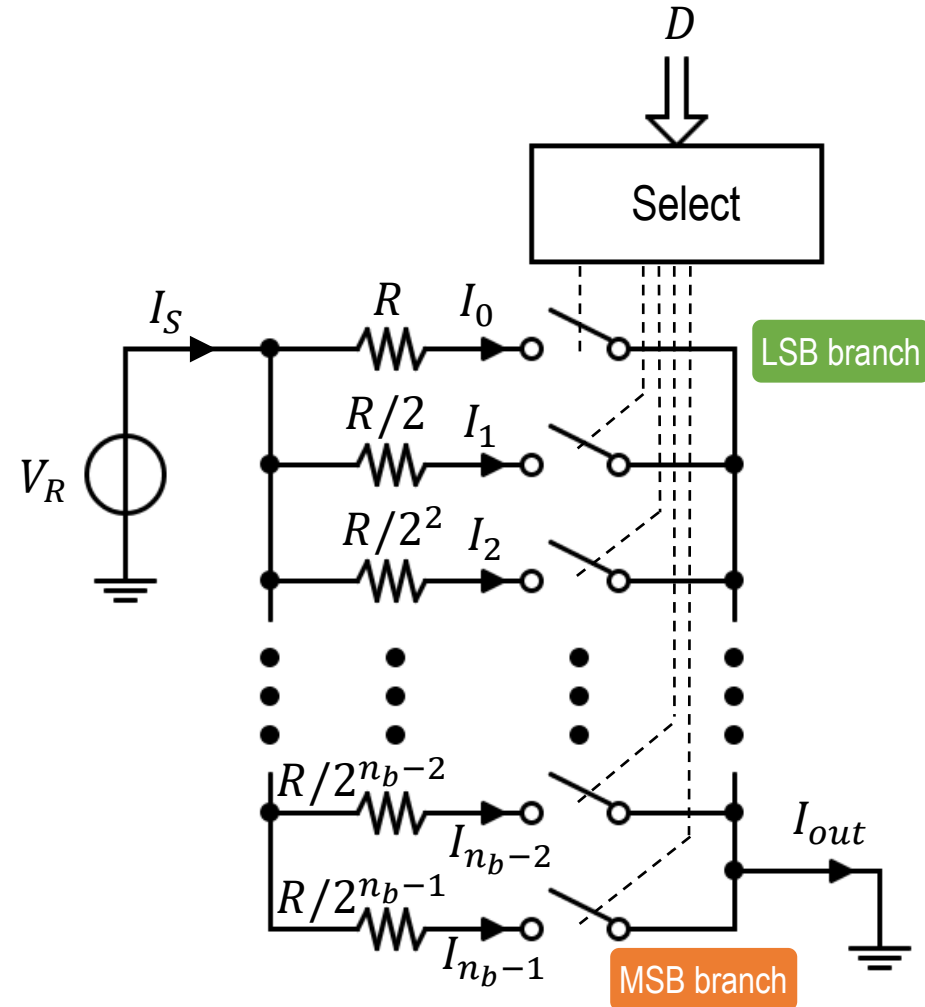
- It is the output resistance of the entire converter
- It changes with the input digital value D



Note that I mention here two different output resistances: one is the output resistance R_S of the voltage reference V_R , the other is the output resistance R_{out} of the entire circuit.

Digital to Analog Conversion

- DAC Circuit with Weighted Quantities
 - Reference voltage V_R
 - Same circuit as before
 - Resistances are still in ratios that are powers of 2
 - The MSB branch is always the one with the smallest resistance
 - It corresponds to the highest current
 - The LSB branch is the one with the highest resistance
 - It corresponds to the the smallest current
 - All currents are different
 - $I_i = 2^i \frac{V_R}{R}, \quad \forall i \in [0, n_b - 1]$
 - Analog output is a current I_{out}
 - Dynamic range between 0 and $\approx 2^{n_b} \frac{V_R}{R}$



Digital to Analog Conversion

- DAC Circuit with Weighted Quantities

- Reference voltage V_R
- Elementary units are currents across different resistances

- There are n_b resistances
- All currents are different

- $I_i = \frac{V_R}{2^i R}, \quad \forall i \in [0, n_b - 1]$

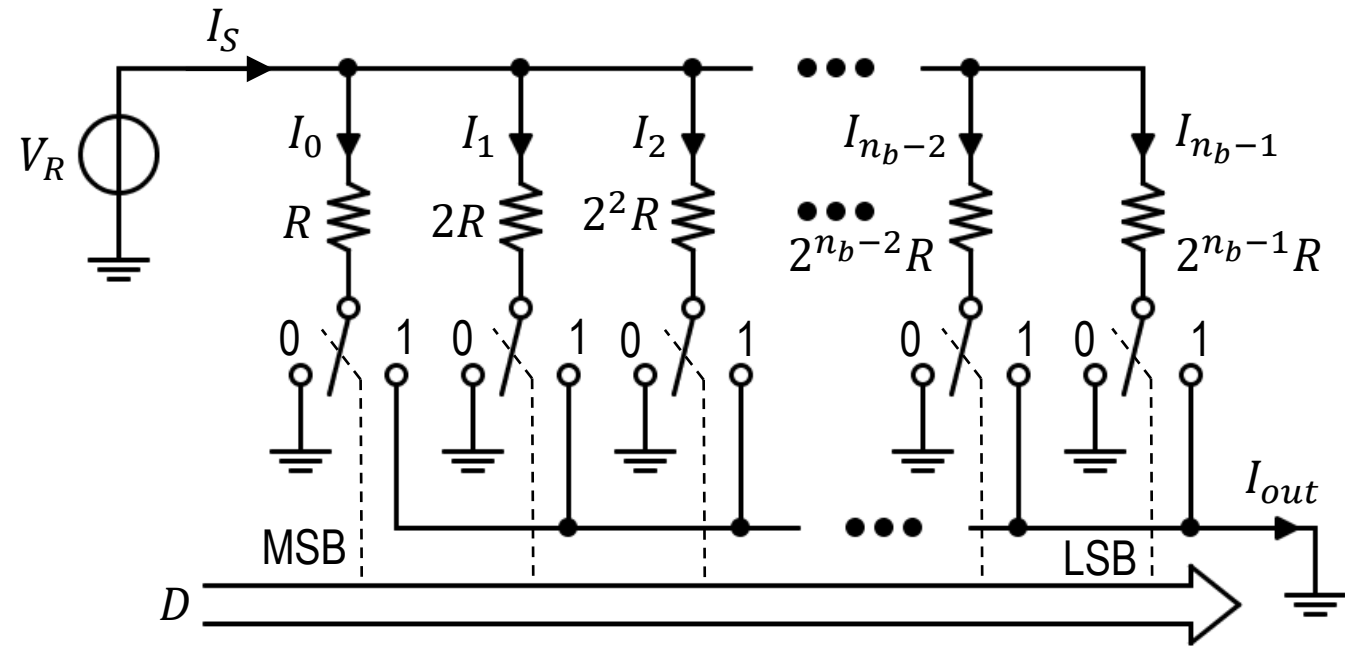
- Linear on the number of bits

- Analog output is a current I_{out}

- Dynamic range between 0 and $\approx 2 \frac{V_R}{R}$

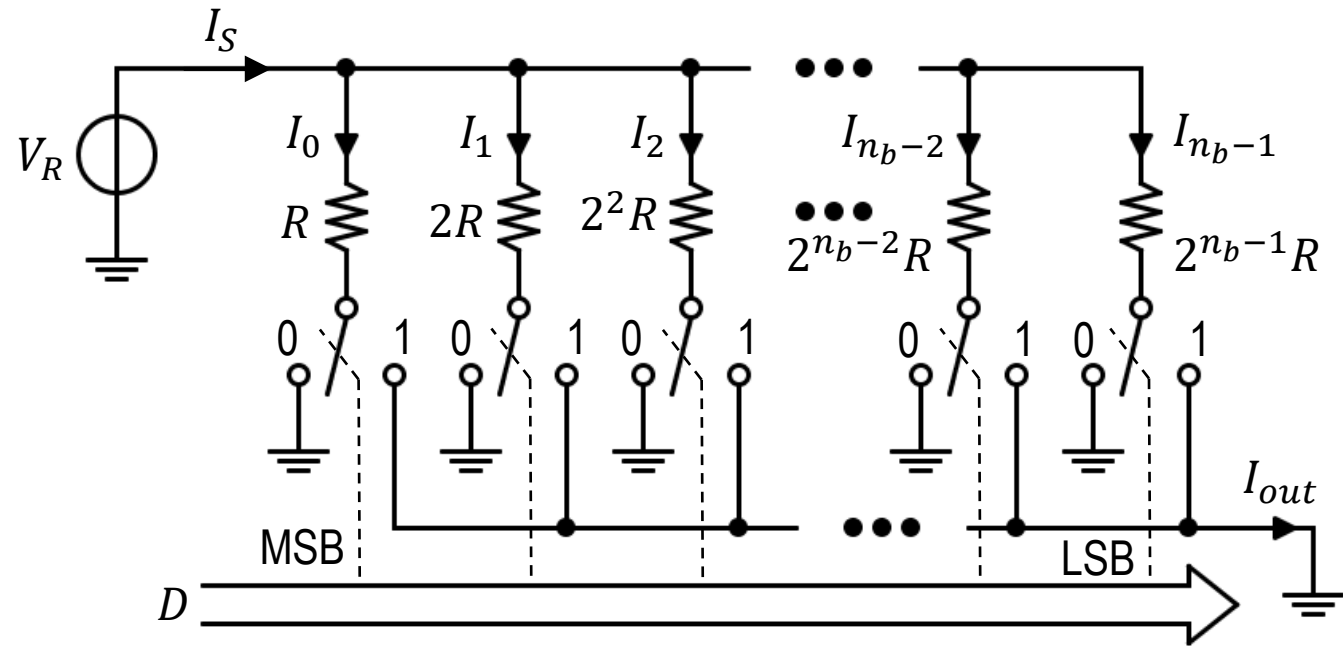
- Elementary units are summed by controlling n_b switches

- Controlling signal is the digital input D
 - Sum obtained using the Kirchoff law



Digital to Analog Conversion

- DAC Circuit with Weighted Quantities
 - Modified circuit using current switches
 - Both node voltages at the switch bottom terminal are at 0 V
 - Reference voltage V_R
 - Output current I_S of the voltage reference is constant
 - It is about $I_S \approx 2 \frac{V_R}{R}$
 - Voltage reference is easier to design
 - The voltage drop across the output resistance R_S of the voltage reference V_R is constant
 - Output resistance R_{out}
 - It changes with the input digital value D



Digital to Analog Conversion

- DAC Circuit with Weighted Quantities

- Modified circuit using voltage switches

- Node voltages at the switch bottom terminal are either 0 V or V_R

- Reference voltage V_R

- Output current I_S of the voltage reference is not constant

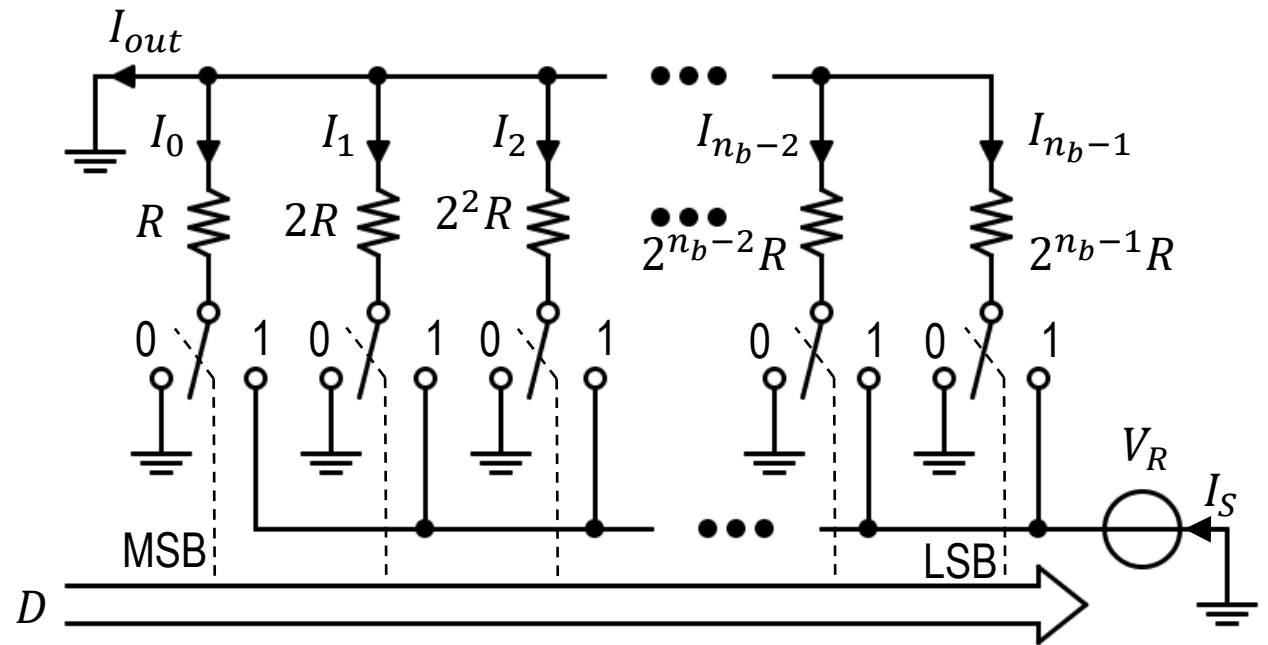
- Dynamic range between 0 and $\approx 2 \frac{V_R}{R}$

- Voltage reference difficult to design

- Output resistance R_{out}

- It is constant
 - It is the parallel of all resistances

- $R_{out} \approx \frac{R}{2}$

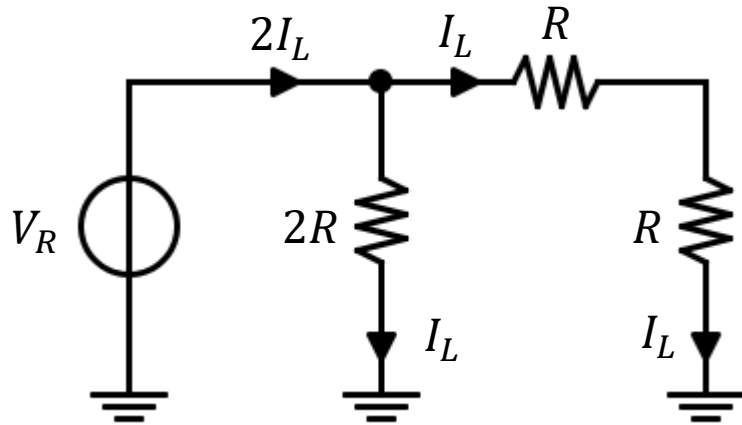


Digital to Analog Conversion

- DAC Circuit with Weighted Quantities
 - There are n_b resistances
 - Their values are different
 - Smallest resistance is R
 - MSB branch
 - Highest resistance is $2^{n_b-1}R$
 - LSB branch
 - Dynamic range of resistance values is exponential on the number of bits
 - Examples
 - $n_b = 8$ bit, $2^{n_b-1} = 128$
 - $R_{min} = 1 \text{ k}\Omega$, $R_{max} = 128 \text{ k}\Omega$
 - $n_b = 12$ bit, $2^{n_b-1} = 2048$
 - $R_{min} = 1 \text{ k}\Omega$, $R_{max} = 2 \text{ M}\Omega$
 - $n_b = 16$ bit, $2^{n_b-1} = 32768$
 - $R_{min} = 1 \text{ k}\Omega$, $R_{max} = 32.8 \text{ M}\Omega$
 - $R_{min} = 100 \text{ }\Omega$, $R_{max} = 3.3 \text{ M}\Omega$

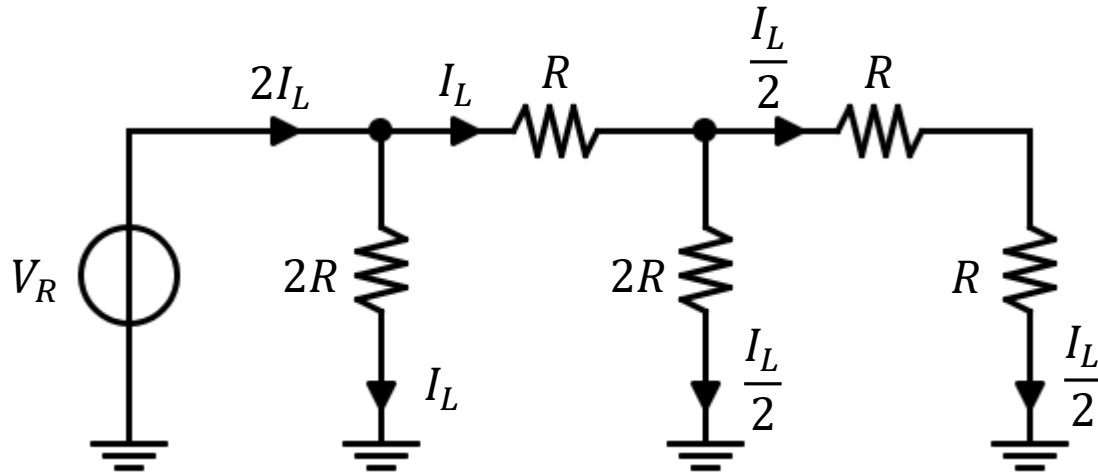
Digital to Analog Conversion

- DAC Circuit with Ladder Network
 - Split a current in two equal parts
 - Use a current divider with equal resistances on the two branches
 - $I_L = \frac{V_R}{2R}$



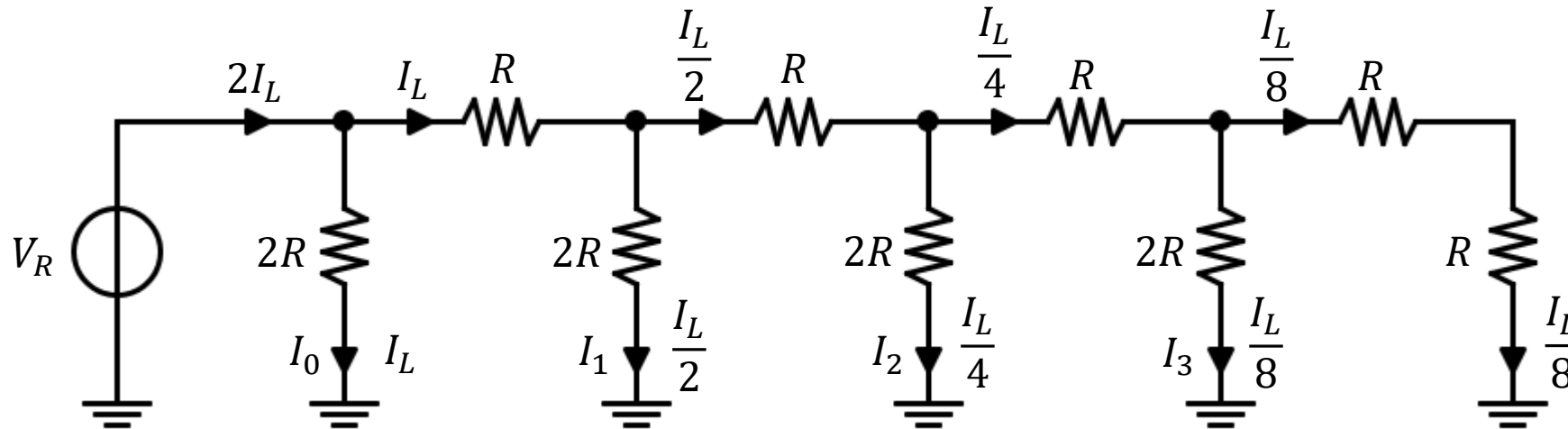
Digital to Analog Conversion

- DAC Circuit with Ladder Network
 - Repeat splitting in two equal parts
 - Use a current divider with equal resistances on the two branches
 - $I_L = \frac{V_R}{2R}$



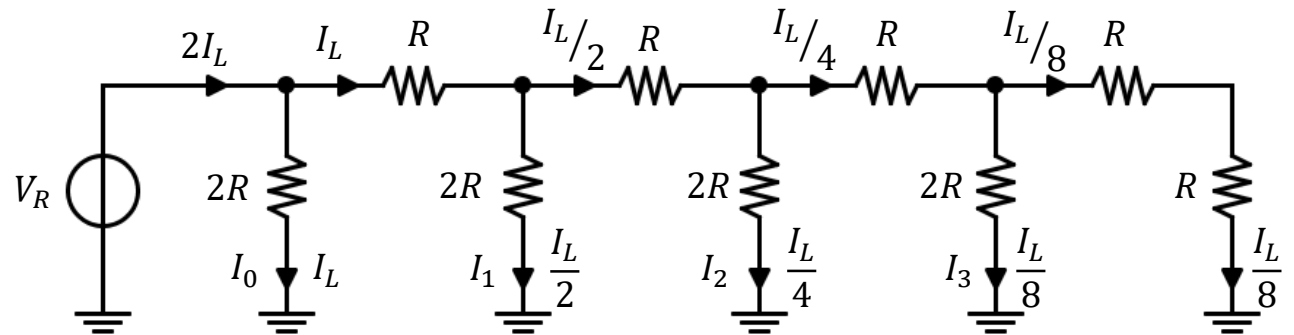
Digital to Analog Conversion

- DAC Circuit with Ladder Network
 - Continue splitting in two equal parts
 - Keep using a current divider with equal resistances on the two branches
 - $I_L = \frac{V_R}{2R}$, $I_i = \frac{I_L}{2^i}$



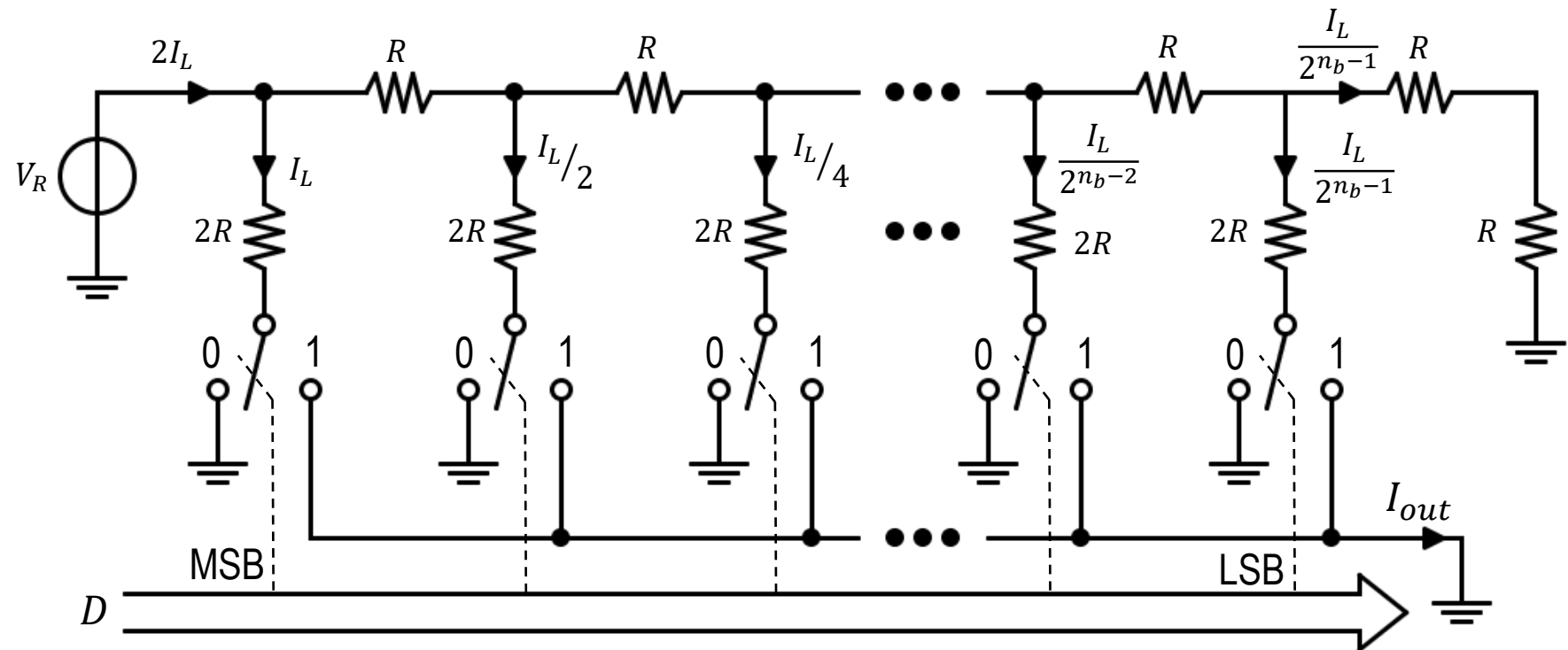
Digital to Analog Conversion

- DAC Circuit with Ladder Network
 - Generates a set of currents
 - Ratio between currents is a power of 2
 - The last (right-hand side in the picture) branch is not used
 - Number of resistances is linear on the number of bits
 - $N_R = 2 \cdot n_b + 1$
 - Twice as many as in the weighted resistances converter
 - Dynamic range of resistance values is independent of the number of bits
 - Values are R and $2R$
 - Same technology, same behavior
 - Ladder network is modular
 - Any number of bits



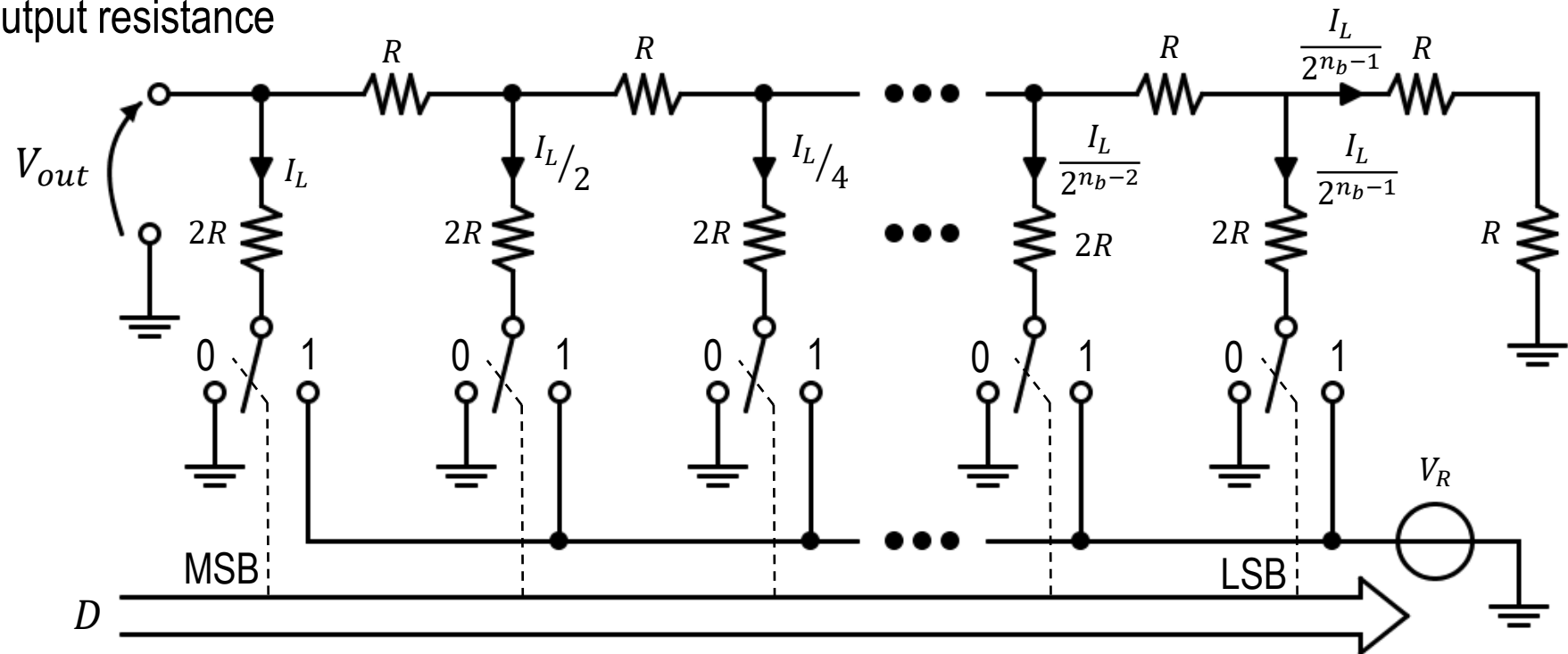
Digital to Analog Conversion

- DAC Circuit with Ladder Network
 - Converter using current switches
 - Steer current between equipotential nodes
 - Output is a current
 - Output has a varying equivalent output resistance
 - To get a voltage output, use a transresistance amplifier



Digital to Analog Conversion

- DAC Circuit with Ladder Network
 - Converter using voltage switches
 - Exchange voltage reference and output short circuit
 - Same transfer function
 - Constant equivalent output resistance
 - The short circuit output can be turned into a voltage output



Digital to Analog Conversion

- DAC Circuit Errors

- Changes in the voltage reference V_R
 - Systematic error in the weight network
 - Gain error
- Leakage current of switches when turned off
 - Non zero current even when $D = 0$
 - Offset error
- Offset of operational amplifier
 - If used as a transresistance amplifier
 - Offset error

Digital to Analog Conversion

- DAC Circuit Errors

- Non zero equivalent ON resistance of switches (R_{DSon})
 - Modifies the total branch resistance
 - Modifies the branch current
 - Similar to branch resistance tolerance
 - If constant $\rightarrow$ gain error
 - In weighted networks
 - Small effect on high resistances (LSBs)
 - Larger effect on small resistances (MSBs)
 - In uniform networks
 - Same effect on all branches
 - Sums up with uniform quantities
 - If not constant $\rightarrow$ non linear errors

Digital to Analog Conversion

- DAC Circuit Errors

- Branch relative error B_{Err} in weighted networks
 - The resistance value is not ideal
 - Each branch contributes to the output only when the corresponding bit is at 1
 - Branches have different weight B_W
 - MSB weight is $\frac{1}{2}$
 - MSB-1 weight is $\frac{1}{4}$
 - ...
 - $\varepsilon_{out} = B_{Err} \cdot B_W$
 - The same relative error in different branches causes different error at the output
 - Higher effect on MSBs
 - Differential non linearity
- MSB branches should be more precise

Digital to Analog Conversion

- DAC Circuit Errors

- Branch error B_{Err} in weighted networks examples

- Error on MSB branch

- Real contribution higher than ideal
 - Branch resistance lower than ideal
 - Raised upper half of characteristic
 - Branch error 10%: $\rightarrow$ output error 5%

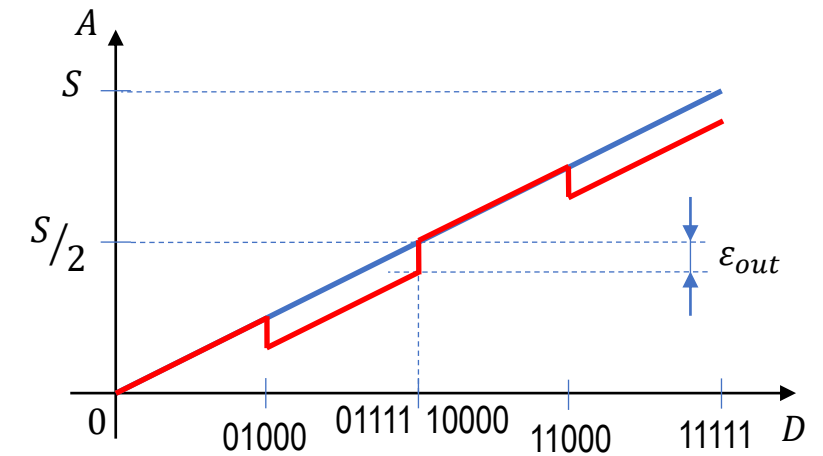
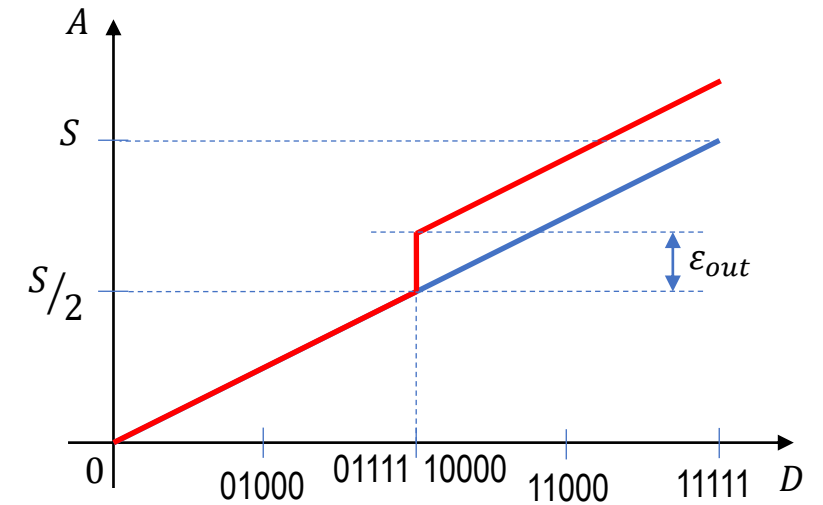
- Error on MSB - 1 branch

- Real contribution lower than ideal
 - Branch resistance higher than ideal
 - Lowered odd quarters (1, 3) of characteristic
 - Branch error 10%: $\rightarrow$ output error 2.5%

- Error on MSB - 2 branch

- Branch error 10%: $\rightarrow$ output error 1.25%

- ...



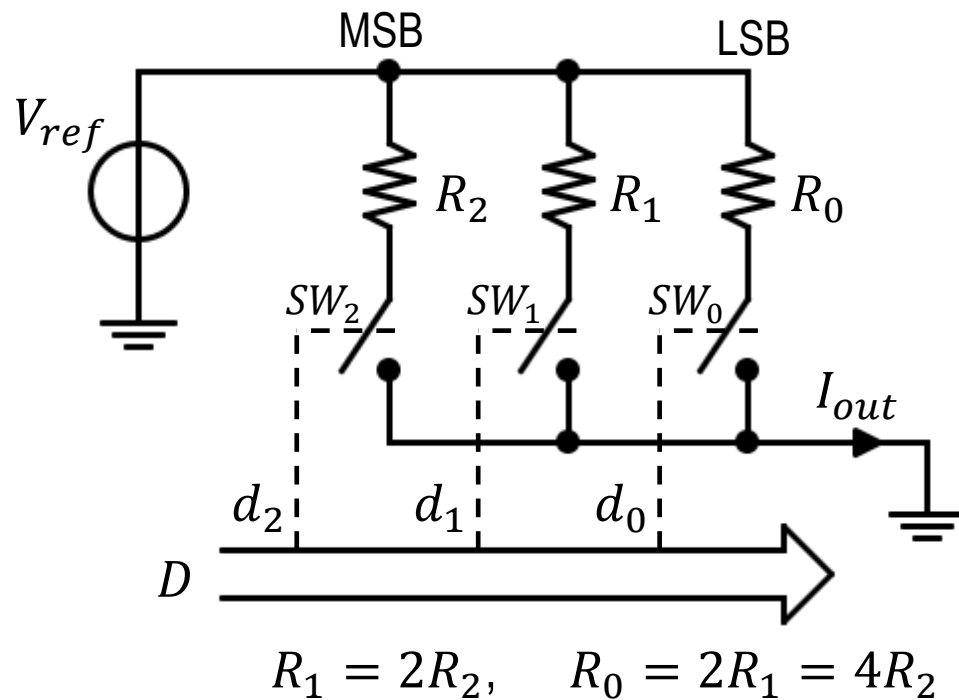
Digital to Analog Conversion

- DAC Circuit Errors
 - Branch error B_{Err} in uniform networks
 - The resistance value is not ideal
 - Branches have the same weight (1 LSB)
 - The same relative error in different branches causes the same error at the output
 - Intrinsically monotonic output
 - Sum of elements with the same sign
 - Errors of all branches sum at the output
 - Integral non linearity
 - All branches can have the same precision

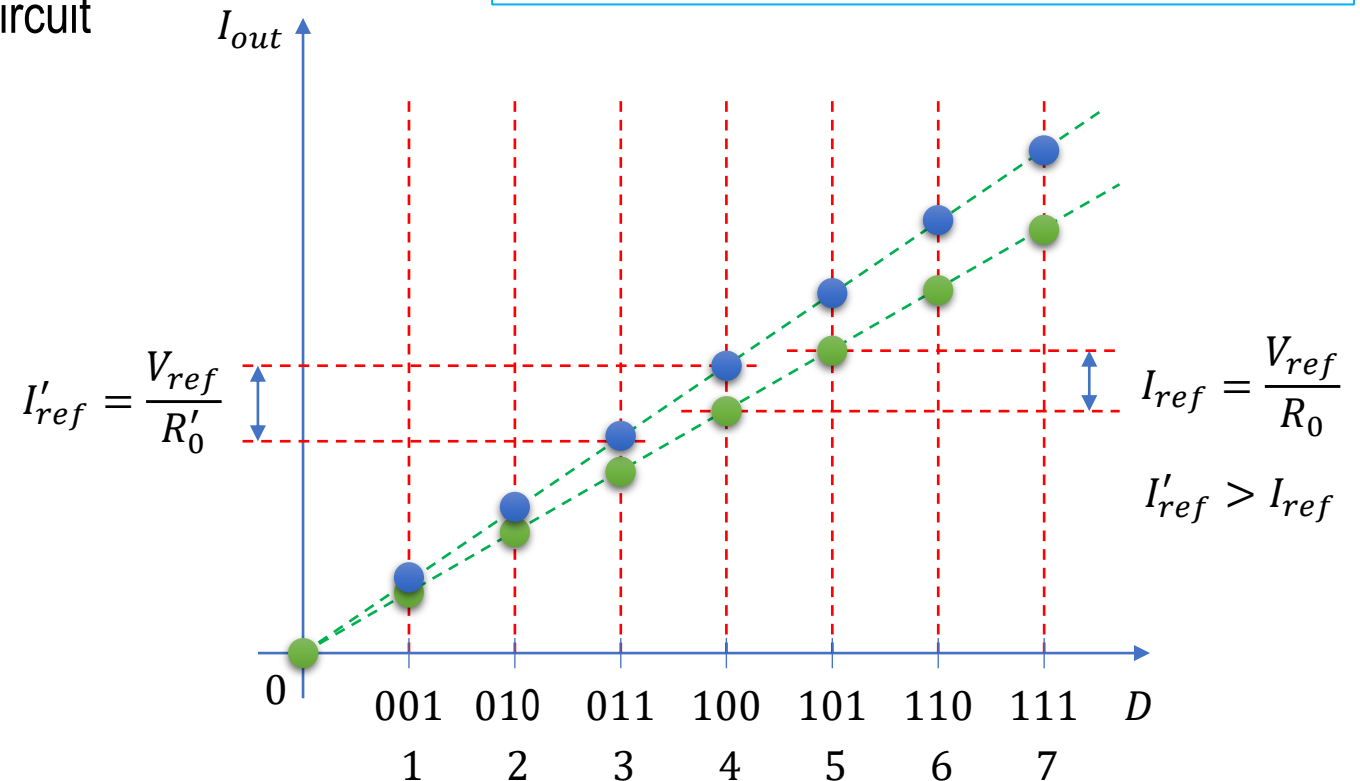
Digital to Analog Conversion

- DAC Circuit Errors example

- All resistances have an error of -2%
 - Not a tolerance, exactly the same relative error in all of them
 - Example with weighted resistance DAC circuit



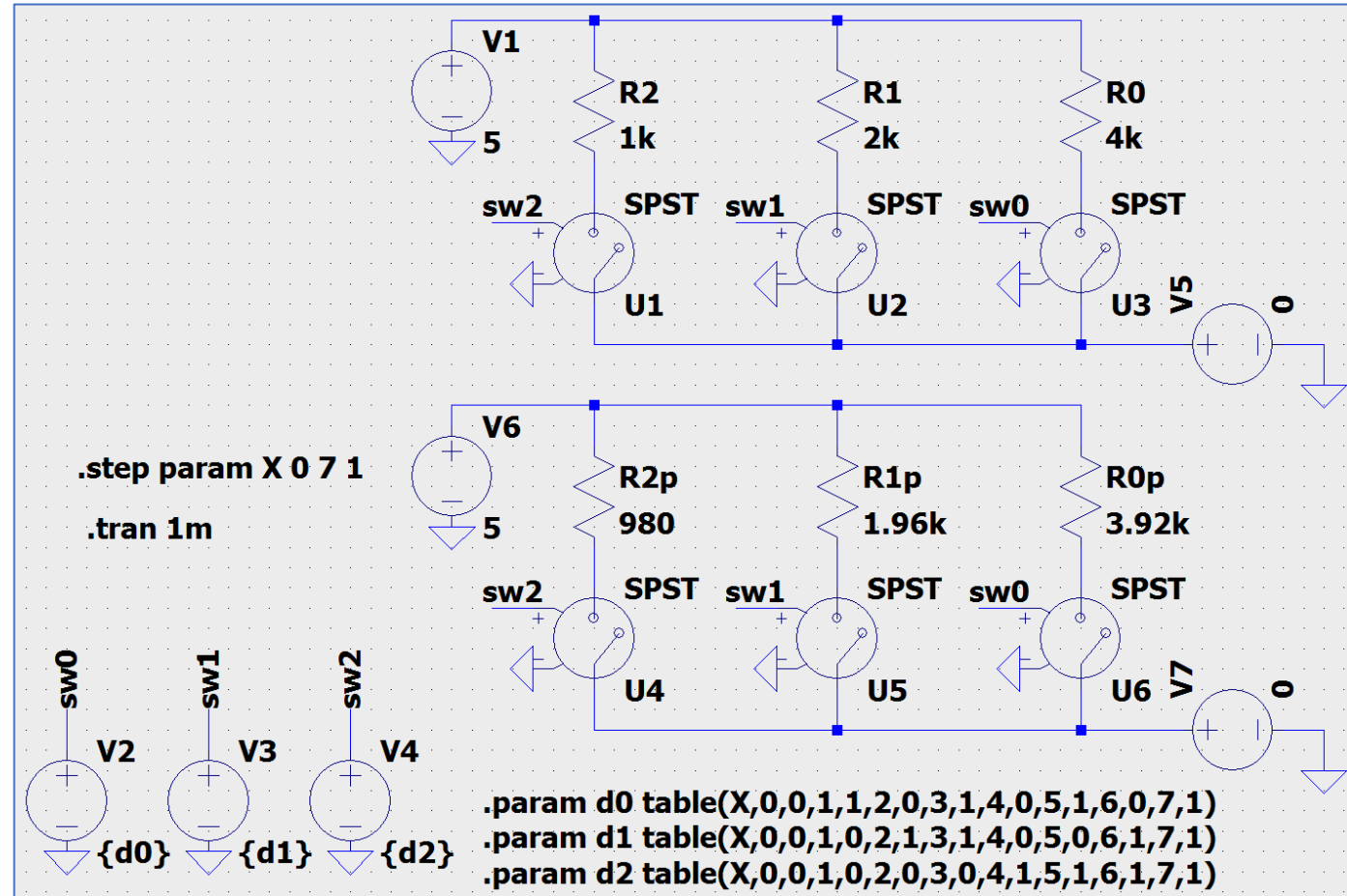
$$\begin{aligned} R'_2 &= 0,98 \cdot R_2 \\ R'_1 &= 0,98 \cdot R_1 = 2R'_2 \\ R'_0 &= 0,98 \cdot R_0 = 2R'_1 = 4R'_2 \end{aligned}$$



Digital to Analog Conversion

- DAC Circuit Errors example
 - All resistances have an error of -2%
 - Not a tolerance, exactly the same relative error in all of them
 - Example with weighted resistance DAC circuit

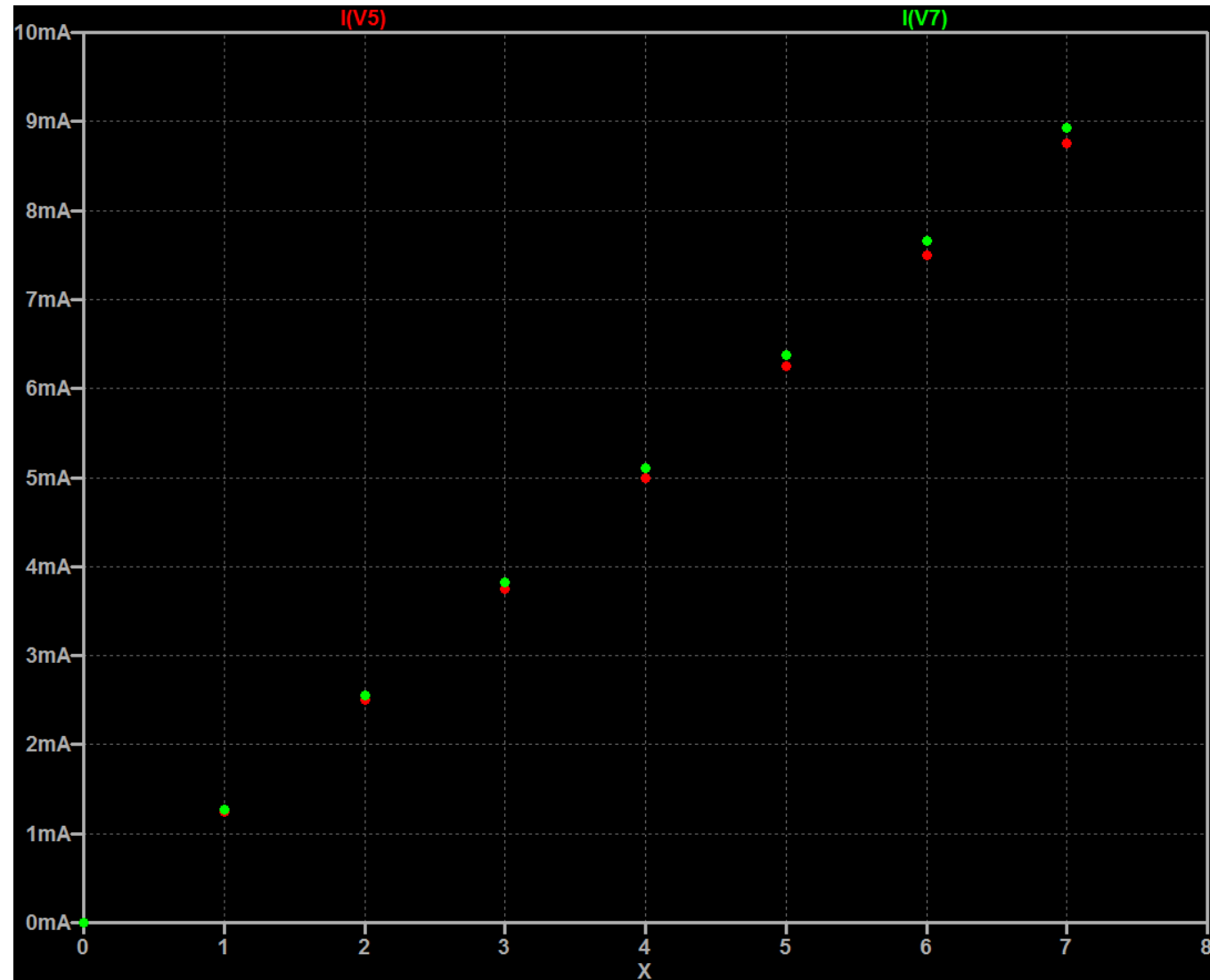
- The voltage sources $V2$, $V3$ e $V4$ represent the input bits and drive the SPST switches
- The X parameter goes from 0 to 7 with step 1
- The table defines parameters $d0$, $d1$ e $d2$ as a function of X
- I inserted voltage sources $V5$ and $V7$ to measure the short circuit current
 - Being $0V$, they behave as short circuits
 - It is possible to select their current to show in the simulation



Digital to Analog Conversion

- DAC Circuit Errors example
 - All resistances have an error of -2%
 - Not a tolerance, exactly the same relative error in all of them
 - Example with weighted resistance DAC circuit

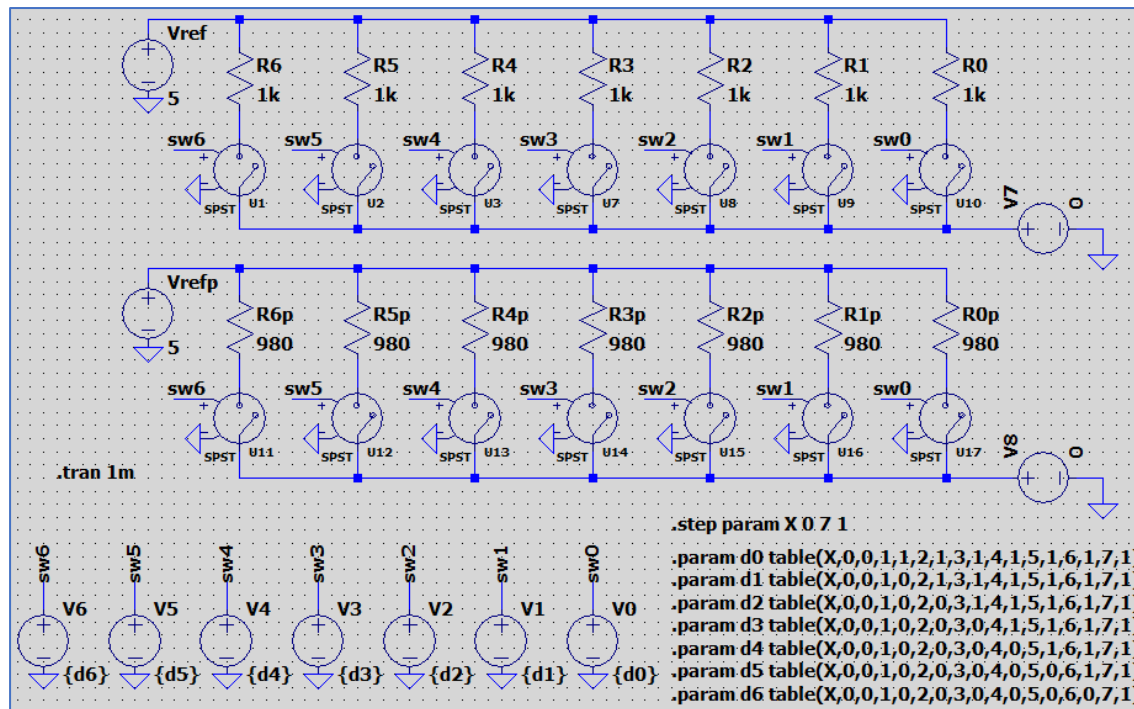
- Simulation time is not important
 - All sources are in DC, I used 1ms
 - You can try a DC simulation
- Steps in LTSpice waveform display
 - Select the X parameter as the independent variable on the horizontal axis
 - Modify the horizontal axis to go from 0 to 8 with steps of 1
 - Modify the vertical axis to go from 0mA to 10mA
 - Click with the right mouse button on the graph and select View → Mark Data Points



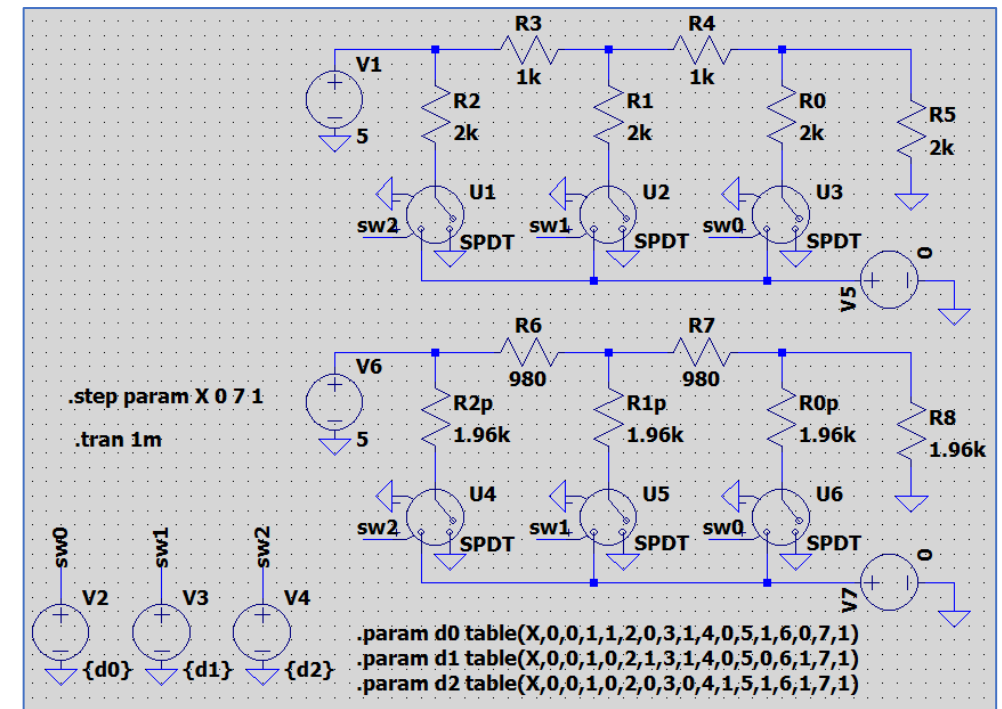
Digital to Analog Conversion

- DAC Circuit Errors example
 - All resistances have an error of -2%
 - Examples with uniform resistance and ladder network DAC circuits
 - Same simulation results

Uniform resistances

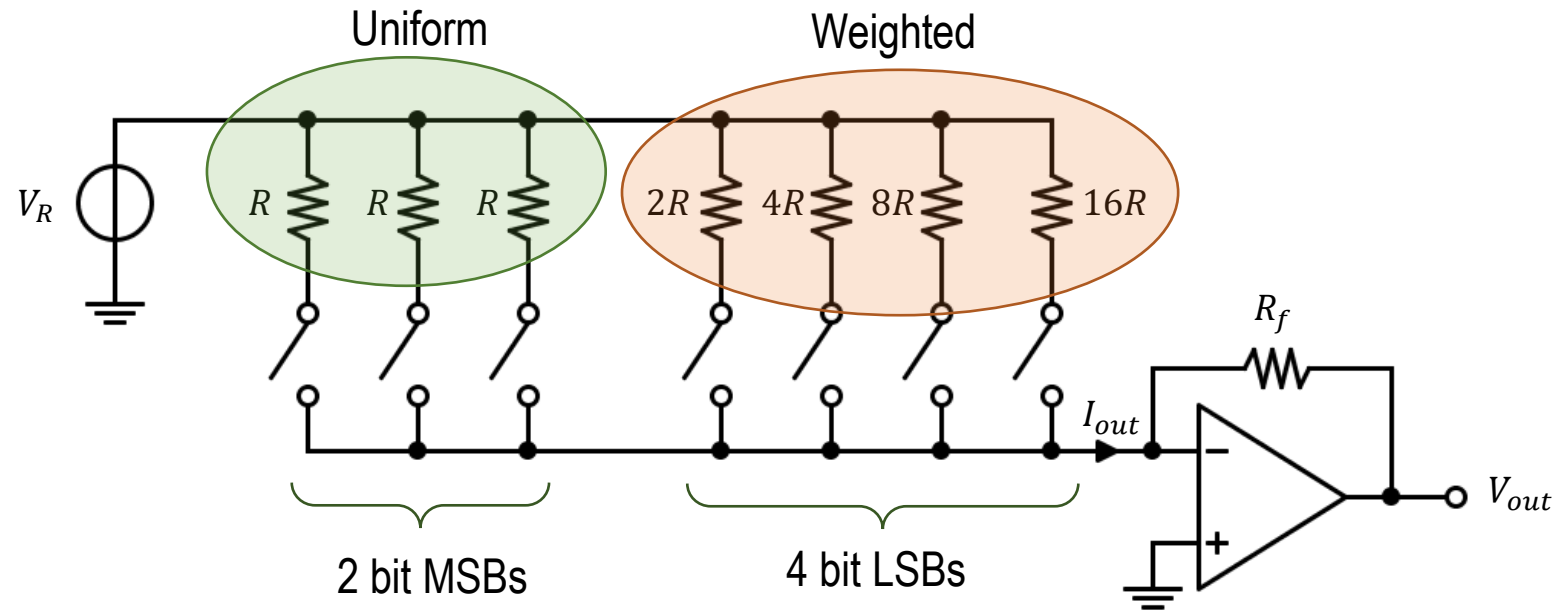


Ladder network



Digital to Analog Conversion

- DAC Circuit Mixed Structures
 - Use uniform branches for MSBs
 - Use weighted branches for LSBs
 - Benefits of both structures
 - Simple
 - No need for high precision
- Example
 - 6-bit DAC
 - 2 MSBs with uniform quantities
 - 4 LSBs with weighted quantities



Digital to Analog Conversion

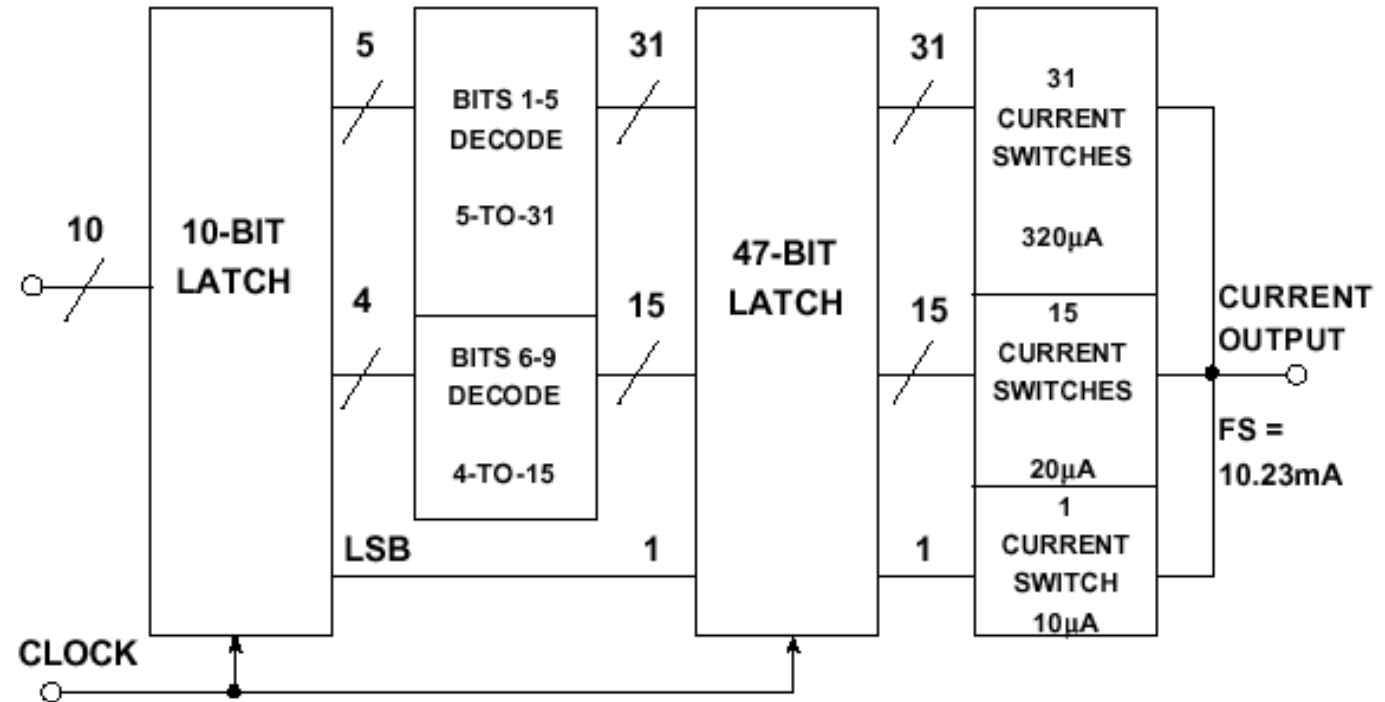
- DAC Circuit Split Structures

- Uniform quantities given a certain weight
- Different weight between split blocks

- Example

- 10-bit DAC
- LSB with uniform quantities
 - Elementary current is 10 μA
- 4 following bits with uniform quantities
 - Elementary current is 20 μA
 - It is twice ($2^1 = 2$) greater than LSB
 - 15 branches
- 5 MSBs with uniform quantities
 - Elementary current is 320 μA
 - It is 32 ($2^5 = 32$) times greater than LSB
 - 31 branches

AD9850 10-BIT CMOS CURRENT SWITCH DAC CORE



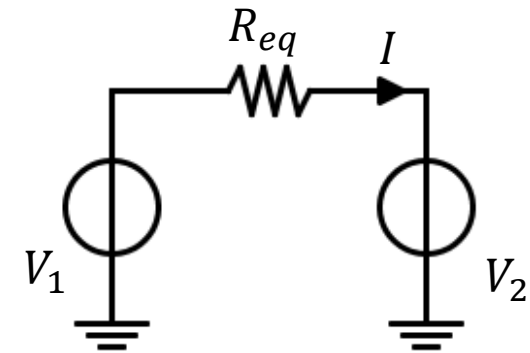
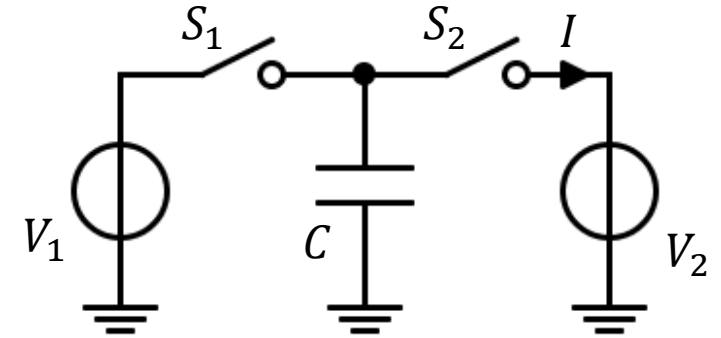
Digital to Analog Conversion

- DAC Circuit with Switched Capacitors

- Resistors are expensive in Integrated Circuits
 - Capacitors are much cheaper

- Switched capacitor circuit

- Switches S_1 and S_2 are in opposite phases
 - Driven alternatively with period T , frequency f
- When S_1 is closed and S_2 is open
 - $Q_1 = C \cdot V_1$
- When S_1 is open and S_2 is closed
 - $Q_2 = C \cdot V_2$
- Charge difference ΔQ is the variation of the quantity of charge transferred from V_1 to V_2
 - $\Delta Q = Q_1 - Q_2 = C(V_1 - V_2)$
- Current I is the charge transferred from V_1 to V_2 in a second
 - $I = \frac{\Delta Q}{T} = \frac{C(V_1 - V_2)}{T} = Cf(V_1 - V_2) = \frac{V_1 - V_2}{R_{eq}}$ with $R_{eq} = \frac{1}{Cf}$



Digital to Analog Conversion

- DAC Circuit with Switched Capacitors

- Values of the capacitors

- MSB

- High current, low resistance
 - High capacitance

- LSB

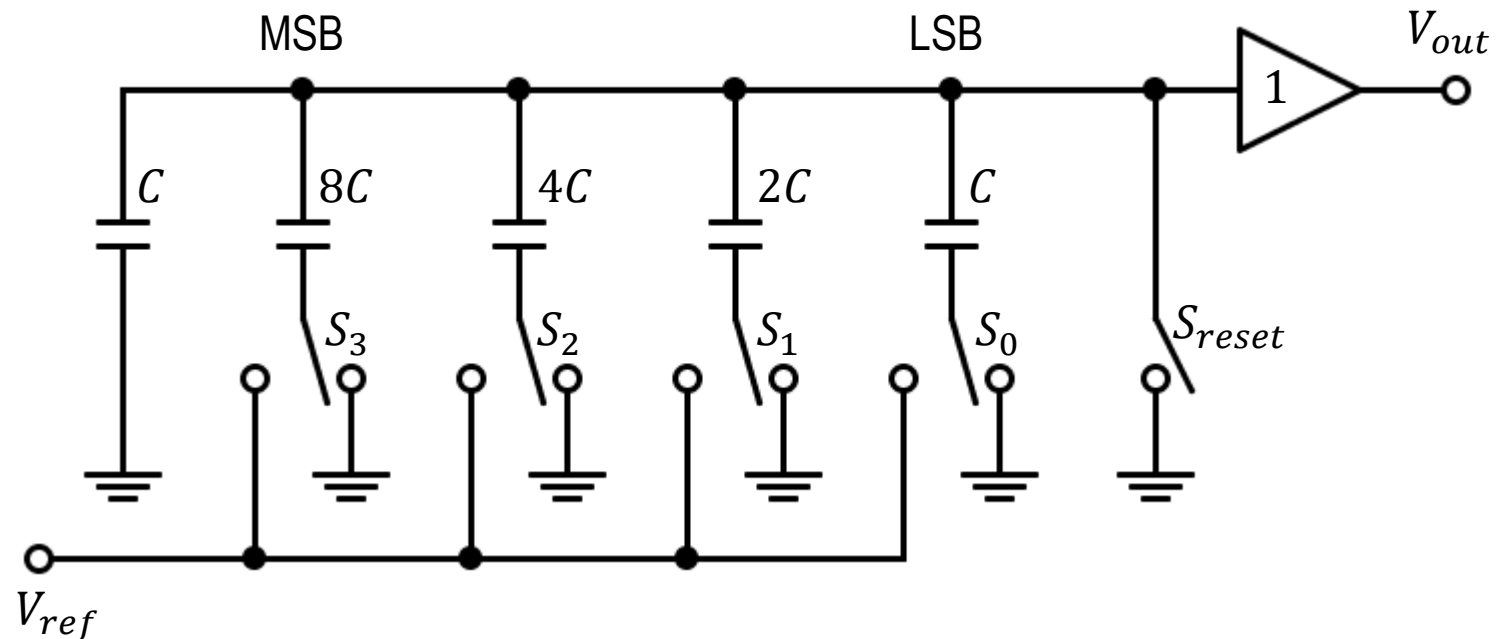
- Low current, high resistance
 - Low capacitance

- Phase 1

- S_{reset} closed
 - S_i towards GND

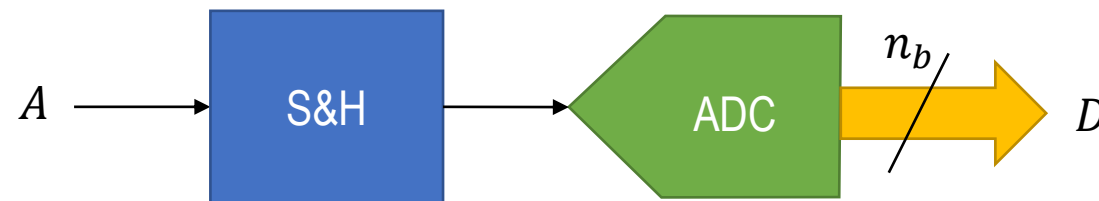
- Phase 2

- S_{reset} open
 - S_i towards GND if $d_i = 0$
 - S_i towards V_{ref} if $d_i = 1$



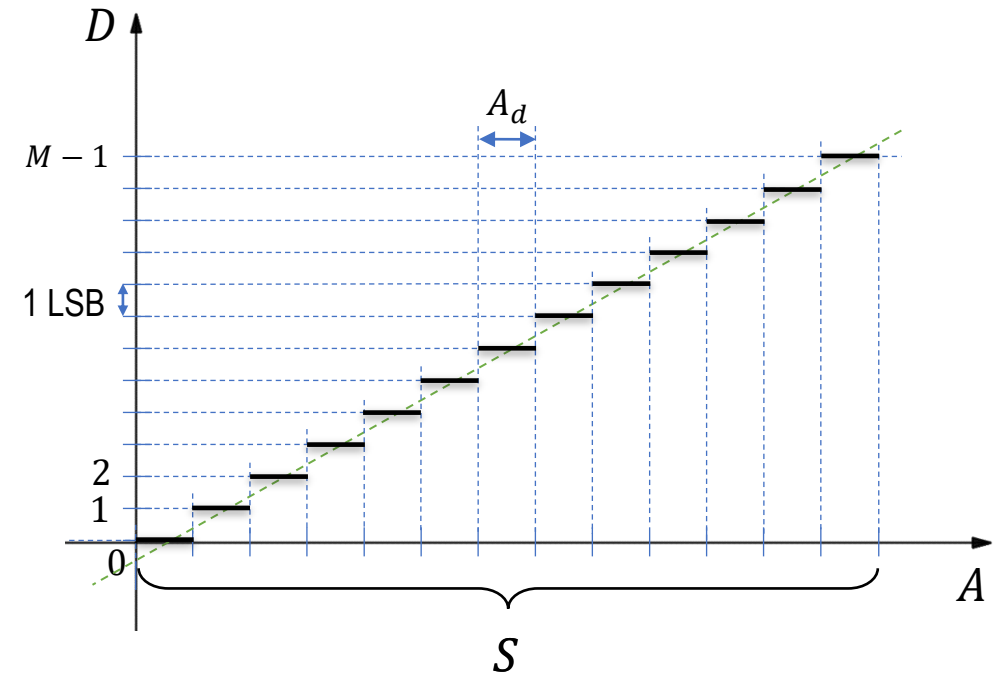
Analog to Digital Conversion

- Sampling and quantization
 - From continuous time to discrete time
 - Sample & Hold
 - Analog signal kept constant during the entire quantization phase
 - Not identical to Dirac comb multiplication
 - Spectrum altered, can be compensated
 - From continuous values to discrete values
 - On n_b binary wires
 - Analog to Digital Converter (ADC) circuit



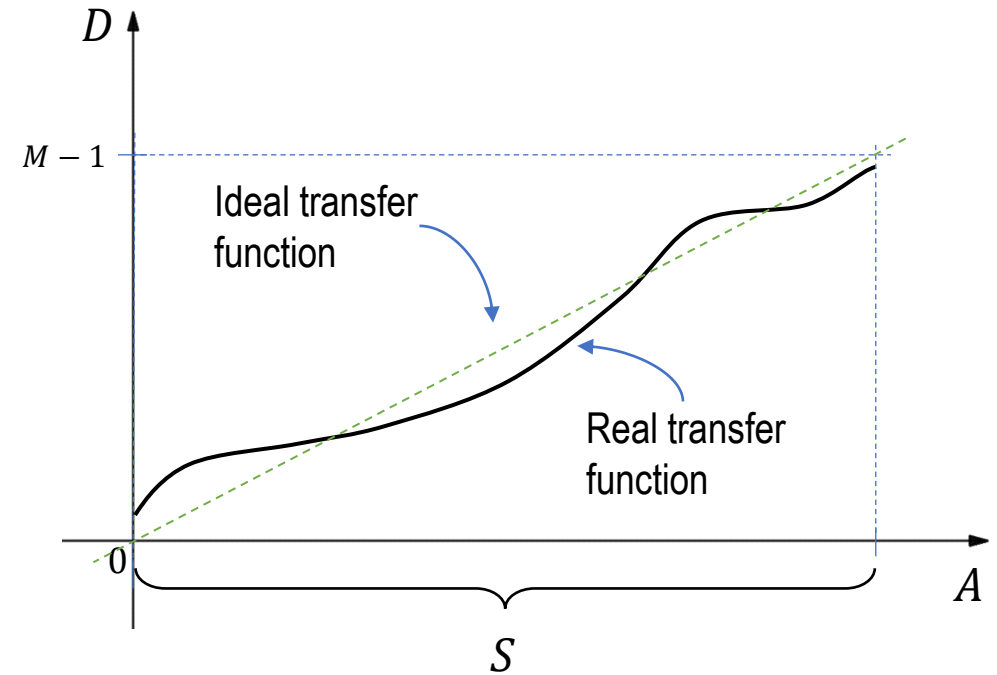
Analog to Digital Conversion

- ADC Transfer Function
 - Computed statically
 - Constant input
 - Steady state behavior
 - Input is analog
 - Dynamic range S
 - Output is discrete
 - M values
 - $M = 2^{n_b}$
 - Transfer function is a staircase
 - With a constant A_d the steps are aligned
 - Uniform conversion
 - With a large M the steps sequence becomes a continuous straight line



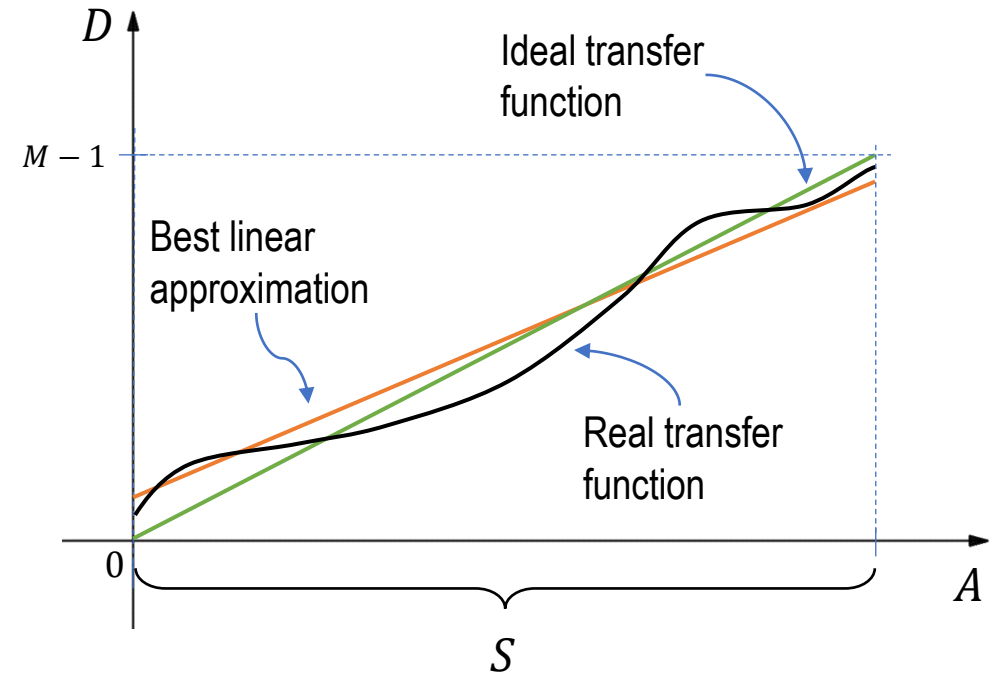
Analog to Digital Conversion

- ADC Transfer Function
 - A real ADC shows deviations from the ideal transfer function
- Static errors
 - Constant input
 - Errors appear in the $D(A)$ transfer function
- Dynamic errors
 - Transient behavior
 - Errors appear in $D(t)$
 - Delay of the converted value



Analog to Digital Conversion

- ADC Transfer Function
 - Static errors
 - The real transfer function is not a perfect straight line
 - Find the best linear approximation of the real transfer function
 - Compare the best linear approximation with the ideal transfer function
 - → Linear errors
 - Compare the real transfer function with the best linear approximation
 - → Non linear errors



Analog to Digital Conversion

- ADC Transfer Function Linear Errors

- Compare the best linear approximation to the ideal transfer function

- Offset error ε_o

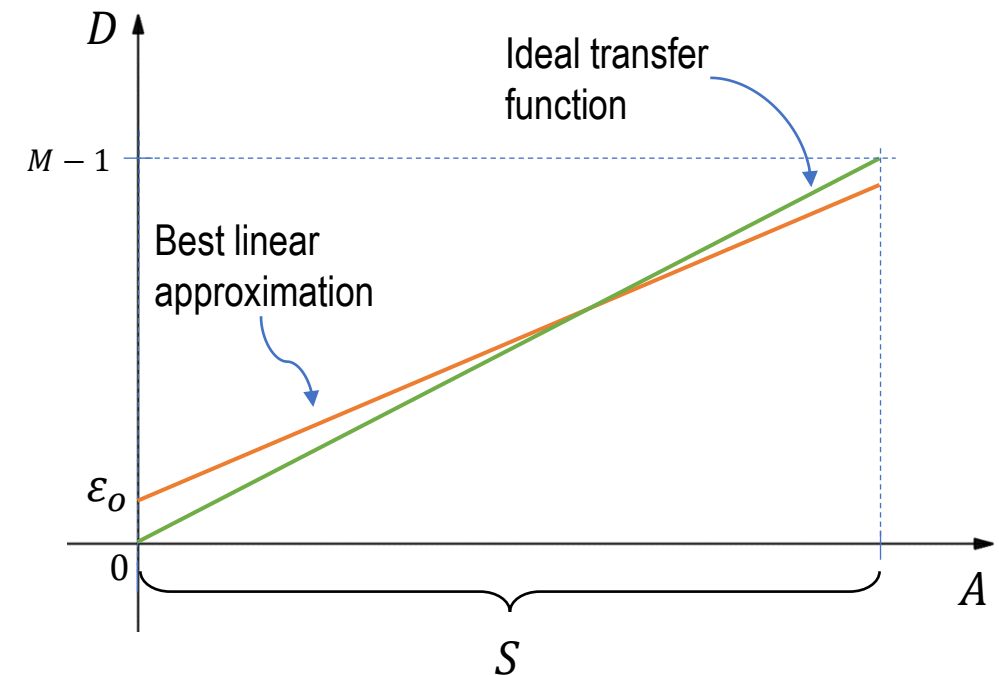
- Intercept of approximating line with D axis
- Can be compensated by adding a constant

- Gain error ε_g

- Difference among the ideal slope and the best linear approximation slope
- Can be compensated by gain correction

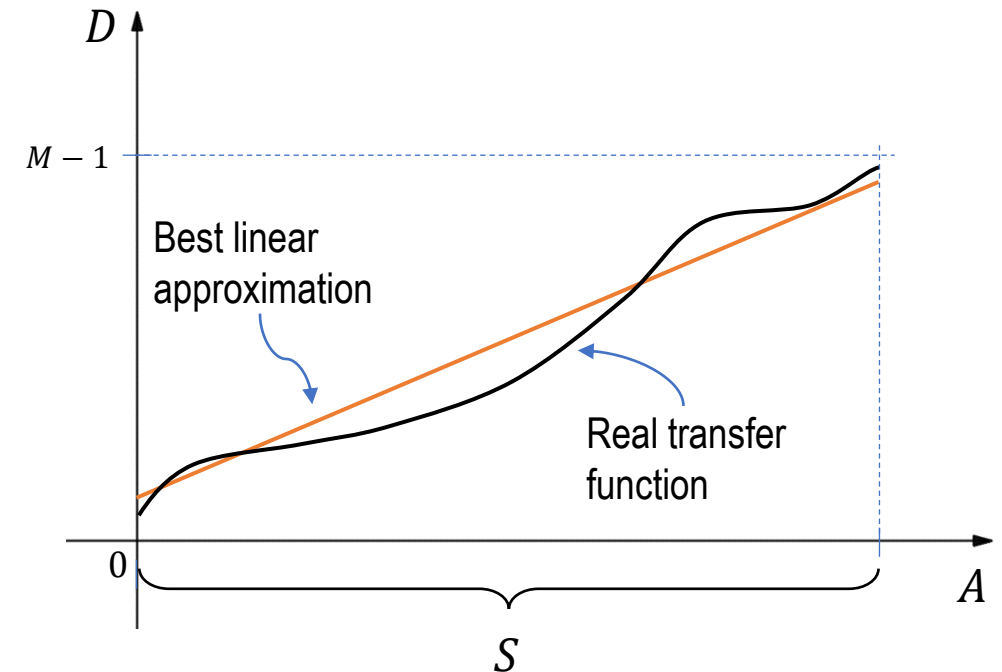
- Compensation is expensive

- Need to characterize and compensate each single unit separately
- Compensation in the digital domain



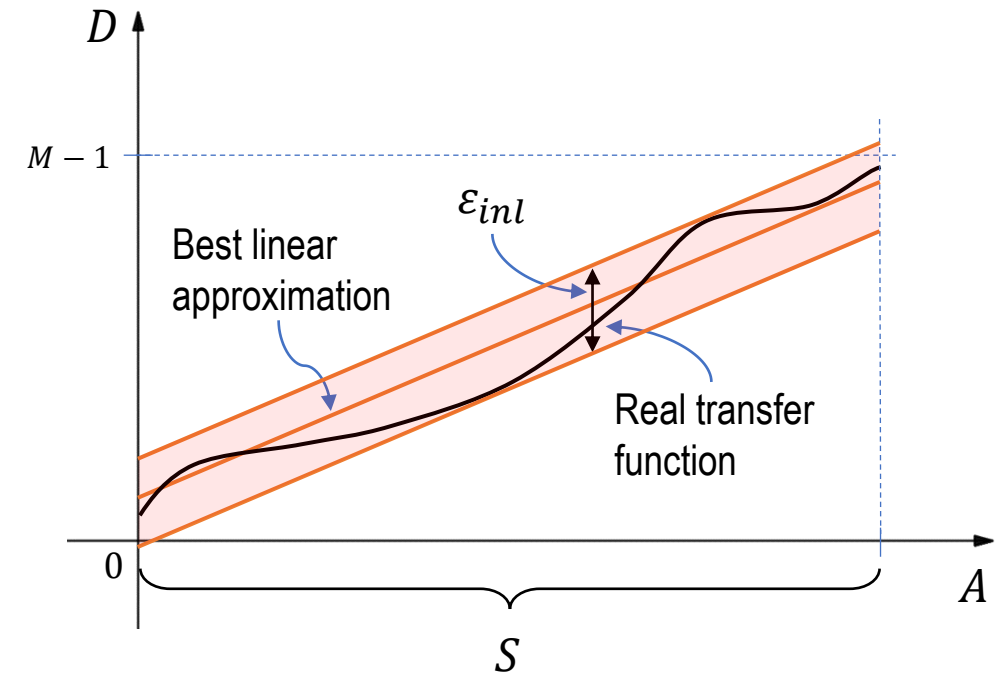
Analog to Digital Conversion

- ADC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Linear errors are factored out
 - Integral non linearity
 - Maximum deviation of the real transfer function from the best linear approximation
 - Global value
 - Differential non linearity
 - Amplitude of analog value steps
 - Local value
 - Need to give 2^{n_b} different data
 - Often only an upper bound is given



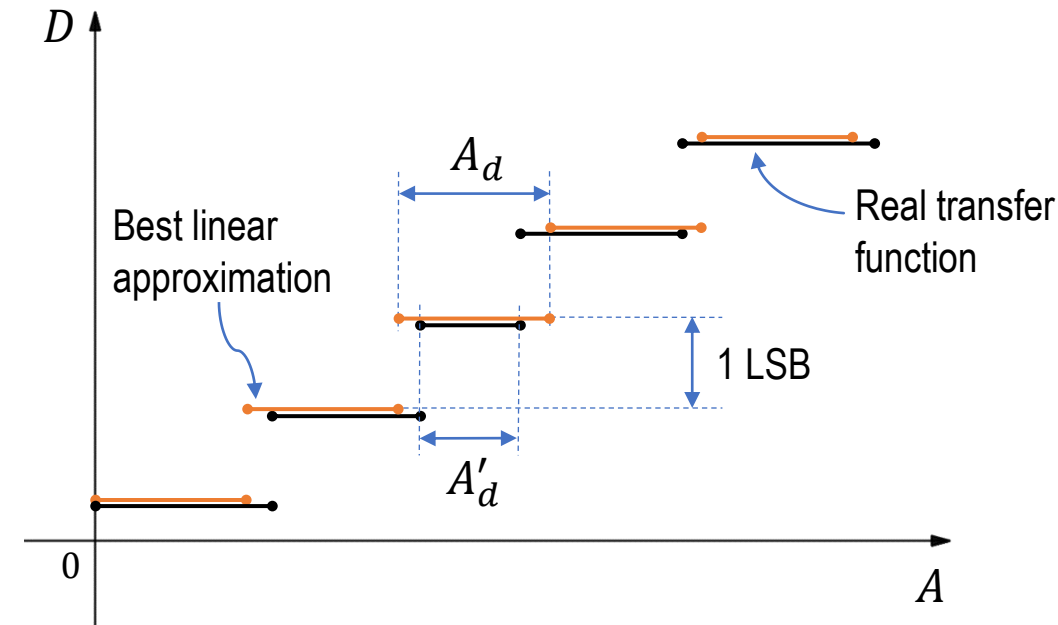
Analog to Digital Conversion

- ADC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Integral non linearity
 - Maximum deviation of the real transfer function from the best linear approximation
 - Global value
 - Measured along the vertical (digital) axis
 - As a fraction of an LSB
 - Cannot be compensated



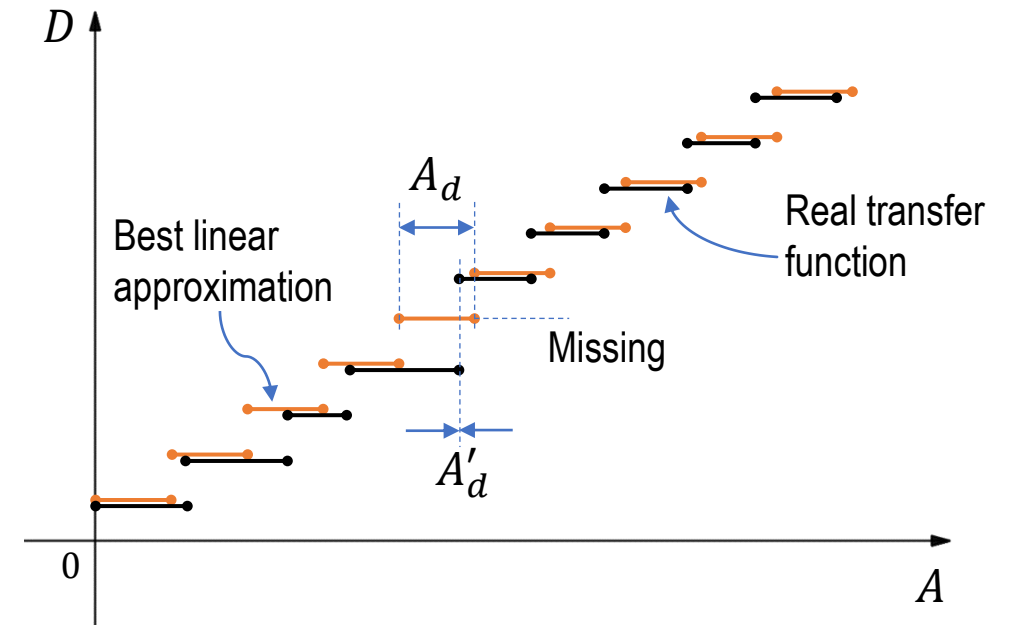
Analog to Digital Conversion

- ADC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Differential non linearity
 - Amplitude of analog value steps
 - Local value
 - Need to give 2^{n_b} different data
 - Often only an upper bound is given
 - $A'_d \neq A_d$
 - $\varepsilon_{dnl} = A_d - A'_d$
 - Measured along the horizontal (analog) axis
 - As an absolute value with the unit of measure of the analog quantity
 - As a fraction of an LSB
 - Cannot be compensated



Analog to Digital Conversion

- ADC Transfer Function Non Linear Errors
 - Compare the real transfer function to the best linear approximation
 - Differential non linearity
 - Amplitude of analog value steps
 - Missing code
 - $A'_d = 0$
 - $\varepsilon_{dnl} = A_d - A'_d = 1 \text{ LSB}$
 - A given digital value never appears



Analog to Digital Conversion

- ADC Transfer Function Dynamic Errors
 - Conversion Time T_{conv}
 - Time it takes to the ADC output to be stable after the start of conversion
 - In most cases
 - The ADC receives a “Conversion Start” (CS) command
 - After T_{conv} the ADC raises an “End Of Conversion” (EOC) flag
 - Can be expressed as a conversion frequency
 - $f_{conv} = \frac{1}{T_{conv}}$
 - But it is different from the sampling frequency f_S
 - It directly affects the sampling period T_S
 - It must be $T_S > T_{conv}$
 - Unless a pipeline is used
 - It sets an upper bound to the sampling frequency f_S
 - $f_S < f_{conv}$

Analog to Digital Conversion

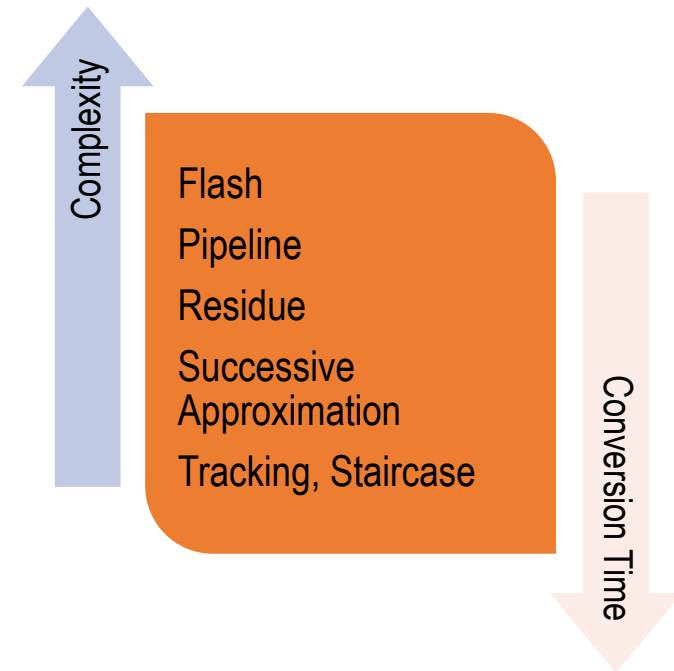
- ADC Transfer Function Dynamic Errors
 - Tracking rate SR
 - Some ADCs can follow a (slowly) changing analog signal
 - Tracking converters
 - Conversion is continuous
 - As soon as one sample is obtained, the next conversion is started
 - Don't need Conversion Start and End Of Conversion signals
 - Dynamic behavior specified by the maximum track rate
 - It corresponds to a slew rate
 - $SR = \frac{dV(t)}{dt}$

Analog to Digital Conversion

- ADC Circuits
 - Input is an analog electrical quantity
 - A voltage
 - A current
 - Output is a digital signal
 - A binary value on n_b wires, with $M = 2^{n_b}$ possible different combinations
 - Expressed as an integer D from 0 to $M - 1$
 - Circuit
 - Compare the input signal to a number of thresholds
 - Identify the interval of the input signal within the input dynamic range
 - Select the corresponding digital value

Analog to Digital Conversion

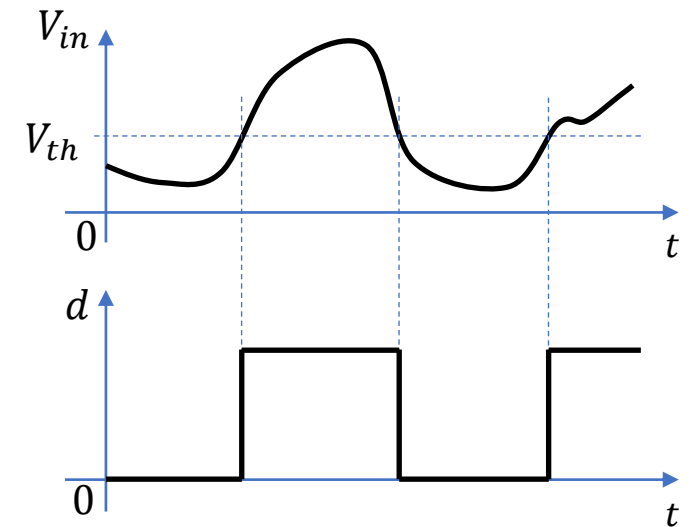
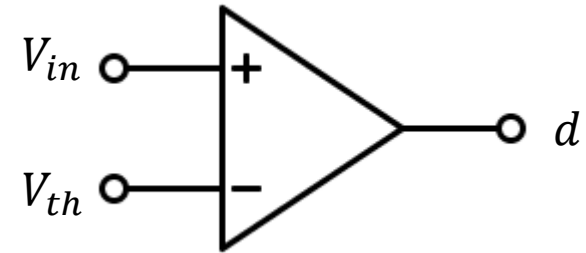
- ADC Circuits
 - Various types of circuits
 - Classified based on
 - Complexity
 - Number of comparators in the circuit
 - Better if fewer comparators
 - Cheaper, smaller area
 - Conversion time T_{conv}
 - Or conversion rate f_{conv} , inverse of T_{conv}
 - Better if low T_{conv}
 - High speed, high f_{conv}
 - High speed converters are more complex
 - High speed and high resolution are expensive



Analog to Digital Conversion

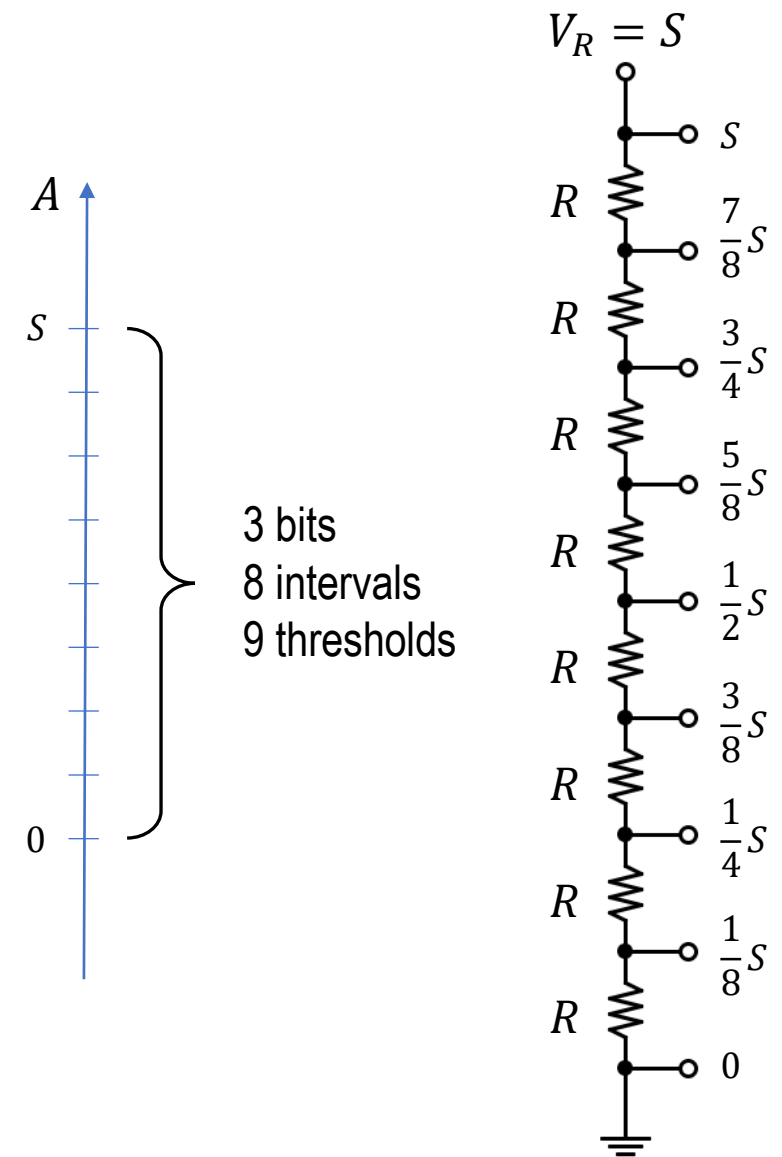
- ADC Circuits

- Compare the analog input signal with a number of thresholds
 - Each couple of adjacent thresholds identifies a digital value
 - There are $2^{n_b} + 1$ possible thresholds
 - Comparators implemented using operational amplifiers
- Different techniques differ for the following reasons
 - Number of thresholds compared concurrently
 - All thresholds at once in parallel (Flash)
 - One at a time (Tracking, SAR, Residue)
 - Multiple at a time (Multibit Residue)
 - Number of samples converted at once
 - One at a time (Flash, Tracking, SAR, Residue)
 - Several at once (Pipeline)



Analog to Digital Conversion

- ADC Circuits
 - Generate thresholds for the comparators
 - Each threshold is an analog quantity
 - Uniform conversion
 - Voltage divider with multiple identical resistances
 - Absolute value of resistances not critical
 - Ratio between resistance is important
 - Easy to control in integrated circuits
 - Non uniform conversion (logarithmic, ...)
 - Voltage divider with different resistances
 - Alternative
 - Use a logic circuit and Digital-to-Analog Converters to generate analog thresholds



Analog to Digital Conversion

- ADC Circuits

- 1-bit converter

- 2 intervals ($0 \div S/2, S/2 \div S$)
 - 3 thresholds ($0, S/2, S$)

- Don't need to compare against

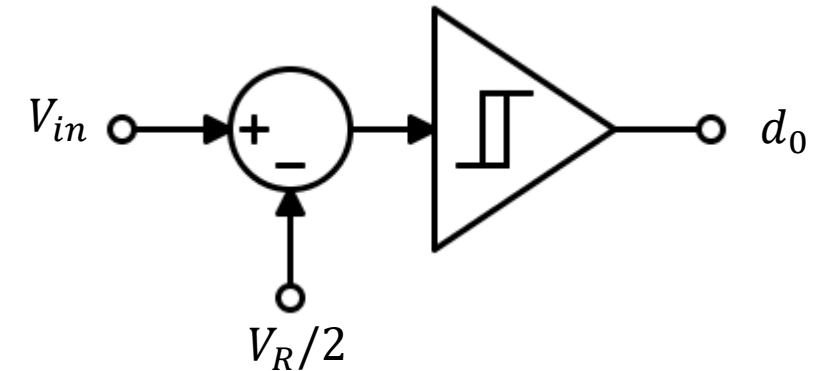
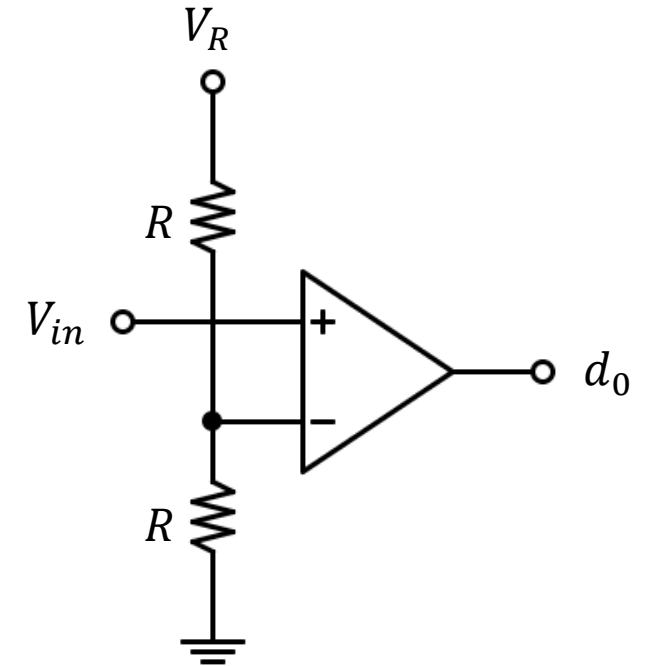
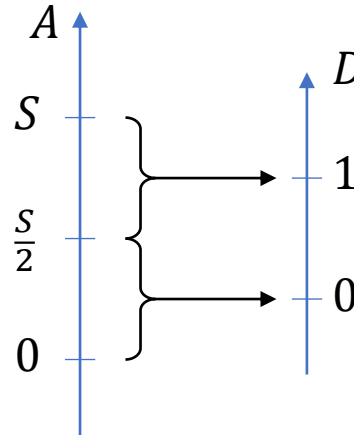
- Lowest threshold (0)
 - Highest threshold (S)
 - Saturation if analog input signal is outside the input dynamic range

- Compare against

- Threshold at $S/2$

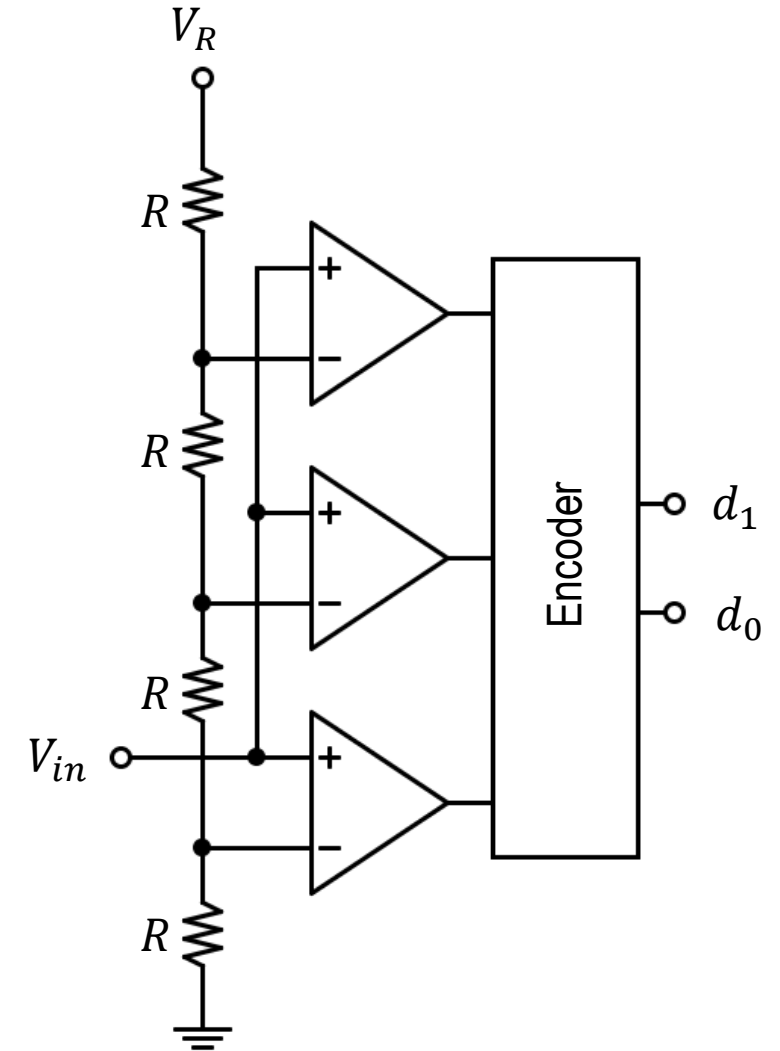
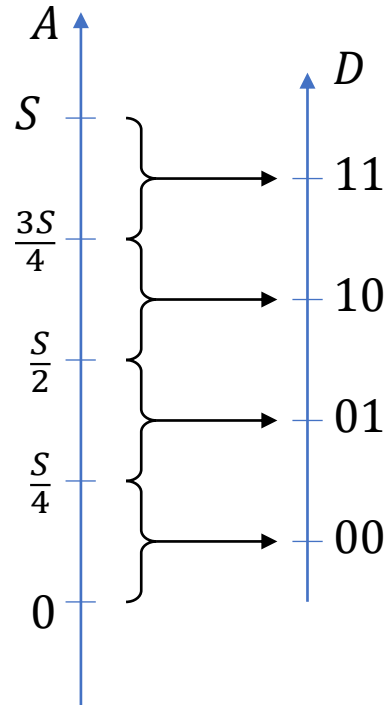
- Conversion time

- One conversion cycle



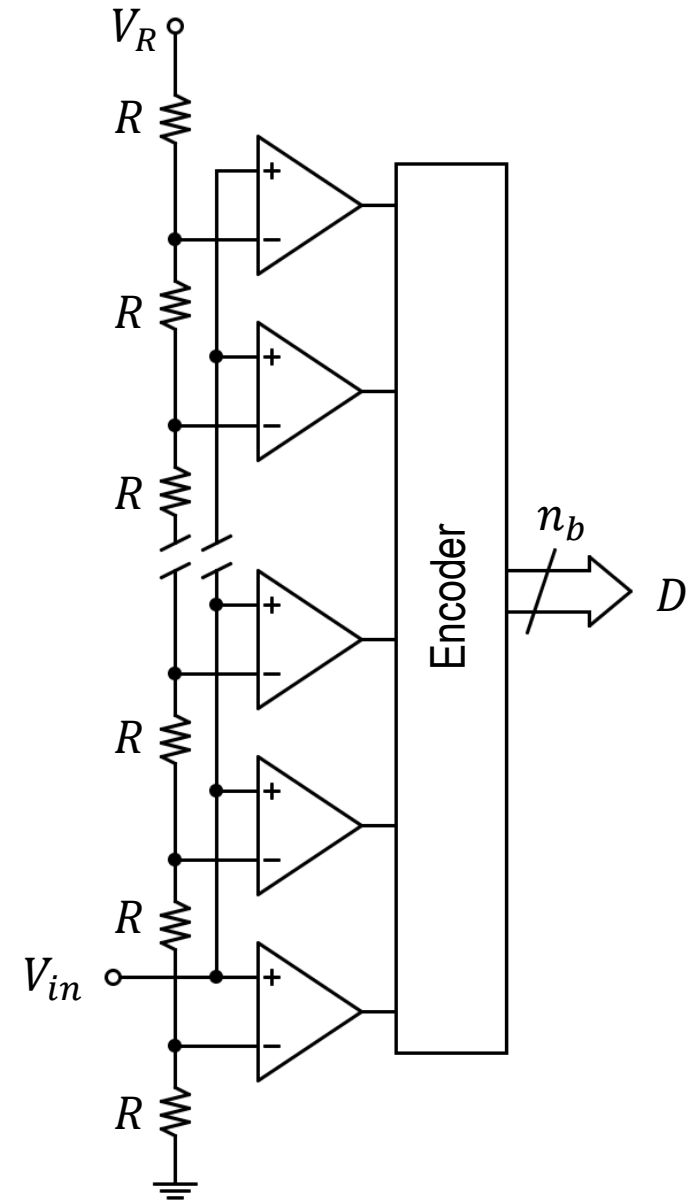
Analog to Digital Conversion

- ADC Circuits
 - 2-bit converter
 - 4 intervals
 - 5 thresholds
 - Don't need to compare against
 - Lowest threshold (0)
 - Highest threshold (S)
 - In practice compare against
 - 3 thresholds
 - Result is not in binary
 - Need an encoder
 - Conversion time
 - One conversion cycle



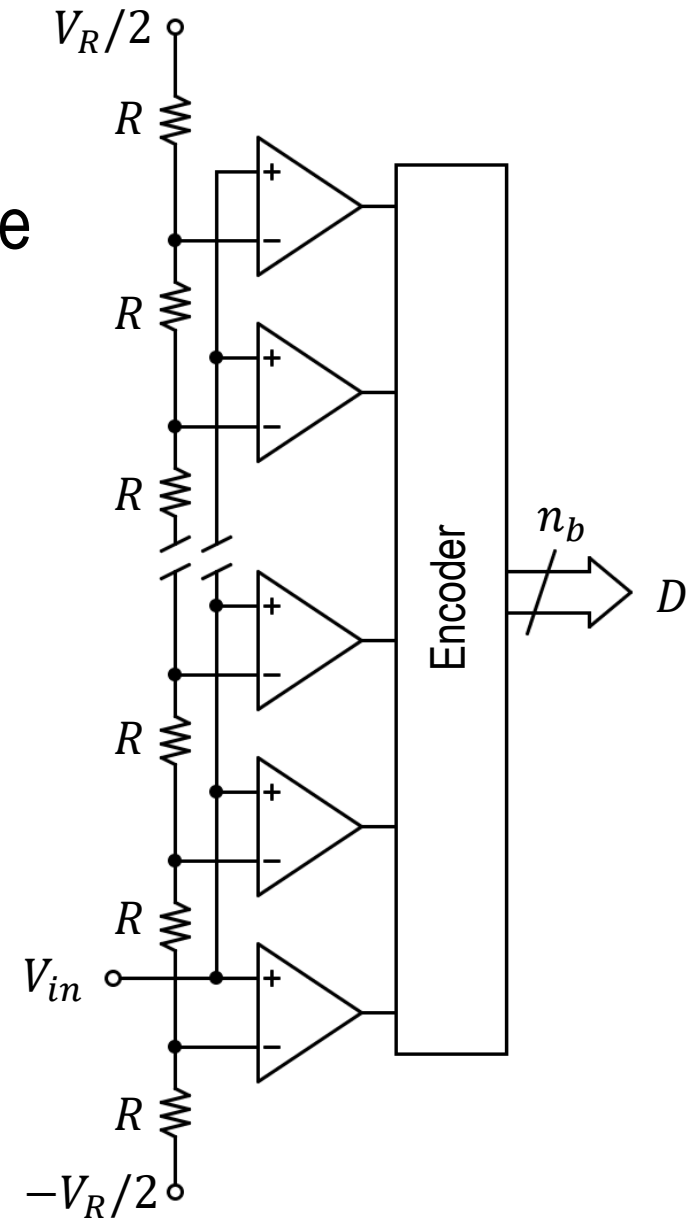
Analog to Digital Conversion

- Flash AD Converter
 - n_b -bit converter
 - 2^{n_b} intervals
 - $2^{n_b} - 1$ thresholds to compare
 - Don't compare against S and 0
 - Thermometric output
 - Need an encoder to generate the digital output
 - $2^{n_b} - 1 \rightarrow n_b$
 - Complexity
 - $2^{n_b} - 1$ comparators
 - Conversion time
 - One conversion cycle



Analog to Digital Conversion

- Flash AD Converter with symmetric input dynamic range
 - n_b -bit converter
 - 2^{n_b} intervals
 - $2^{n_b} - 1$ thresholds to compare
 - Don't compare against $S/2$ and $-S/2$
 - Thermometric output
 - Need an encoder to generate the digital output
 - $2^{n_b} - 1 \rightarrow n_b$
 - Complexity
 - $2^{n_b} - 1$ comparators
 - Conversion time
 - One conversion cycle



Analog to Digital Conversion

- Flash AD Converter
 - Encoder exploits many don't care
 - In the thermometric output many configurations are impossible

2 bits conversion

x_2	x_1	x_0	$d_1 d_0$
0	0	0	00
0	0	1	01
0	1	0	-
0	1	1	10
1	0	0	-
1	0	1	-
1	1	0	-
1	1	1	11

$$d_1 = x_1$$

$$d_0 = x_2 + \overline{x_1}x_0$$

3 bits conversion

x_6	x_5	x_4	x_3	x_2	x_1	x_0	$d_2 d_1 d_0$
0	0	0	0	0	0	0	000
0	0	0	0	0	0	1	001
0	0	0	0	0	1	0	-
0	0	0	0	0	1	1	010
0	0	0	0	1	0	0	-
0	0	0	0	1	0	1	-
0	0	0	0	1	1	0	-
0	0	0	0	1	1	1	011
0	0	0	1	0	0	0	-
0	0	0	1	0	0	1	-
0	0	0	1	0	1	0	-
0	0	0	1	0	1	1	-
0	0	0	1	1	0	0	-
0	0	0	1	1	0	1	-
0	0	0	1	1	1	0	-
0	0	0	1	1	1	1	100
0	0	1	0	0	0	0	-
0	0	1	0	0	0	1	-
0	0	1	0	0	1	0	-
0	0	1	0	0	1	1	-
0	0	1	0	1	0	0	-
0	0	1	0	1	0	1	-
0	0	1	0	1	1	0	-
0	0	1	0	1	1	1	-
0	0	1	1	0	0	0	-
0	0	1	1	0	0	1	-
0	0	1	1	0	1	0	-
0	0	1	1	0	1	1	-
0	0	1	1	1	0	0	-
0	0	1	1	1	0	1	-
0	0	1	1	1	1	0	-
0	0	1	1	1	1	1	101

$$d_2 = x_3$$

$$d_1 = x_5 + \overline{x_3}x_1$$

x_6	x_5	x_4	x_3	x_2	x_1	x_0	$d_2 d_1 d_0$
0	1	0	0	0	0	0	-
0	1	0	0	0	0	1	-
0	1	0	0	0	1	0	-
0	1	0	0	0	1	1	-
0	1	0	0	1	0	0	-
0	1	0	0	1	0	1	-
0	1	0	0	1	1	0	-
0	1	0	0	1	1	1	-
0	1	0	1	0	0	0	-
0	1	0	1	0	0	1	-
0	1	0	1	0	1	0	-
0	1	0	1	0	1	1	-
0	1	0	1	1	0	0	-
0	1	0	1	1	0	1	-
0	1	0	1	1	1	0	-
0	1	0	1	1	1	1	-
0	1	1	0	0	0	0	-
0	1	1	0	0	0	1	-
0	1	1	0	0	1	0	-
0	1	1	0	0	1	1	-
0	1	1	0	1	0	0	-
0	1	1	0	1	0	1	-
0	1	1	0	1	1	0	-
0	1	1	0	1	1	1	-
0	1	1	1	0	0	0	-
0	1	1	1	0	0	1	-
0	1	1	1	0	1	0	-
0	1	1	1	0	1	1	-
0	1	1	1	1	0	0	-
0	1	1	1	1	0	1	-
0	1	1	1	1	1	0	-
0	1	1	1	1	1	1	110

x_6	x_5	x_4	x_3	x_2	x_1	x_0	$d_2 d_1 d_0$
1	0	0	0	0	0	0	-
1	0	0	0	0	0	1	-
1	0	0	0	0	1	0	-
1	0	0	0	0	1	1	-
1	0	0	0	1	0	0	-
1	0	0	0	1	0	1	-
1	0	0	0	1	1	0	-
1	0	0	0	1	1	1	-
1	0	0	1	0	0	0	-
1	0	0	1	0	0	1	-
1	0	0	1	0	1	0	-
1	0	0	1	0	1	1	-
1	0	0	1	1	0	0	-
1	0	0	1	1	0	1	-
1	0	0	1	1	1	0	-
1	0	0	1	1	1	1	-
1	0	1	0	0	0	0	-
1	0	1	0	0	0	1	-
1	0	1	0	0	1	0	-
1	0	1	0	0	1	1	-
1	0	1	0	1	0	0	-
1	0	1	0	1	0	1	-
1	0	1	0	1	1	0	-
1	0	1	0	1	1	1	-
1	0	1	1	0	0	0	-
1	0	1	1	0	0	1	-
1	0	1	1	0	1	0	-
1	0	1	1	0	1	1	-
1	0	1	1	1	0	0	-
1	0	1	1	1	0	1	-
1	0	1	1	1	1	0	-
1	0	1	1	1	1	1	-

$$d_0 = x_6 + \overline{x_5}x_4 + \overline{x_3}x_2 + \overline{x_1}x_0$$

x_6	x_5	x_4	x_3	x_2	x_1	x_0	$d_2 d_1 d_0$
1	1	0	0	0	0	0	-
1	1	0	0	0	0	1	-
1	1	0	0	0	1	0	-
1	1	0	0	0	1	1	-
1	1	0	0	1	0	0	-
1	1	0	0	1	0	1	-
1	1	0	0	1	1	0	-
1	1	0	0	1	1	1	-
1	1	0	1	0	0	0	-
1	1	0	1	0	0	1	-
1	1	0	1	0	1	0	-
1	1	0	1	0	1	1	-
1	1	0	1	1	0	0	-
1	1	0	1	1	0	1	-
1	1	0	1	1	1	0	-
1	1	0	1	1	1	1	-
1	1	1	0	0	0	0	-
1	1	1	0	0	0	1	-
1	1	1	0	0	1	0	-
1	1	1	0	0	1	1	-
1	1	1	0	1	0	0	-
1	1	1	0	1	0	1	-
1	1	1	0	1	1	0	-
1	1	1	0	1	1	1	-
1	1	1	1	0	0	0	-
1	1	1	1	0	0	1	-
1	1	1	1	0	1	0	-
1	1	1	1	0	1	1	-
1	1	1	1	1	0	0	-
1	1	1	1	1	0	1	-
1	1	1	1	1	1	0	-
1	1	1	1	1	1	1	111

Analog to Digital Conversion

- Flash AD Converter

- Complexity

- $2^{n_b} - 1$ comparators
 - Exponential on the number of bits
 - Very expensive for high n_b
 - Not practical for $n_b > 8$

- Conversion time

- One conversion cycle
 - Cycle time t_{conv} depends on
 - Delay of the comparator t_{comp}
 - Delay of the encoder t_{enc}
 - $t_{conv} > t_{comp} + t_{enc}$, $f_{conv} < \frac{1}{t_{comp} + t_{enc}}$
 - High performance, exploits parallelism
 - Independent of the number of bits

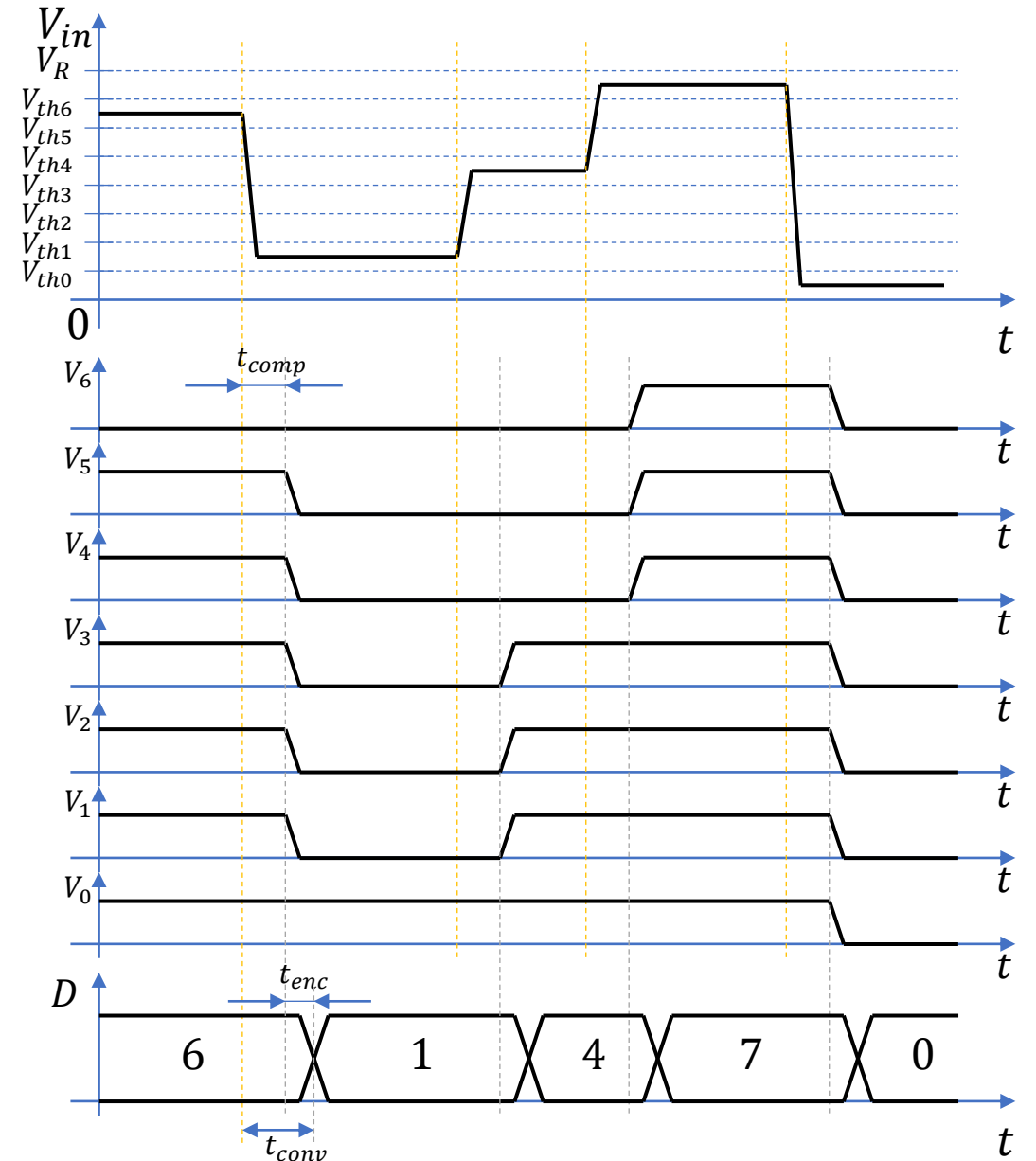
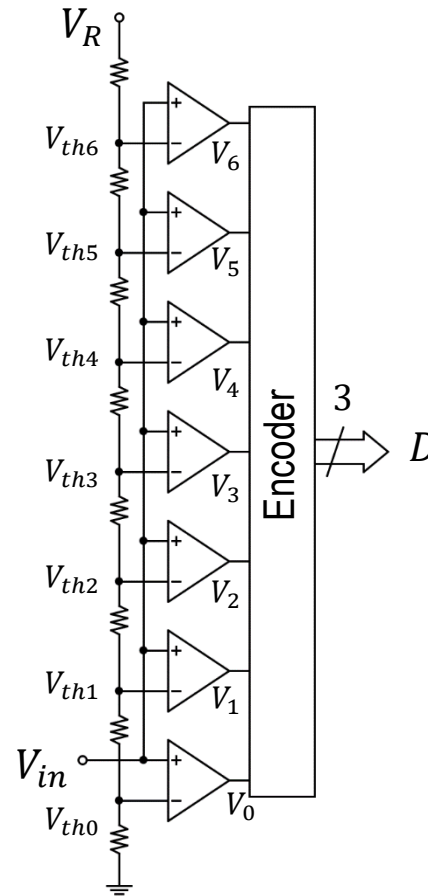
n_b	n_{comp}
3	7
4	15
5	31
6	63
7	127
8	255
9	511
...	...
16	65535

Converter	Complexity	Conversion time
Flash	$\mathcal{O}(2^{n_b})$	$\mathcal{O}(1)$

Analog to Digital Conversion

- Flash AD Converter example waveforms

- Flash converter on 3 bits
- Input dynamic range between 0 and V_R
 - 9 total thresholds
 - 7 effective thresholds
 - V_{th0} to V_{th6}
- Encoder 7-to-3
- Input signal constant during the conversion
 - See Sample&Hold



Analog to Digital Conversion

- Flash AD Converter errors

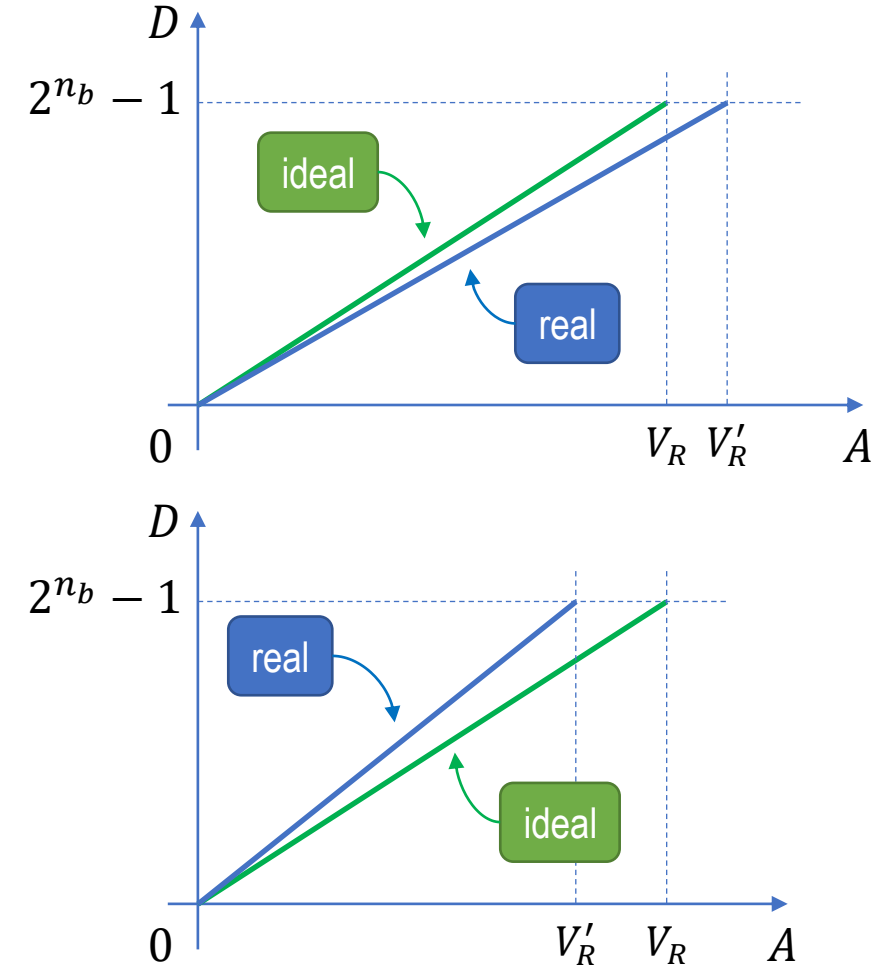
- Reference voltage V_R higher or lower than ideal value
 - Intervals larger or smaller than ideal
 - Thresholds not in the ideal position
 - Gain error

- Expression of the characteristic

$$D = GA \quad G = \frac{2^{n_b} - 1}{V_R} \quad G' = \frac{2^{n_b} - 1}{V'_R}$$

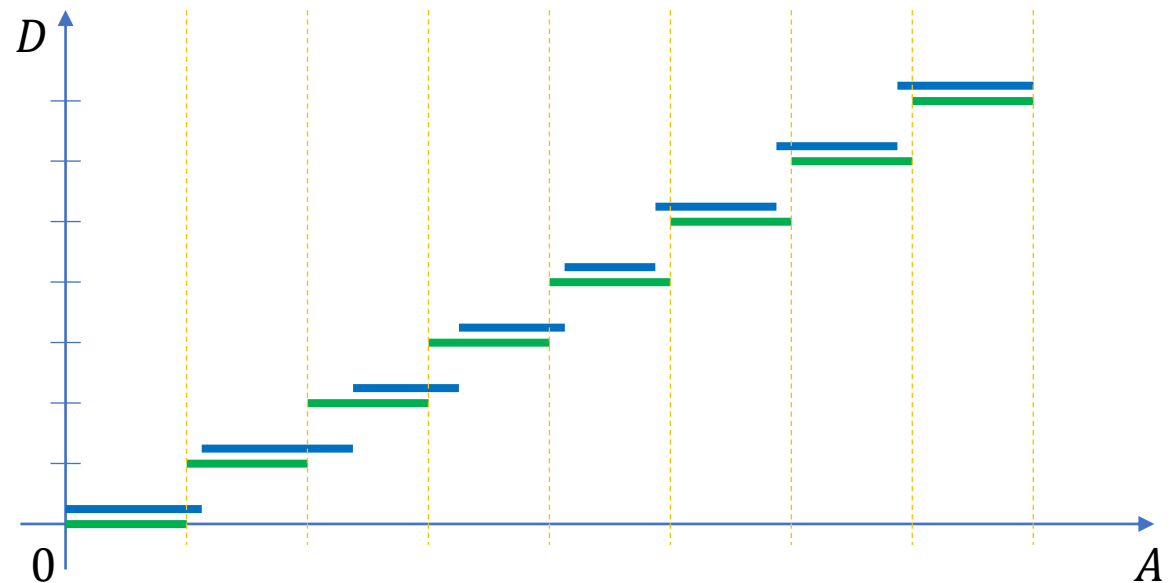
- Expression of the relative gain error

$$\varepsilon_G = \frac{G' - G}{G} = \frac{\frac{2^{n_b} - 1}{V'_R} - \frac{2^{n_b} - 1}{V_R}}{\frac{2^{n_b} - 1}{V_R}} = \frac{V_R - V'_R}{V'_R} = \varepsilon_V$$



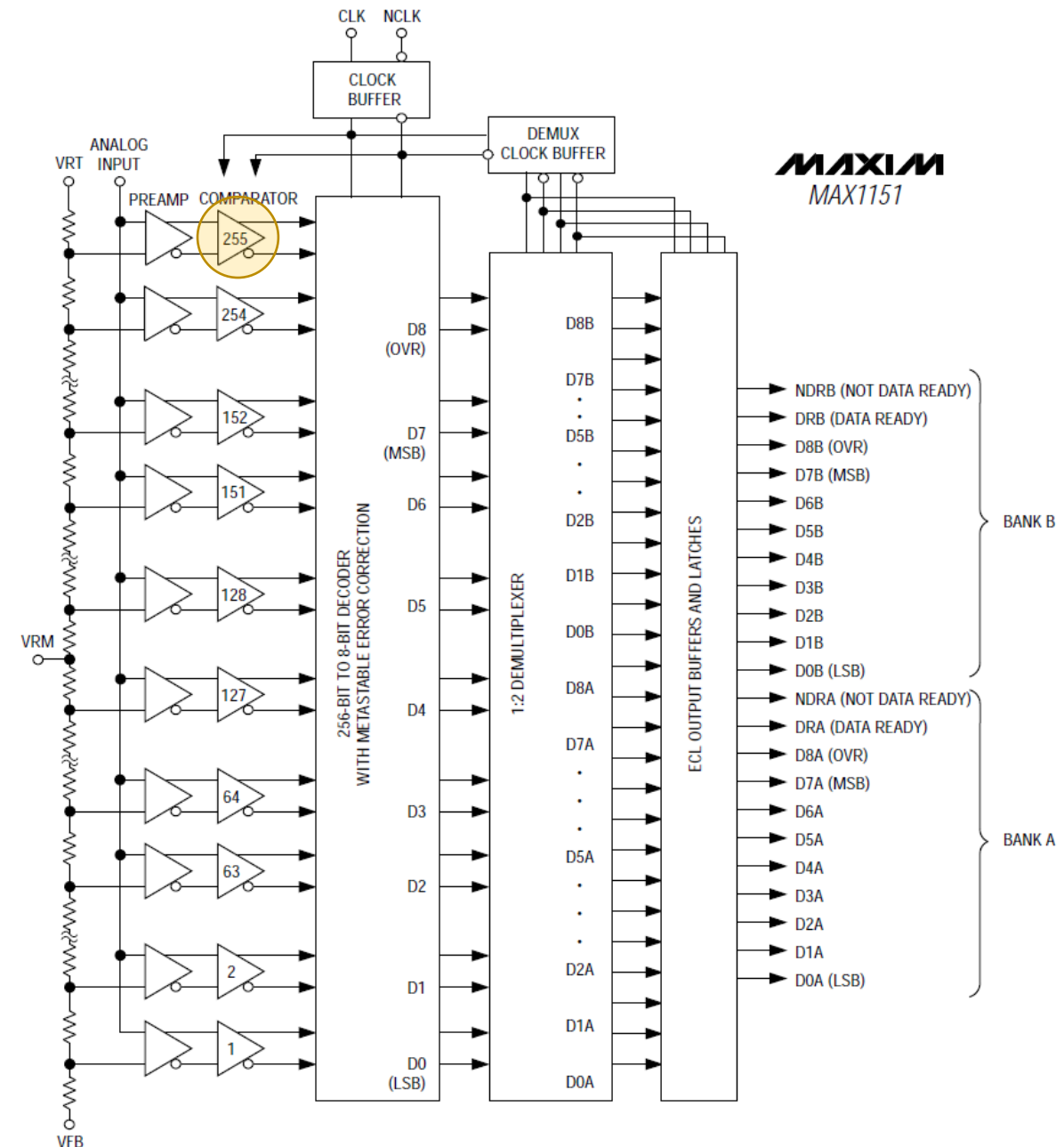
Analog to Digital Conversion

- Flash AD Converter errors
 - Equal resistances but value different from ideal
 - No errors in transfer characteristic
 - Different current in voltage divider
 - Resistances with tolerances
 - Differential non linearity errors
 - Integral non linearity error



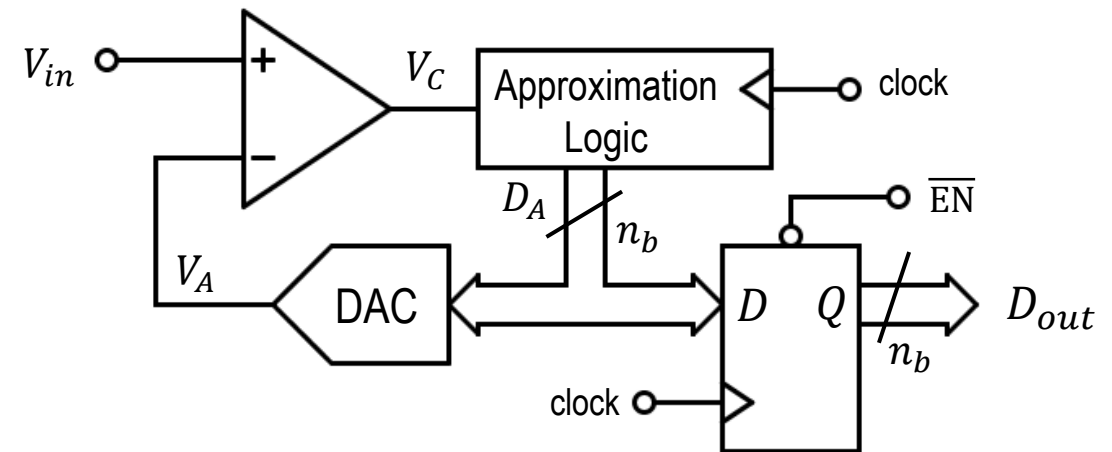
Analog to Digital Conversion

- Flash AD Converter example
 - MAXIM MAX1151
 - 8-bits, 750 Msp/s Flash ADC
 - Thresholds and input analog signal preamplified before comparators
 - There are 255 comparators
 - The digital output is alternatively given on D0A – D7A and on D0B – D7B to reach high performance



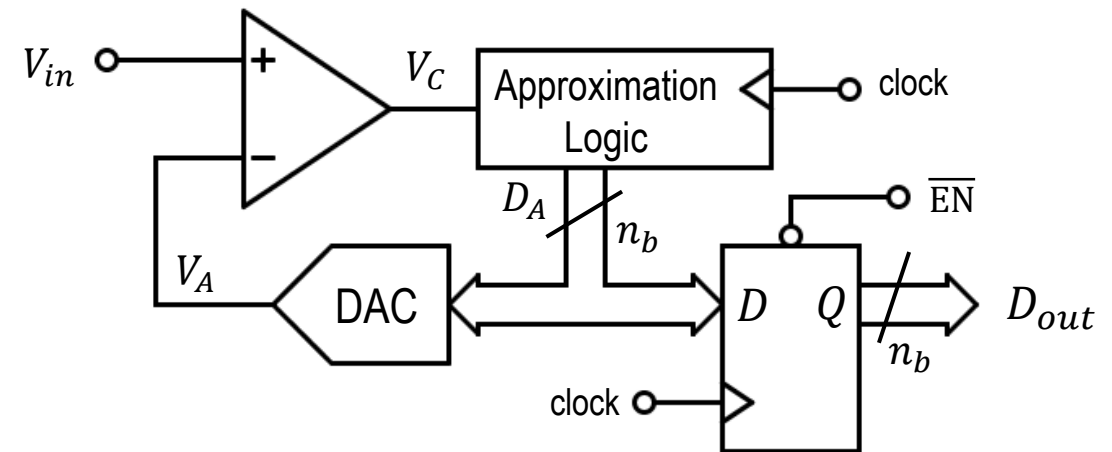
Analog to Digital Conversion

- ADC Circuits Feedback Converter
 - Based on a feedback loop
 - Comparator, Approximation Logic, DAC on n_b bits
 - Input signal to be converted compared to the output of the DAC
 - The DAC produces the threshold for comparison
 - DAC is driven by the approximation logic
 - The outcome of the comparison guides the approximation logic
 - Process iterated a number of times to reach the desired accuracy of the result
 - The logic network modifies D_A till V_A becomes the best approximation of V_{in}
 - Within the n_b -bit resolution
 - At the end of the process, the result is latched on an output register



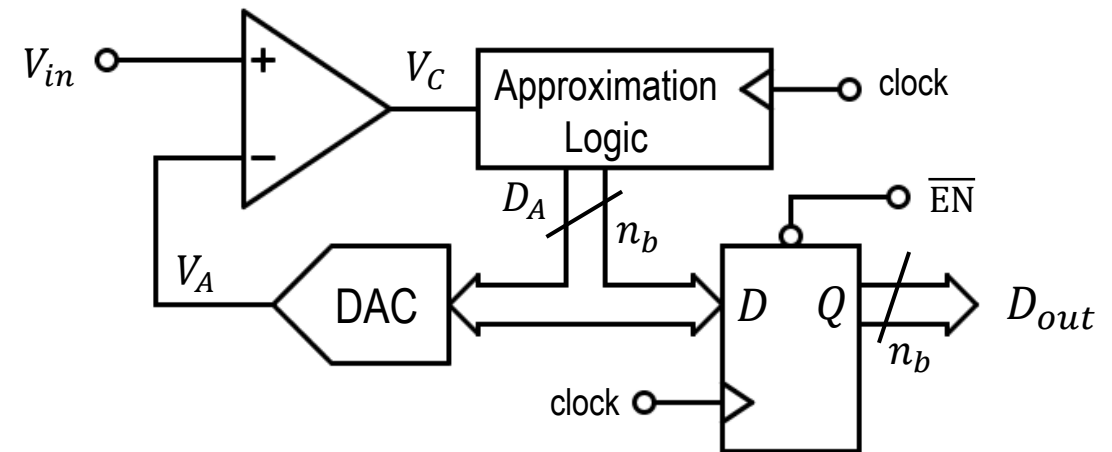
Analog to Digital Conversion

- ADC Circuits Feedback Converter
 - Basic scheme applies to
 - Staircase
 - Tracking
 - Integrating
 - Differential
 - Successive Approximation
 - Difference is the approximation algorithm
 - 1 LSB steps for the staircase, tracking, integrating and differential ADCs
 - Variable steps (starting from MSB) for the successive approximation ADC



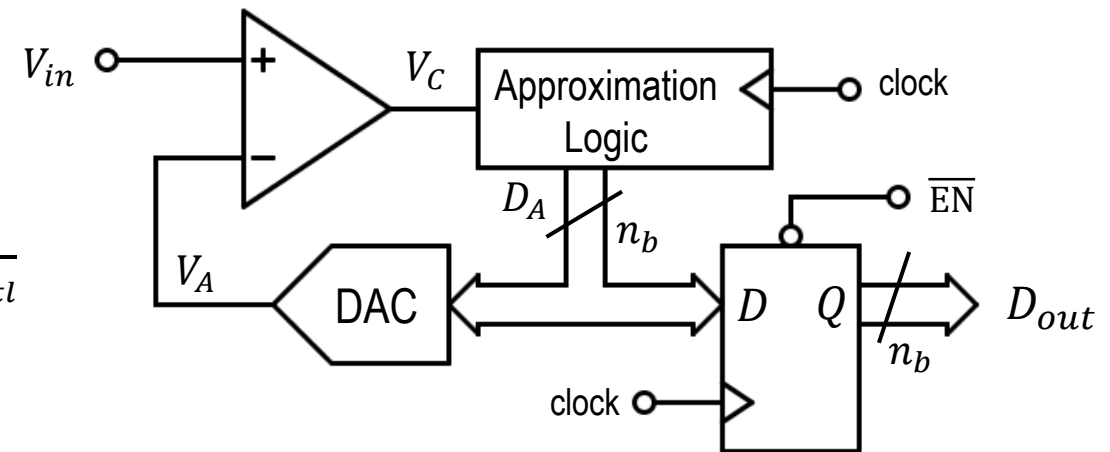
Analog to Digital Conversion

- ADC Circuits Feedback Converter
 - Complexity
 - Comparator
 - A single one
 - Independently of the number of bits
 - Main contribution to total complexity
 - DAC
 - Size linear on the number of bits
 - But small with respect to the comparator
 - Approximation logic is small
 - The algorithm does not significantly impact complexity
 - Neglected with respect to the other contributions
 - Total complexity independent of the number of bits



Analog to Digital Conversion

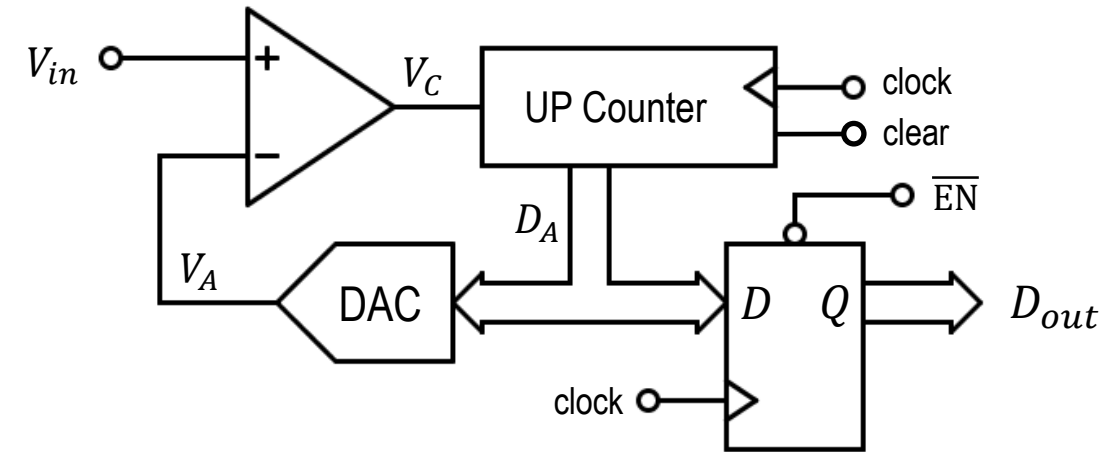
- ADC Circuits Feedback Converter
 - Conversion time
 - Sequential decisions
 - One threshold compared at a time
 - One conversion cycle for each comparison
 - Number of decisions n_{cycles} depends on the approximation logic
 - See next slides for details
 - Cycle time t_{ck} depends on the feedback loop delays
 - Delay of the comparator t_{comp}
 - Delay of the approximation logic t_{logic}
 - Settling time of the DAC t_{stl}
 - $t_{ck} > t_{comp} + t_{logic} + t_{stl}$, $f_{ck} < \frac{1}{t_{comp} + t_{logic} + t_{stl}}$
 - Dominated by the delay of the comparator
 - Conversion time $t_{conv} = n_{cycles} \cdot t_{ck}$



Analog to Digital Conversion

- Staircase AD Converter

- Approximation logic is an UP counter
 - Counter cleared at conversion start
 - Incremented at each conversion cycle
 - Until the DAC output V_A crosses the input V_{in}
 - The comparator output goes from High to Low
 - Disables the counter until the next start
 - Enables the output latch
- Complete dynamic range S explored in 2^{n_b} steps
 - Full conversion requires 2^{n_b} sequential decisions in the worst case
 - Conversion time
 - $t_{conv} = 2^{n_b} \cdot t_{ck}$
 - Exponential on the number of bits



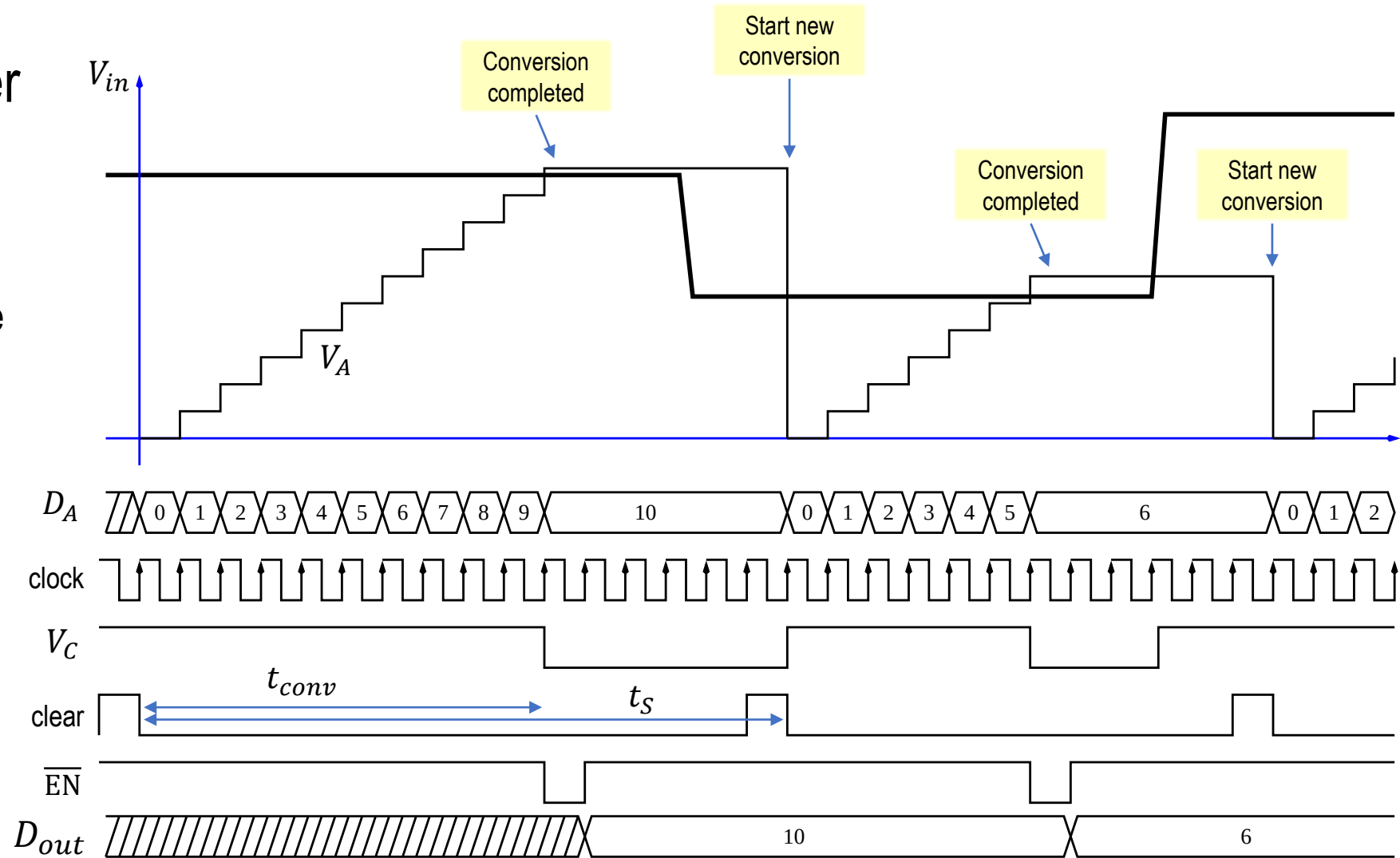
Converter	Complexity	Conversion time
Staircase	$\mathcal{O}(1)$	$\mathcal{O}(2^{n_b})$

Analog to Digital Conversion

- Staircase AD Converter

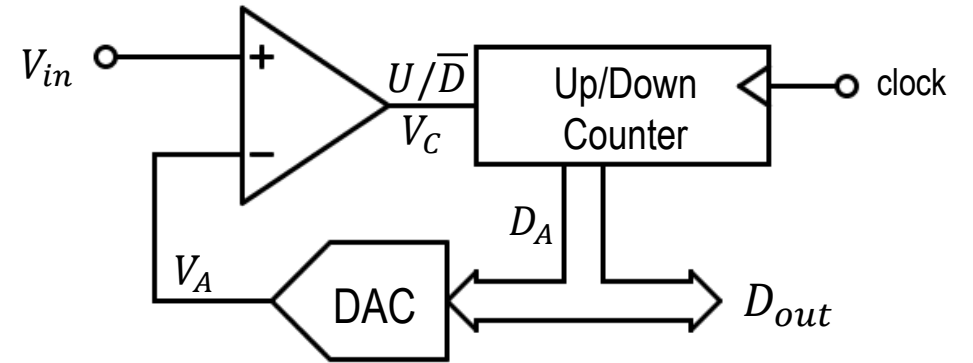
- Example

- Signal kept constant during conversion
 - Additional logic on the counter and latch inputs to correctly control
 - Clear
 - Enable
 - Run
 - t_{conv} depends on the input signal
 - $t_S > t_{conv}$



Analog to Digital Conversion

- Tracking AD Converter
 - Approximation logic is an UP/DOWN counter
 - Start a new conversion from the previous sample
 - It is potentially faster than always starting from 0, if the input signal does not change too rapidly
 - Counter never cleared
 - Counter incremented or decremented at each conversion cycle
 - UP if $V_{in} > V_A$
 - DOWN if $V_{in} < V_A$
 - The comparator output drives the $U/\overline{D}$ input of the counter
 - The converter provides a new output at each cycle
 - No need of an output latch
 - $D_{out} \equiv D_A$
 - The result may or may not be accurate
 - Depends on the relative Slew Rates of the converter and of the input signal



Analog to Digital Conversion

- Tracking AD Converter

- Converter Slew Rate

- 1 LSB at each clock cycle

- $1 \text{ LSB} = A_d = \frac{S}{2^{n_b}}$

- Clock period t_{ck}

- $SR_{track} = \frac{A_d}{t_{ck}} = \frac{S}{2^{n_b} t_{ck}} = \frac{S}{2^{n_b}} f_{ck}$

- Signal Slew Rate

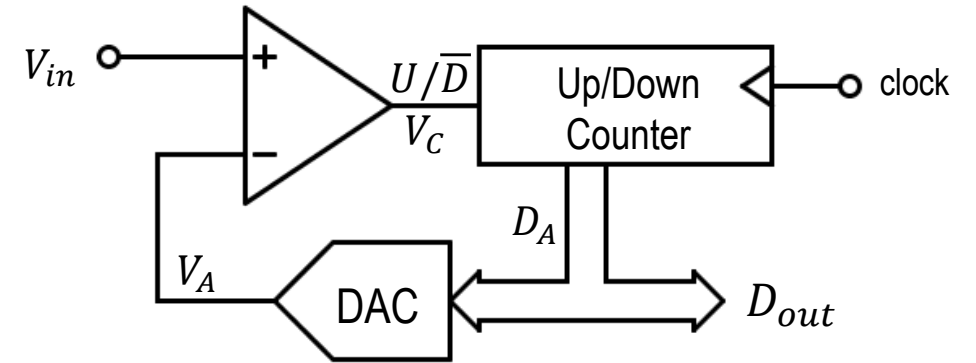
- For a sinusoid $V_{in} = \frac{S}{2} \sin(2\pi f_{in} t)$

- $SR_{in} = \left. \frac{dV_{in}}{dt} \right|_{max} = S\pi f_{in}$

- Comparison

- $SR_{track} > SR_{in}, \frac{A_d}{t_{ck}} > SR_{in}$

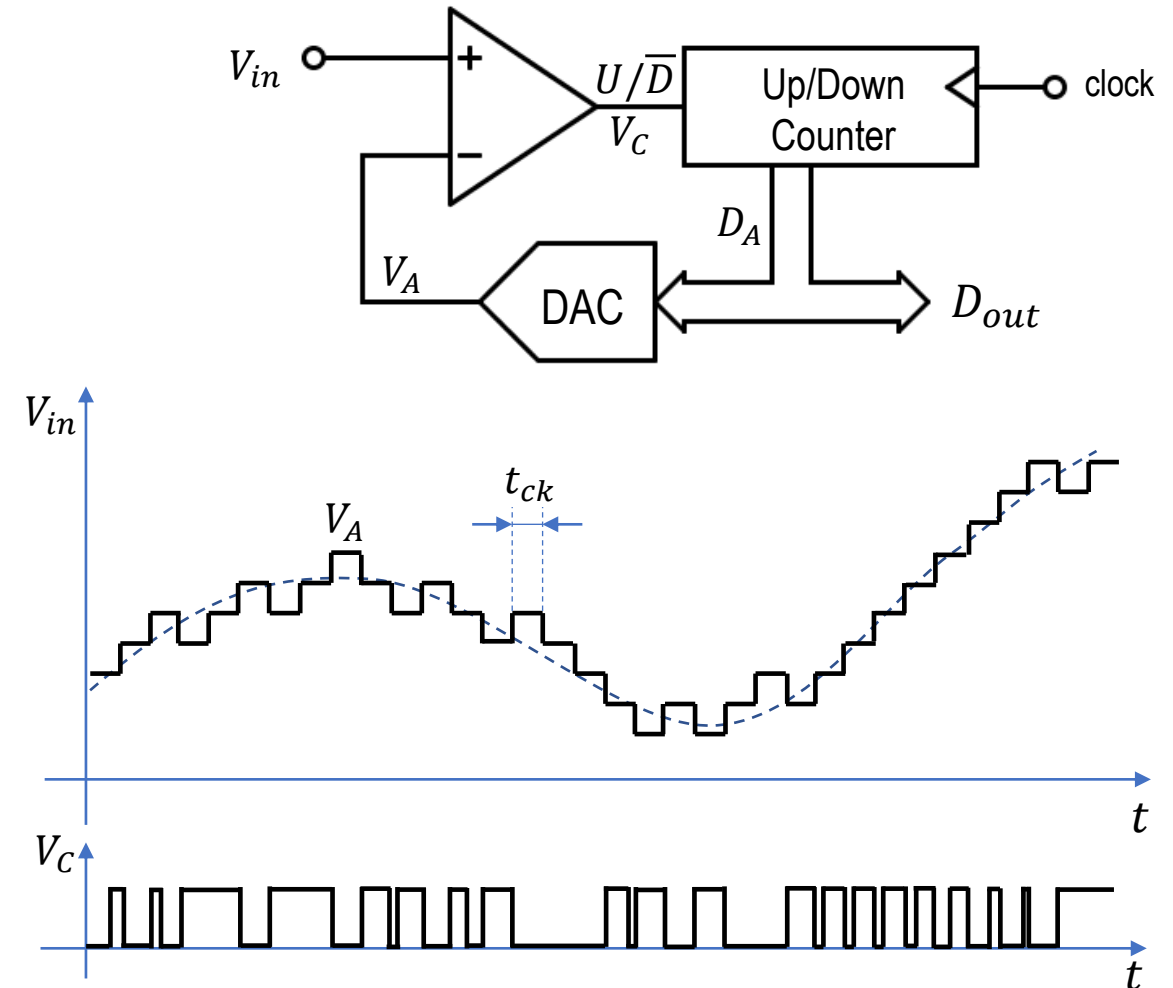
- $t_{ck} < \frac{A_d}{SR_{in}} = \frac{A_d}{S\pi f_{in}} = \frac{1}{2^{n_b} \pi f_{in}}, f_{ck} > 2^{n_b} \pi f_{in}$



Analog to Digital Conversion

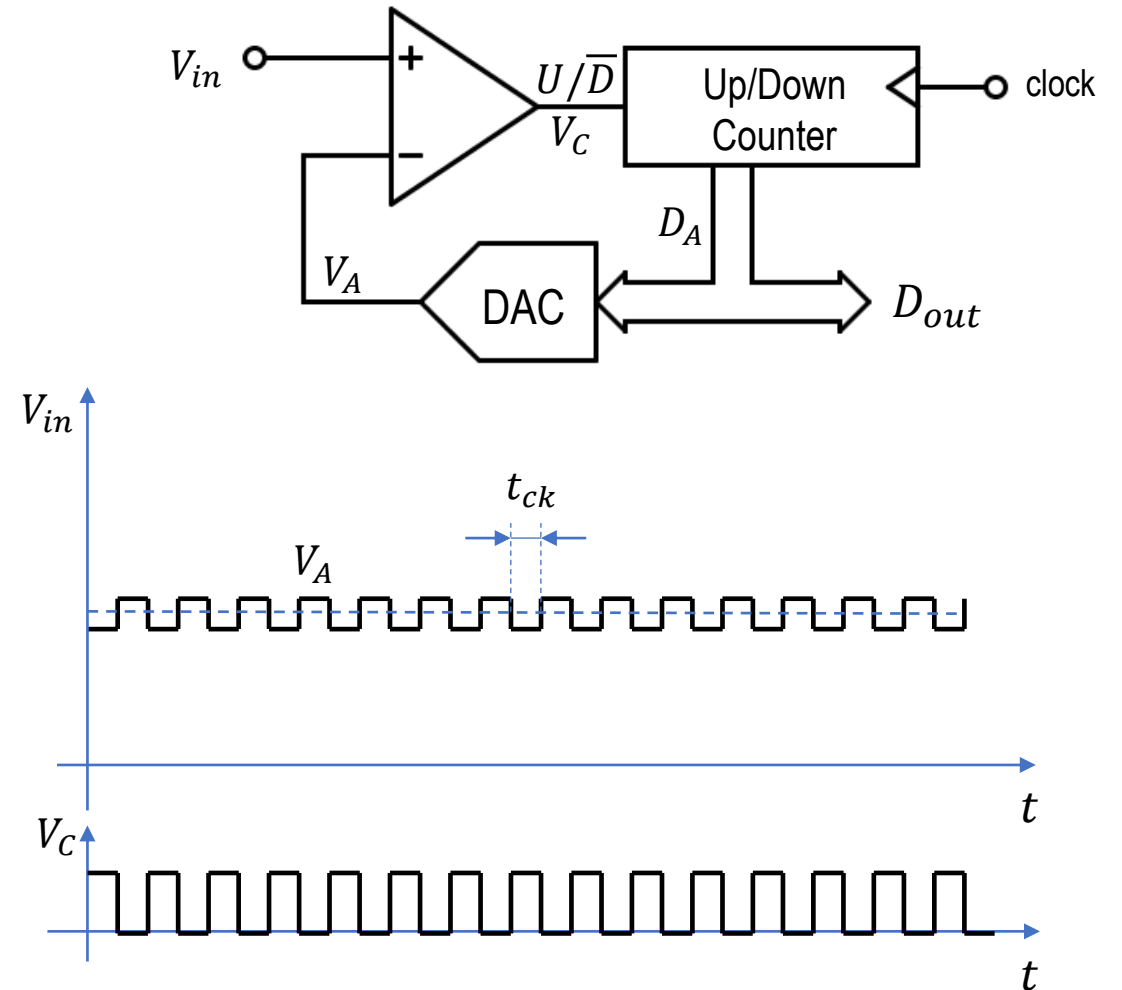
- Tracking AD Converter

- If $SR_{track} > SR_{in}$
 - One correct sample at each cycle
 - Conversion time comparable to Flash Converter
 - Much better than the Staircase Converter
 - Strong requirement on the input signal Slew Rate
 - In these conditions, the sampling frequency f_S and the clock frequency f_{ck} are the same
- Example 1
 - $n_b = 8, S = 10 \text{ V}, f_{ck} = 10 \text{ MHz}$
 - $f_{in} < \frac{f_{ck}}{2^{n_b} \pi} = \frac{10 \cdot 10^6}{2^8 \cdot \pi} = 12.4 \text{ kHz}$
- Example 2
 - $n_b = 16, S = 10 \text{ V}, f_{ck} = 10 \text{ MHz}$
 - $f_{in} < \frac{f_{ck}}{2^{n_b} \pi} = \frac{10 \cdot 10^6}{2^{16} \cdot \pi} = 48.6 \text{ Hz}$



Analog to Digital Conversion

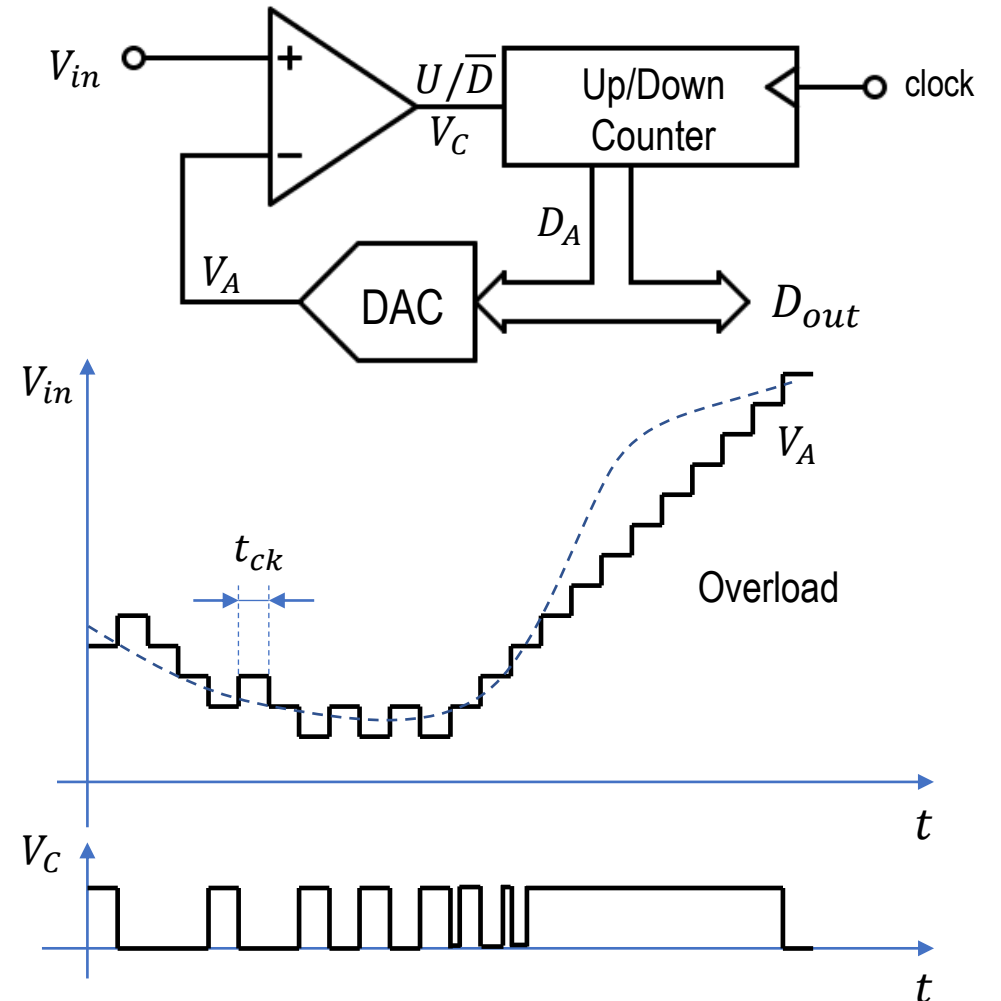
- Tracking AD Converter
 - If input is constant
 - $SR_{track} > SR_{in} = 0$
 - Output alternates between two adjacent values
 - Flickr noise
 - High frequency contribution to the spectrum
 - Staircase converter gives a constant output



Analog to Digital Conversion

- Tracking AD Converter
 - If $SR_{track} < SR_{in}$
 - Overload
 - The output digital signal cannot track the fast input
 - Complete dynamic range S explored in 2^{n_b} steps
 - Output may never be correct if input constantly changes
 - Example
 - Input is a square wave
 - Extremely high slew rate at transitions
 - Conversion time in worst case
 - $t_{conv} = 2^{n_b} \cdot t_{ck}$
 - Exponential on the number of bits

Converter	Complexity	Conversion time
Tracking	$\mathcal{O}(1)$	$\mathcal{O}(2^{n_b})$



Analog to Digital Conversion

- Tracking AD Converter

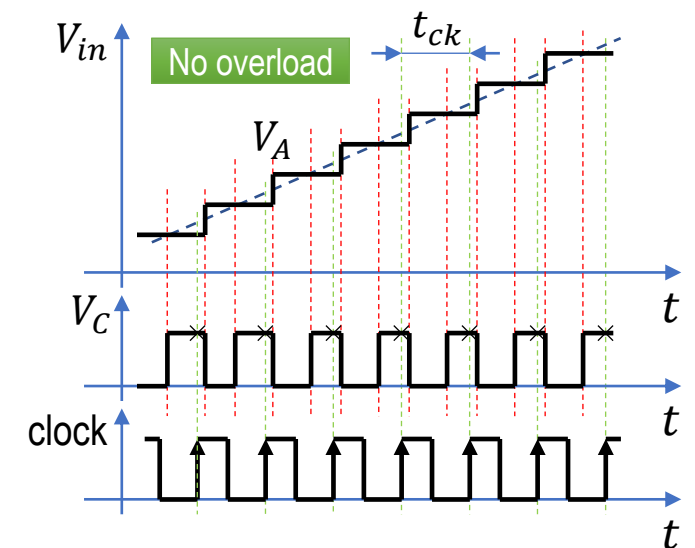
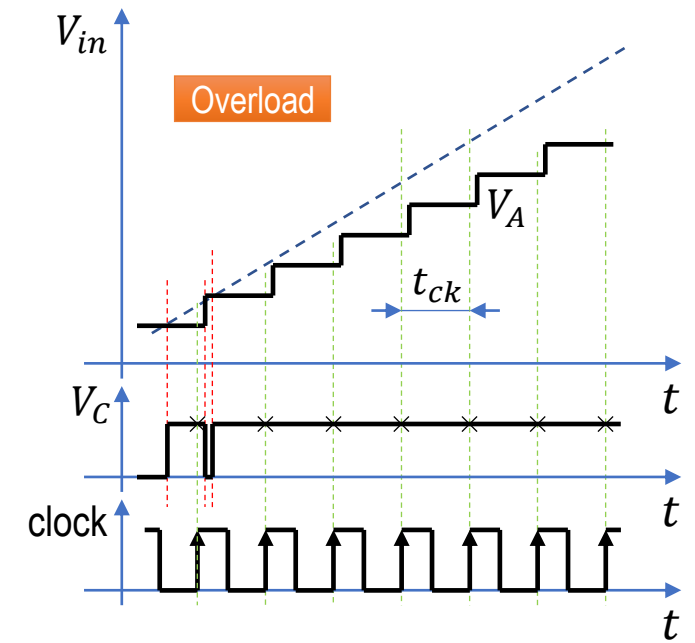
- If $SR_{track} < SR_{in}$

- How to detect overload

- The counter counts in the same direction for several cycles, and ...
 - ... the output of the comparator is constant in those cycles
 - Counting in the same direction for several cycles is not sufficient
 - The input signal may be increasing or decreasing monotonically, but with an acceptable slew rate
 - The output of the comparator shows transitions when V_A crosses the input signal V_{in}

- What to do in case of overload

- Count up or down by more than 1 LSB in a clock cycle
 - It increases SR_{track}
 - Risk of overshooting the input signal
 - Go back to 1 LSB steps once overload is finished

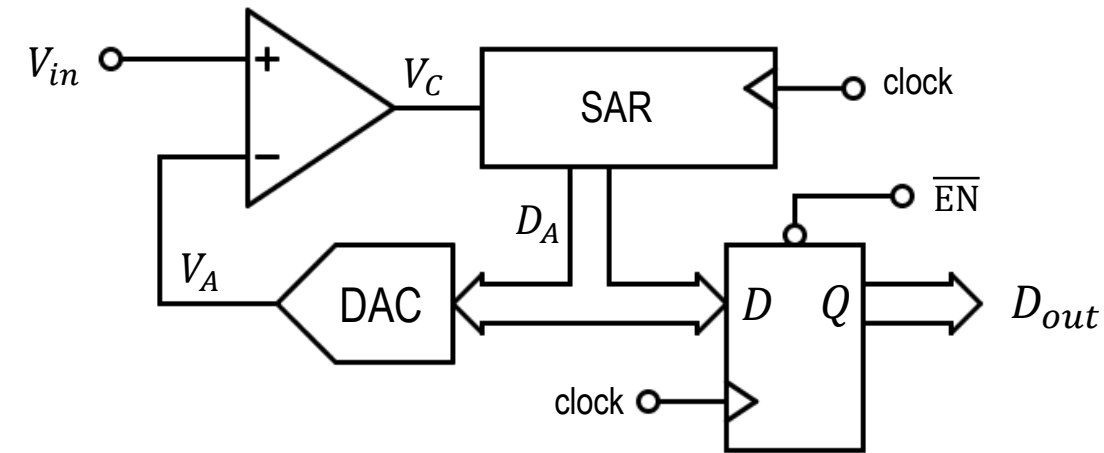


Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter

- Approximation logic uses a *Divide&Conquer* technique

- Divide input dynamic range in two halves
 - Detect where the input signal belongs
 - Bottom half MSB = 0
 - Upper half MSB = 1
 - Further divide the identified half in two halves
 - Detect where the input signal belongs
 - Bottom half MSB-1 = 0
 - Upper half MSB-1 = 1
 - Iterate for the required number of bits
 - It follows a binary decision tree
 - Number of sequential decisions is linear on the number of bits



Converter	Complexity	Conversion time
SAR	$\mathcal{O}(1)$	$\mathcal{O}(n_b)$

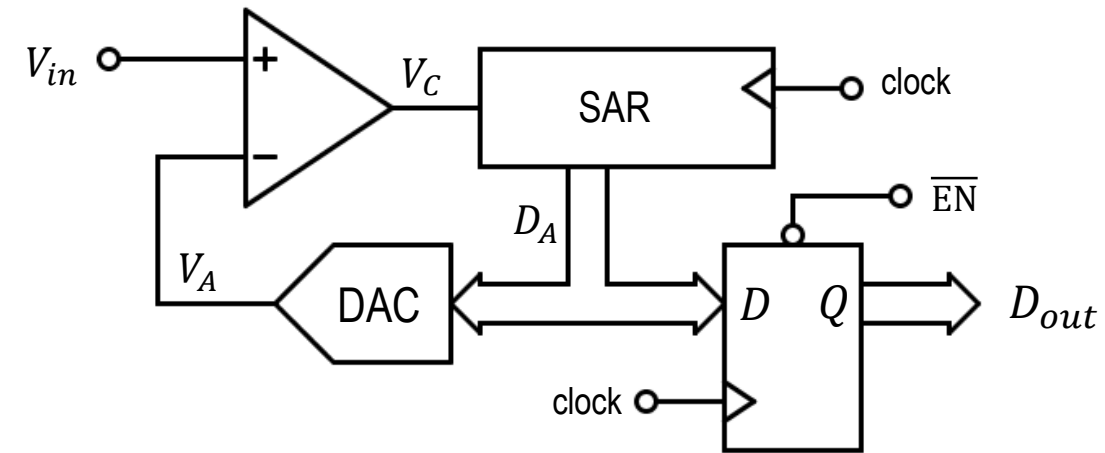
Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter

- Approximation logic uses a *Divide&Conquer* technique

- Informal description

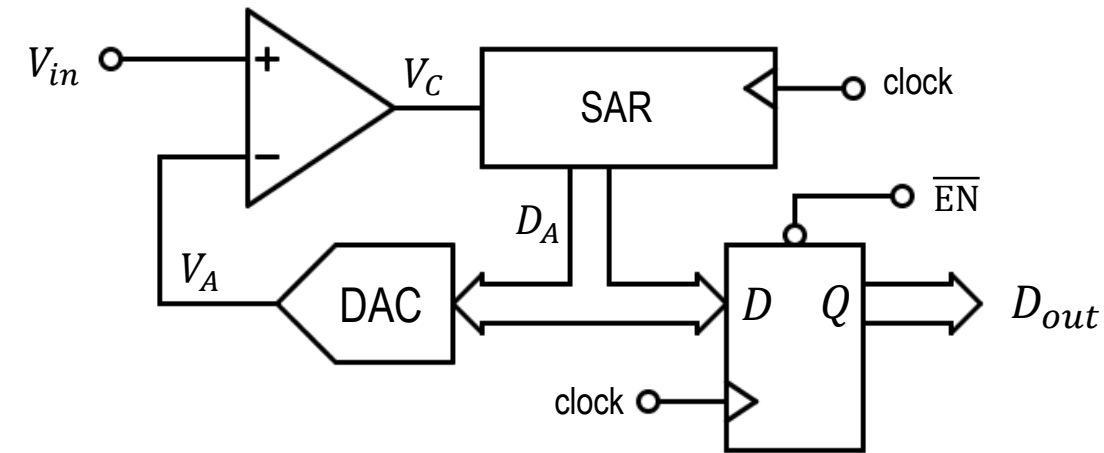
- The SAR should guess a value within a certain range
 - It tries a first guess in the middle of the range
 - The comparator tells the SAR whether the value to be guessed is higher or lower than the first guess
 - The SAR then excludes one half of the range and refines the guess in the middle of the remaining half
 - Again, the comparator tells the SAR whether the value to be guessed is higher or lower
 - The process iterates multiple times
 - It stops when the required precision is reached



Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter

- Approximation logic uses a *Divide&Conquer* technique
 - Informal description is similar to a game
 - Think a number between 1 and 100 (think 38)
 - Is it 50? No, it is lower
 - Is it 25? No, it is higher
 - Is it 37? No, it is higher
 - Is it 44? No, it is lower
 - Is it 40? No, it is lower
 - Is it 38? Yes, you are right!!!
 - Game started in base 10
 - But then divide in two halves, getting fractions (not shown)
 - The SAR always plays in base 2



Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter

- Approximation logic uses a *Divide&Conquer* technique

- Informal description is similar to a game

- The SAR always plays in base 2

- Each time a different bit position is considered

- Starts from the MSB and moves sequentially towards the LSB

- The bit at the current position is set to 1

- All most significant bits are already correct

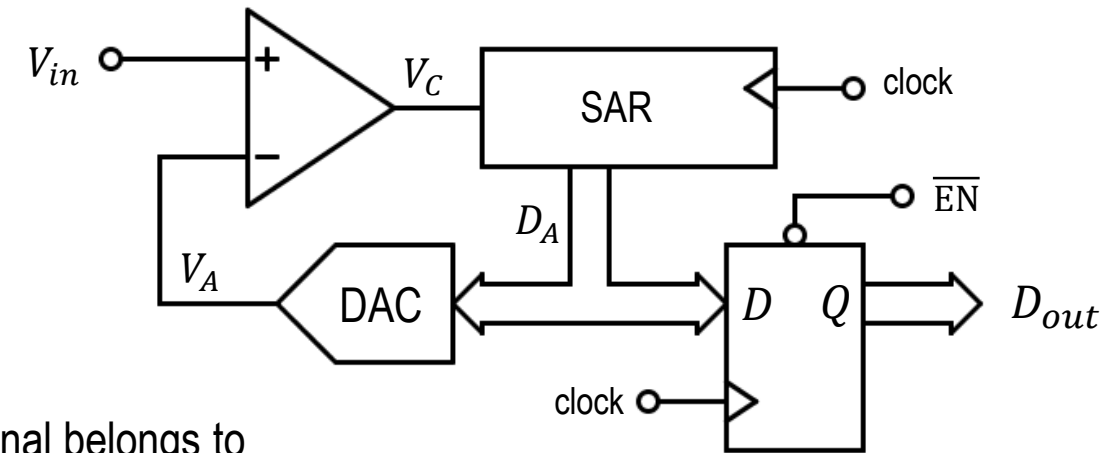
- All less significant bits are set to 0

- The SAR logic assumes that the input signal belongs to the upper half of the considered dynamic range

- The DAC generates the analog threshold using the current digital approximation

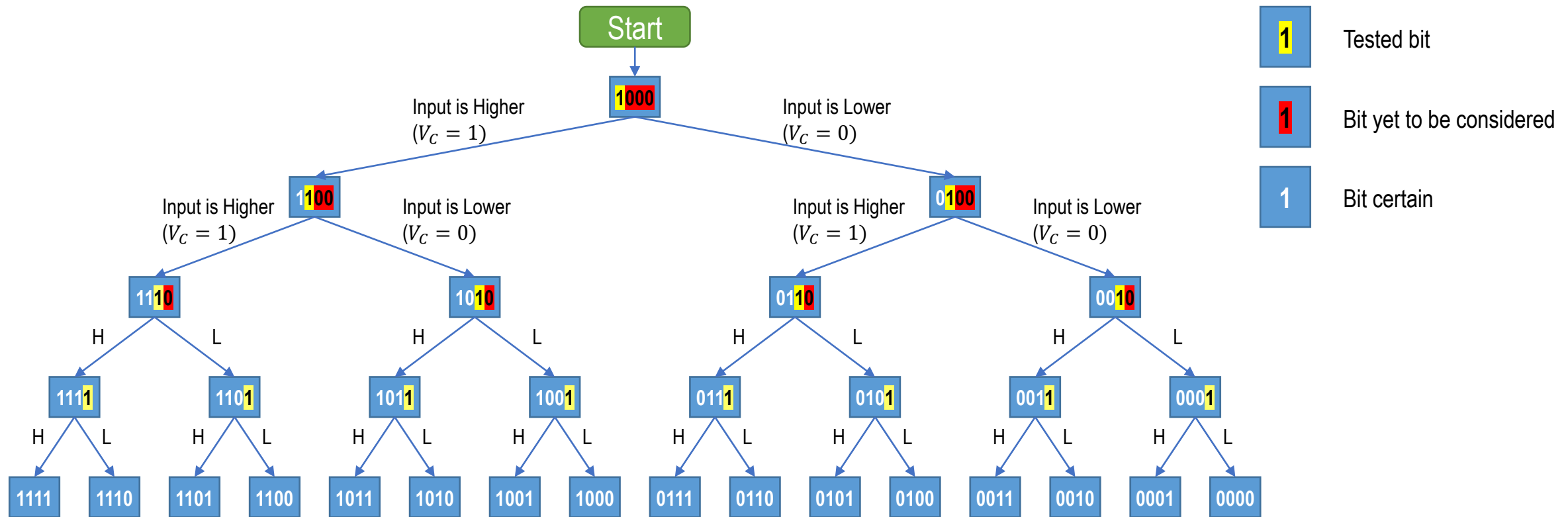
- The comparator confirms or reverses the hypothesis

- The bit at the current position can be set to the correct value



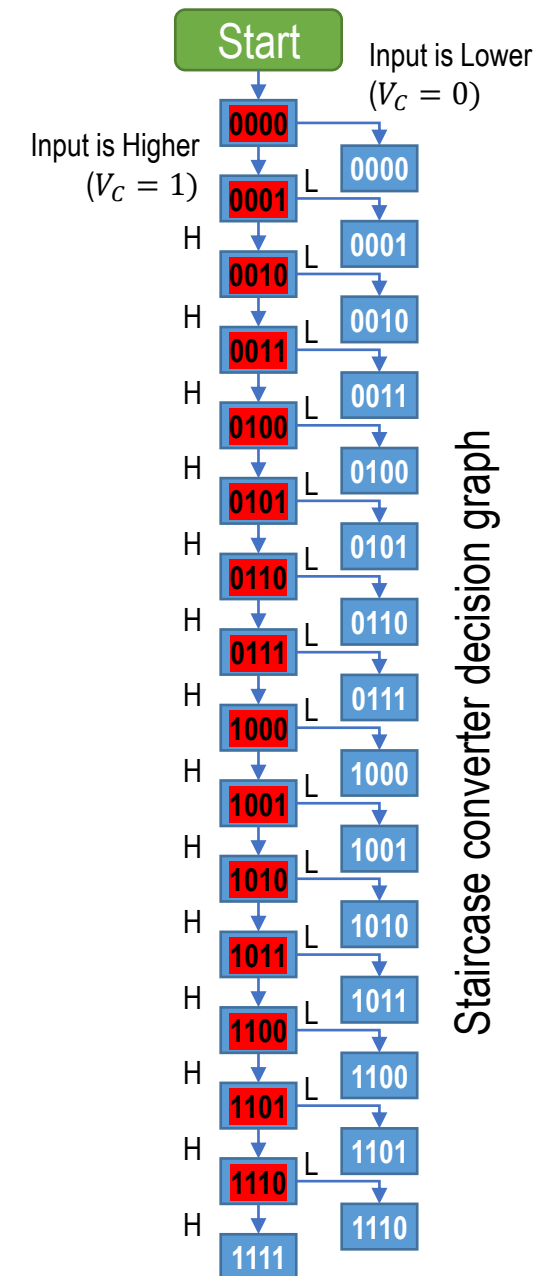
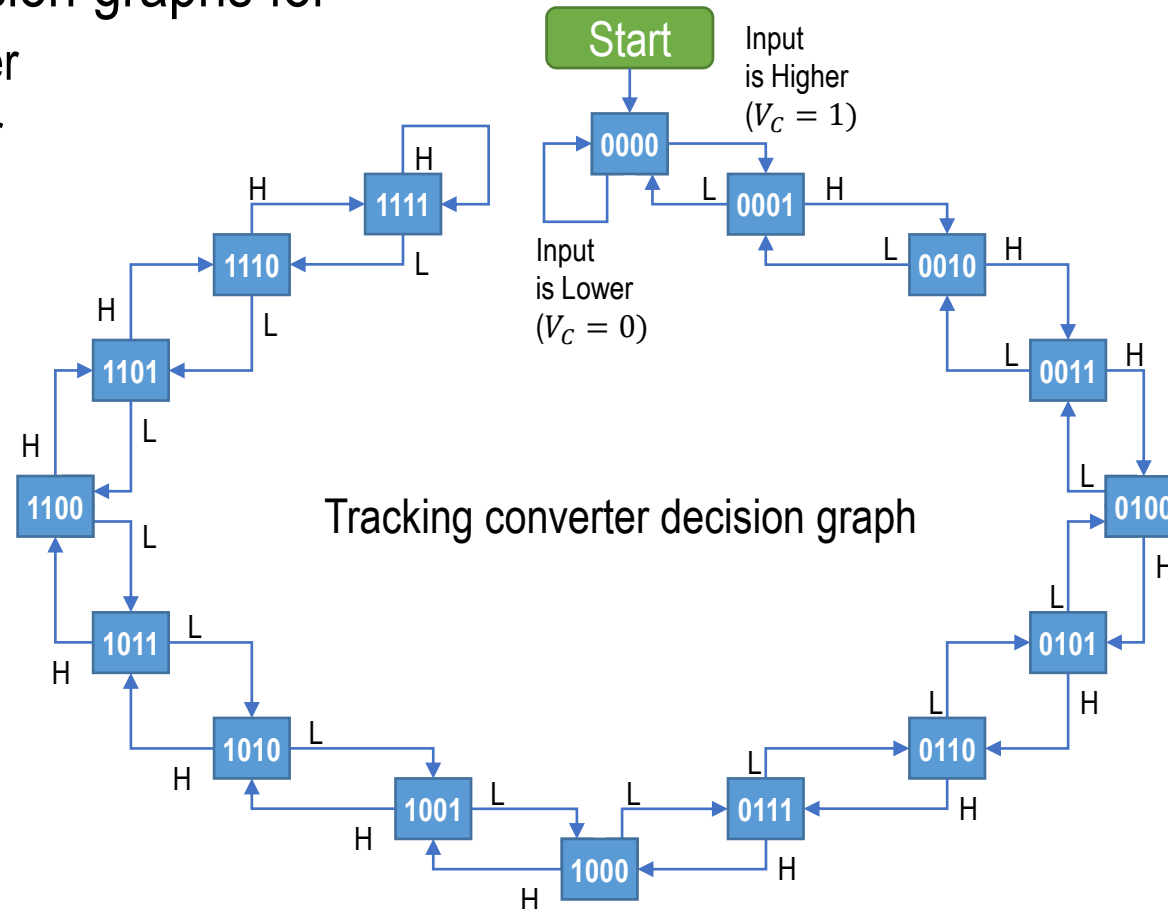
Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter
 - Approximation logic uses a *Divide&Conquer* technique
 - Decision tree



Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter
 - Compare with decision graphs for
 - Staircase converter
 - Tracking converter

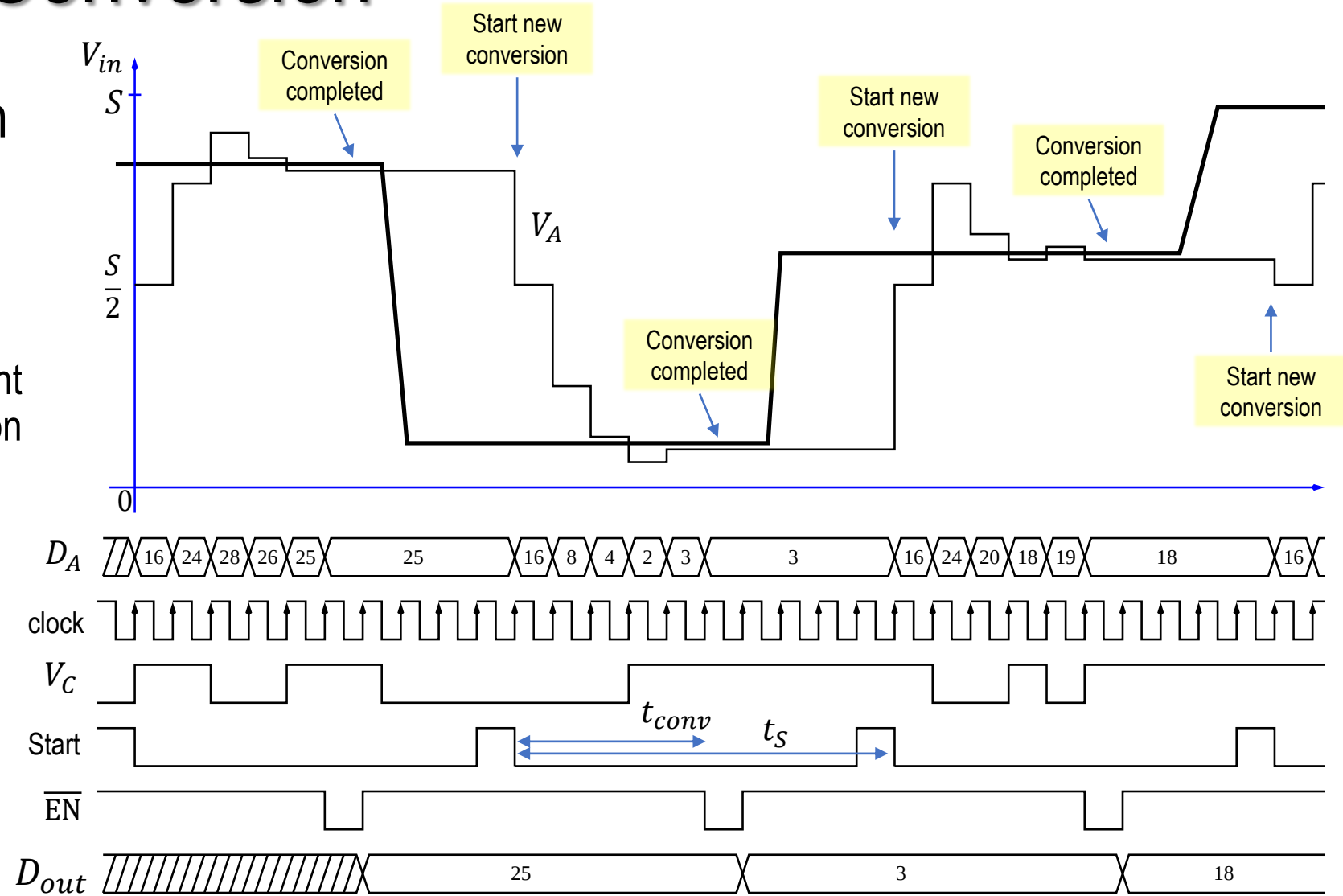


Analog to Digital Conversion

- Successive Approximation Register (SAR) AD Converter

- Example

- Input signal is kept constant during the entire conversion
- t_{conv} is 5 clock cycles
- $t_S > t_{conv}$



Analog to Digital Conversion

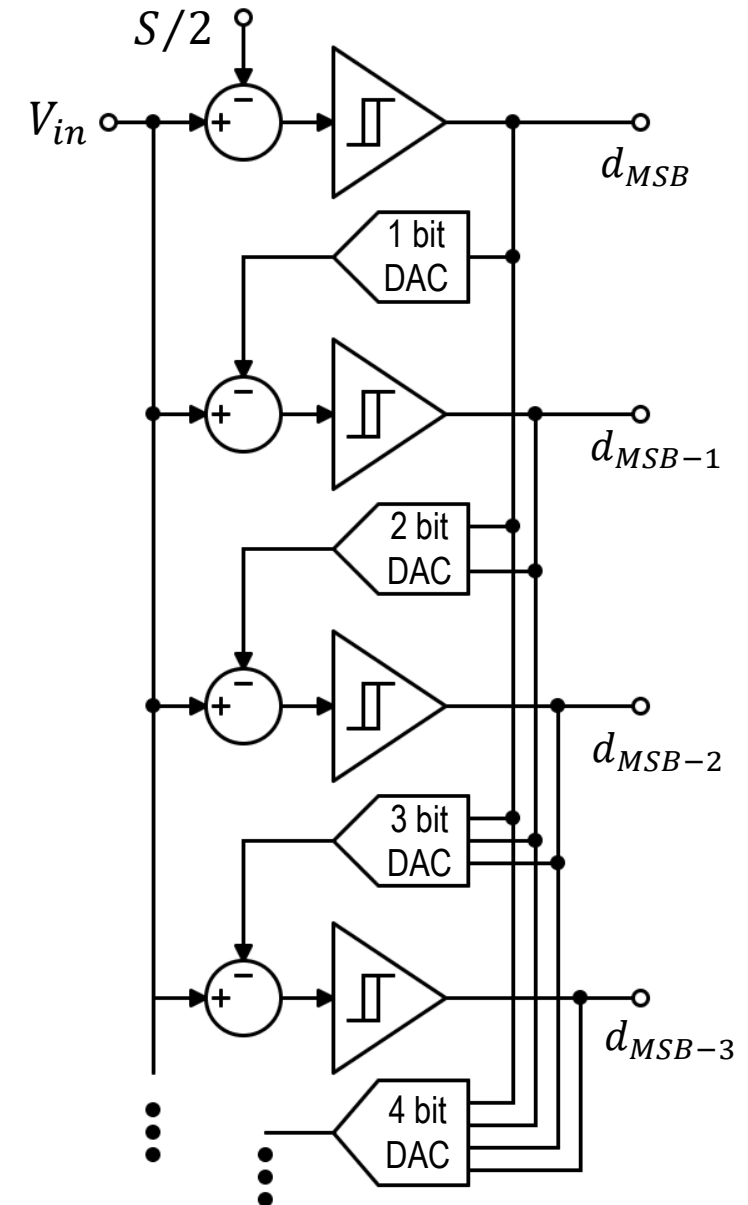
- ADC Circuits
 - Initial evaluation
 - Flash is the fastest
 - But the most complex
 - Staircase is the slowest
 - But cheap
 - Tracking is potentially fast
 - And cheap
 - Strong requirement on input signal slew rate
 - SAR is faster than staircase and tracking
 - And it is also cheap (same complexity)
 - But slower than Flash
 - It is a good tradeoff
 - Other solutions?

Converter	Complexity	Conversion time
Flash	2^{n_b}	1
...	n_b	1
...	1	1
...	n_b	n_b
SAR	1	n_b
Tracking	1	2^{n_b}
Staircase	1	2^{n_b}

Analog to Digital Conversion

- Residue AD Converter

- SAR is a promising architecture because it gives a good tradeoff
- SAR implements a loop of n_b iterations
 - Unroll the loop
 - Use a separate comparator for each bit
 - Comparison threshold for the MSB is $S/2$
 - Comparison threshold for the MSB-1 is $S/4$ or $3S/4$
 - It depends on the value of the MSB
 - Comparison threshold for the MSB-2 is $S/8$ or $3S/8$ or $5S/8$ or $7S/8$
 - It depends on the value of the MSB and of the MSB-1
 - ...
 - There are n_b single bit stages in sequence
 - It is more expensive (complexity n_b)
 - Decisions are sequential
 - Total conversion time equal to a standard SAR converter



Analog to Digital Conversion

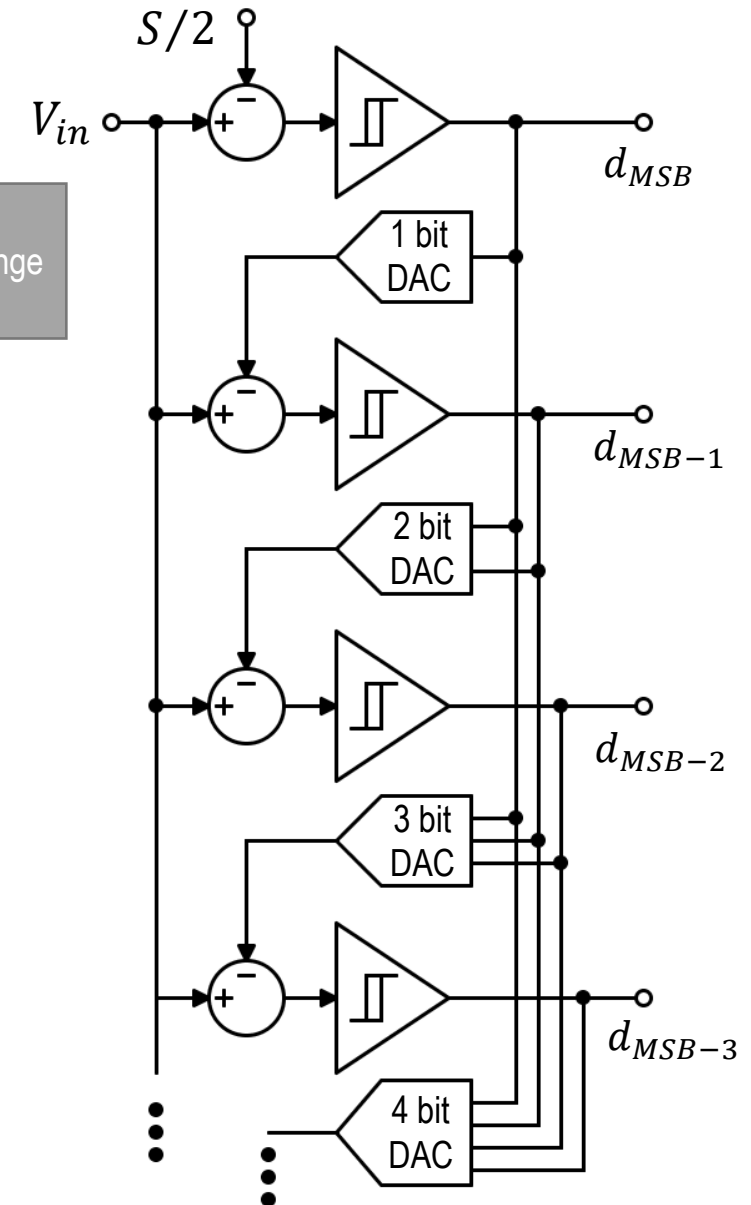
- Residue AD Converter

- SAR implements a loop of n_b iterations

- Comparisons in the unrolled ADCs are

- Stage1: MSB $\leftarrow V_{in} > \frac{S}{2}$
 - $2R_{MSB+1} > \frac{S}{2}$
 - Stage2: MSB - 1 $\leftarrow V_{in} > \frac{S}{2} \text{MSB} + \frac{S}{4}$
 - $2 \left(V_{in} - \frac{S}{2} \text{MSB} \right) > \frac{S}{2}$
 - $2 \left(2R_{MSB+1} - \frac{S}{2} \text{MSB} \right) > \frac{S}{2}$
 - $2R_{MSB} > \frac{S}{2}$
 - Stage3: MSB - 2 $\leftarrow V_{in} > \frac{S}{2} \text{MSB} + \frac{S}{4} (\text{MSB} - 1) + \frac{S}{8}$
 - $2 \left[2 \left(V_{in} - \frac{S}{2} \text{MSB} \right) - \frac{S}{2} (\text{MSB} - 1) \right] > \frac{S}{2}$
 - $2 \left(2R_{MSB} - \frac{S}{2} (\text{MSB} - 1) \right) > \frac{S}{2}$
 - $2R_{MSB-1} > \frac{S}{2}$

Assume V_{in} has a positive dynamic range between 0 and S



Analog to Digital Conversion

- Residue AD Converter

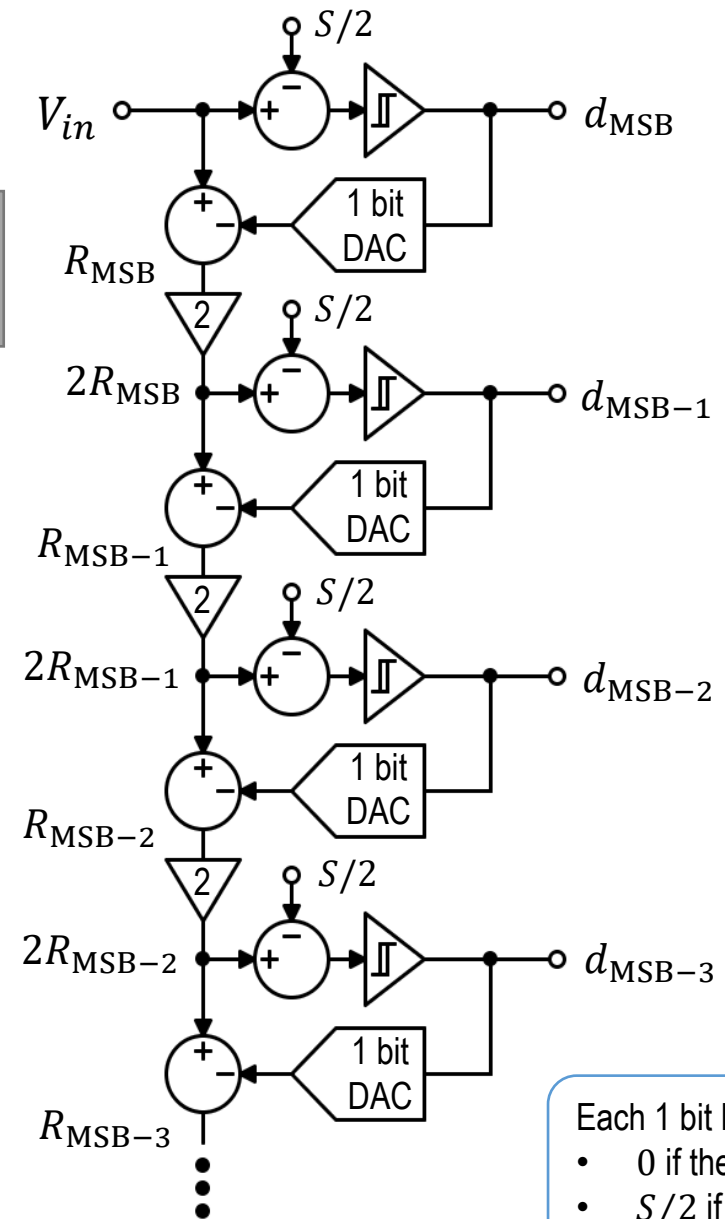
- In the general case, given bit d_i , with $i \in [0, n_b - 1]$, and with
 - $2R_{n_b} = V_{in}$
 - $R_i = 2R_{i+1} - \frac{S}{2}d_i$, with $i \in [0, n_b - 1]$
- Then conditions become
 - $2R_{i+1} > \frac{S}{2} \quad ? \quad \begin{cases} \text{True: } d_i \leftarrow 1 \\ \text{False: } d_i \leftarrow 0 \end{cases}$
- The quantity R_i is the *Residue* for bit d_i
 - It is the difference between V_{in} and its approximation on $n_b - i$ bits
 - Each residue is amplified to the full range S
 - Because each residue in the chain is multiplied by 2 to fill the input range of the following stage
- Let's turn these definitions into a circuit
 - Technique is also known as Subranging AD Converter

Analog to Digital Conversion

• Residue AD Converter

- $d_{n_b-1} \leftarrow 2R_{n_b} > \frac{S}{2}$
 - $d_{\text{MSB}} \leftarrow V_{in} > \frac{S}{2}$
- $R_{n_b-1} = 2R_{n_b} - \frac{S}{2}d_{\text{MSB}}$
 - $R_{\text{MSB}} = V_{in} - \frac{S}{2}d_{\text{MSB}}$
- $d_{n_b-2} \leftarrow 2R_{n_b-1} > \frac{S}{2}$
 - $d_{\text{MSB}-1} \leftarrow 2R_{\text{MSB}} > \frac{S}{2}$
- $R_{n_b-2} = 2R_{n_b-1} - \frac{S}{2}d_{\text{MSB}-1}$
 - $R_{\text{MSB}-1} = 2R_{\text{MSB}} - \frac{S}{2}d_{\text{MSB}-1}$
- ...

Assume V_{in} has a positive dynamic range between 0 and S



Each 1 bit DAC produces

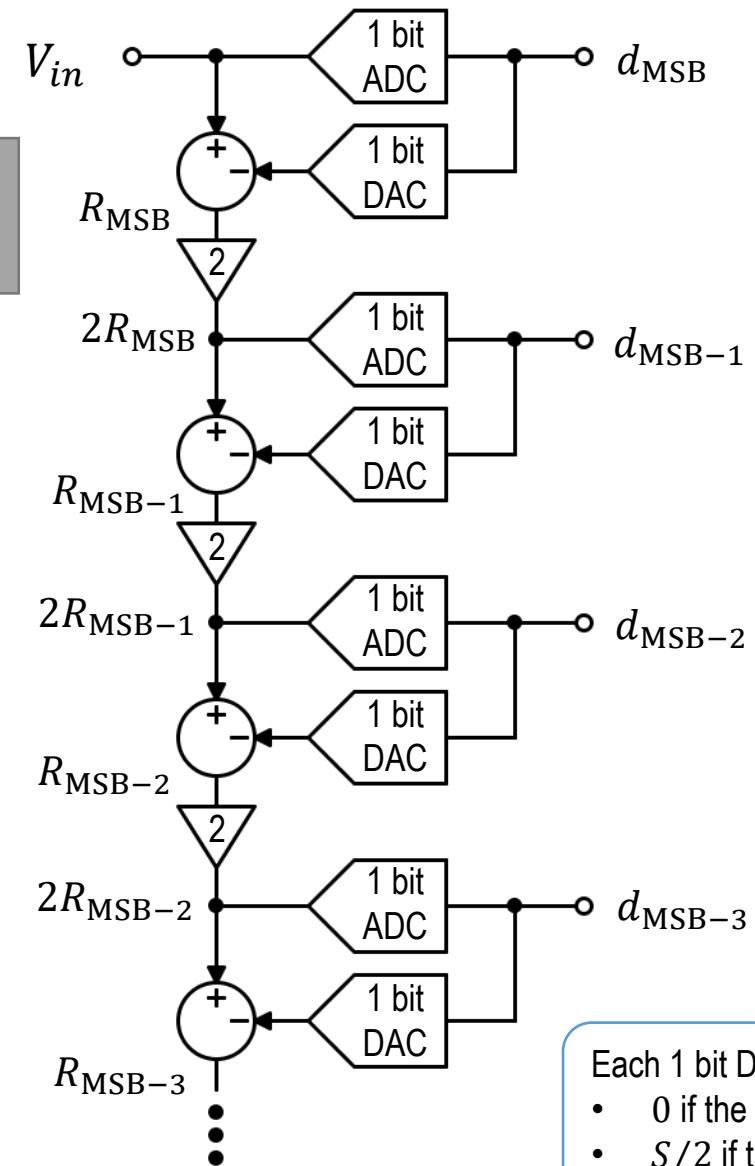
- 0 if the input is 0
- $S/2$ if the input is 1

Analog to Digital Conversion

• Residue AD Converter

- $d_{n_b-1} \leftarrow 2R_{n_b} > \frac{S}{2}$
 - $d_{\text{MSB}} \leftarrow V_{in} > \frac{S}{2}$
- $R_{n_b-1} = 2R_{n_b} - \frac{S}{2}d_{\text{MSB}}$
 - $R_{\text{MSB}} = V_{in} - \frac{S}{2}d_{\text{MSB}}$
- $d_{n_b-2} \leftarrow 2R_{n_b-1} > \frac{S}{2}$
 - $d_{\text{MSB}-1} \leftarrow 2R_{\text{MSB}} > \frac{S}{2}$
- $R_{n_b-2} = 2R_{n_b-1} - \frac{S}{2}d_{\text{MSB}-1}$
 - $R_{\text{MSB}-1} = 2R_{\text{MSB}} - \frac{S}{2}d_{\text{MSB}-1}$
- ...

Assume V_{in} has a positive dynamic range between 0 and S

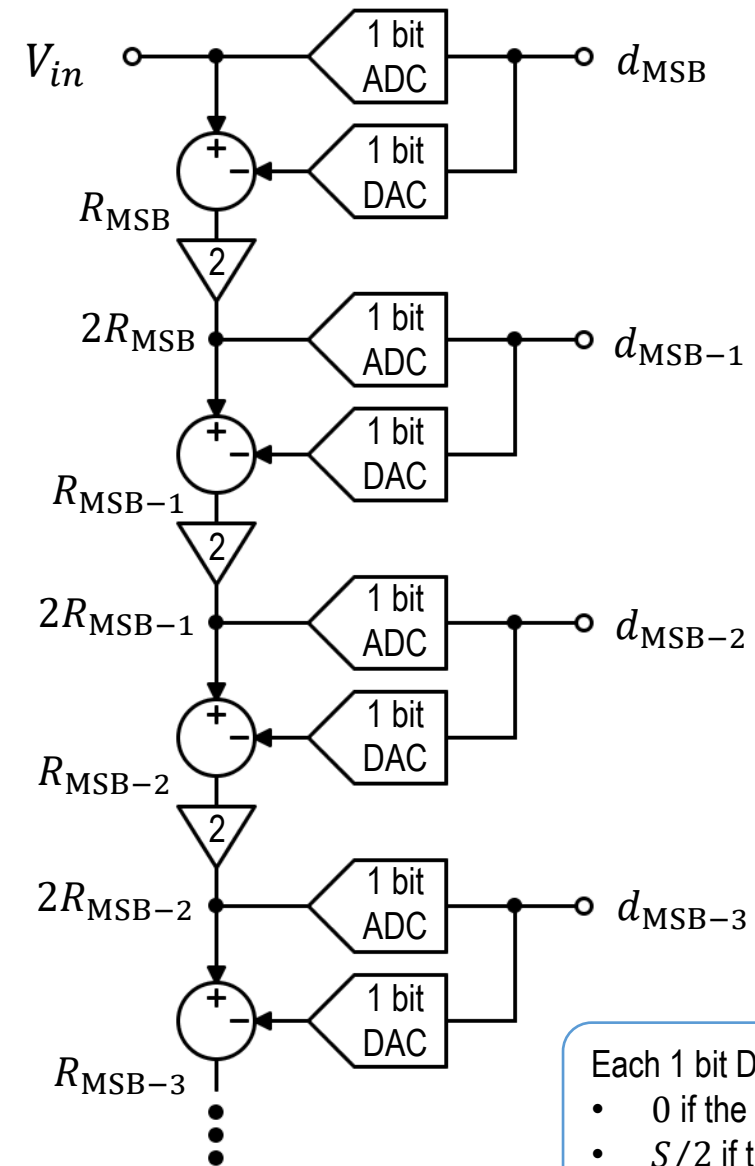


Each 1 bit DAC produces

- 0 if the input is 0
- $S/2$ if the input is 1

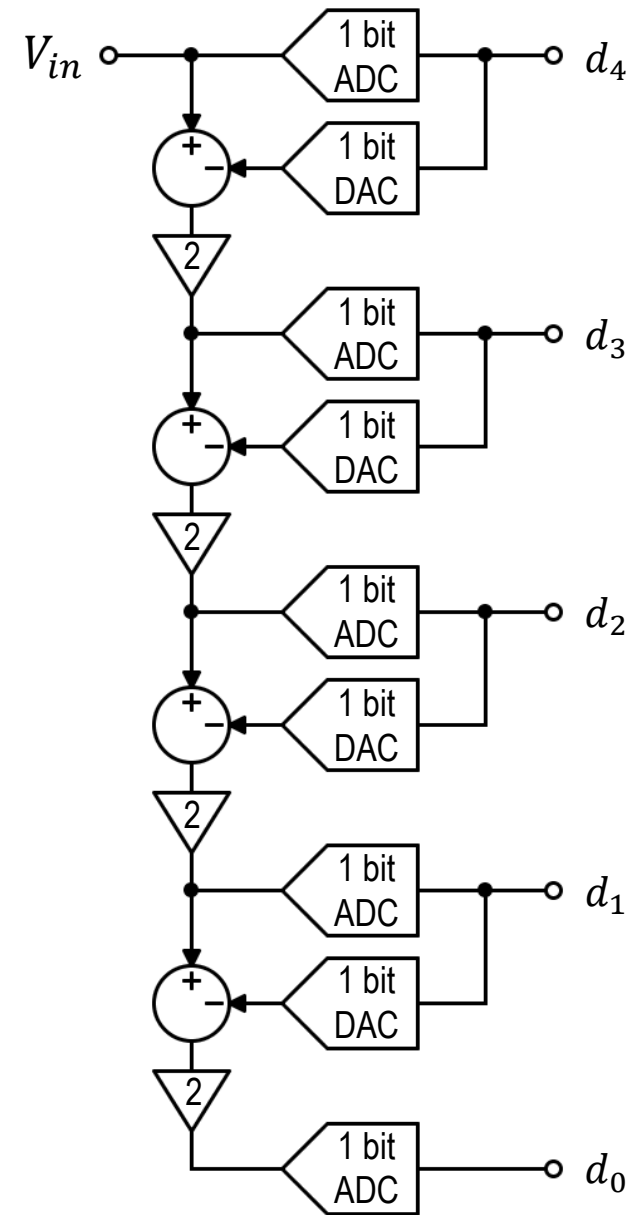
Analog to Digital Conversion

- Residue AD Converter
 - A 1-bit ADC is a single comparator
 - The adder/subtractor and the amplifier can be merged into a single block
 - A single operational amplifier $V_{out} = 2(V^+ - V^-)$
 - A 1-bit DAC is a simple switch
 - Connects either $S/2$ or ground to the output
 - Each stage is identical
 - Modular design, easy to extend to any number of bits
 - Complexity
 - Linear on the number of bits
 - Conversion time
 - Linear on the number of bits



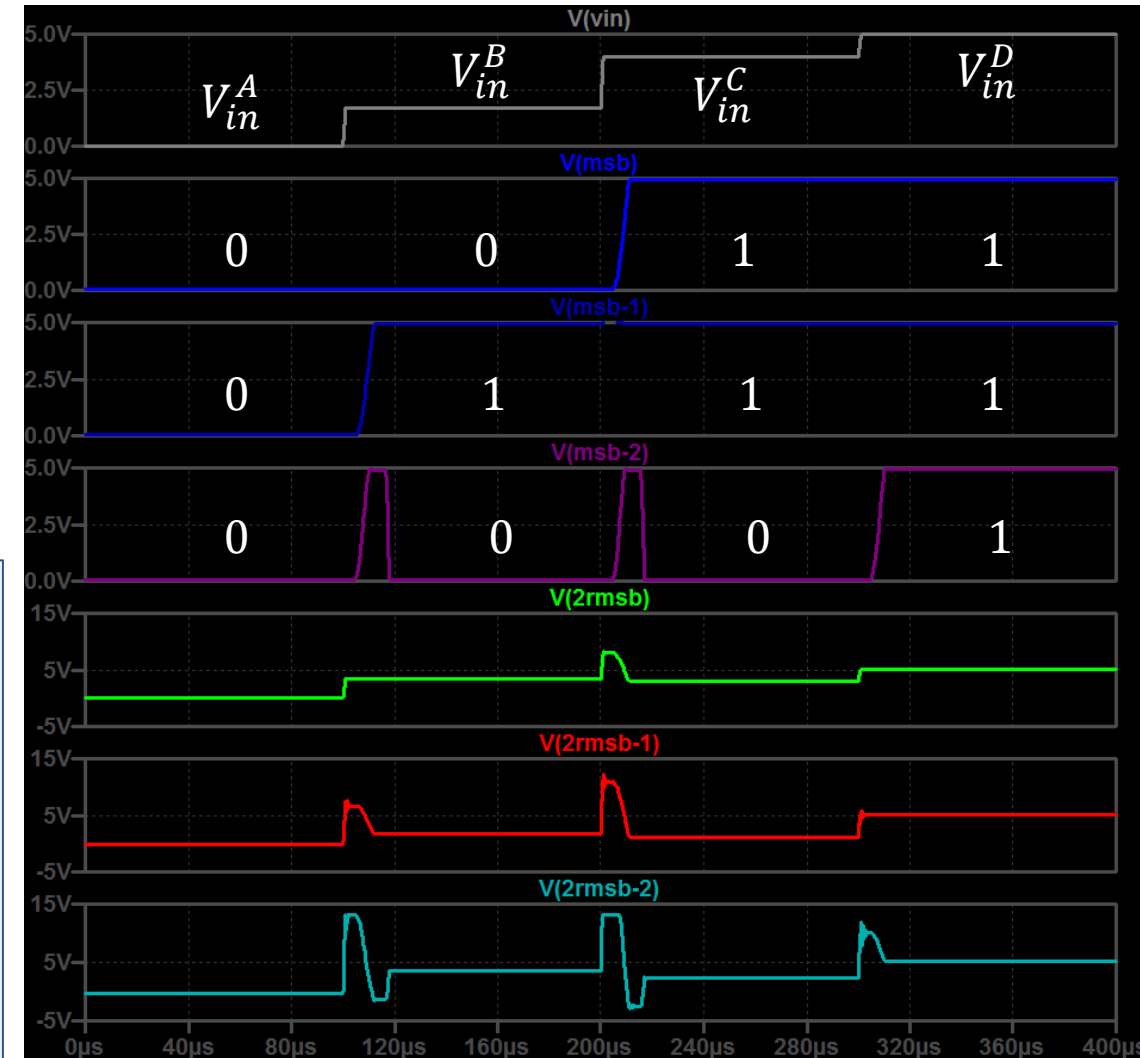
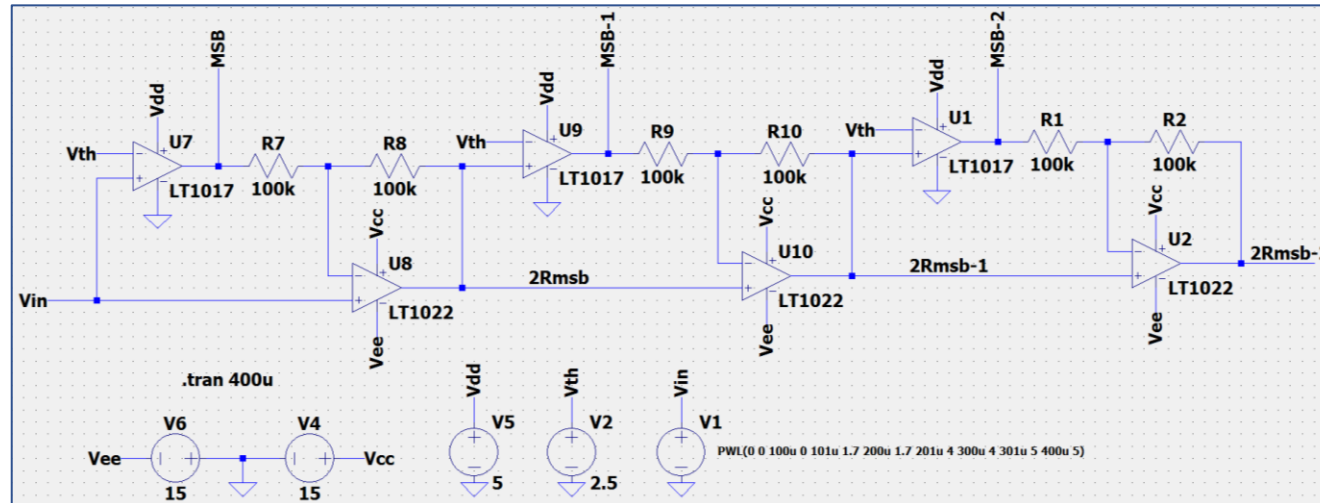
Analog to Digital Conversion

- Residue AD Converter example
 - 5-bit Residue AD Converter
 - Five stages
 - All stages are identical
 - Except the last one (LSB) that does not compute the residue
 - Delay for each stage
 - $t_{stage} = t_{ADC} + t_{DAC}$
 - $t_{stage} = t_{flash1} + t_{stl}$
 - Similar to one cycle of the SAR converter
 - Last stage (LSB) does not include t_{stl}
 - Stages are sequential
 - Total delay is the sum of all delays
 - $t_{conv} = (n_b - 1)(t_{flash1} + t_{stl}) + t_{flash1}$
 - $t_{conv} = n_b \cdot t_{flash1} + (n_b - 1) \cdot t_{stl}$



Analog to Digital Conversion

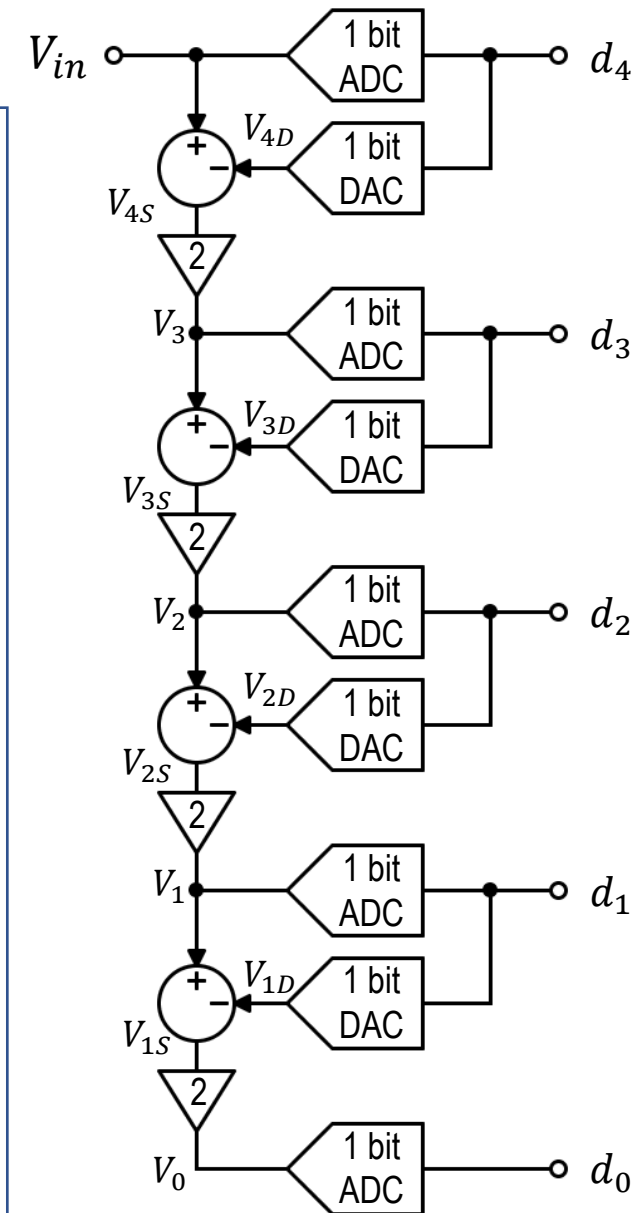
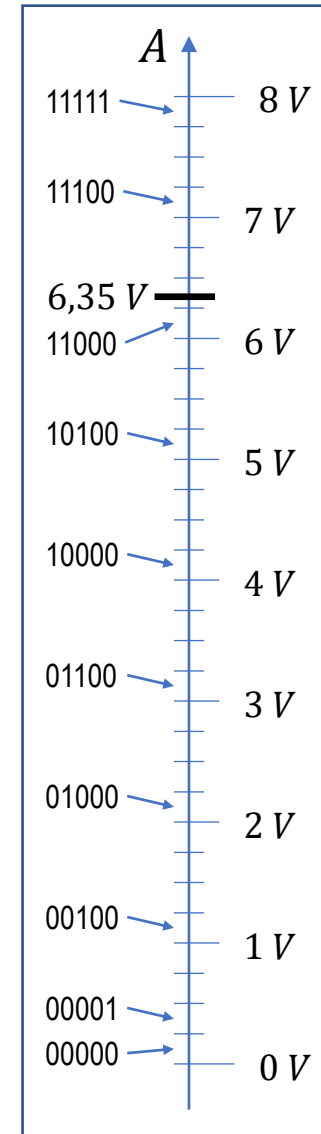
- Residue AD Converter
 - LTSpice example on 3 bits
 - $V_{in}^A = 0V$, $V_{in}^B = 1.7V$, $V_{in}^C = 4V$, $V_{in}^D = 5V$
 - $D^A = 000$, $D^B = 010$, $D^C = 110$, $D^D = 111$
 - There are some glitches while the data propagates along the converter



Analog to Digital Conversion

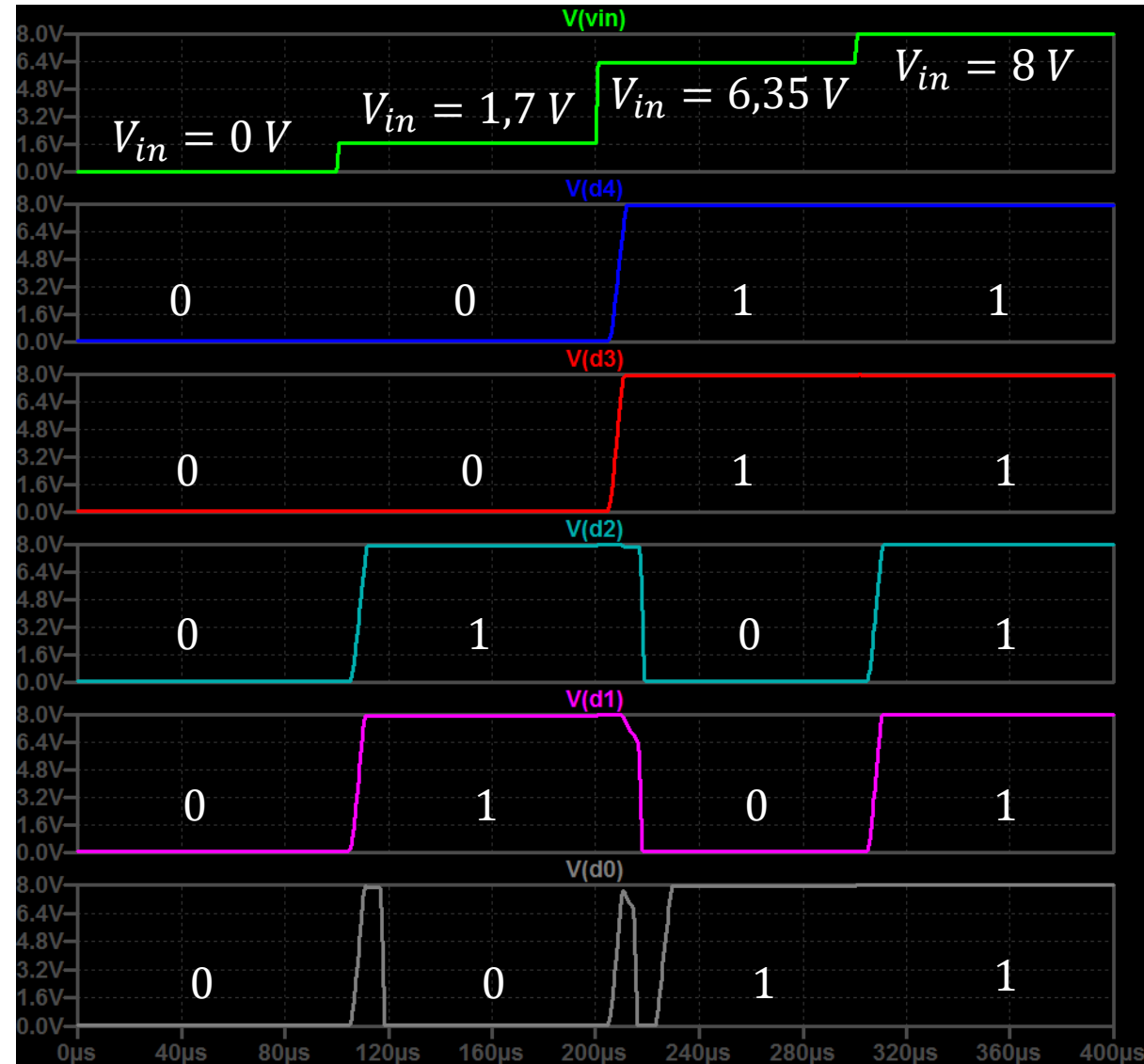
- Residue AD Converter example

- $V_{in} = 6.35 \text{ V}$, input dynamic range between 0 and 8 V
 - $d_4 = 1$, $V_{4D} = 4 \text{ V}$
 - $V_{4S} = 6.35 - 4 = 2.35 \text{ V}$, $V_3 = 4.7 \text{ V}$
 - $d_3 = 1$, $V_{3D} = 4 \text{ V}$
 - $V_{3S} = 4.7 - 4 = 0.7 \text{ V}$, $V_2 = 1.4 \text{ V}$
 - $d_2 = 0$, $V_{2D} = 0 \text{ V}$
 - $V_{2S} = 1.4 - 0 = 1.4 \text{ V}$, $V_1 = 2.8 \text{ V}$
 - $d_1 = 0$, $V_{1D} = 0 \text{ V}$
 - $V_{1S} = 2.8 - 0 = 2.8 \text{ V}$, $V_0 = 5.6 \text{ V}$
 - $d_0 = 1$



Analog to Digital Conversion

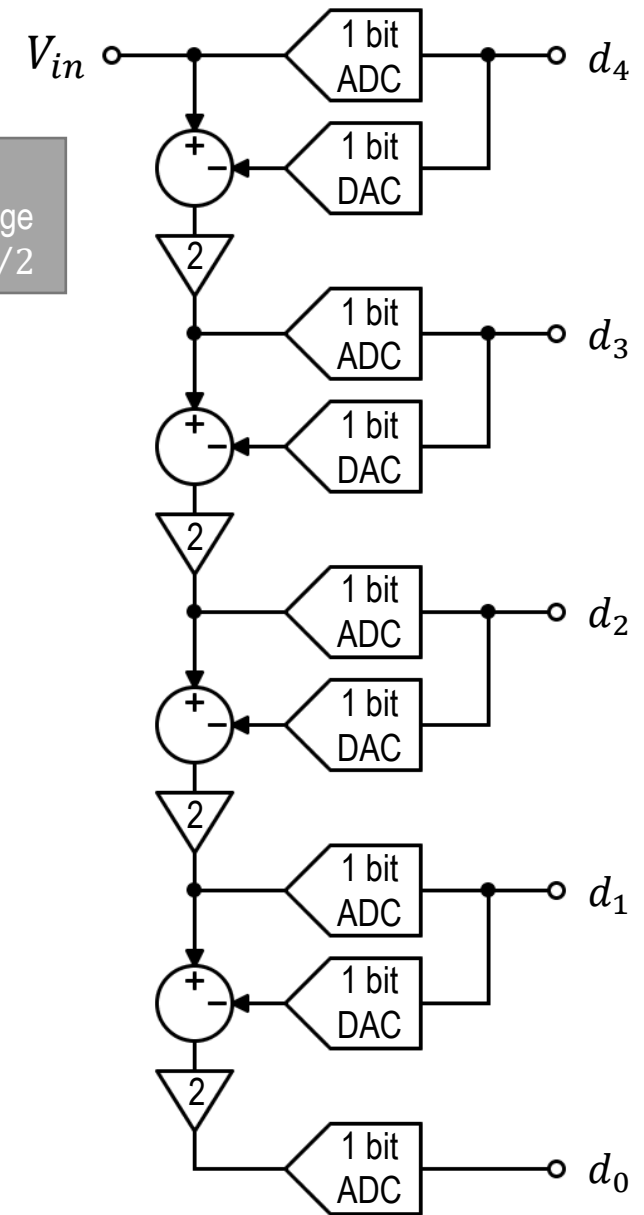
- Residue AD Converter example
 - $V_{in} = 6.35\text{ V}$, input dynamic range between 0 and 8 V
 - $d_4 = 1$, $V_{4D} = 4\text{ V}$
 - $V_{4S} = 6.35 - 4 = 2.35\text{ V}$, $V_3 = 4.7\text{ V}$
 - $d_3 = 1$, $V_{3D} = 4\text{ V}$
 - $V_{3S} = 4.7 - 4 = 0.7\text{ V}$, $V_2 = 1.4\text{ V}$
 - $d_2 = 0$, $V_{2D} = 0\text{ V}$
 - $V_{2S} = 1.4 - 0 = 1.4\text{ V}$, $V_1 = 2.8\text{ V}$
 - $d_1 = 0$, $V_{1D} = 0\text{ V}$
 - $V_{1S} = 2.8 - 0 = 2.8\text{ V}$, $V_0 = 5.6\text{ V}$
 - $d_0 = 1$



Analog to Digital Conversion

- Residue AD Converter
 - Symmetric dynamic range for V_{in}
 - The ADC should also have a symmetric input dynamic range
 - The DAC produces
 - $-S/4$ if the input is 0
 - $+S/4$ if the input is 1
 - The residue is between $-S/4$ and $+S/4$
 - When amplified by 2, the input to next stage has a symmetric dynamic range between $-S/2$ and $+S/2$
 - The adder/subtractor and/or the amplifier may change the dynamic range if needed
 - To match the dynamic range of the ADC
 - To match the dynamic range of the DAC

Assume V_{in} has a symmetric dynamic range between $-S/2$ and $S/2$



Analog to Digital Conversion

- Residue AD Converter
 - Better complexity with respect to a Flash AD Converter
 - But worse conversion time due to the sequential decisions
 - Worse complexity with respect to a SAR AD Converter
 - But same conversion time
 - Residue is not in any way better than SAR
 - No benefits
 - But it is a starting point for other architectures

Converter	Complexity	Conversion time
Flash	2^{n_b}	1
...	n_b	1
...	1	1
Residue	n_b	n_b
SAR	1	n_b
Tracking	1	2^{n_b}
Staircase	1	2^{n_b}

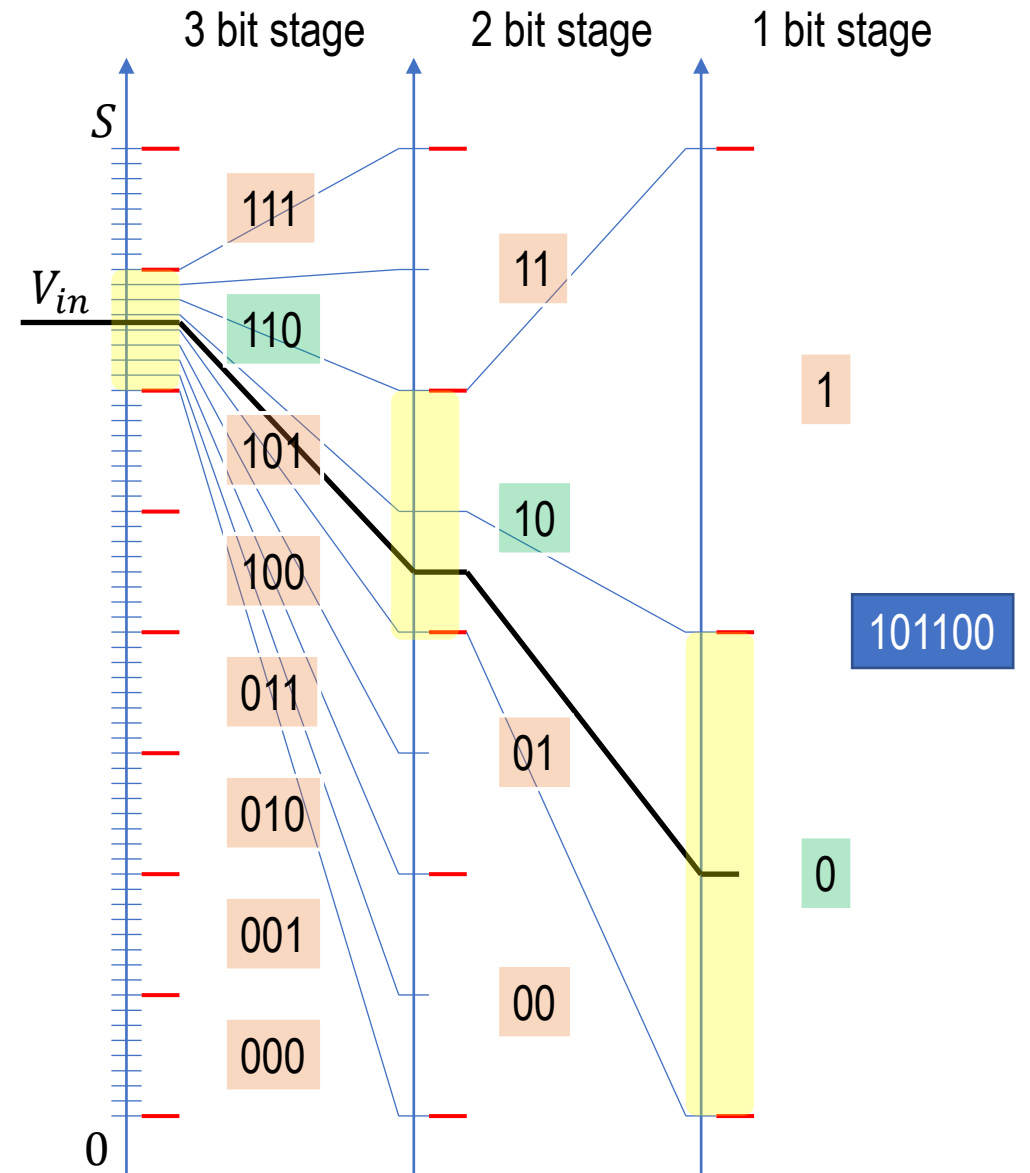
Analog to Digital Conversion

- Residue AD Converter
 - Any error in residue evaluation is propagated to the following stages
 - Residue must be evaluated with a resolution corresponding to the residual bit number
 - ADC precision
 - DAC precision
 - Amplifier
 - First stage ADC and DAC (1-bit)
 - Need n_b -bit precision
 - Precision decreases towards LSBs

Converter	Complexity	Conversion time
Flash	2^{n_b}	1
...	n_b	1
...	1	1
Residue	n_b	n_b
SAR	1	n_b
Tracking	1	2^{n_b}
Staircase	1	2^{n_b}

Analog to Digital Conversion

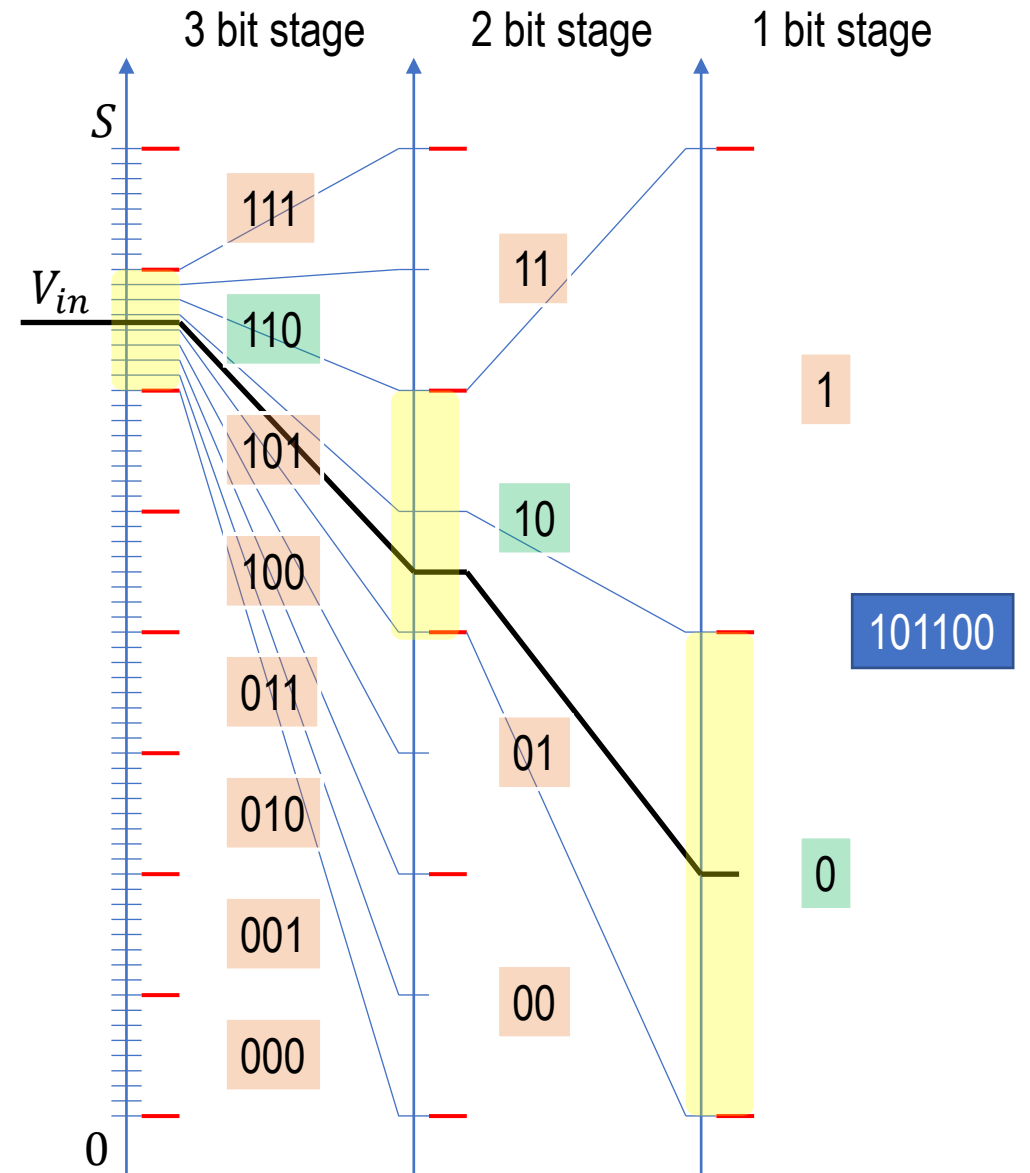
- Multibit Residue AD Converter
 - The residue can be evaluated on multiple bits in a single stage
 - Multiple thresholds compared at each stage
 - But not all thresholds like in a full Flash converter
 - Previous example compares one threshold in each stage
 - Each stage j can evaluate q_j bits
 - Stages are not necessarily identical
 - Comparators are replaced by a q_j -bit ADC
 - The approximation is built using a q_j -bit DAC
 - The amplifier gain is 2^{q_j}
 - Different speed/complexity tradeoff
 - Higher speed, with the same number of comparators
 - Less comparators for the same speed



Analog to Digital Conversion

- Multibit Residue AD Converter

- The residue can be evaluated on multiple bits in a single stage
 - Each q_j -bit ADC works on an input dynamic range of S
 - It divides the input dynamic range into 2^{q_j} intervals
 - Each interval is $A_d^j = \frac{S}{2^{q_j}}$
 - The error (i.e. the residue) is in the range
$$\varepsilon_j \in \left(-\frac{1}{2} \frac{S}{2^{q_j}}, +\frac{1}{2} \frac{S}{2^{q_j}} \right)$$
- To return it to the S dynamic range for the next stage it must be amplified
 - The amplifier gain is 2^{q_j}



Analog to Digital Conversion

- Multibit Residue AD Converter Example
 - 12-bit AD Converters using different techniques
 - Compare different solutions
 - Evaluate complexity
 - Count the number of comparators
 - Evaluate conversion time
 - Comparators with delay t_{comp}
 - DACs with settling time t_{DAC}
 - Regardless of the number of bits
 - Normalize with respect to t_{comp} and neglecting t_{DAC}
 - 12-bit Flash AD Converter
 - Complexity $C_{flash12} = 2^{12} - 1 = 4095$
 - Conversion time $T_{flash12} = t_{comp}$
 - Normalized conversion time $\bar{T}_{flash12} = 1$

Analog to Digital Conversion

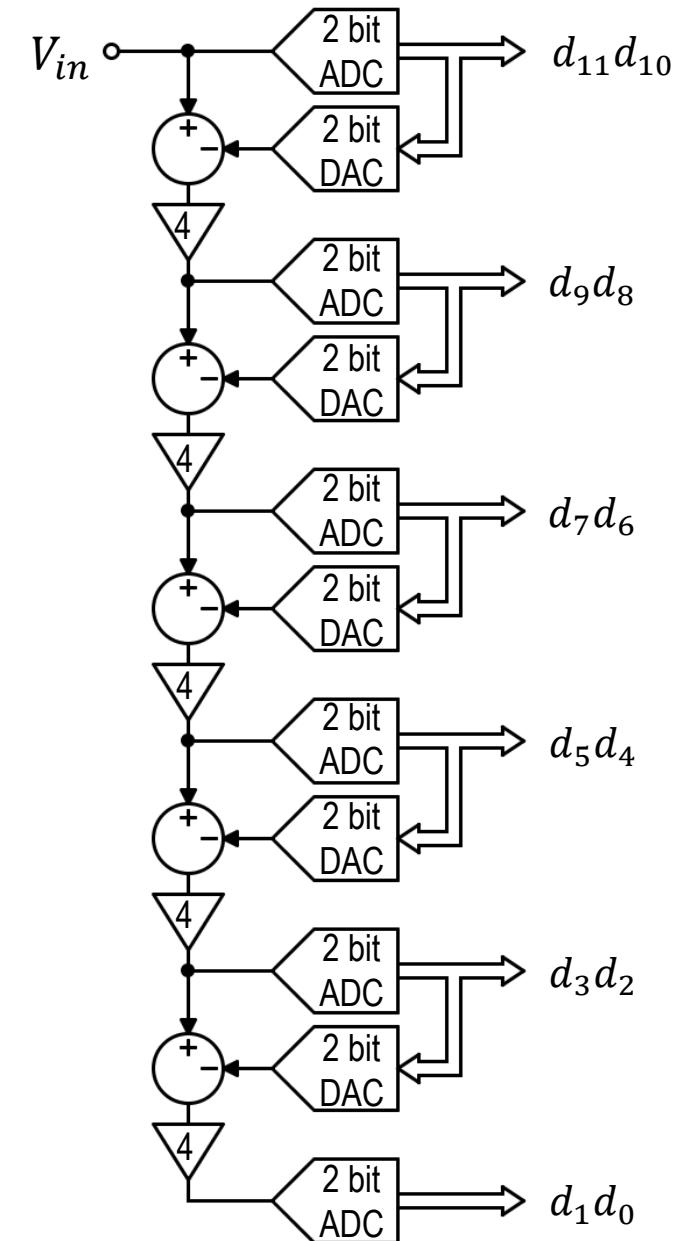
- Multibit Residue AD Converter Example
 - 12-bit SAR AD Converter
 - Complexity $C_{SAR12} = 1$
 - Conversion time $T_{SAR12} = 12 \cdot (t_{comp} + t_{DAC})$
 - Normalized conversion time $\bar{T}_{SAR12} = 12$
 - 12-bit Staircase/Tracking AD Converter
 - Complexity $C_{Stair} = 1$
 - Conversion time $T_{Stair} = 2^{12} \cdot t_{comp} + (2^{12} - 1) \cdot t_{DAC}$
 - Normalized conversion time $\bar{T}_{Stair} = 4096$
 - 12-bit Residue AD Converter with 1-bit stages
 - Complexity $C_{Res12x1} = 12$
 - Conversion time $T_{Res12x1} = 12 \cdot t_{comp} + 11 \cdot t_{DAC}$
 - Normalized conversion time $\bar{T}_{Res12x1} = 12$

Analog to Digital Conversion

- Multibit Residue AD Converter Example

- 12-bit Residue AD Converter with 2-bit stages ($q_j = 2$)

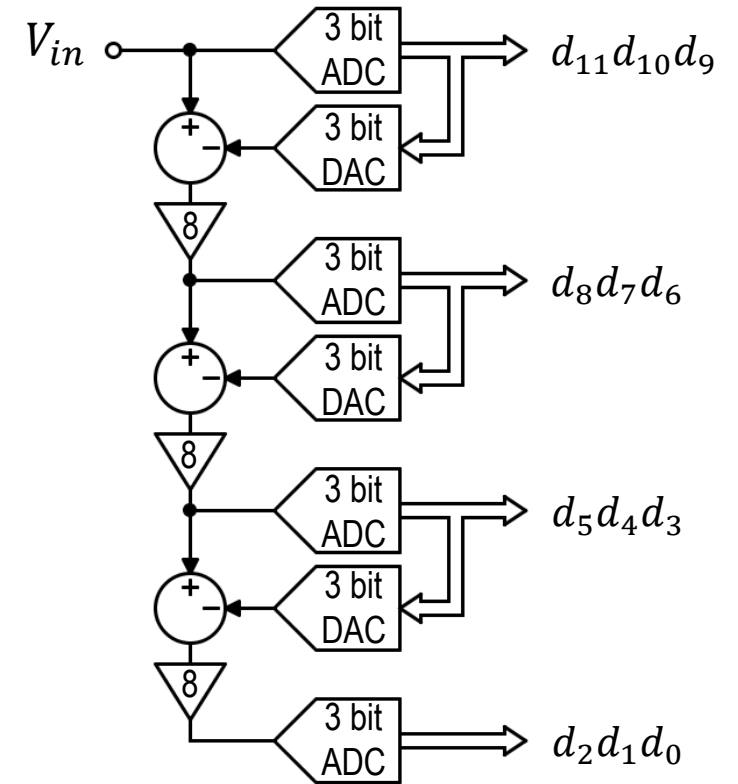
- All stages are identical
- ADC in each stage is a Flash converter on 2 bits
 - Complexity of a single stage is $C_{stage} = 2^2 - 1 = 3$
 - Conversion delay of each stage is $T_{stage} = t_{comp} + t_{DAC}$
 - Identical to single bit stages
- There are $n_{stage} = \frac{12}{2} = 6$ stages
- Gain of the amplifiers between each stage is $2^{q_j} = 2^2 = 4$
- Complexity $C_{Res6x2} = n_{stage} \cdot C_{stage} = 6 \cdot 3 = 18$
- Conversion time $T_{Res6x2} = 6 \cdot t_{comp} + 5 \cdot t_{DAC}$
 - Normalized conversion time $\bar{T}_{Res6x2} = 6$



Analog to Digital Conversion

- Multibit Residue AD Converter Example

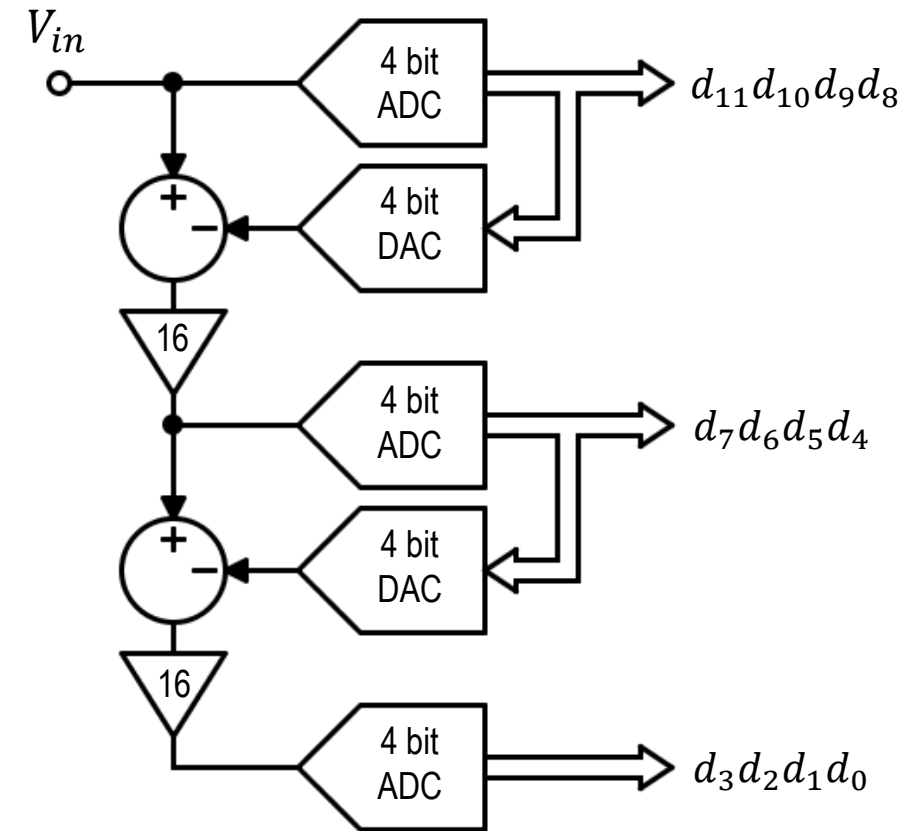
- 12-bit Residue AD Converter with 3-bit stages ($q_j = 3$)
 - All stages are identical
 - ADC in each stage is a Flash converter on 3 bits
 - Complexity of a single stage is $C_{stage} = 2^3 - 1 = 7$
 - Conversion delay of each stage is $T_{stage} = t_{comp} + t_{DAC}$
 - Identical to single bit and 2-bit stages
 - There are $n_{stage} = \frac{12}{3} = 4$ stages
 - Gain of the amplifiers between each stage is $2^{q_j} = 2^3 = 8$
 - Complexity $C_{Res4x3} = n_{stage} \cdot C_{stage} = 4 \cdot 7 = 28$
 - Conversion time $T_{Res4x3} = 4 \cdot t_{comp} + 3 \cdot t_{DAC}$
 - Normalized conversion time $\bar{T}_{Res4x3} = 4$



Analog to Digital Conversion

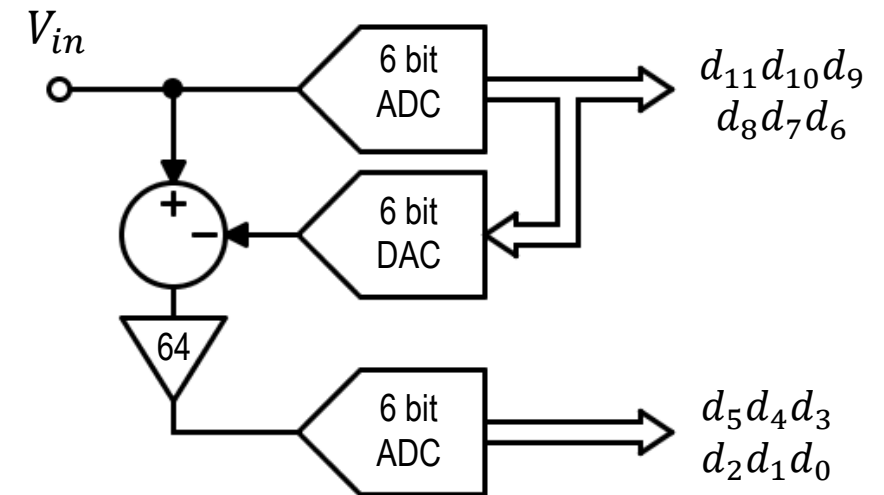
- Multibit Residue AD Converter Example

- 12-bit Residue AD Converter with 4-bit stages ($q_j = 4$)
 - All stages are identical
 - ADC in each stage is a Flash converter on 4 bits
 - Complexity of a single stage is $C_{stage} = 2^4 - 1 = 15$
 - Conversion delay of each stage is $T_{stage} = t_{comp} + t_{DAC}$
 - Identical to single bit, 2-bit and 3-bit stages
 - There are $n_{stage} = \frac{12}{4} = 3$ stages
 - Gain of the amplifiers between each stage is $2^{q_j} = 2^4 = 16$
 - Complexity $C_{Res3x4} = n_{stage} \cdot C_{stage} = 3 \cdot 15 = 45$
 - Conversion time $T_{Res3x4} = 3 \cdot t_{comp} + 2 \cdot t_{DAC}$
 - Normalized conversion time $\bar{T}_{Res3x4} = 3$



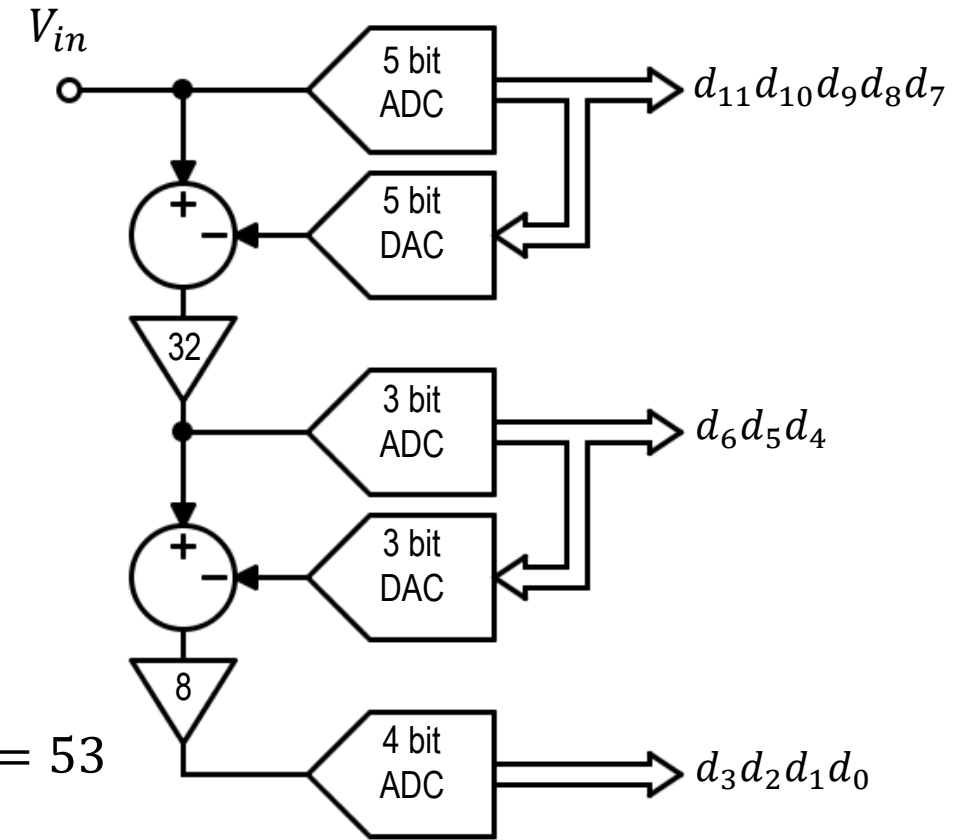
Analog to Digital Conversion

- Multibit Residue AD Converter Example
 - 12-bit Residue AD Converter with 6-bit stages ($q_j = 6$)
 - All stages are identical
 - ADC in each stage is a Flash converter on 6 bits
 - Complexity of a single stage is $C_{stage} = 2^6 - 1 = 63$
 - Conversion delay of each stage is $T_{stage} = t_{comp} + t_{DAC}$
 - Identical to single bit, 2-bit, 3-bit and 4-bit stages
 - There are $n_{stage} = \frac{12}{6} = 2$ stages
 - Gain of the amplifiers between each stage is $2^{q_j} = 2^6 = 64$
 - Complexity $C_{Res2x6} = n_{stage} \cdot C_{stage} = 2 \cdot 63 = 126$
 - Conversion time $T_{Res2x6} = 2 \cdot t_{comp} + t_{DAC}$
 - Normalized conversion time $\bar{T}_{Res2x6} = 2$



Analog to Digital Conversion

- Multibit Residue AD Converter Example
 - 12-bit Residue AD Converter with 3 stages
 - Stages are different
 - ADC in each stage is a Flash converter on 3, 4 or 5 bits
 - Complexity of stages is
 - $C_{st1} = 2^5 - 1 = 31$
 - $C_{st2} = 2^3 - 1 = 7$
 - $C_{st3} = 2^4 - 1 = 15$
 - Conversion delay of each stage is $T_{stage} = t_{comp} + t_{DAC}$
 - Identical to single bit, 2-bit, 3-bit, 4-bit and 6-bit stages
 - Complexity $C_{Res534} = C_{st1} + C_{st2} + C_{st3} = 31 + 7 + 15 = 53$
 - Conversion time $T_{Res534} = 3 \cdot t_{comp} + 2 \cdot t_{DAC}$
 - Normalized conversion time $\bar{T}_{Res534} = 3$



Analog to Digital Conversion

- Multibit Residue AD Converter Example

- Summary

- Multiple solutions with different tradeoffs
 - High complexity, low conversion time
 - Low complexity, high conversion time
 - Middle complexity, middle conversion time
 - Values in the table are normalized and approximated
 - When implementing the converter, a more precise analysis is necessary
 - In general a K stages residue converter with q_j bits per stage
 - Residue $K \times q_j$
 - Complexity $\mathcal{O}\left(\frac{n_b}{q_j} \cdot 2^{q_j}\right)$
 - Conversion time $\mathcal{O}\left(\frac{n_b}{q_j}\right)$

Converter	Complexity	Conversion time
Flash 12	4095	1
Residue 2x6	126	2
Residue 5,3,4	53	3
Residue 3x4	45	3
Residue 4x3	28	4
Residue 6x2	18	6
Residue 12x1	12	12
SAR	1	12
Staircase/Tracking	1	4096

Analog to Digital Conversion

- Multibit Residue AD Converters

- Summary

- Full design space

- Use stages with different number of bits
 - Use SAR converters instead of Flash converters in some stages

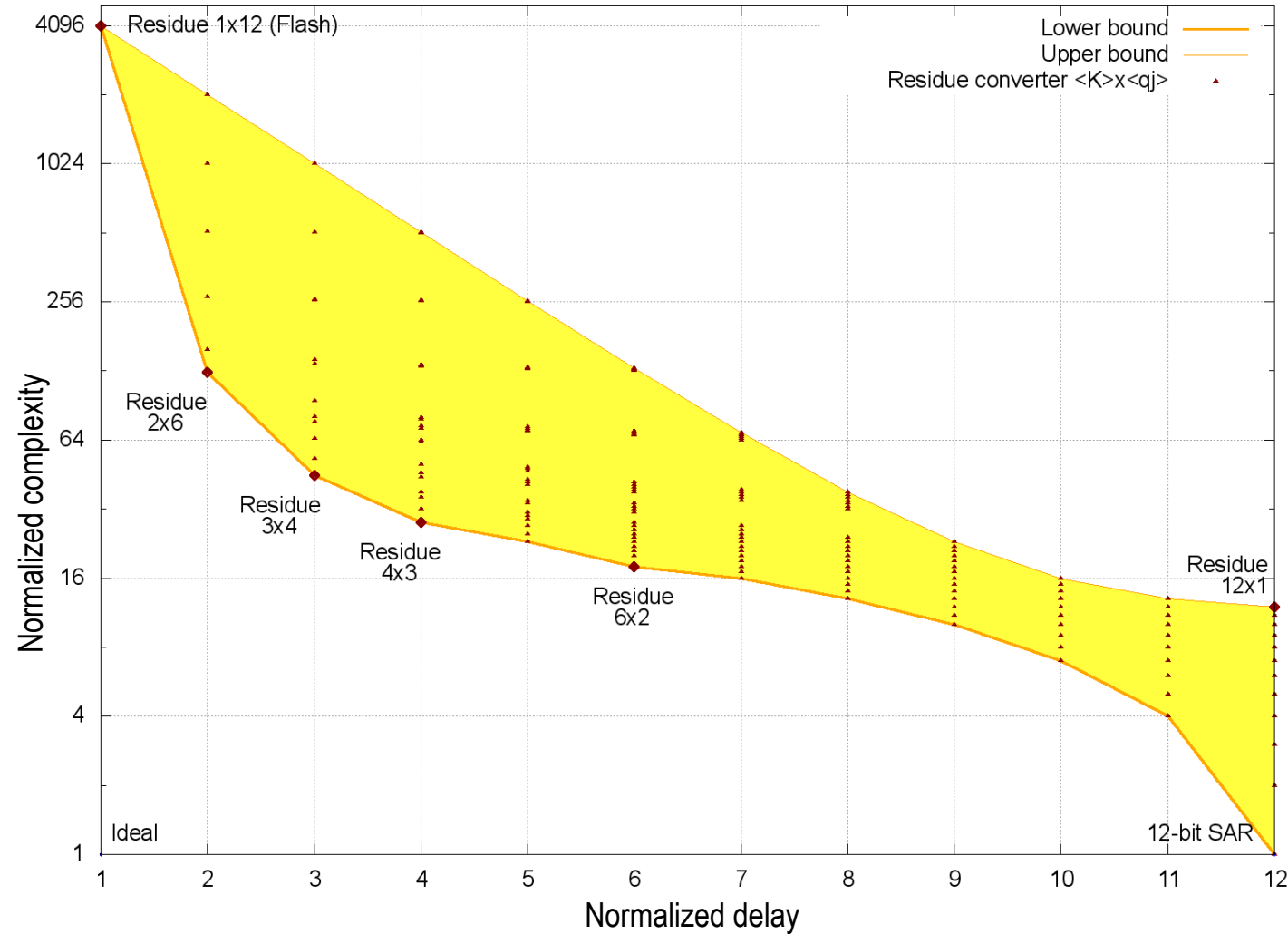
- The design space does not show the Staircase AD Converter

- Horizontal axis is linear

- The vertical axis is logarithmic

- In all analyzed cases a single sample is considered at any given time

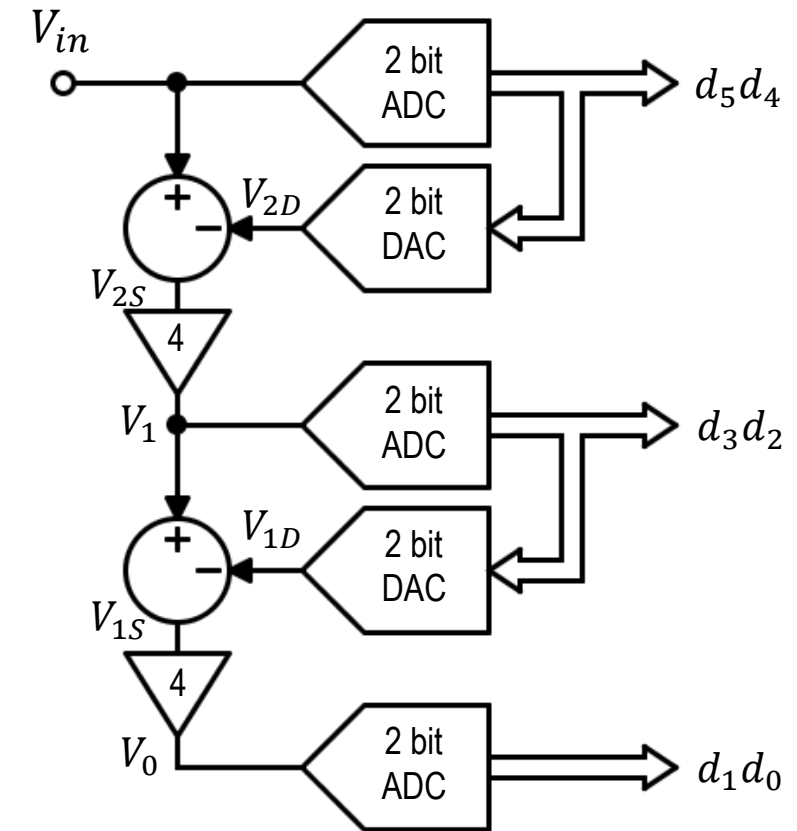
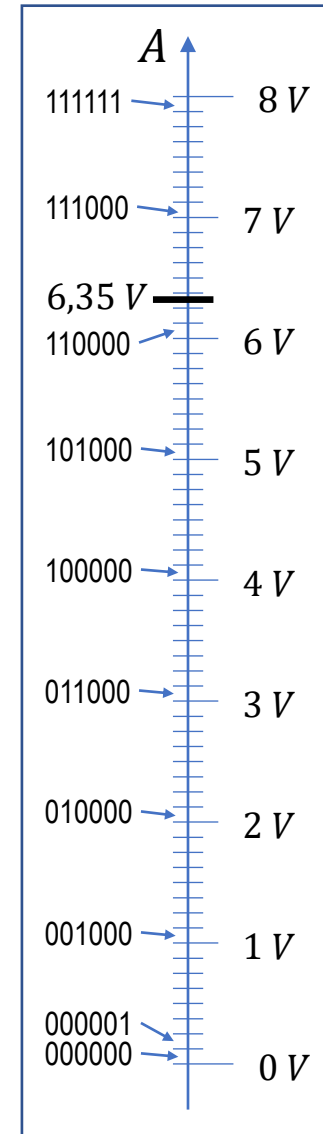
- Conversion time is also the latency between start and end of conversion
 - Throughput is the reciprocal of latency in all cases



Analog to Digital Conversion

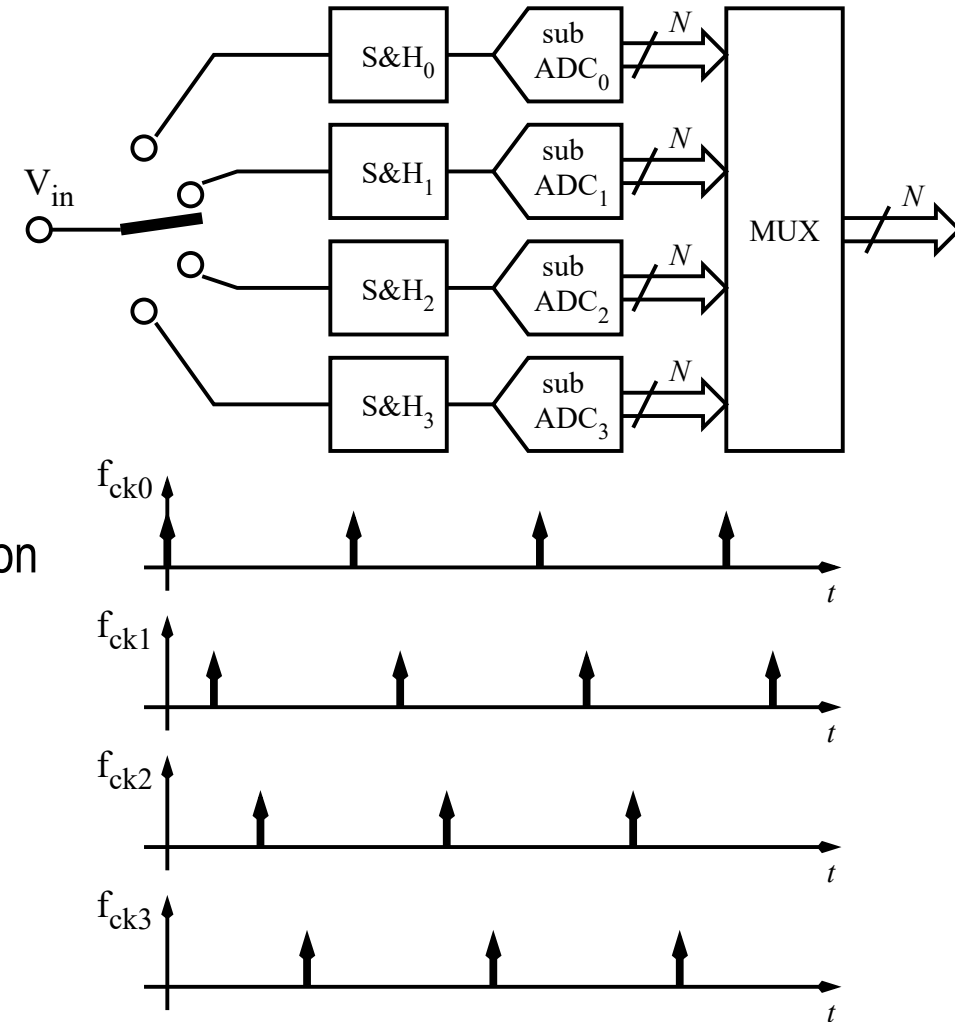
- Multibit Residue AD Converter Example

- $V_{in} = 6.35\text{ V}$, input dynamic range between 0 and 8 V, 6 bits
 - $d_5d_4 = 11$, $V_{2D} = 6\text{ V}$
 - $V_{2S} = 6.35 - 6 = 0.35\text{ V}$, $V_1 = 1.4\text{ V}$
 - $d_3d_2 = 00$, $V_{1D} = 0\text{ V}$
 - $V_{1S} = 1.4 - 0 = 1.4\text{ V}$, $V_0 = 5.6\text{ V}$
 - $d_1d_0 = 10$



Analog to Digital Conversion

- Parallel AD Converters
 - Process multiple samples at the same time
 - Parallel architectures
- Time interleaved AD Converter
 - U conversion chains on the same input signal
 - Each including a Sample&Hold and an AD Converter on n_b bits
 - Switch from one chain to the next while conversion is still going on
 - Switching rate is faster than conversion rate
 - Switching rate corresponds to sampling rate
 - A multiplexer working at the switching rate selects the output
 - Conversion latency equal to the latency of a single chain
 - Throughput is U times better than the reciprocal of latency
 - Complexity is U times worse than a single chain

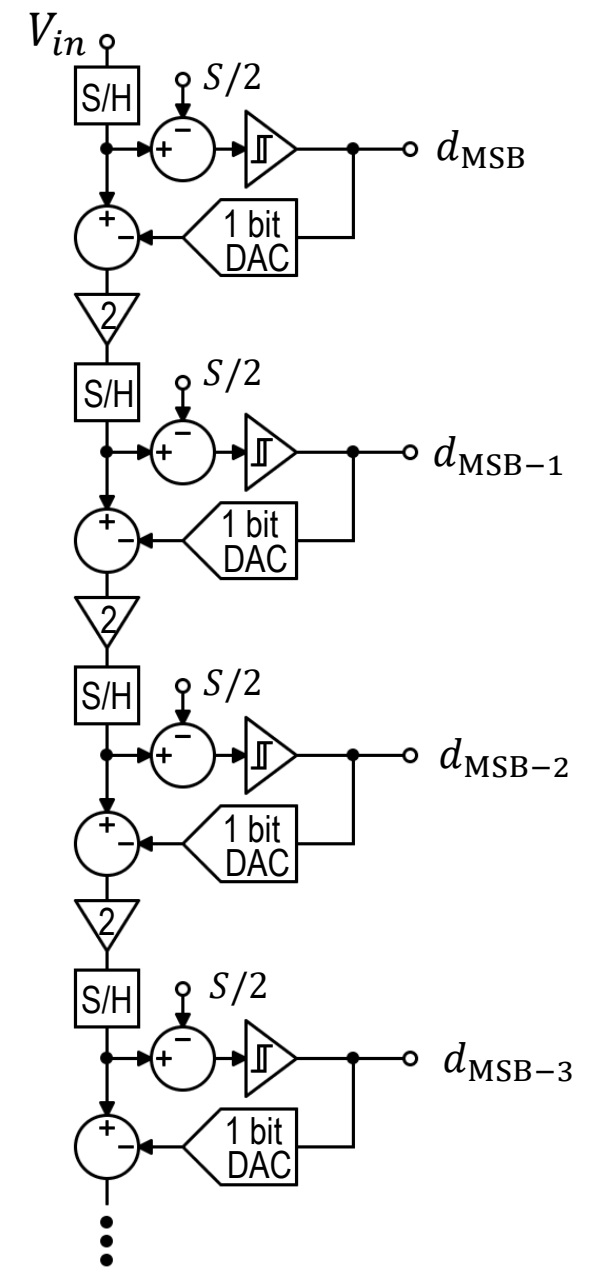
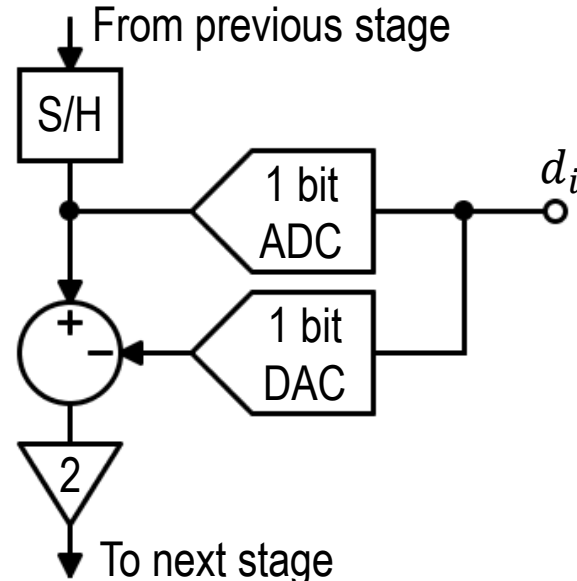
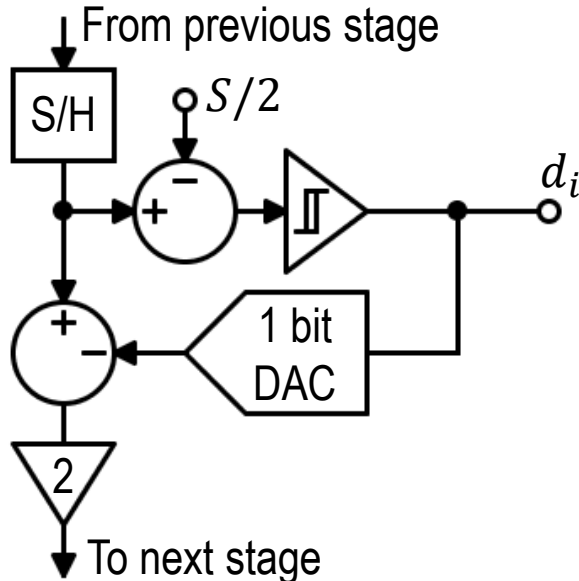


Analog to Digital Conversion

- Pipeline AD Converter
 - Residue AD Converters have multiple stages
 - Stages operate sequentially
 - They are not in parallel
 - Residues need to be propagated
 - Create a pipeline between stages
 - Each stage operates on a different sample
 - Need analog memory elements to separate stages
 - Sample & Hold circuit
 - Latency of a conversion is not improved
 - It even get worse because of the presence of the additional memory elements
 - Throughput is improved
 - One full output at each cycle once the pipeline is full
 - Cycle time set by the slowest stage of the pipeline

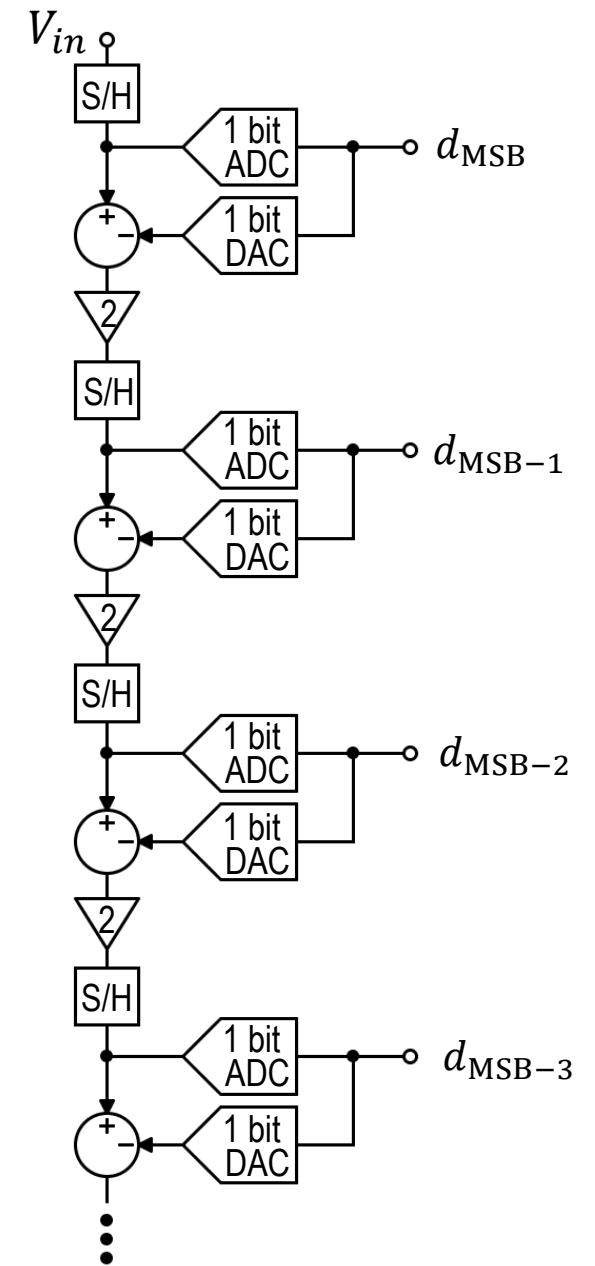
Analog to Digital Conversion

- Pipeline AD Converter Example
 - Start from a Residue AD Converter with 1-bit stages
 - Add a Sample&Hold circuit between each stage
 - Driven at the sampling frequency
 - The first stage is also preceded by a Sample&Hold circuit



Analog to Digital Conversion

- Pipeline AD Converter Example
 - Each stage works on independent chunks of analog information
 - The residues from different samples
 - Decisions on different bits can be taken concurrently
 - Digital outputs in a given instant apply to different samples
 - Need to reconstruct a full sample in the digital domain
 - Use delay line (chain of Flip-Flops) of different length on the various outputs
 - All stages are identical
 - Modular design
 - Can be extended to any number of bits



Analog to Digital Conversion

- Pipeline AD Converter Example

- Delay and rate of the Residue AD Converter with 1-bit stages

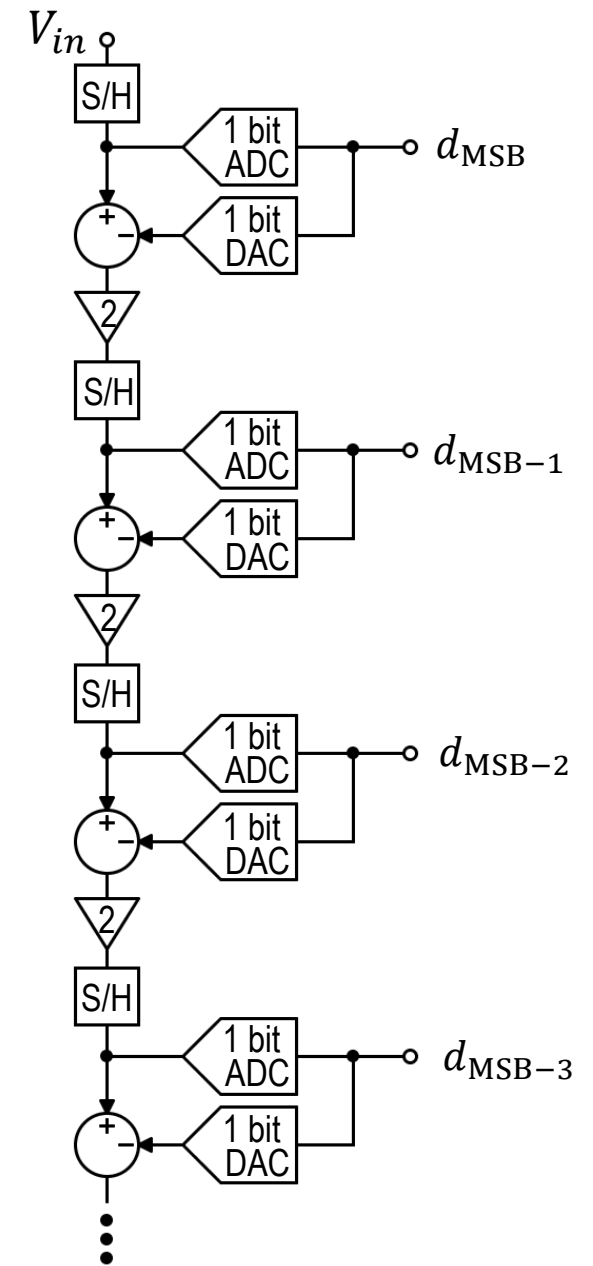
- Delay of a stage $T_{stage} = t_{comp} + t_{DAC}$
- Total delay of converter $T_{Res} = n_b \cdot T_{stage}$
- Maximum conversion rate $f_{convRes} = \frac{1}{T_{Res}} = \frac{1}{n_b} \frac{1}{T_{stage}}$

- Delay and rate of the Pipeline AD Converter with 1-bit stages

- Delay of a stage $T_{stage} = t_{comp} + t_{DAC} + t_{SH}$
- Total latency of converter $T_{Pipe} = n_b \cdot T_{stage}$
- Sample&Hold are activated with period T_{stage}
- Maximum conversion rate $f_{convPipe} = \frac{1}{T_{stage}}$

- Ratio between converters

- $\frac{f_{convPipe}}{f_{convRes}} = n_b$



Analog to Digital Conversion

- Pipeline AD Converter Example

- Comparison between 4-bit Residue and Pipeline AD Converter

- Input samples are A, B, C, D, ...
- One cycle is the conversion delay of a single stage
- Residue converter produces 3 output conversions in 12 cycles
- Pipeline converter produces 9 output conversions in 12 cycles
 - When pipeline is full it produces 12 output conversions in 12 cycles

Res	1	2	3	4	5	6	7	8	9	10	11	12
V_{in}	A	A	A	A	B	B	B	B	C	C	C	C
d_3	d_3^A	d_3^A	d_3^A	d_3^A	d_3^B	d_3^B	d_3^B	d_3^B	d_3^C	d_3^C	d_3^C	d_3^C
d_2	x	d_2^A	d_2^A	d_2^A	x	d_2^B	d_2^B	d_2^B	x	d_2^C	d_2^C	d_2^C
d_1	x	x	d_1^A	d_1^A	x	x	d_1^B	d_1^B	x	x	d_1^C	d_1^C
d_0	x	x	x	d_0^A	x	x	x	d_0^B	x	x	x	d_0^C

Pipe	1	2	3	4	5	6	7	8	9	10	11	12
V_{in}	A	B	C	D	E	F	G	H	I	J	K	L
d_3	d_3^A	d_3^B	d_3^C	d_3^D	d_3^E	d_3^F	d_3^G	d_3^H	d_3^I	d_3^J	d_3^K	d_3^L
d_2	x	d_2^A	d_2^B	d_2^C	d_2^D	d_2^E	d_2^F	d_2^G	d_2^H	d_2^I	d_2^J	d_2^K
d_1	x	x	d_1^A	d_1^B	d_1^C	d_1^D	d_1^E	d_1^F	d_1^G	d_1^H	d_1^I	d_1^J
d_0	x	x	x	d_0^A	d_0^B	d_0^C	d_0^D	d_0^E	d_0^F	d_0^G	d_0^H	d_0^I

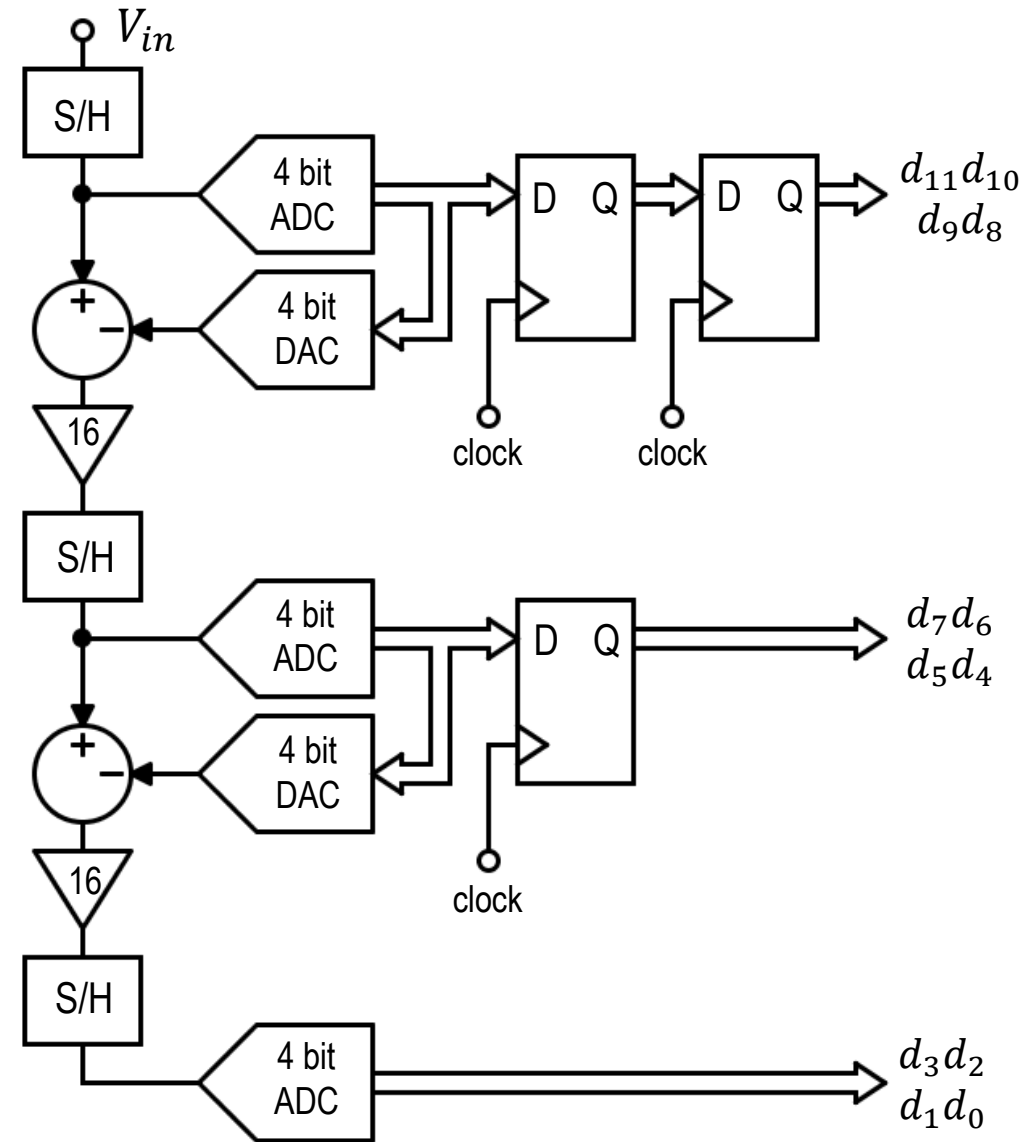
Analog to Digital Conversion

- Pipeline AD Converter
 - Conversion time
 - Latency $L_{Pipe} = \mathcal{O}(n_b)$
 - Equivalent Time $T_{Pipe} = \mathcal{O}(1)$
 - Same conversion time as the Flash converter
 - Complexity
 - Same as the Residue AD converter
 - $C_{Pipe} = \mathcal{O}(n_b)$
 - Have to consider the S&H complexity as well
 - The dependency does not change

Converter	Complexity	Conversion time
Flash	2^{n_b}	1
Pipeline	n_b	1
...	1	1
Residue	n_b	n_b
SAR	1	n_b
Tracking	1	2^{n_b}
Staircase	1	2^{n_b}

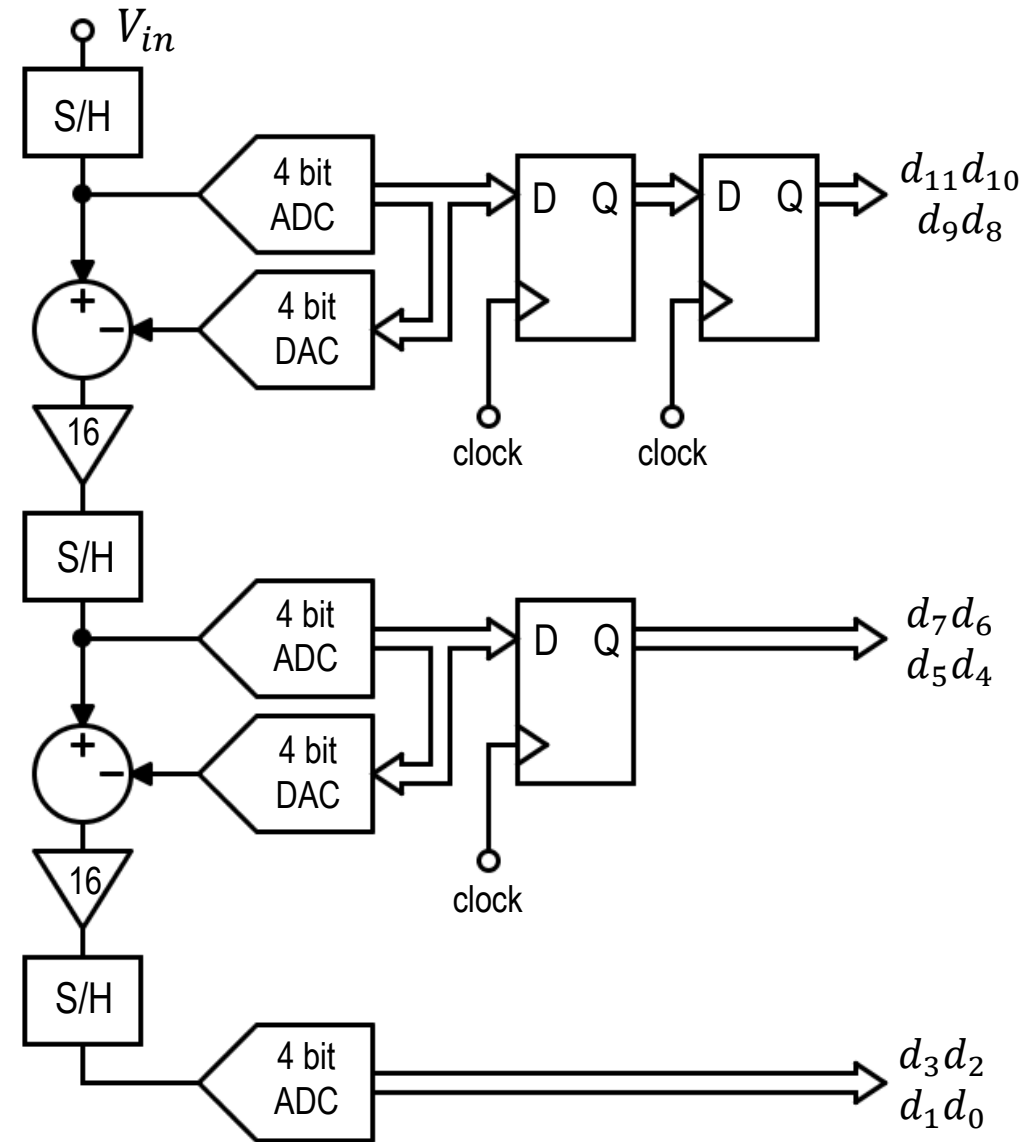
Analog to Digital Conversion

- Multibit Pipeline AD Converter
 - Start from a Multibit Residue AD Converter
 - Assume Flash converters are used in each stage
 - Residue 4x3 in the picture
 - Add analog memory elements between stages
 - Complexity
 - Similar to the Residue AD Converter
 - Add the Sample&Hold circuits
 - Conversion time
 - One full conversion at each cycle
 - Stage delay does not depend on the number of bits
 - If Flash ADCs are used in each stage
 - Same performance regardless of the number of bits per stage



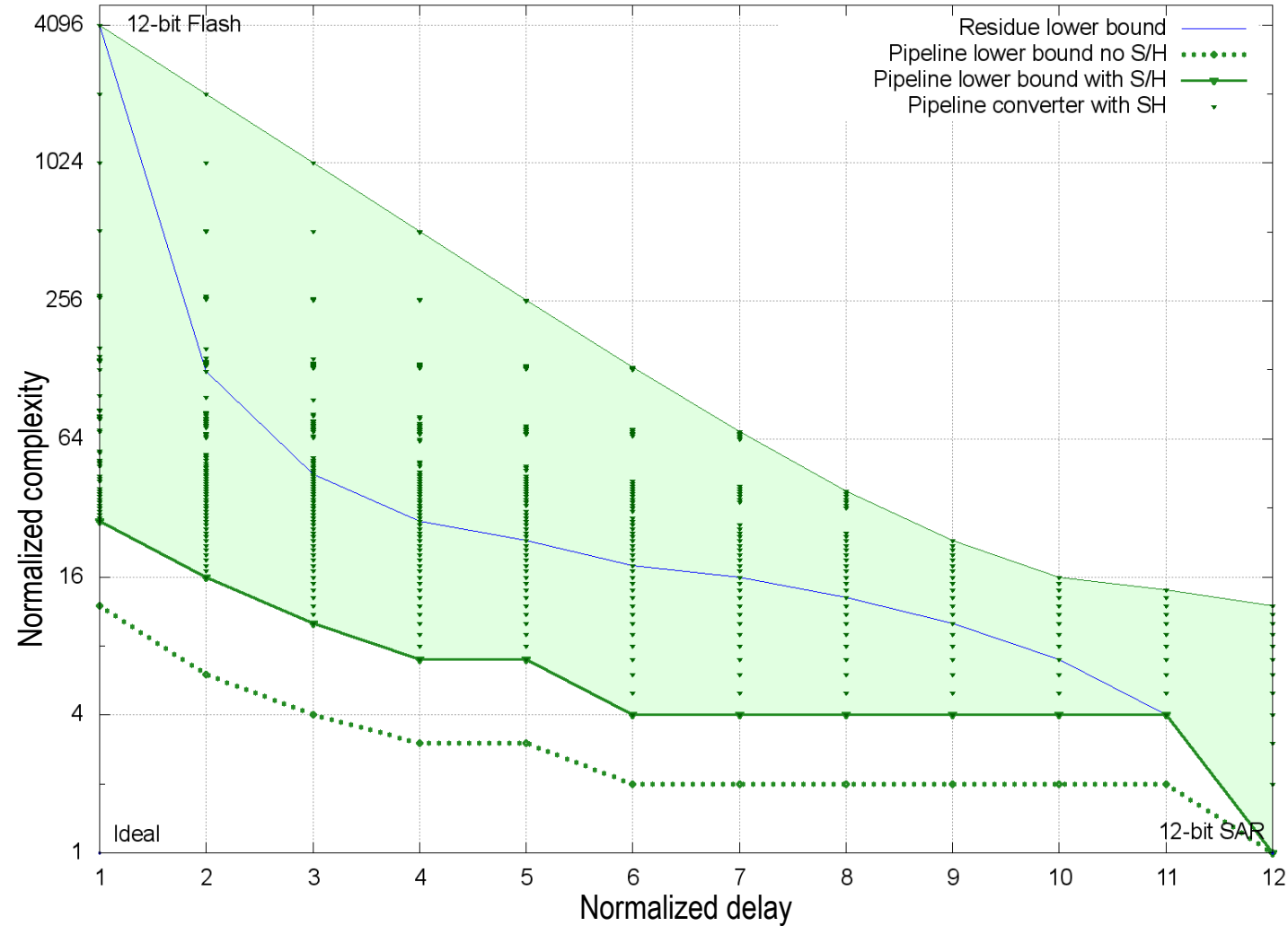
Analog to Digital Conversion

- Multibit Pipeline AD Converter
 - Stages can be different
 - Different number of bits
 - Different kind of AD Converter
 - Flash converter on q_j bits
 - Fast but complex
 - SAR converter on q_j bits
 - Slow but simple
 - If only Flash converters are used
 - The equivalent conversion time is always one cycle
 - If a mixed implementation is used
 - The slowest stage sets the maximum conversion rate
 - Balance stages to have similar delays



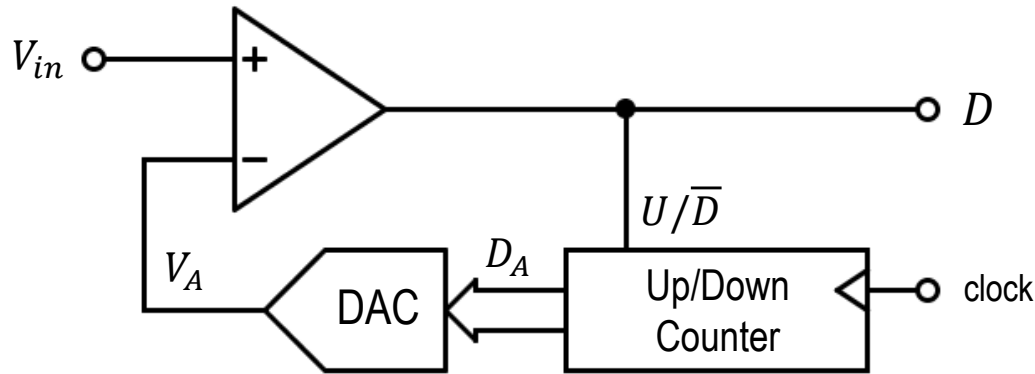
Analog to Digital Conversion

- Multibit Pipeline AD Converter
 - Full design space
 - Stages with different number of bits
 - Stages with different implementation
 - Extended towards lower complexities compared to the previous design space
 - Include also costs due to Sample&Hold circuits
- Multibit technique applicable to SAR and other architectures
 - Additional degrees of freedom
 - Further extends the design space



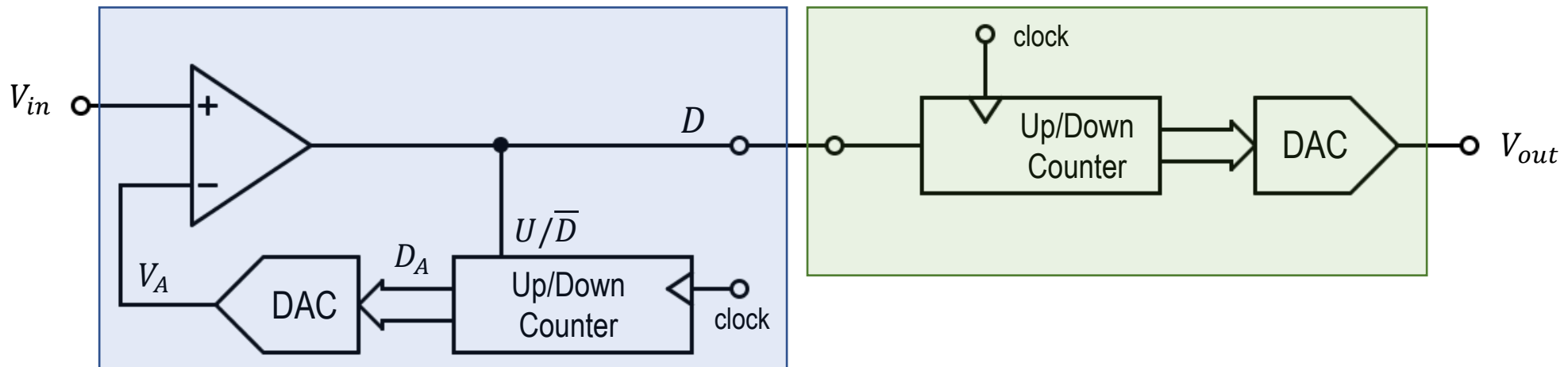
Analog to Digital Conversion

- Special AD Converters
 - The Tracking AD Converter is a differential converter
 - The serial bit flow from the comparator output represents the sign of $V_{in} - V_A$
 - Current value – previous value
 - It is a binary value
 - Logic 1: the current value is higher than the previous one
 - Logic 0: the current value is lower than the previous one



Analog to Digital Conversion

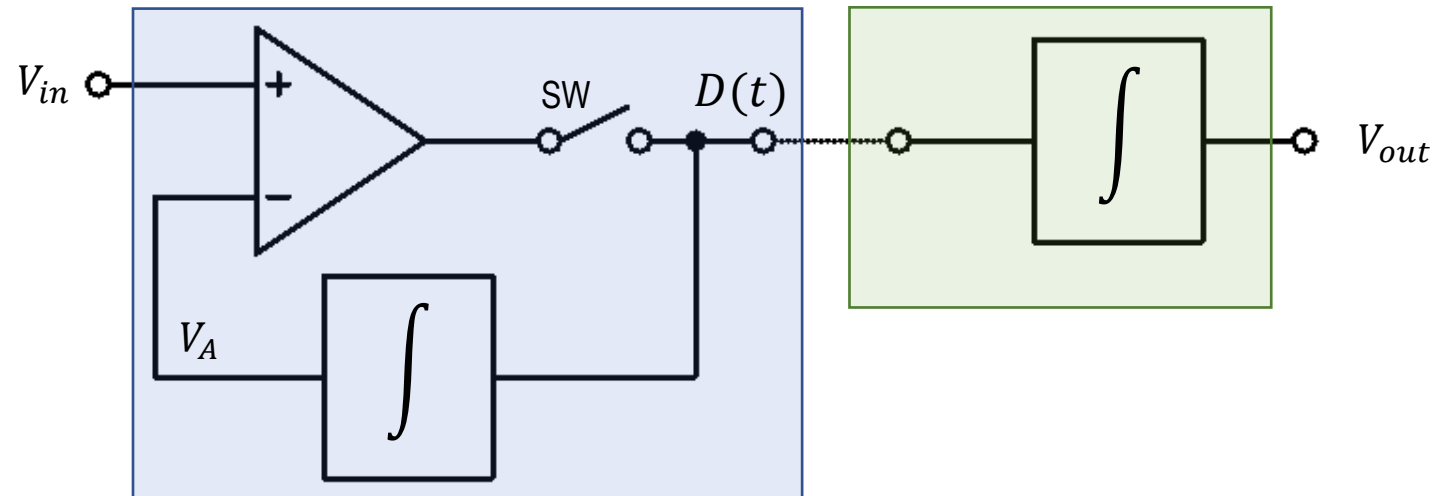
- Delta (Δ) AD Converter
 - The comparator output is the output of the converter
 - 1 bit AD conversion
 - Serial flow of uniform bits
 - Dequantization and reconstruction requires a similar structure
 - Up/Down counter and DAC
 - Anti-aliasing filter and reconstruction filter not shown



Analog to Digital Conversion

- Delta (Δ) AD Converter

- The sequence Up/Down Counter and DAC can be replaced by an integrator
 - Analog circuit
 - Digital output $D(t)$ is a sequence of positive or negative pulses
 - Otherwise the signal is 0
 - Pulses are integrated
 - The integrator output makes a positive or negative step of amplitude γ
 - The value γ plays the same role as the value A_d in a Tracking AD Converter
 - The pulse rate is $f_{ck} = \frac{1}{T_{ck}}$
 - The frequency f_{ck} is the one that drives the switch SW
 - It corresponds to the sampling frequency



Analog to Digital Conversion

- Delta (Δ) AD Converter

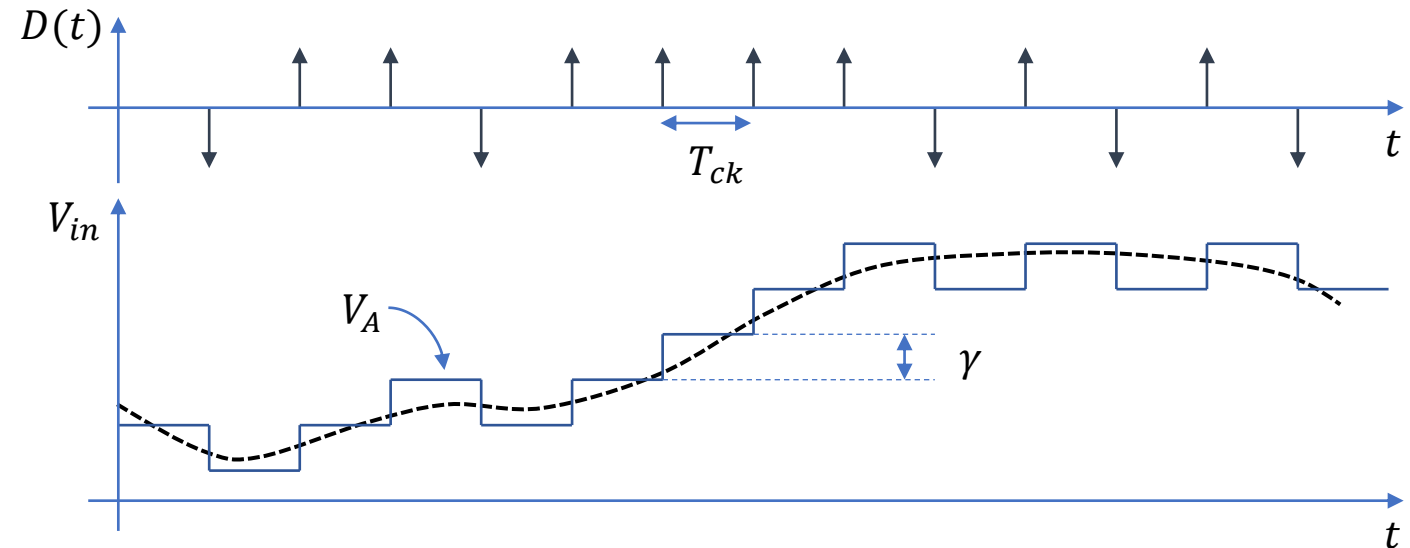
- Example

- Input analog signal characterized by its slew rate SR_{in}
 - For a sinusoidal signal it is $SR_{in} = V_P \omega$

- The step γ depends on
 - The duration of the pulse
 - The characteristics of the integrator

- The slew rate SR_{Δ} of the converter is given by

- $SR_{\Delta} = \frac{\gamma}{T_{ck}}$



Analog to Digital Conversion

- Delta (Δ) AD Converter

- Dynamic range

- Minimum input signal dynamic range

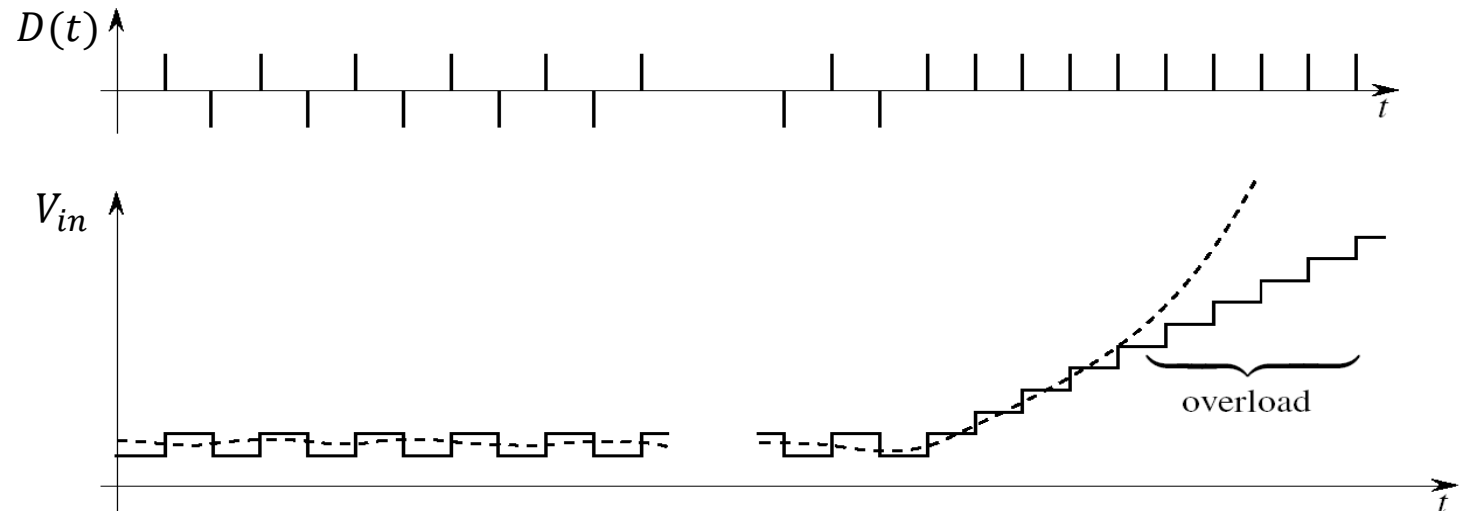
- $V_{min} = \frac{\gamma}{2}$
 - Idle noise

- Maximum input signal slew rate

- $SR_{max} = \frac{\gamma}{T_{ck}}$
 - Overload

- Input signal dynamic range is limited

- $\frac{\gamma}{2} < V_P < \frac{\gamma}{\omega T_{ck}}$



Analog to Digital Conversion

- Delta (Δ) AD Converter
 - Input signal dynamic range is limited
 - $\frac{\gamma}{2} < V_P < \frac{\gamma}{\omega T_{ck}}, \quad \frac{\gamma}{2} < V_P < \frac{\gamma}{\omega} f_{ck}$
 - Need a small value for γ
 - It defines the input signal quantization
 - Conversion is on 1 bit
 - Same quantization as a Tracking AD converter on n_b bits with $\gamma = \frac{S}{2^{n_b}}$
 - The effective number of bits is therefore $n_b = \log_2 \frac{S}{\gamma}$
 - Need a high value for f_{ck}
 - It corresponds to the sampling frequency
 - Much higher than the Nyquist frequency
 - OVERSAMPLING

Analog to Digital Conversion

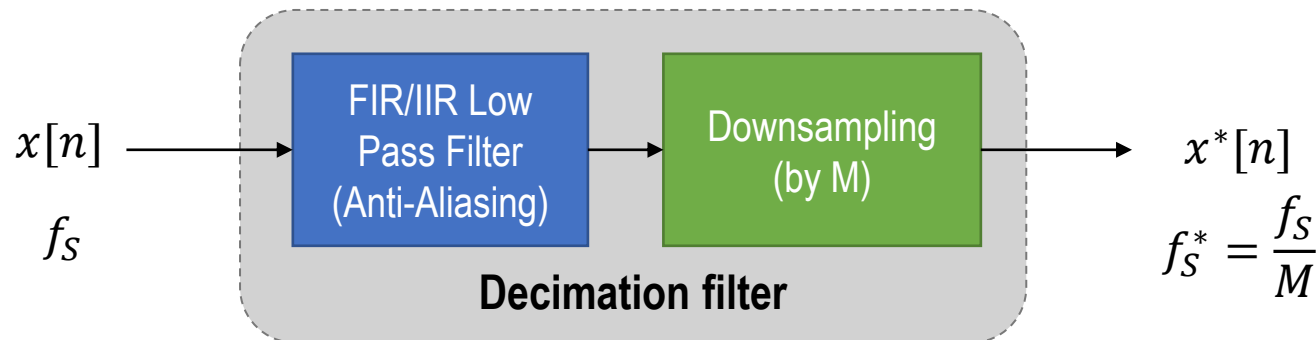
- Delta (Δ) AD Converter

- Example

- Input signal with $S = 5$ V, $f_{in} = 4$ kHz, equivalent quantization of $n_b = 8$ bits
 - $\gamma = \frac{S}{2^{n_b}} = \frac{5}{256} = 0.0195 = 19.5$ mV
 - $\omega_{in} = 2\pi f_{in} = 2\pi \cdot 4000 = 25133$ rad/s
 - $V_{Pmax} = 2.5$ V
 - Nyquist frequency $f_{Ny} = 2f_{in} = 2 \cdot 4000 = 8$ kHz
 - From the upper dynamic range condition (overload)
 - $f_{ck} > \frac{V_{Pmax} \cdot \omega}{\gamma} = \frac{2.5 \cdot 25133}{19.5 \cdot 10^{-3}} = 3.2$ MHz
 - The sampling frequency for the Delta AD Converter is 2.6 decades higher than the Nyquist frequency

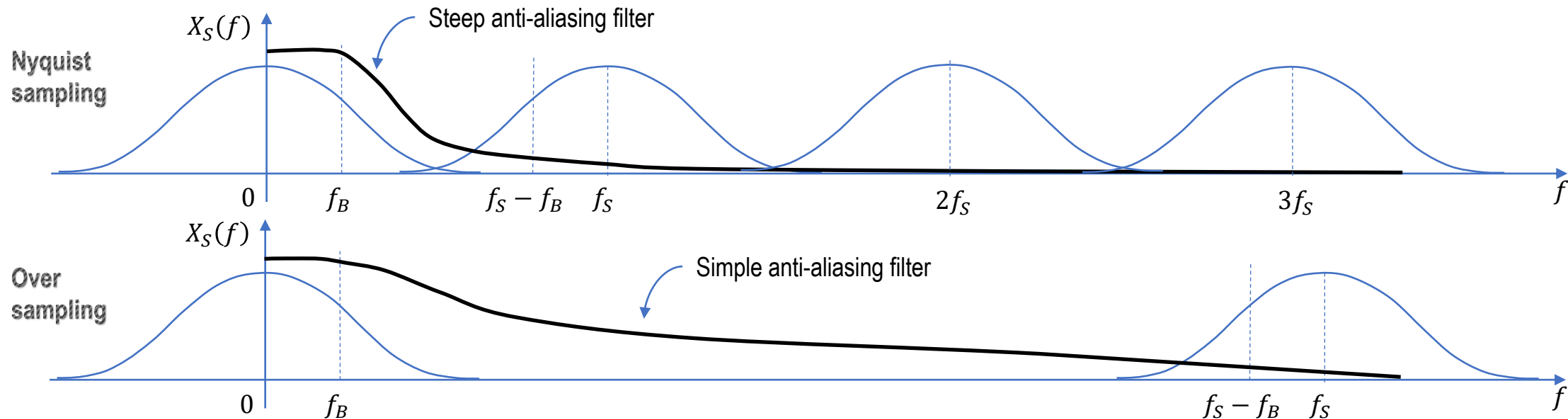
Analog to Digital Conversion

- Delta (Δ) AD Converter
 - Oversampling
 - It sends replicas in the sampled signal spectrum very far away
 - Reduce inband noise
 - Relaxed specifications of the anti-aliasing input filter
 - It generates a very high bit rate for the output stream
 - There are many more samples per second
 - The digital domain should handle the high bit rate
 - Complexity moved from the analog domain to the digital domain
 - Decimation filters in the digital domain to reduce the bit rate



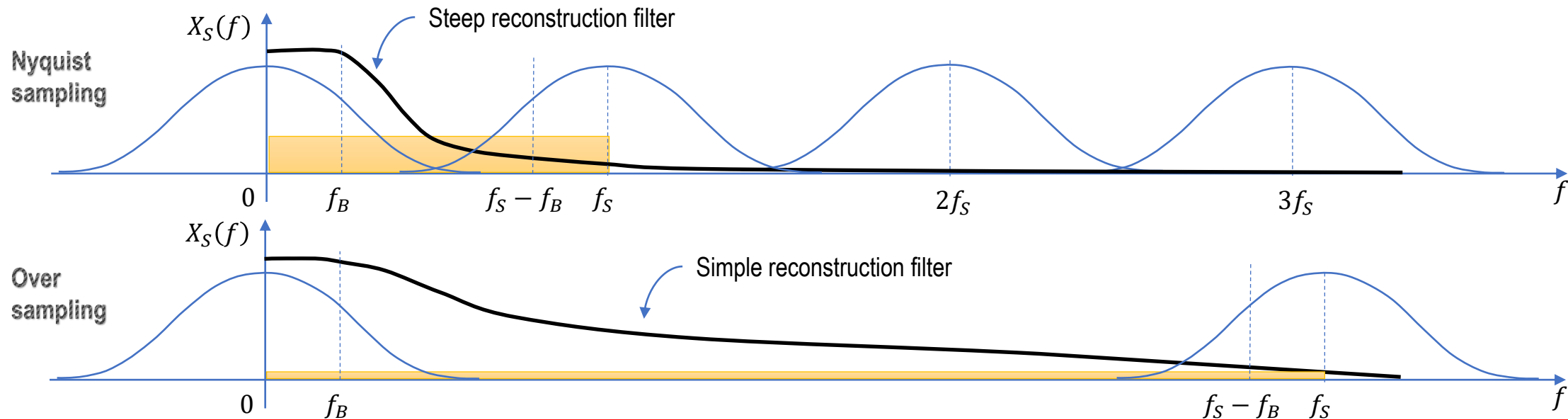
Analog to Digital Conversion

- Delta (Δ) AD Converter
 - Oversampling and aliasing noise
 - It sends replicas in the sampled signal spectrum very far away
 - Same SNR_a , simpler anti-aliasing filter
 - Same anti-aliasing filter, better SNR_a
 - Additional filtering in the digital domain



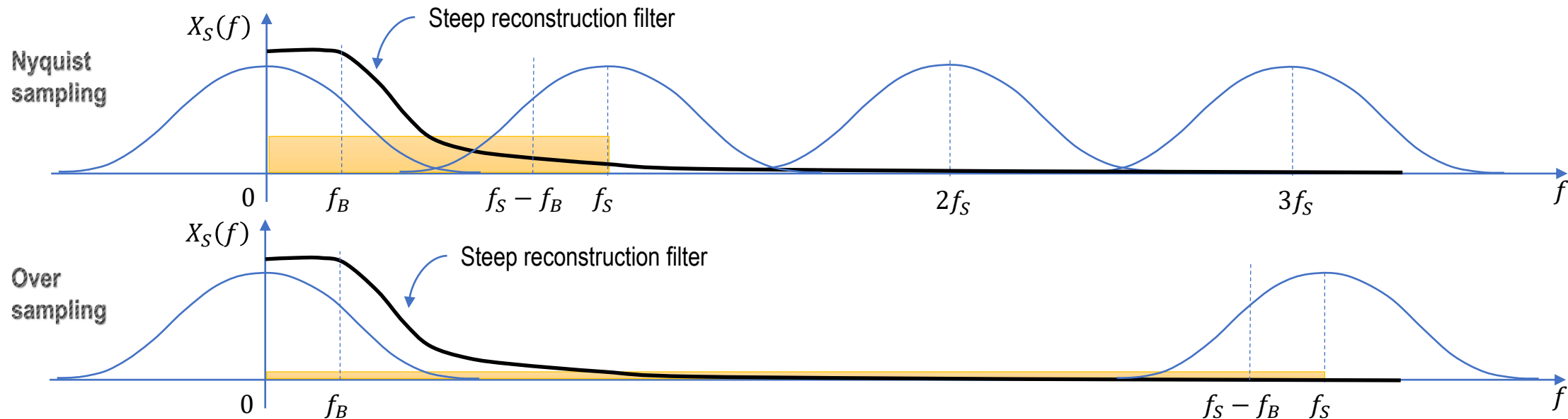
Analog to Digital Conversion

- Delta (Δ) AD Converter
 - Oversampling and quantization noise
 - It sends replicas in the sampled signal spectrum very far away
 - Same quantization noise power (after reconstruction filter), simpler reconstruction filter
 - Same reconstruction filter, reduced quantization noise power (after reconstruction filter)



Analog to Digital Conversion

- Delta (Δ) AD Converter
 - Oversampling and quantization noise
 - It sends replicas in the sampled signal spectrum very far away
 - Same quantization noise power (after reconstruction filter), simpler reconstruction filter
 - Same reconstruction filter, reduced quantization noise power (after reconstruction filter)

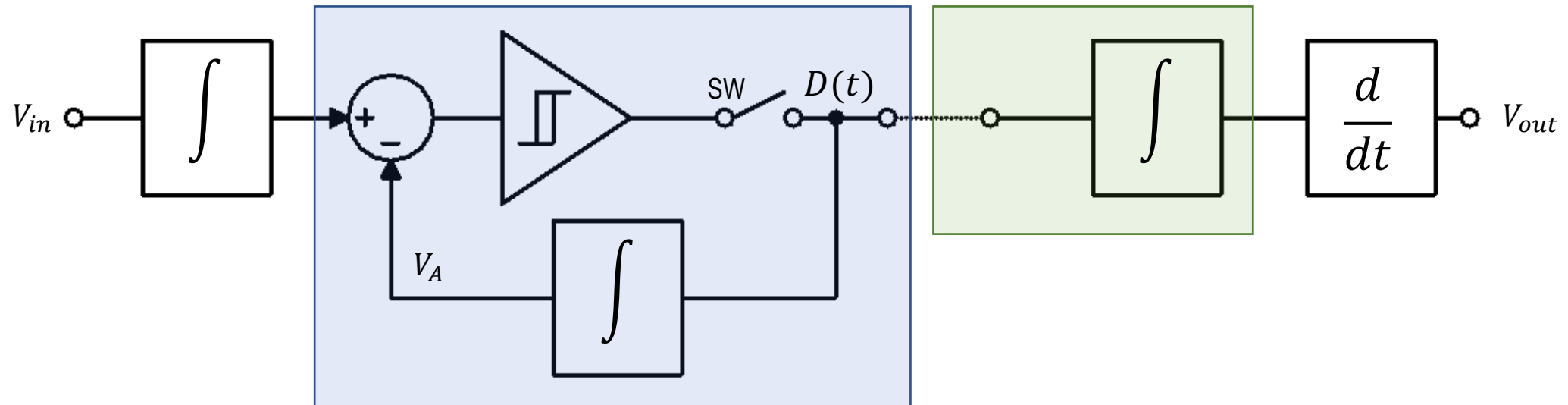


Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - Delta (Δ) AD Converter input signal dynamic range is limited
 - $\frac{V}{2} < V_P < \frac{V}{\omega T_{ck}}, \quad \frac{V}{2} < V_P < \frac{V}{\omega} f_{ck}$
 - Upper bound limited by input signal slew rate
 - Dependency on ω
 - For wider dynamic range
 - Limit the slew rate
 - Integrator on input signal
 - Example: input is a square wave
 - After integration it becomes a triangular wave
 - Slew rate from infinite to a finite value
 - Need to derive the reconstructed signal to recover the original one

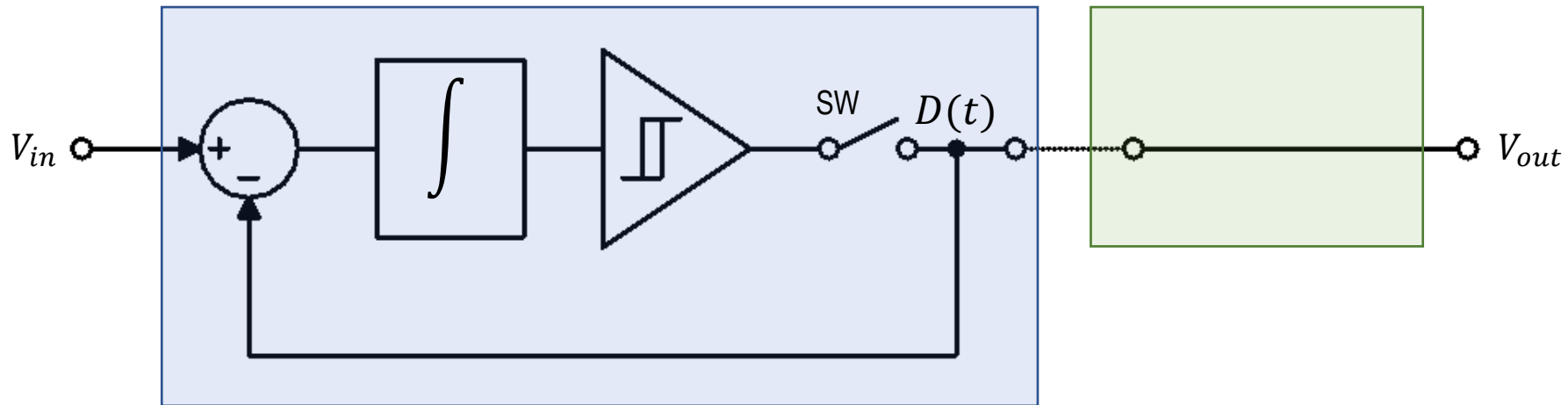
Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - Integrator on input signal
 - Derivator on output signal
 - To correctly rebuild the signal
 - Differential converter in between
 - Anti-aliasing filter before and reconstruction filter afterwards (not shown)



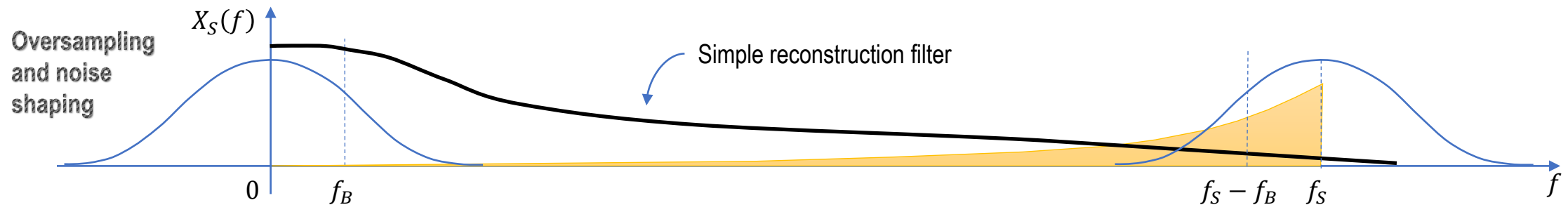
Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - Integration is linear
 - Use a single integrator after the analog adder/subtractor
 - Remove integrator – derivator in the DAC
 - Anti-aliasing filter before and reconstruction filter afterwards (not shown)



Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - Quantization noise is added after integration
 - The transfer function moves the noise power to higher frequencies
 - Noise shaping
 - Noise power spectrum in baseband is reduced
 - Simpler reconstruction filter or further reduction of output noise power



Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - Complete conversion chain
 - Anti aliasing filter
 - Oversampling allows simple filters
 - ADC $\Sigma\Delta$ order 1, 2, ..., N
 - Produces a high speed, non-weighted bit stream
 - Decimator
 - Changes the high speed bit rate in low rate words

Digital processing

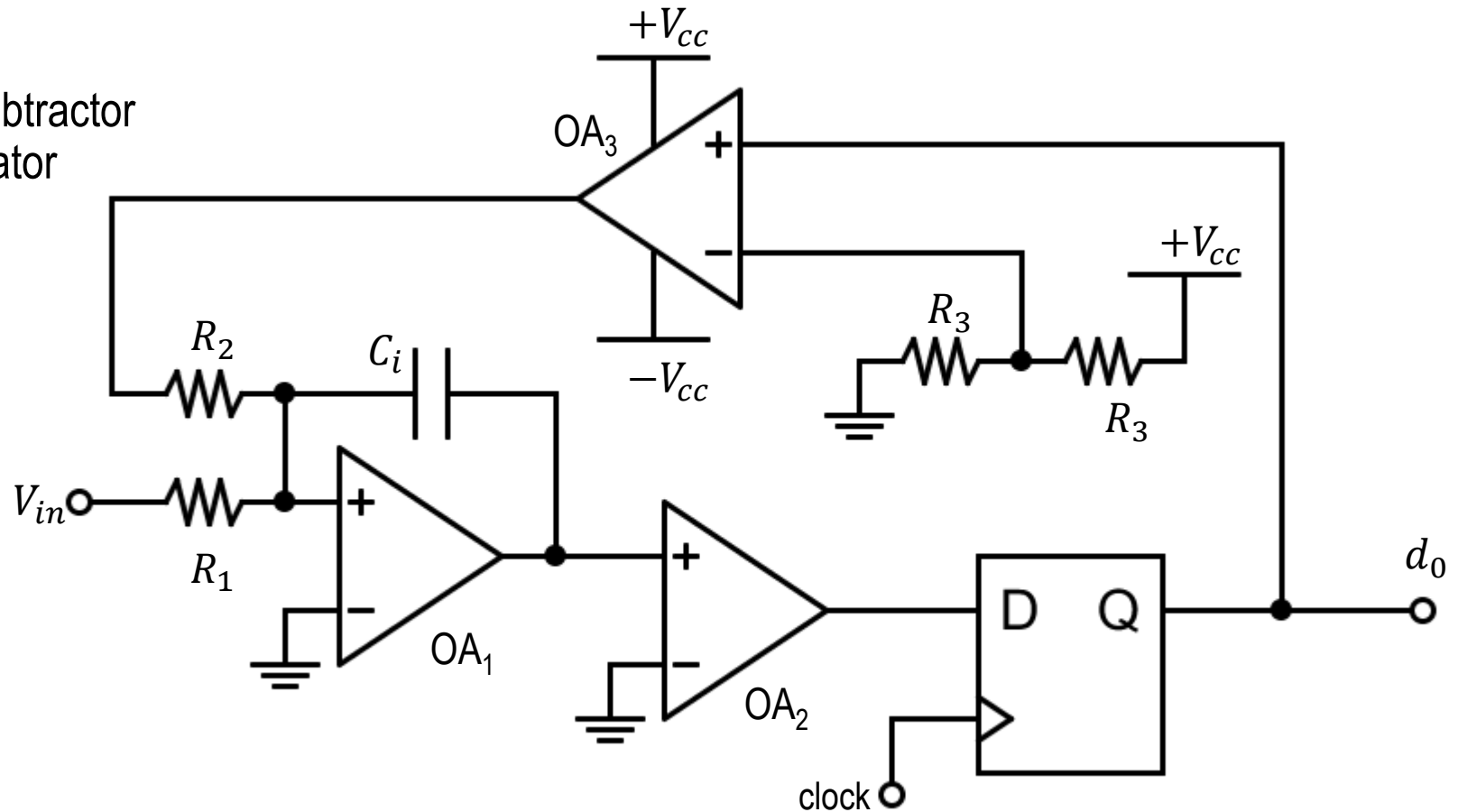
- Interpolator
 - Recreates the high speed serial flow
- $\Sigma\Delta$ DAC
 - Rebuilds analog signal
- Reconstruction filter

Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter

- Example circuit

- OA_1 implements the adder/subtractor followed by the analog integrator
 - OA_2 implements the analog comparator
 - The Flip-Flop implements the switch and generates an output bit on digital levels
 - OA_3 transforms the output bit in the analog range



Analog to Digital Conversion

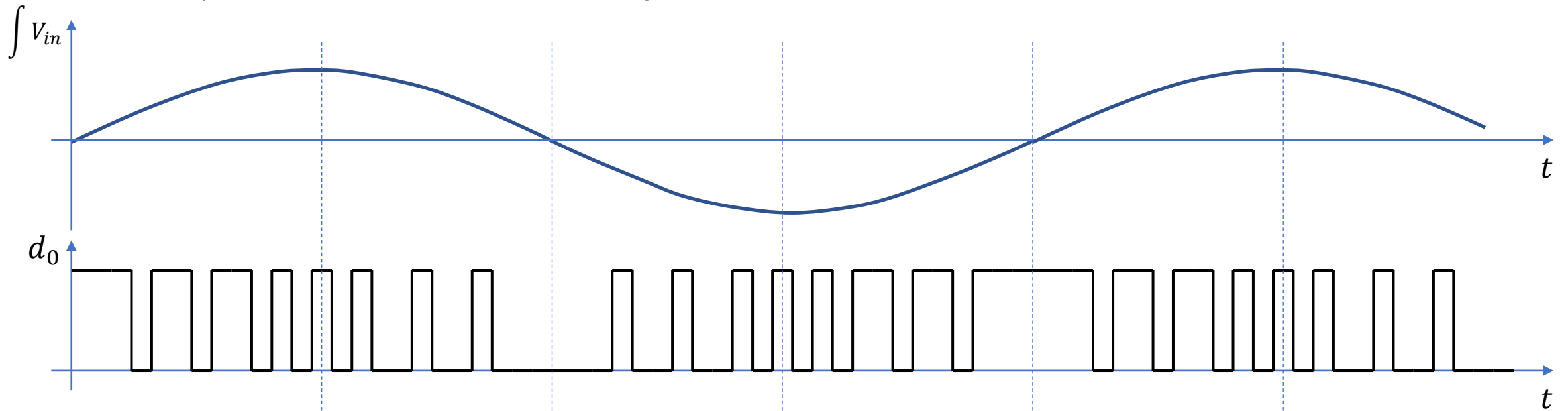
- Sigma-Delta ($\Sigma\Delta$) AD Converter

- Example waveform

- $V_{in}(t) = V_P \cos(\omega t)$

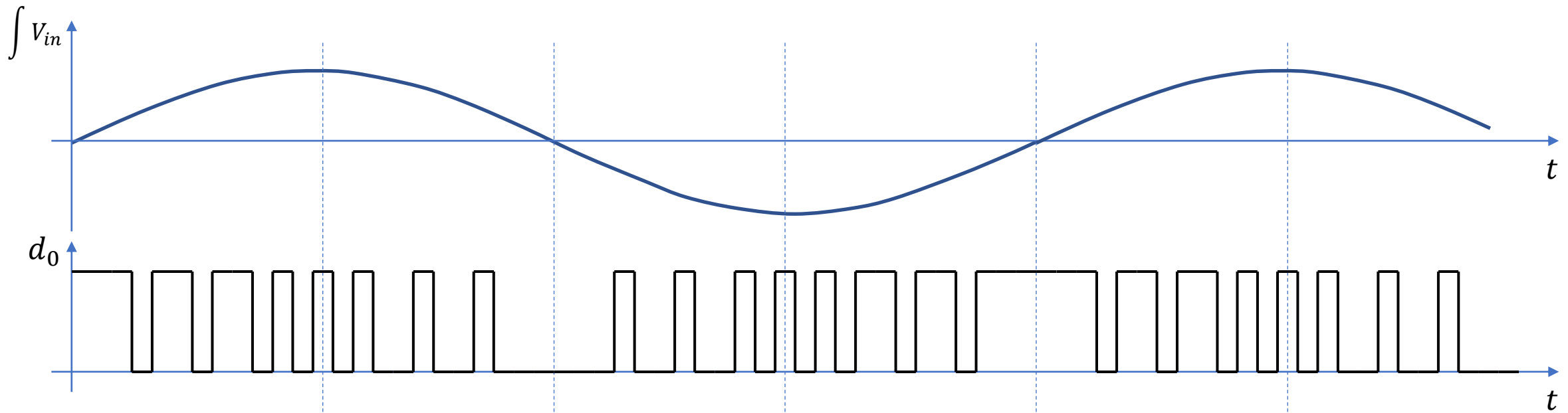
- $\int V_{in}(t) dt = \frac{V_P}{\omega} \sin(\omega t)$

- Apply differential conversion to the integral



Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - Example waveform
 - The converter output looks very similar to a PWM representation of the original input signal (the cosine wave)
 - Applying low pass filtering (reconstruction) gives back a cosine waveform

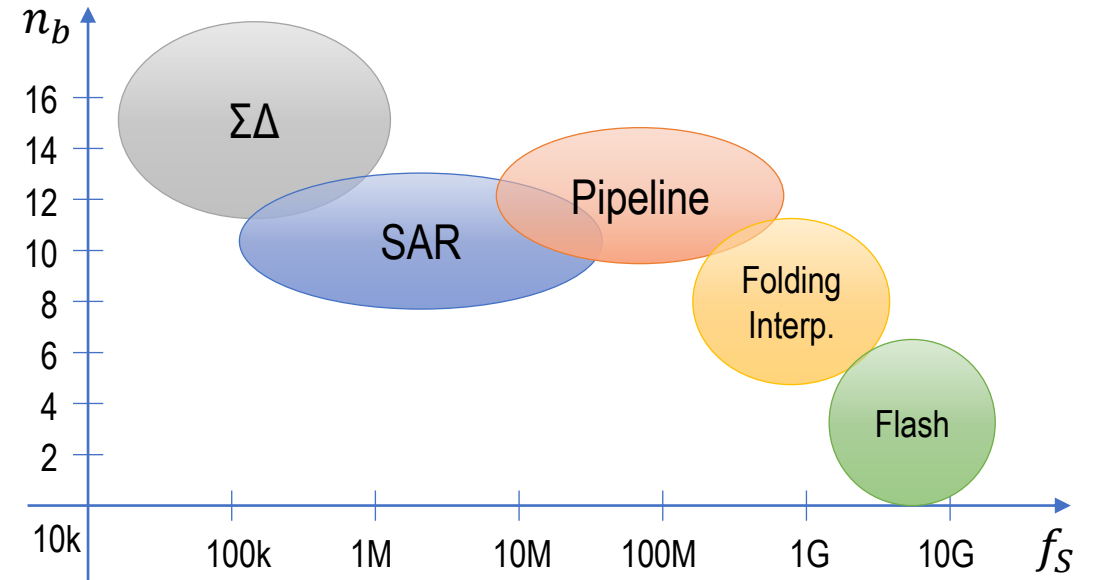


Analog to Digital Conversion

- Sigma-Delta ($\Sigma\Delta$) AD Converter
 - When is it used?
 - Components in differential converters don't need a very high precision
 - Noise is greatly reduced
 - The output bit stream has a very high frequency
 - But in the digital domain a decimator turns the unweighted bits into words with a much lower rate
 - The number of bits used in the words in the digital domain can be high
 - The higher the number of bits of a word, the lower the rate in the digital domain
 - This rate corresponds to the sampling frequency of a classic AD Converter
 - Differential converters used to make equivalent conversions on many bits
 - Many more than a Flash converter
 - More than a SAR converter
 - Example: conversion on 24 or 32 bits
 - The equivalent sampling frequency however is not very high
 - Audio frequencies are typical applications

Analog to Digital Conversion

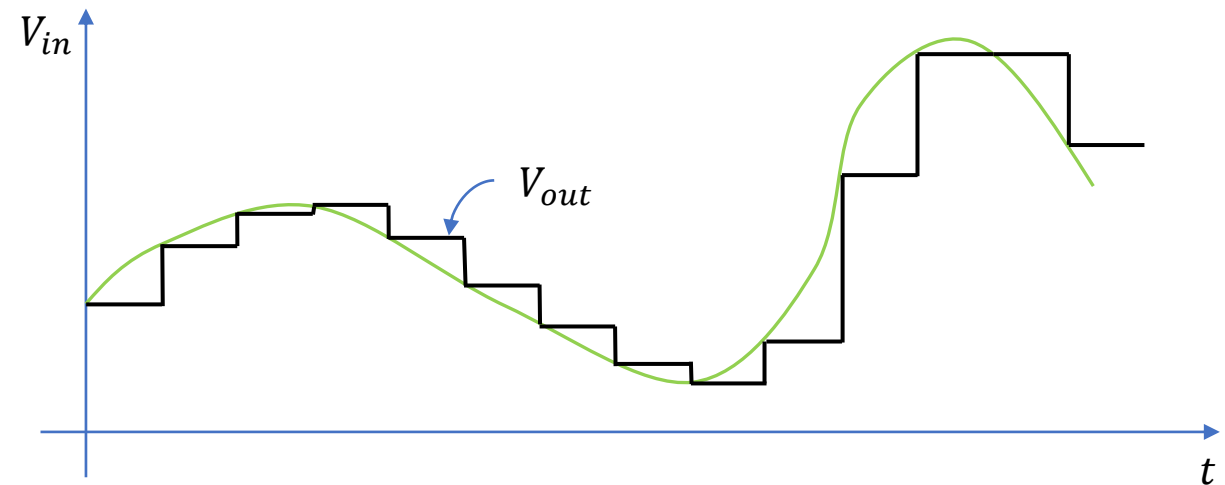
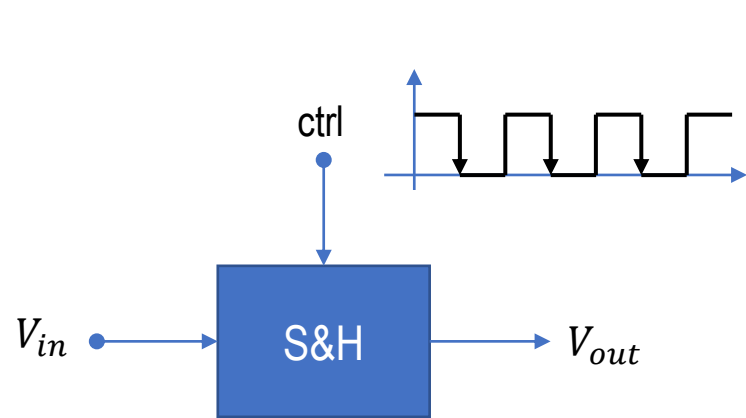
- Summary of converters
 - Sampling frequency
 - Latency
 - Throughput
 - Number of bits
 - Accuracy
 - Costs
 - Area
 - Power Consumption



Architecture	Latency	Speed	Accuracy	Area
Flash	Low	High	Low	High
SAR	Low	Low-Medium	Medium-High	Low
Folding/Interp.	Low	Medium-High	Medium	High
Sigma/Delta	High	Low	High	Medium
Pipeline	High	Medium-High	Medium-High	Medium

Acquisition systems

- Sample & Hold Circuits
 - It is an analog memory device
 - Sample the input signal in a well defined instant
 - Keep the sample constant during the entire AD conversion time
 - The input signal $V_{in}(t)$ changes with time
 - The control signal (clock) specifies the sample instant



Acquisition systems

- Sample & Hold Circuits
 - Cycle divided in four phases
 - Sample
 - The current value of the input signal is stored
 - Ideally, this phase is infinitely fast
 - Hold
 - The stored value is kept on the output
 - The input signal is allowed to change, with no effects on the output
 - Acquisition
 - The output changes to reflect the current value of the input
 - It takes some time for the output to reach the input
 - Low pass effects
 - Slew Rate effects
 - Track
 - The output is identical to the input, following all its changes

Acquisition systems

- Sample & Hold Circuits

- Cycle divided in four phases

- Acquisition

- Rising edge of the control signal
 - Part of the high level of the control signal

- Track

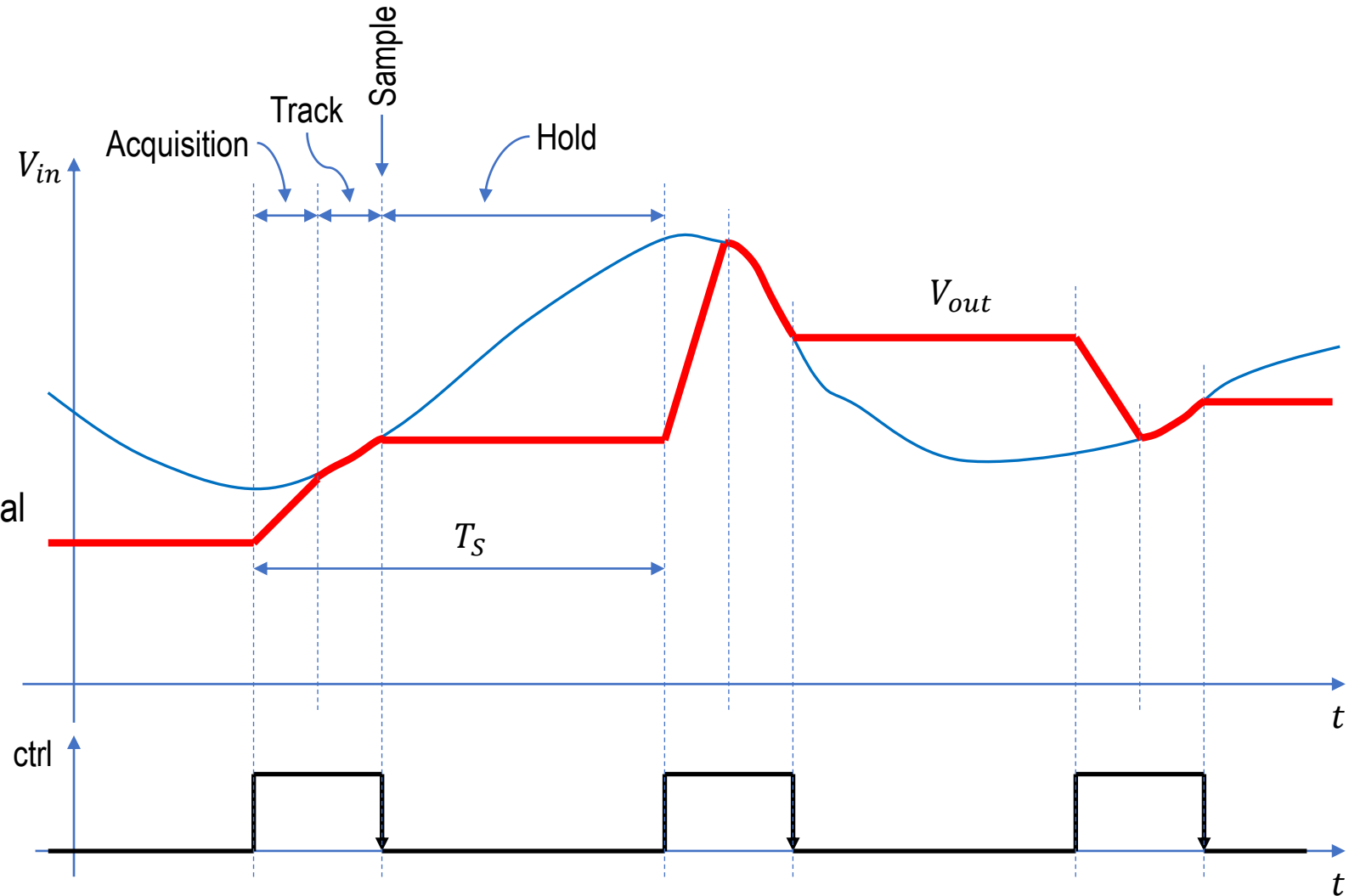
- High level of the control signal
 - Can be very short

- Sample

- Falling edge of the control signal

- Hold

- Low level of the control signal



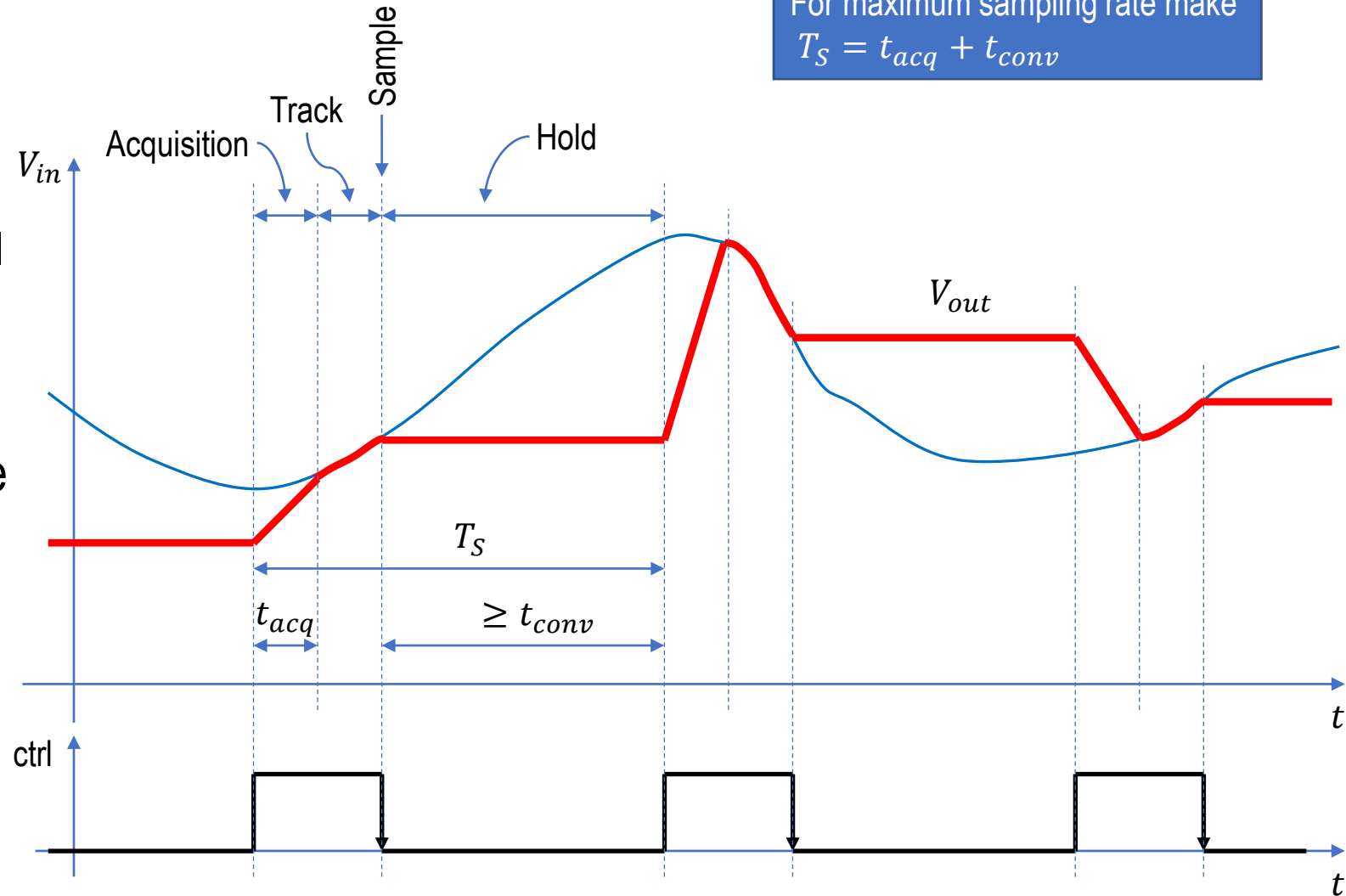
Acquisition systems

- Sample & Hold Circuits

- Quantization applied during the hold phase
 - Hold should be longer or equal to the conversion time t_{conv}
 - For maximum sampling rate make it equal to t_{conv}
- Acquisition time increases the sampling period
 - For maximum sampling rate make it short
- Tracking time increases the sampling period
 - For maximum sampling rate make it of 0 length

$$T_S = t_{acq} + t_{track} + t_{hold}$$

For maximum sampling rate make
 $T_S = t_{acq} + t_{conv}$



Acquisition systems

- Sample & Hold Circuits

- Basic circuit

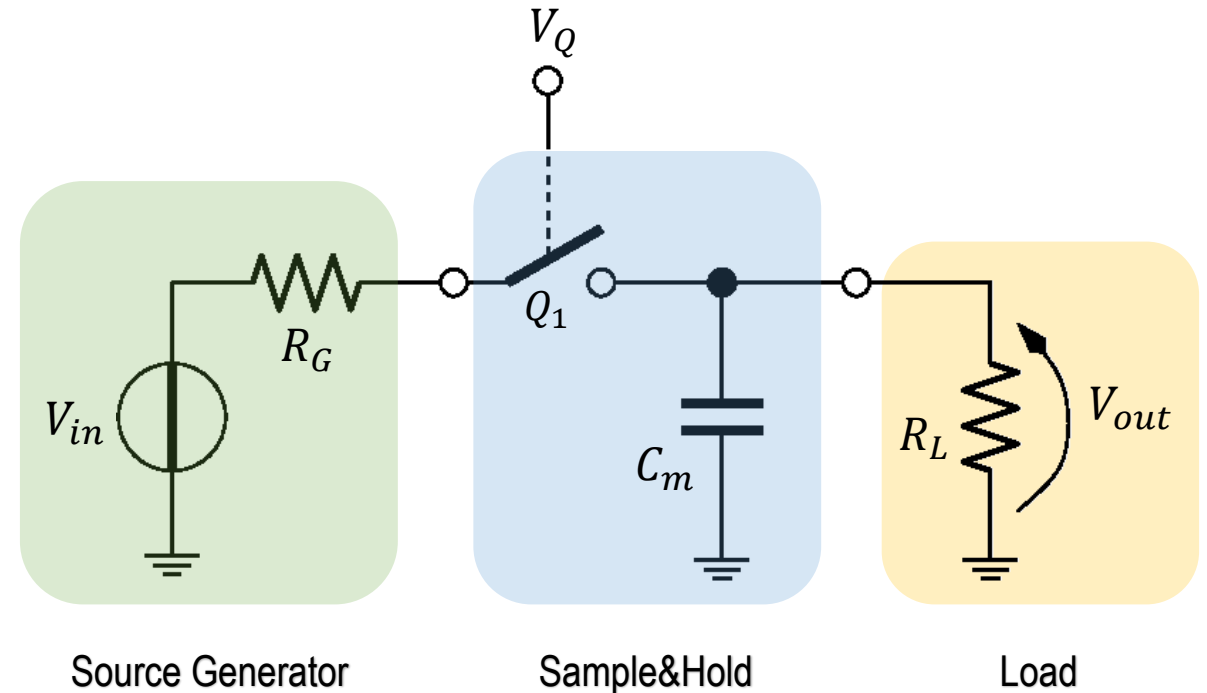
- A switch SW

- Implemented using a transistor
 - Driven by the input control signal
 - Switch closed: Acquisition and Tracking
 - Switch open: Sample and Hold

- A storage capacitor C_m

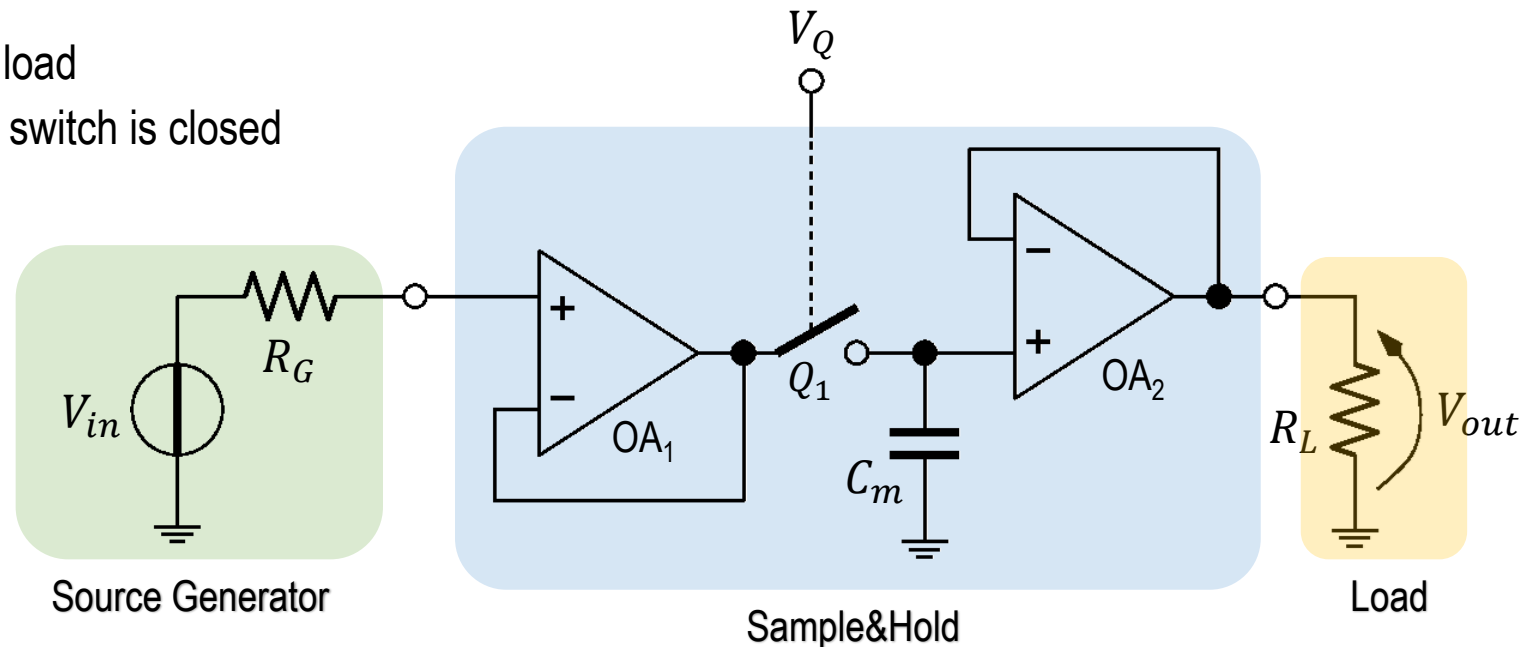
- Problems

- When switch is closed
 - Time constant $\tau = C_m(R_G \parallel R_L)$
 - Voltage divider $V_{out} = \frac{R_L}{R_G + R_L} V_{in}$
 - When switch is open
 - C_m discharges over R_L
 - Output voltage decreases



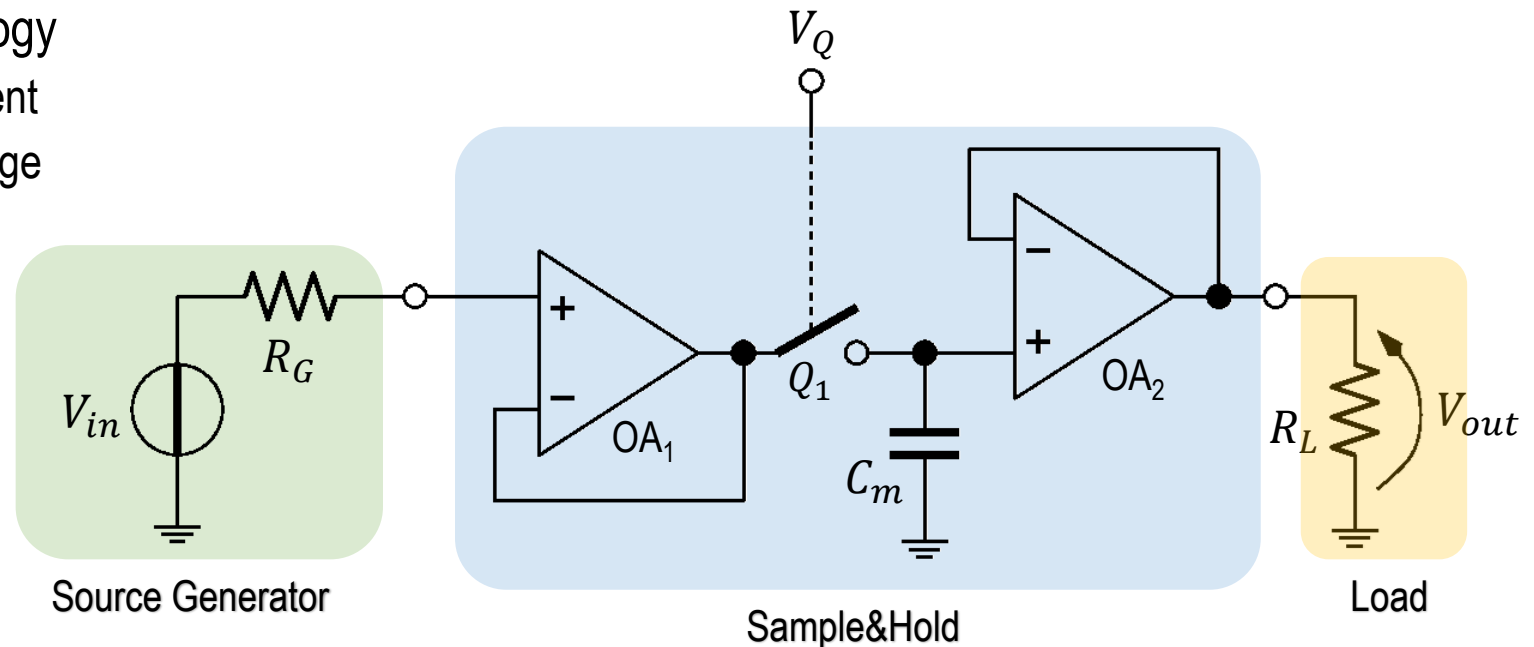
Acquisition systems

- Sample & Hold Circuits
 - Adapt impedances using Voltage Followers
 - High input resistance
 - The source generator works in ideal conditions
 - Small C_m discharge current when the switch is open
 - Low output resistance
 - No voltage divider effect on the load
 - Small time constant τ when the switch is closed
 - Switch closed
 - Charge and discharge of C_m limited by the slew rate of operational amplifier OA_1



Acquisition systems

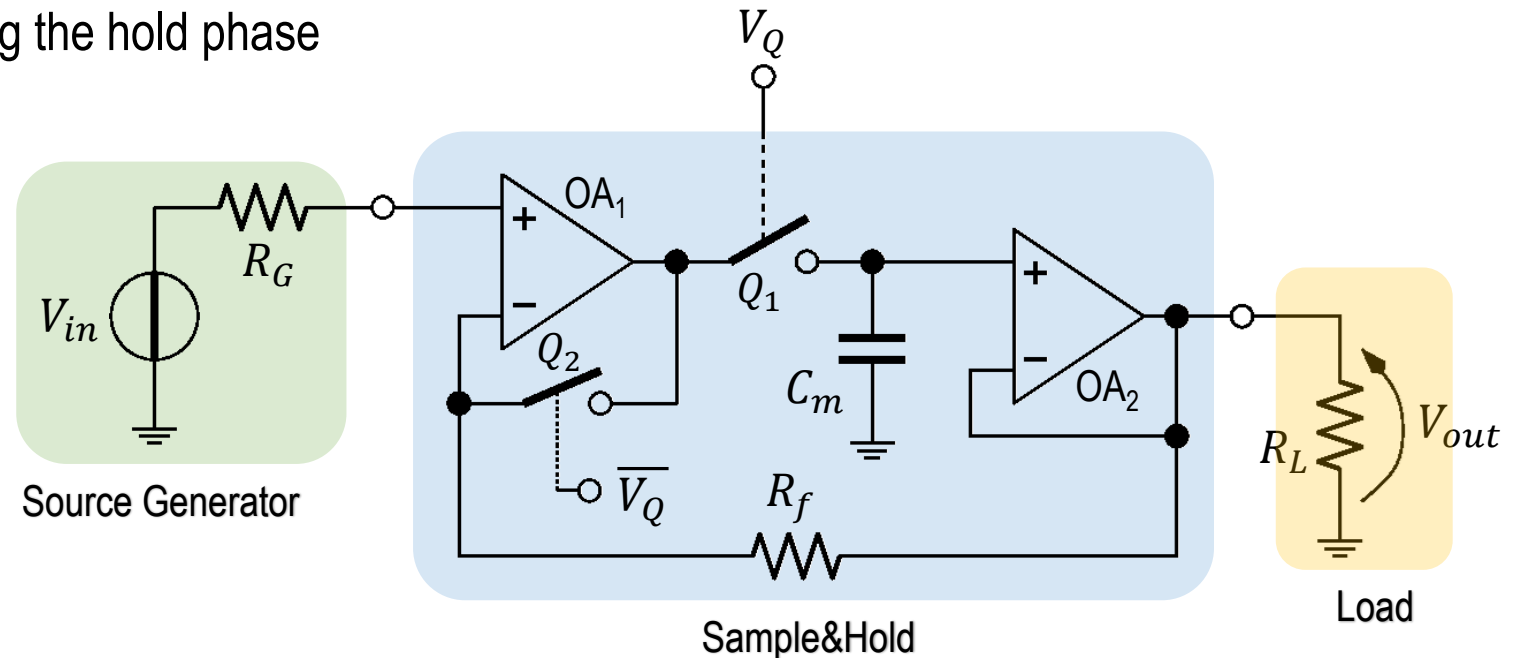
- Sample & Hold Circuits
 - Operational amplifiers are not ideal
 - Output offset
 - $V_{out} = V_{in} \pm V_{off1} \pm V_{off2}$
 - Offset of OA₂ is high
 - Implemented using MOS technology
 - Guarantees low input bias current
 - Small storage capacitor discharge
 - High output offset
 - Solution
 - Entire circuit closed in a single negative feedback loop



Acquisition systems

- Sample & Hold Circuits

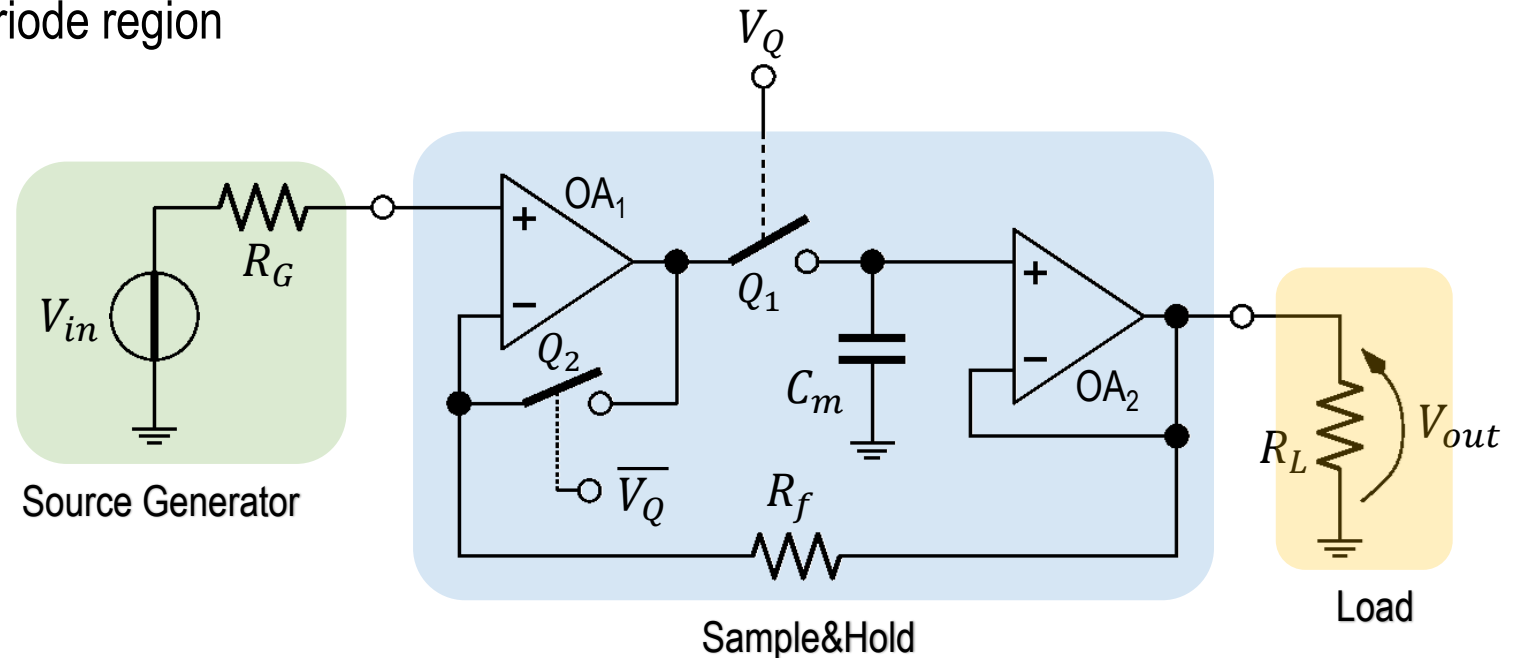
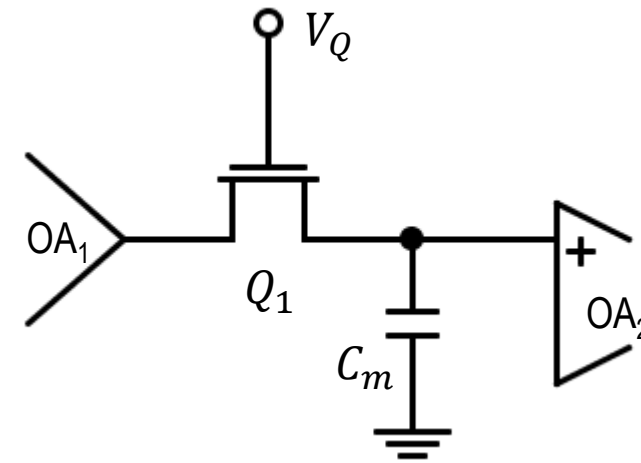
- Switches Q_1 and Q_2 operated alternatively
 - Acquisition and Tracking: Q_1 closed, Q_2 open
 - Sample and Hold: Q_1 open, Q_2 closed
- Switch Q_2 keeps OA_1 with negative feedback when outer loop is open
 - Otherwise it quickly saturates during the hold phase
 - It tracks the input
- Resistance R_f
 - Allows the outputs of OA_2 and OA_1 to be different during the hold phase
- Offset of OA_2
 - Eliminated by the high loop gain



Acquisition systems

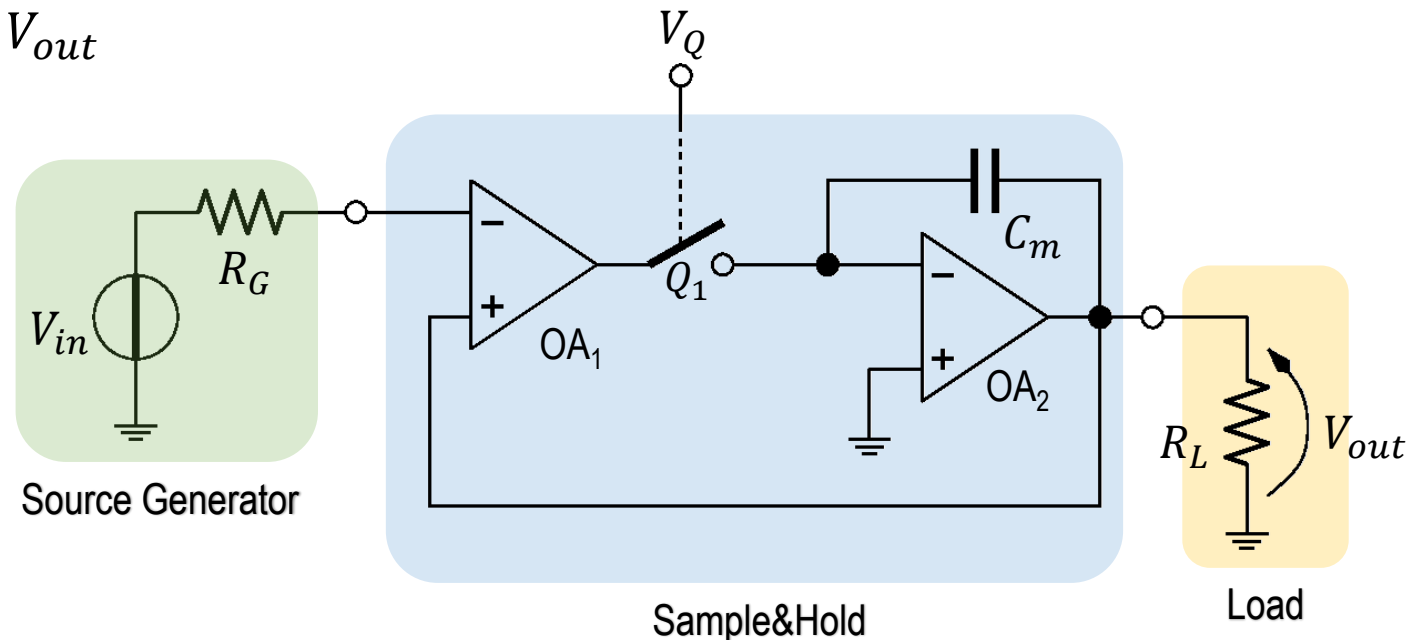
- Sample & Hold Circuits

- Switch Q_1 is difficult to drive
 - It is normally an nMOS transistor
 - Drain and Source are at similar voltage levels
 - Hard to bring V_{GS} to a high voltage
 - The nMOS may not be working in triode region
 - High R_{DSon}
 - Higher τ
- Force nMOS Source to ground level
 - It is the best condition to drive an nMOS transistor



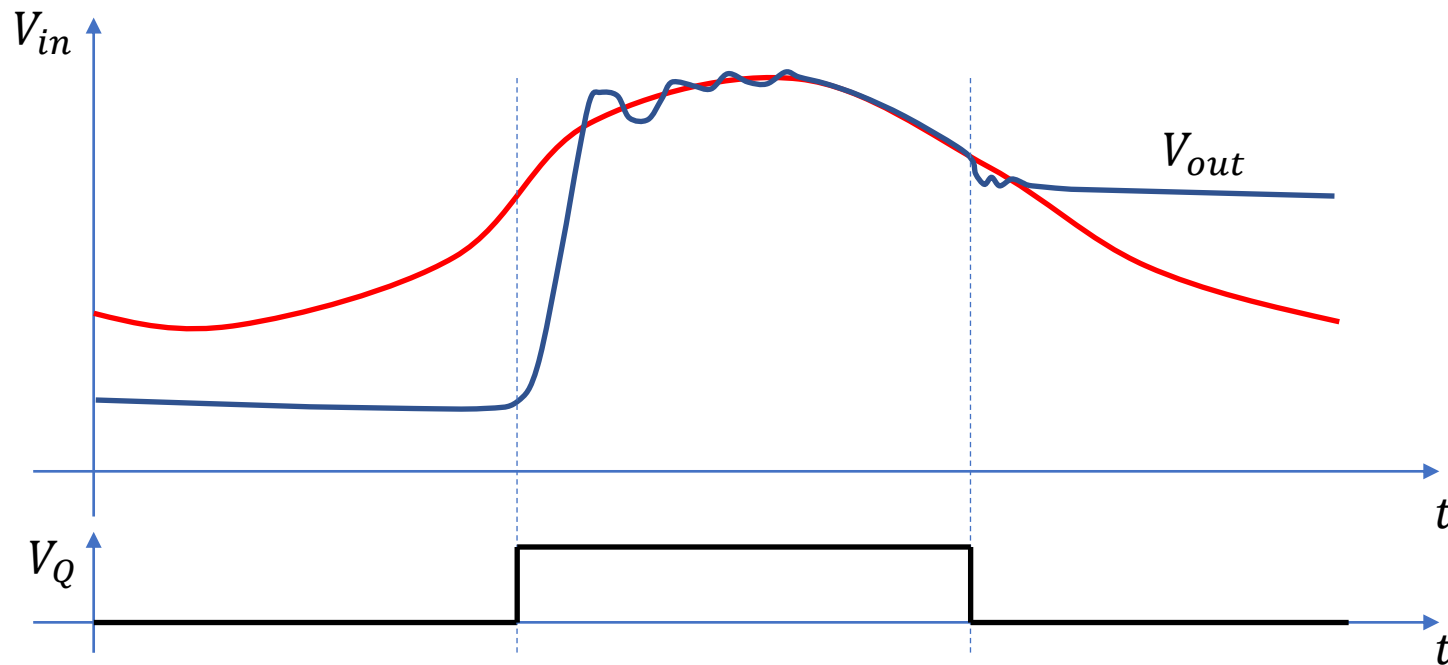
Acquisition systems

- Sample & Hold Circuits
 - Terminal – of OA_2 is a virtual ground
 - The Source of Q_1 is grounded
 - Capacitor C_m
 - One terminal grounded
 - The other terminal at the output voltage V_{out}
 - No current flowing in the capacitor when Q_1 is open (hold phase)
 - The capacitor maintains the output voltage constant
 - Negative feedback
 - OA_1 terminals are inverted
 - Because OA_2 now has an inverting behavior



Acquisition systems

- Sample & Hold Errors
 - Non ideal behaviors lead to voltage output errors
 - Affect the total Signal-to-Noise Ratio of an acquisition system
 - Analyze errors to quantify them
 - Analyze errors to choose components in order to minimize them



Acquisition systems

- Sample & Hold Errors

- Acquisition time t_{acq}

- During the hold phase

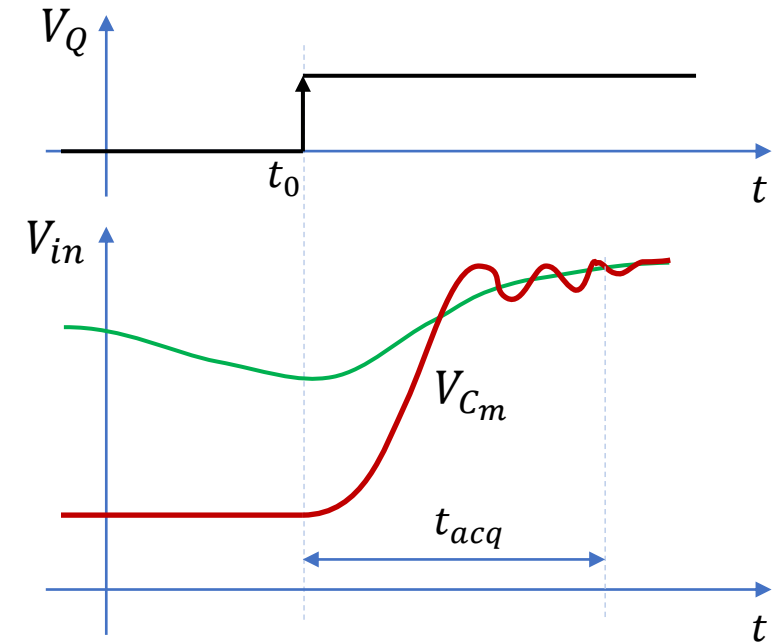
- The capacitor C_m keeps a constant voltage
 - The input signal can change

- At the end of the hold phase, the switch is closed

- Voltage V_Q makes a low to high transition

- The voltage V_{C_m} on the capacitor C_m and the output of the operational amplifier OA_1 differ

- The output of the operational amplifier OA_1 is equal to V_{in}
 - The capacitor C_m is charged or discharged
 - Time constant depends on
 - The operational amplifier OA_1 maximum output current and output resistance
 - The switch ON resistance
 - The value of the capacitor C_m

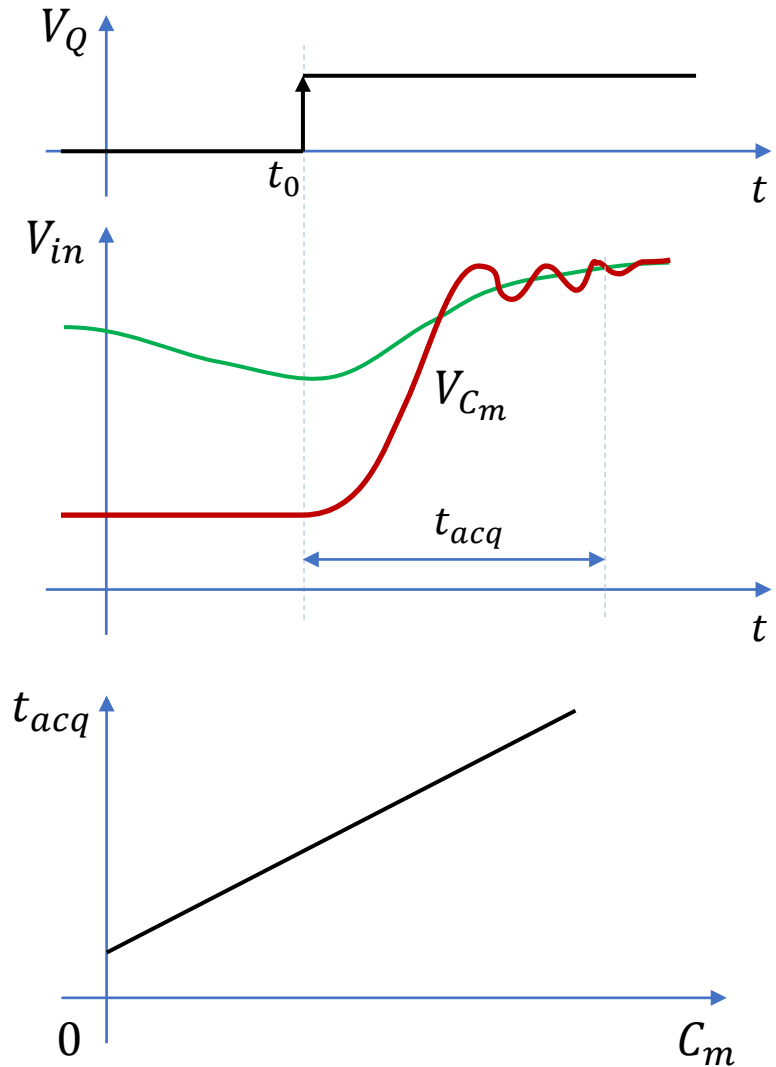


Acquisition systems

- Sample & Hold Errors

- Acquisition time t_{acq}

- The voltage V_{C_m} on the capacitor reaches the input voltage V_{in} in a certain time
 - The transient starts with a linear trend
 - Current is constant due to maximum slew rate
 - Second order damped oscillations follow
 - The transient is considered completed when $V_{C_m} = V_{in} \pm \frac{1}{2}\text{LSB}$
 - Or within a pre-defined tolerance
 - Acquisition time is proportional to the value of the capacitor C_m
 - The higher C_m , the higher t_{acq}
 - The lower C_m , the lower t_{acq}
 - The acquisition time t_{acq} limits the maximum sampling frequency
 - During acquisition a conversion cannot take place

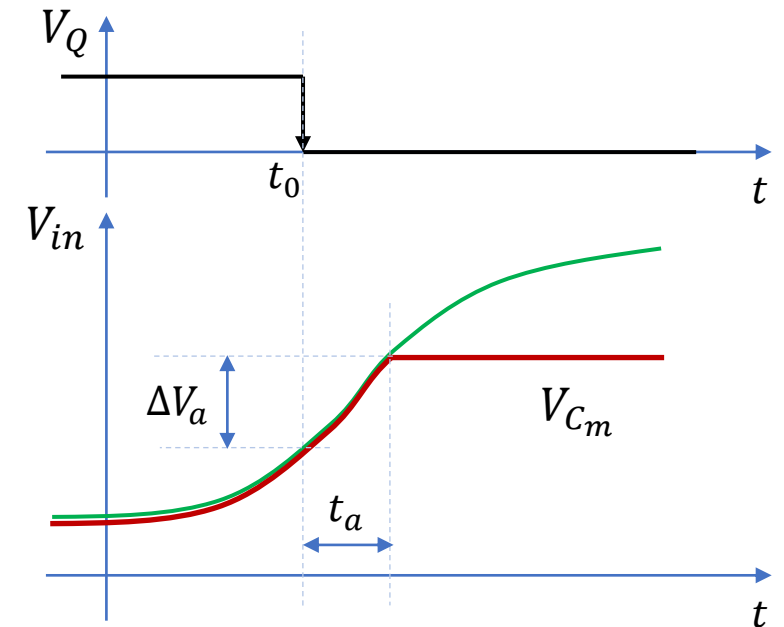


Acquisition systems

- Sample & Hold Errors

- Aperture time t_a

- Sampling corresponds to the high to low transition of the switch control command V_Q
 - Assume the transition occurs at time t_0
 - The transistor implementing the switch responds after a delay t_a
 - But the input signal can change in the meantime
 - Maximum change is
 - $\Delta V_a = SR_{in} \cdot t_a$
 - Error can be compensated
 - If t_a is known, the command can be anticipated



Acquisition systems

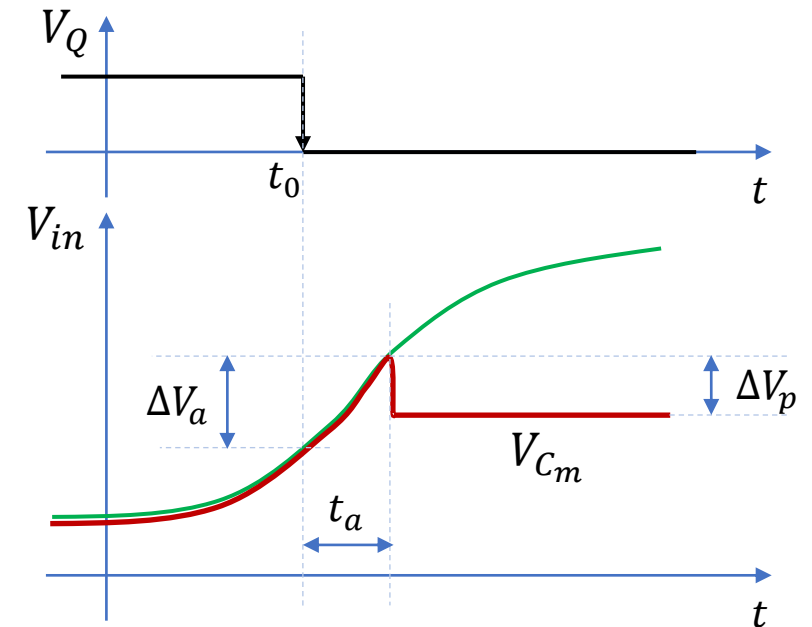
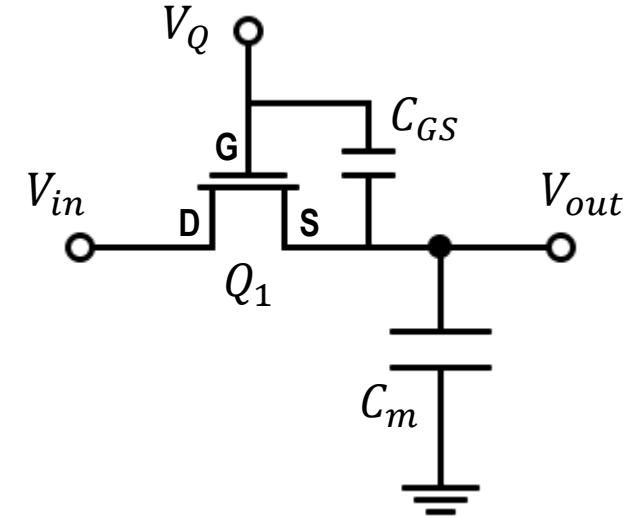
- Sample & Hold Errors
 - Aperture jitter t_j
 - Aperture time t_a is known as an average delay $\overline{t_a}$ with an uncertainty t_j
 - $t_a = \overline{t_a} \pm t_j$
 - Aperture jitter is a random value
 - Cannot be compensated
 - The input signal can change, and the maximum change is
 - $\Delta V_j = SR_{in} \cdot t_j$
 - Example
 - Input sinusoidal signal with $V_P = 5 \text{ V}$ and $f = 5 \text{ kHz}$
 - $t_a = 100 \text{ ns} \pm 5 \text{ ns}$
 - $SR_{in} = V_P \omega = 5 \cdot 2\pi \cdot 5000 = 157 \text{ V/ms}$
 - $\Delta V_a = SR_{in} \cdot \overline{t_a} = 157 \cdot 10^3 \cdot 100 \cdot 10^{-9} = 15.7 \text{ mV}$
 - Compensated by providing the command 100 ns earlier
 - $\Delta V_j = SR_{in} \cdot t_j = 157 \cdot 10^3 \cdot 5 \cdot 10^{-9} = 0.8 \text{ mV}$

Acquisition systems

- Sample & Hold Errors

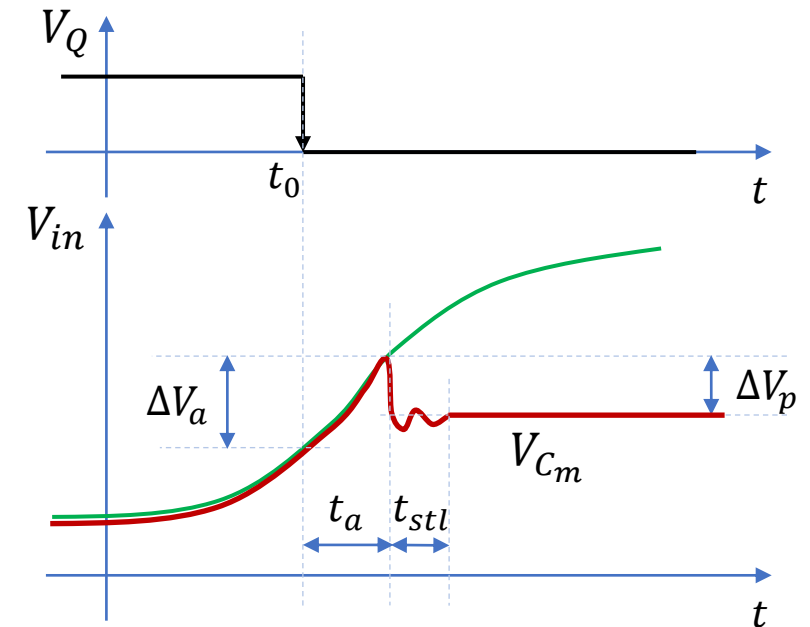
- Pedestal

- The nMOS transistor implementing the switch has several parasitic capacitances
 - The sampling transition on the gate of the transistor induces a charge injection on the storage capacitance
 - $\Delta Q = \Delta V_{GS} \cdot C_{GS}$
 - $\Delta V_p = \frac{\Delta Q}{C_m} = \frac{C_{GS}}{C_m} \Delta V_{GS}$
 - The sampling transition is from high to low
 - From V_{dd} to GND for instance
 - The voltage change ΔV_p is negative
 - Pedestal is inversely proportional to the storage capacitor C_m
 - The higher C_m , the lower ΔV_p
 - The lower C_m , the higher ΔV_p



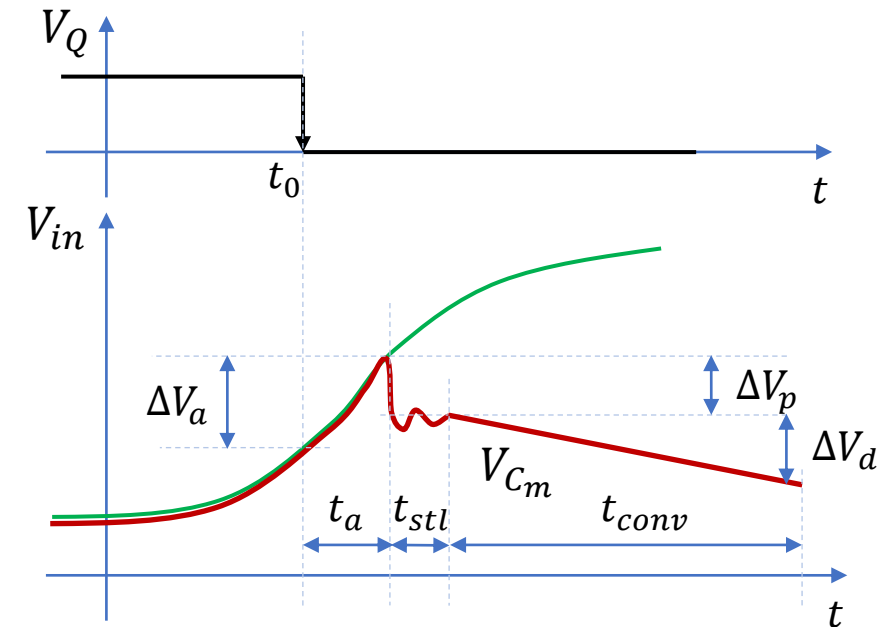
Acquisition systems

- Sample & Hold Errors
 - Settling time
 - After the falling edge due to pedestal
 - Second order oscillations
 - They prevent starting the conversion immediately
 - The output becomes stable after a settling time t_{stl}
 - The settling time limits the maximum sampling frequency
 - But it is small compared to the acquisition time
 - The voltage swing is small
 - The oscillations are small



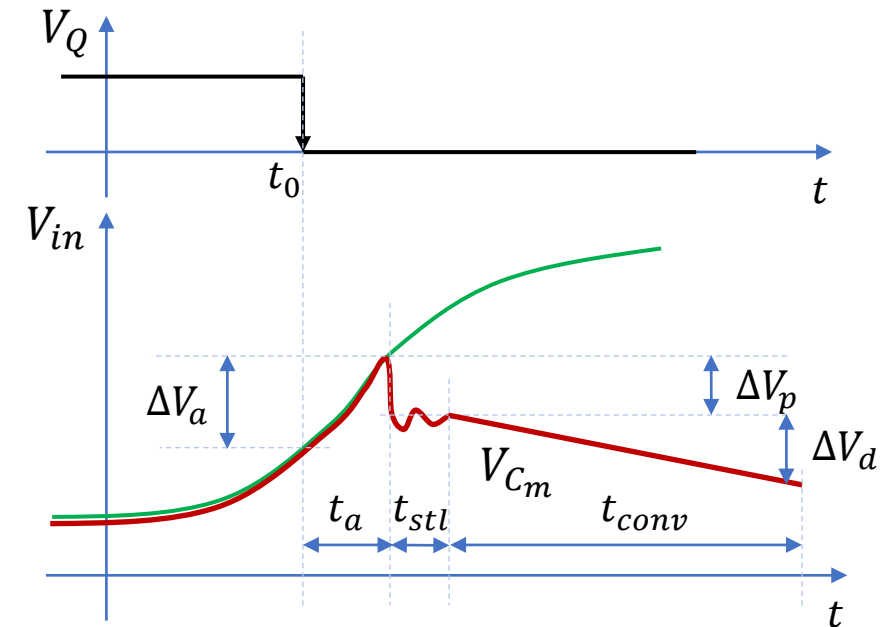
Acquisition systems

- Sample & Hold Errors
 - Droop
 - During the hold phase
 - The switch is not a perfect open circuit
 - Small leakage current
 - A small current flows in the OA₂ operational amplifier input
 - Total current I_{droop}
 - The voltage on the storage capacitor changes linearly according to the law
 - $\frac{dV_{out}}{dt} = \frac{I_{droop}}{C_m}$, $\Delta V_{out}(t) = \frac{I_{droop} \cdot t}{C_m}$
 - The hold phase lasts at least a time equal to the conversion time t_{conv} of the ADC
 - The maximum voltage change is therefore
 - $\Delta V_d = \frac{I_{droop} \cdot t_{conv}}{C_m}$



Acquisition systems

- Sample & Hold Errors
 - Droop
 - Droop is inversely proportional to the storage capacitor C_m
 - The higher C_m , the lower ΔV_d
 - The lower C_m , the higher ΔV_d
 - Make sure that ΔV_d is small
 - Voltage change of the input signal during quantization
 - Should be $\Delta V_d < \frac{1}{2}\text{LSB}$



Acquisition systems

- Sample & Hold Errors

- Feedthrough

- The nMOS transistor implementing the switch has several parasitic capacitances
 - The changes of the input signal induces a charge injection on the storage capacitance

- $\Delta Q = \Delta V_{in} \cdot C_{DS}$

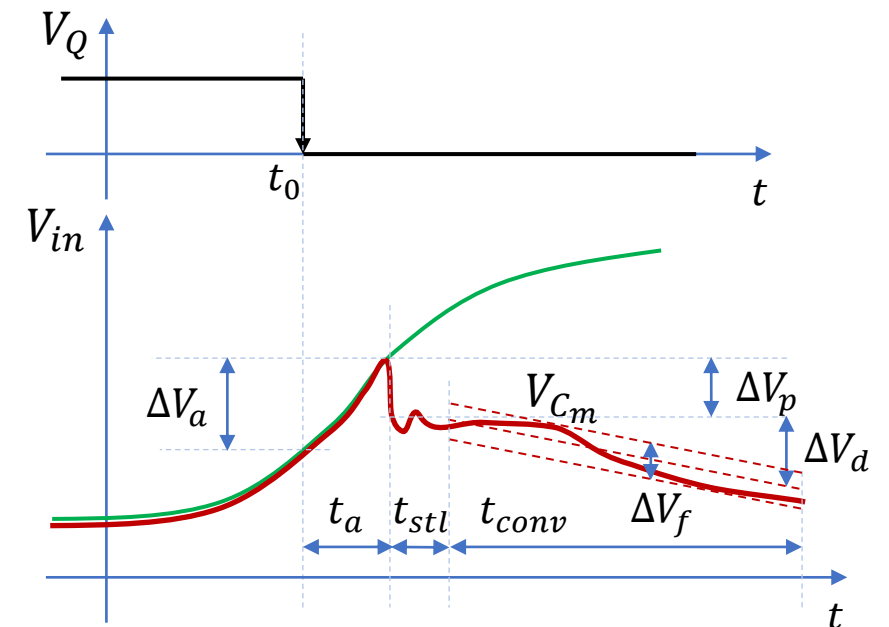
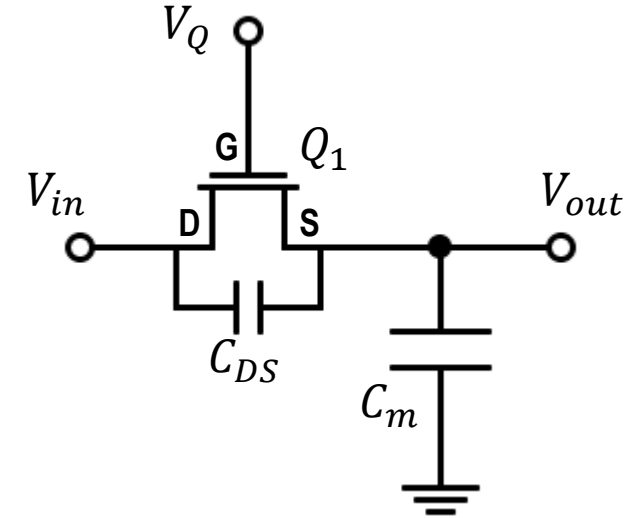
- $\Delta V_f = \frac{\Delta Q}{C_m} = \frac{C_{DS}}{C_m} \Delta V_{in} = \frac{C_{DS}}{C_m} S R_{in} t_{conv}$

- The voltage change ΔV_f follows the input shape

- Feedthrough is inversely proportional to the storage capacitor C_m

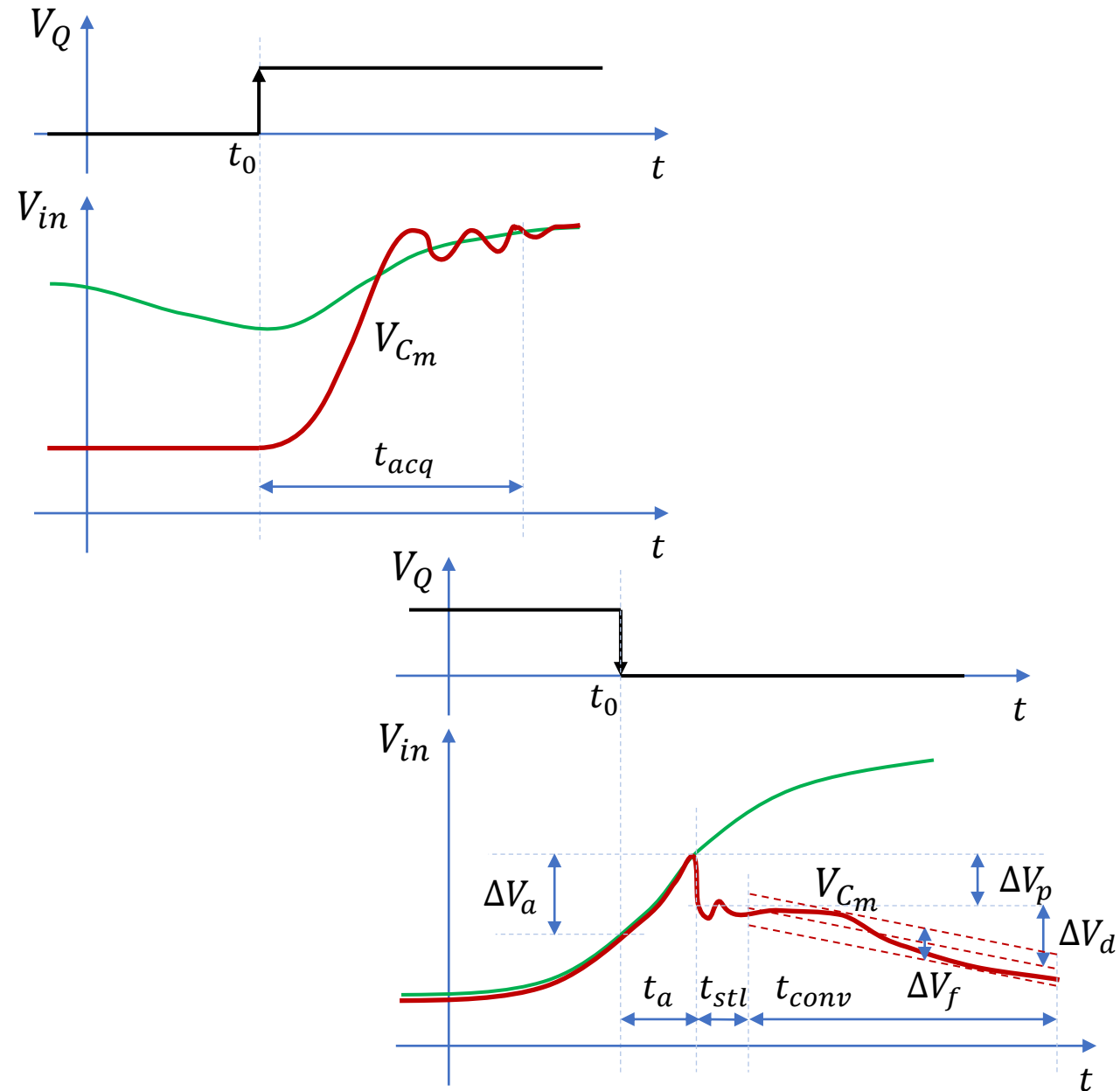
- The higher C_m , the lower ΔV_f

- The lower C_m , the higher ΔV_f



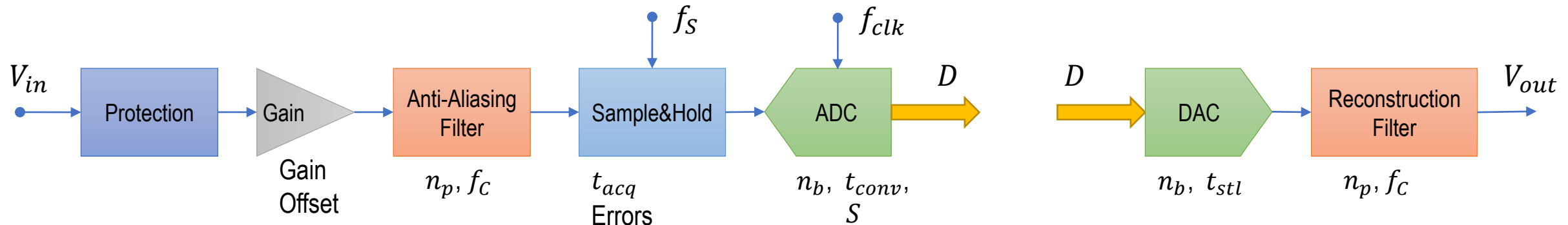
Acquisition systems

- Sample & Hold Errors
 - Total error
 - Sum of the absolute value of all contributions
 - $\Delta V_{out} = \Delta V_j + \Delta V_p + \Delta V_d + \Delta V_f$
 - ΔV_a compensated
 - Can be expressed as a fraction of an LSB
 - Contributes to the total Signal-to-Noise Ratio of the acquisition system
 - Many errors benefit from a large C_m
 - But the acquisition time t_{acq} is worse
 - Choice of C_m is a trade-off between accuracy and performance



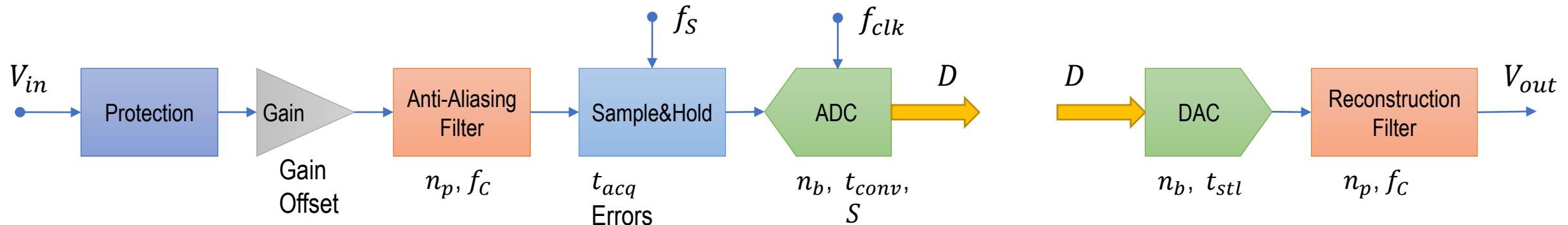
Acquisition systems

- Complete acquisition system
 - Single channel
 - Input signal
 - Input dynamic range S_{in}
 - Useful bandwidth f_B
 - Protection
 - Avoids damaging the acquisition system in case the input signal shows some wrong behavior
 - Gain
 - Adapts the dynamic range S_{in} of the input signal to the dynamic range S of the AD Converter
 - Maximizes the SNR_q given the number of bits n_b



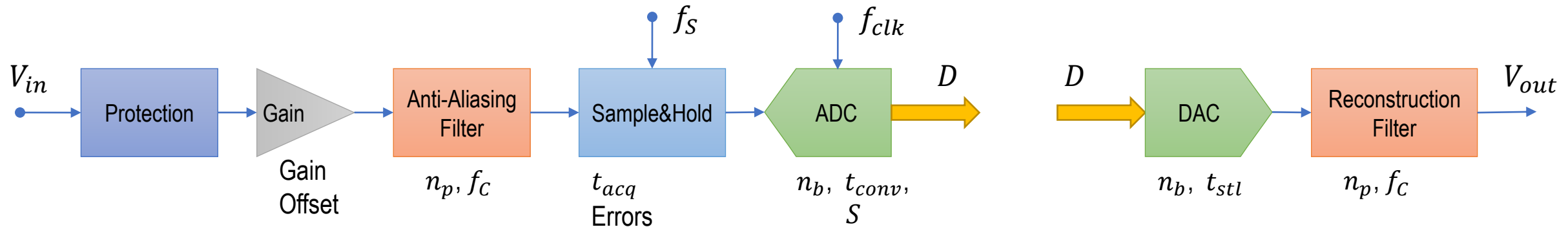
Acquisition systems

- Complete acquisition system
 - Single channel
 - Anti-Aliasing Filter
 - Attenuates the spectrum contribution of the input signal outside the useful bandwidth f_B
 - Maximizes the SNR_a given the sampling frequency f_S
 - Sample & Hold
 - Converts from continuous time to discrete time
 - Switch driven at the sampling frequency f_S



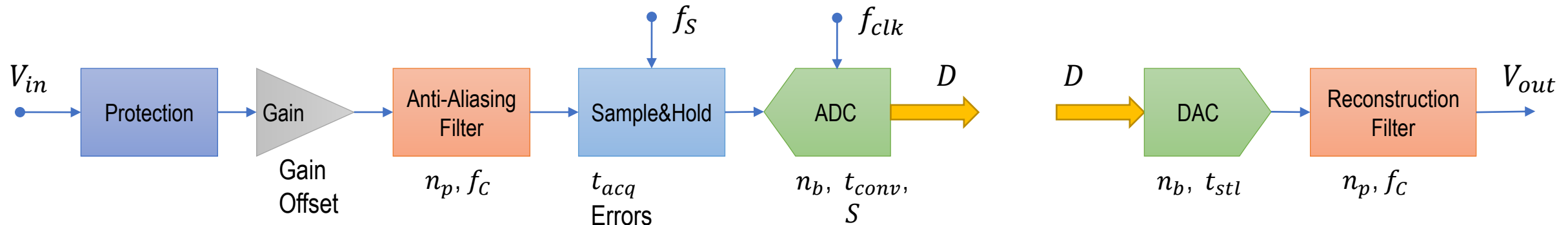
Acquisition systems

- Complete acquisition system
 - Single channel
 - Analog-to-Digital Converter
 - Converts from continuous values to discrete values
 - Input dynamic range is S
 - Conversion is on n_b bits
 - Determines the SNR_q
 - Conversion takes t_{conv}
 - Many kind of ADCs characterized by an input clock frequency f_{clk}



Acquisition systems

- Complete acquisition system
 - Single channel
 - Digital-to-Analog Converter
 - Converts from discrete values to continuous values
 - Conversion is on n_b bits
 - Output signal is stable after a settling time t_{stl}
 - Reconstruction filter
 - Converts from discrete time to continuous time
 - Isolate the baseband alias in the spectrum



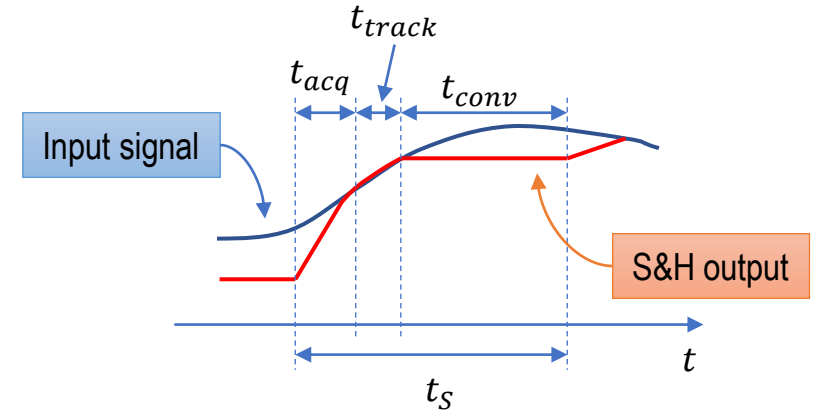
Acquisition systems

- Complete acquisition system

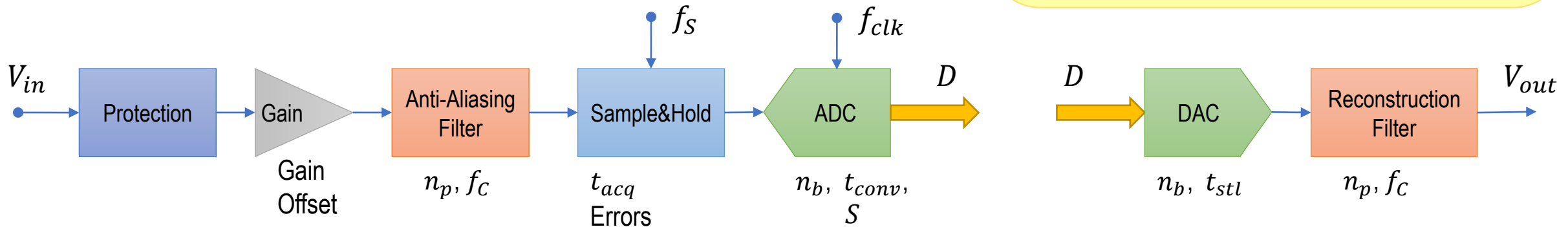
- Single channel

- Sampling frequency

- It should be $f_S > 2f_B$
- It should be $t_S > t_{acq} + t_{conv}$
- SAR, Residue: $t_{conv} = n_b \cdot t_{clk}$
- Staircase: $t_{conv} = 2^{n_b} \cdot t_{clk}$
- Flash, Pipeline: $t_{conv} = t_{clk}$
- Multibit architectures: t_{conv} depends on the actual structure
- Tracking, Delta, Sigma-Delta: oversampling

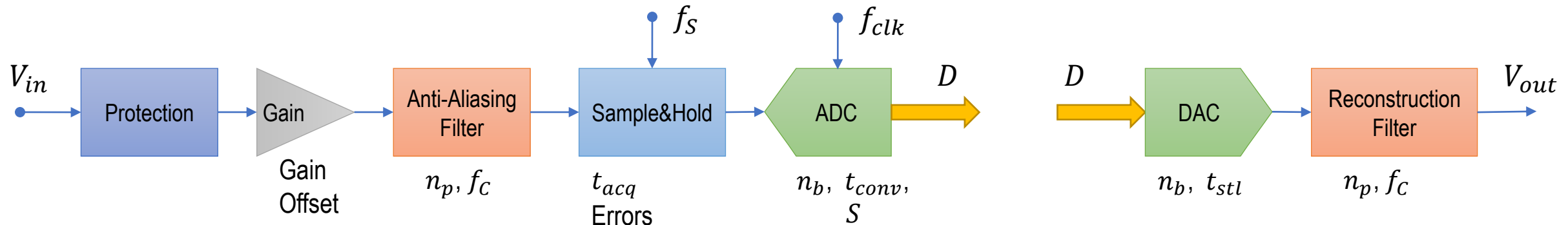


- Minimum sampling period
 - $t_{Smin} = t_{acq} + t_{conv}$
- Normally t_S is higher than the minimum, but not by much to have high performance
 - $t_{track} = t_S - t_{acq} - t_{conv}$
- In the exam usually the minimum is requested



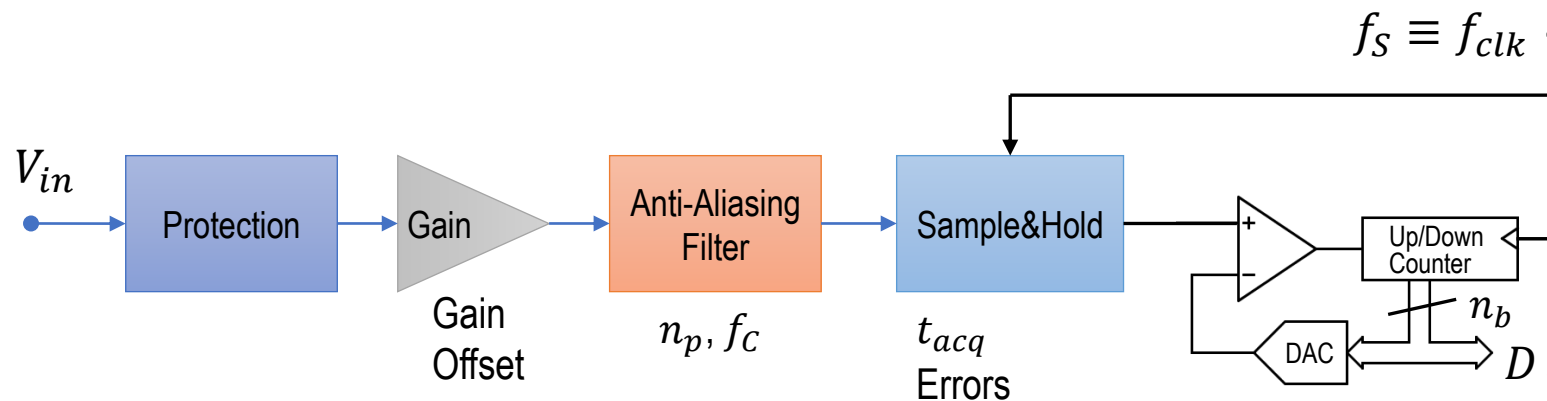
Acquisition systems

- Complete acquisition system
 - Single channel
 - Tracking converter
 - When working without overload (in oversampling) the Sample&Hold is often not needed
 - $t_S = t_{clk}, f_S = f_{clk}$
 - The output of the converter is at a very high frequency
 - If not in oversampling, a Sample&Hold and a register at the output of the converter are used (like the Staircase ADC)
 - Conversion time exponential on the number of bits
 - $t_{conv} = 2^{n_b} \cdot t_{clk}$
 - The clock frequency is comparable to the previous case, but the output of the converter is at a much lower rate



Acquisition systems

- Complete acquisition system
 - Single channel
 - $f_B = 5 \text{ kHz}$, $SR_{in} = 250 \text{ V/ms}$, $f_{Ny} = 2f_B = 10 \text{ kHz}$
 - $S = 10$, $n_b = 8$, $A_d = S/2^{n_b} = 39 \text{ mV}$, $t_{acq} = 100 \text{ ns}$
 - Tracking converter, in oversampling
 - One sample at each clock cycle
 - Sampling frequency equal to the clock frequency



$$t_{clk} = \frac{A_d}{SR_{in}} = 156 \text{ ns}$$
$$f_{clk} = \frac{1}{t_{clk}} = 6,4 \text{ MHz}$$
$$f_s = f_{clk} = 6,4 \text{ MHz}$$
$$t_{comp} + t_{CNT} + t_{DAC} + t_{acq} < t_{clk}$$

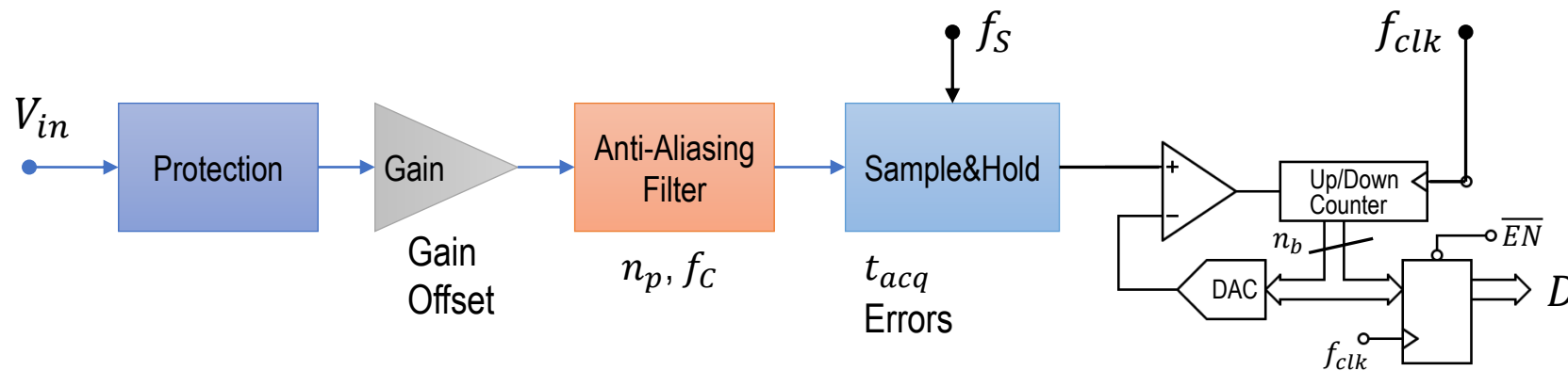
To prevent overloading

The S&H should be fast, so the storage capacitor C_m should be small

The data rate of signal D is high

Acquisition systems

- Complete acquisition system
 - Single channel
 - $f_B = 5 \text{ kHz}$, $SR_{in} = 250 \text{ V/ms}$, $f_{Ny} = 2f_B = 10 \text{ kHz}$
 - $S = 10$, $n_b = 8$, $A_d = S/2^{n_b} = 39 \text{ mV}$, $t_{acq} = 100 \text{ ns}$
 - Tracking converter or staircase converter
 - Sampling frequency and clock frequency are now different
 - Sampling frequency is much lower
 - The S&H specifications can be relaxed

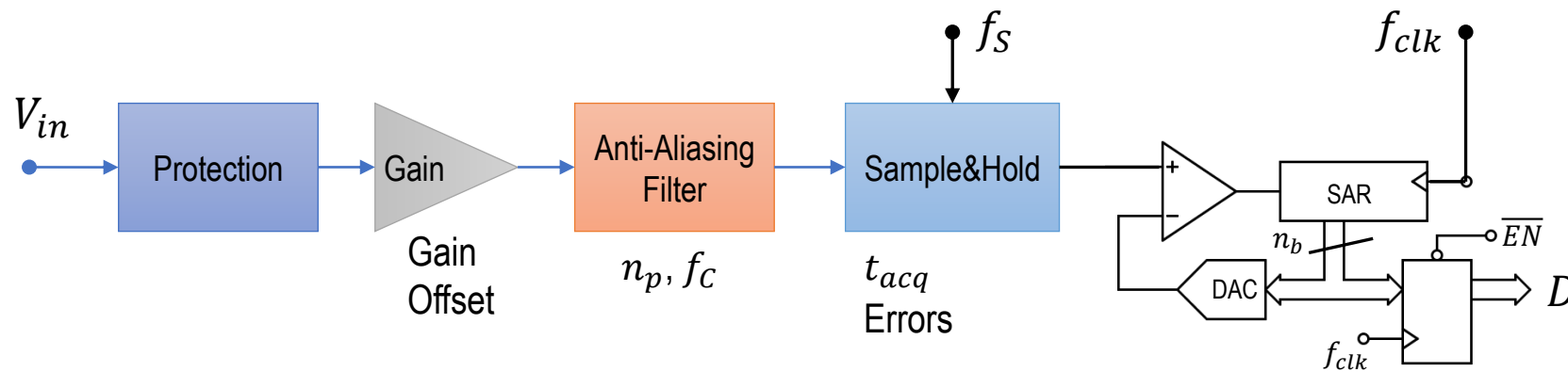


$$\begin{aligned} f_s &= 3f_{Ny} = 30 \text{ kHz} \\ t_s &= \frac{1}{f_s} = 33,3 \mu\text{s} \\ t_{conv} &= t_s - t_{acq} = 33,2 \mu\text{s} \\ t_{clk} &= \frac{t_{conv}}{2^{n_b}} = 130 \text{ ns} \\ f_{clk} &= \frac{1}{t_{clk}} = 7,7 \text{ MHz} \\ t_{comp} + t_{CNT} + t_{DAC} &< t_{clk} \end{aligned}$$

The data rate of signal D is now lower

Acquisition systems

- Complete acquisition system
 - Single channel
 - $f_B = 5 \text{ kHz}$, $SR_{in} = 250 \text{ V/ms}$, $f_{Ny} = 2f_B = 10 \text{ kHz}$
 - $S = 10$, $n_b = 8$, $A_d = S/2^{n_b} = 39 \text{ mV}$, $t_{acq} = 100 \text{ ns}$
 - SAR converter
 - Sampling frequency and clock frequency are different
 - Clock frequency is lower compared to the previous example
 - S&H, comparator, DAC and logic specifications can be relaxed

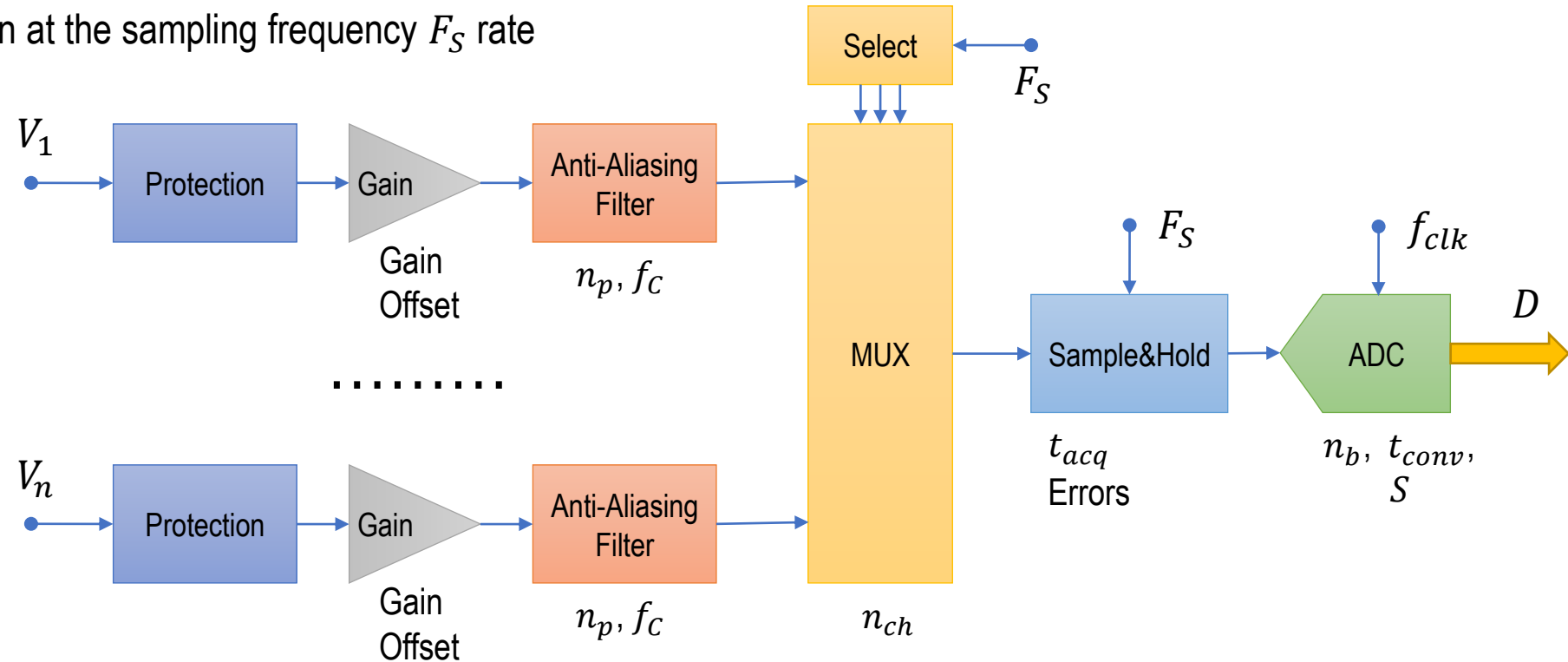


$$\begin{aligned} f_s &= 3f_{Ny} = 30 \text{ kHz} \\ t_s &= \frac{1}{f_s} = 33,3 \mu\text{s} \\ t_{conv} &= t_s - t_{acq} = 33,2 \mu\text{s} \\ t_{clk} &= \frac{t_{conv}}{n_b} = 4,15 \mu\text{s} \\ f_{clk} &= \frac{1}{t_{clk}} = 241 \text{ kHz} \\ t_{comp} + t_{CNT} + t_{DAC} &< t_{clk} \end{aligned}$$

The data rate of signal D is now lower

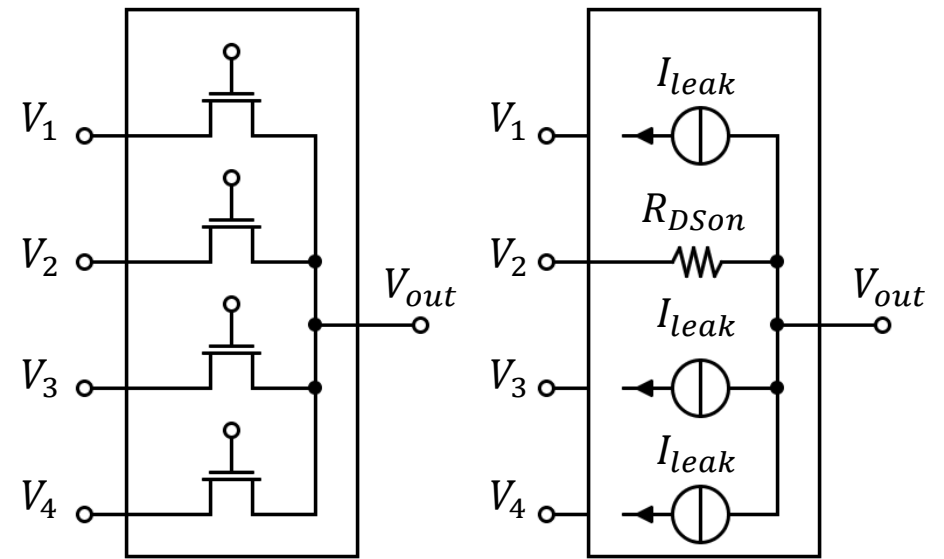
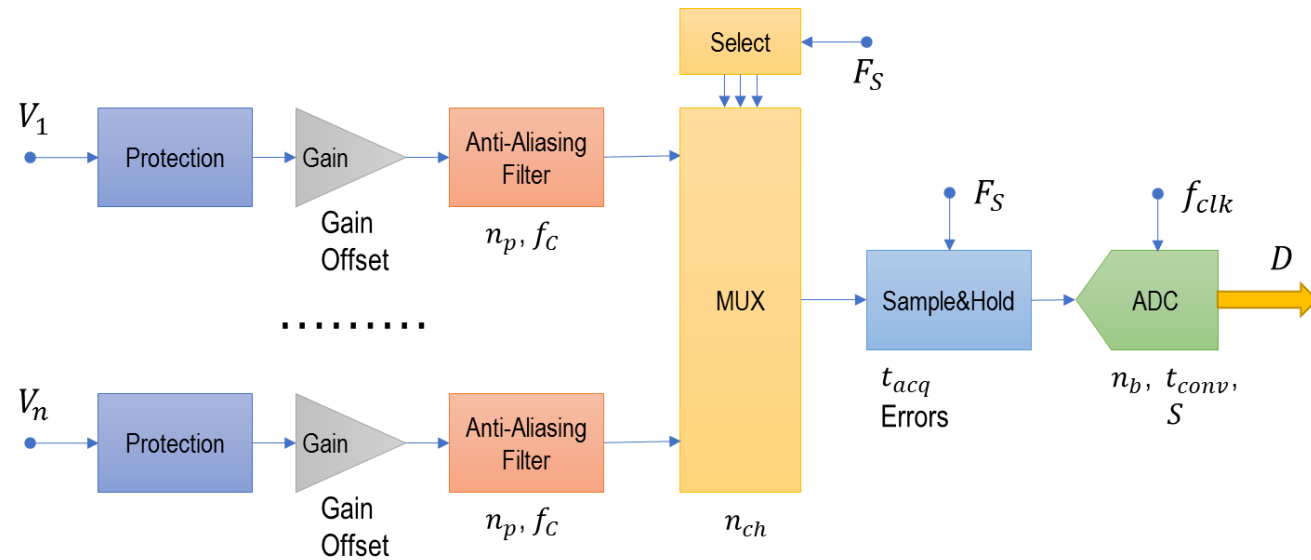
Acquisition systems

- Complete acquisition system
 - Multiple channels
 - Share expensive components (S&H, ADC) among multiple acquisition chains
 - n_{ch} inputs, 1 output analog multiplexer
 - Change selection at the sampling frequency F_S rate



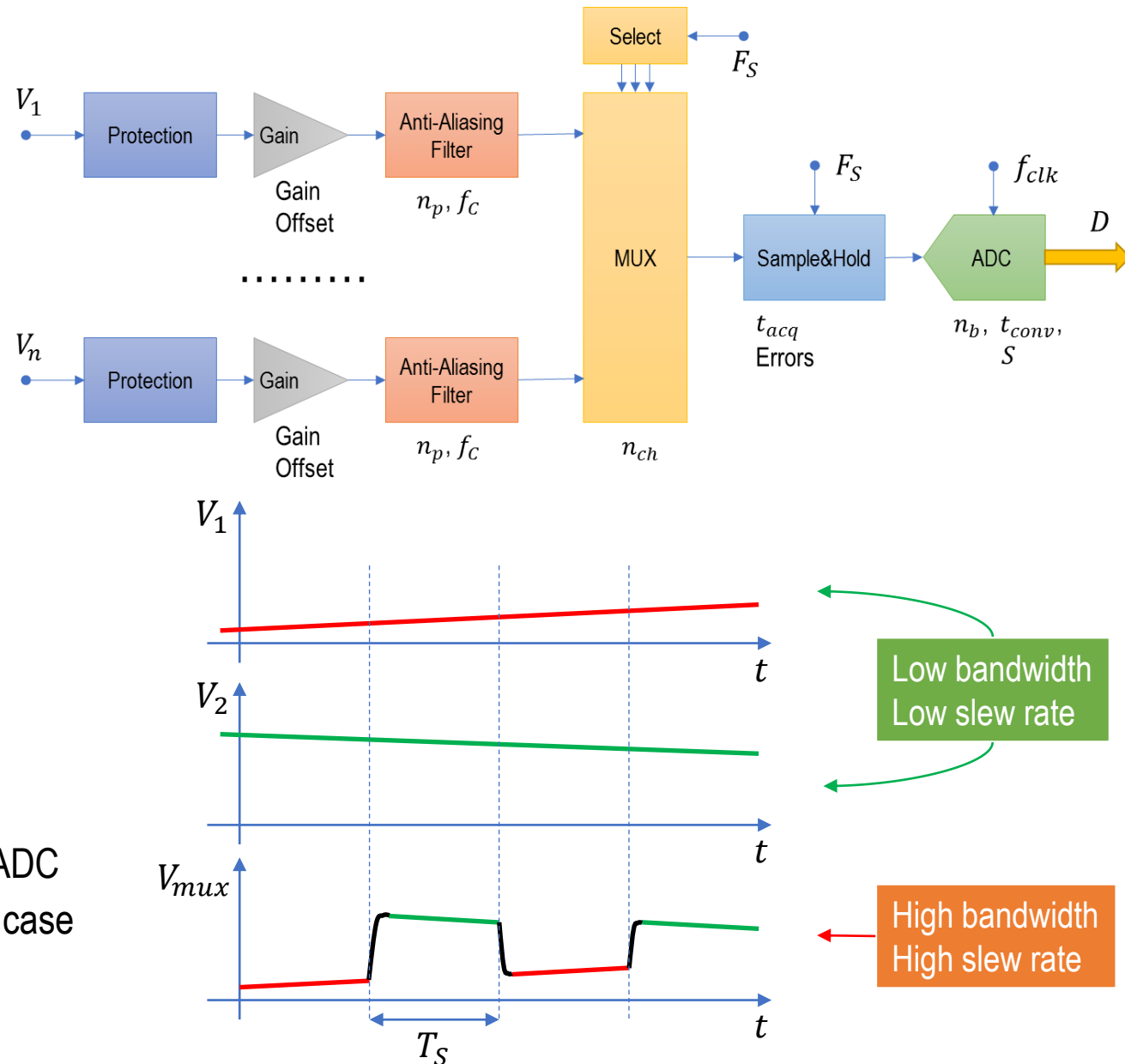
Acquisition systems

- Complete acquisition system
 - Multiple channels
 - Multiplexer alternatively selects each input channel in sequence
 - Errors in the multiplexer
 - The selected channel finds the ON resistance of the active pass transistor (or transmission gate)
 - All unselected channel have a small leakage current that produce an offset error
 - When switching from one channel to the next, there is a delay due to aperture and closing times of the transistors
 - When switching, the output voltage may take some time to reach the new value
 - Charge and discharge of parasitic capacitances
 - Similar to the acquisition time of a S&H



Acquisition systems

- Complete acquisition system
 - Multiple channels
 - The frequency content at the output of the multiplexer is very different from the spectra of the input signals
 - There are many high frequency contributions due to switching from one signal to another
 - Cannot share the anti-aliasing filter between different channels
 - The slew rate at the output of the multiplexer is potentially very high
 - Even with very slow input signals
 - Cannot use a Tracking AD Converter as the ADC
 - Or wait an exponential time in the worst case

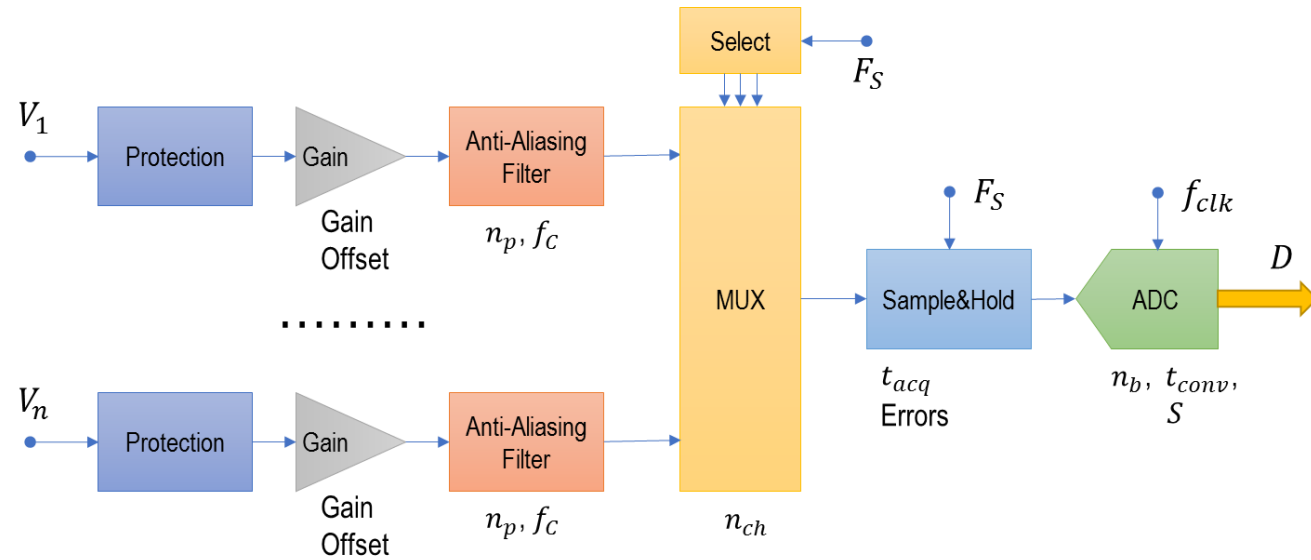


Acquisition systems

- Complete acquisition system

- Multiple channels

- The sampling frequency F_S should be higher than the required sampling frequency for a single channel
 - Assume all channels have the same useful bandwidth f_B
 - Let's sample each of them at $f_S = k \cdot f_B$, with $k > 2$
 - Then the sampling frequency F_S of the S&H and ADC chain is
 - $F_S = n_{ch} \cdot f_S$
 - $T_S = \frac{1}{F_S} = \frac{1}{n_{ch} \cdot f_S} = \frac{t_S}{n_{ch}}$
 - The multiplexer selects the first channel for a time T_S
 - Then another, again for T_S , and so on for all input channels
 - After $t_S = n_{ch} \cdot T_S$ the multiplexer selects again the first channel



Acquisition systems

- Complete acquisition system

- Multiple channels

- If channels have different useful bandwidths f_{Bi}

- Choose $f_B = \max_i(f_{Bi})$

- Oversample some of the channels
 - Conservative solution, more expensive

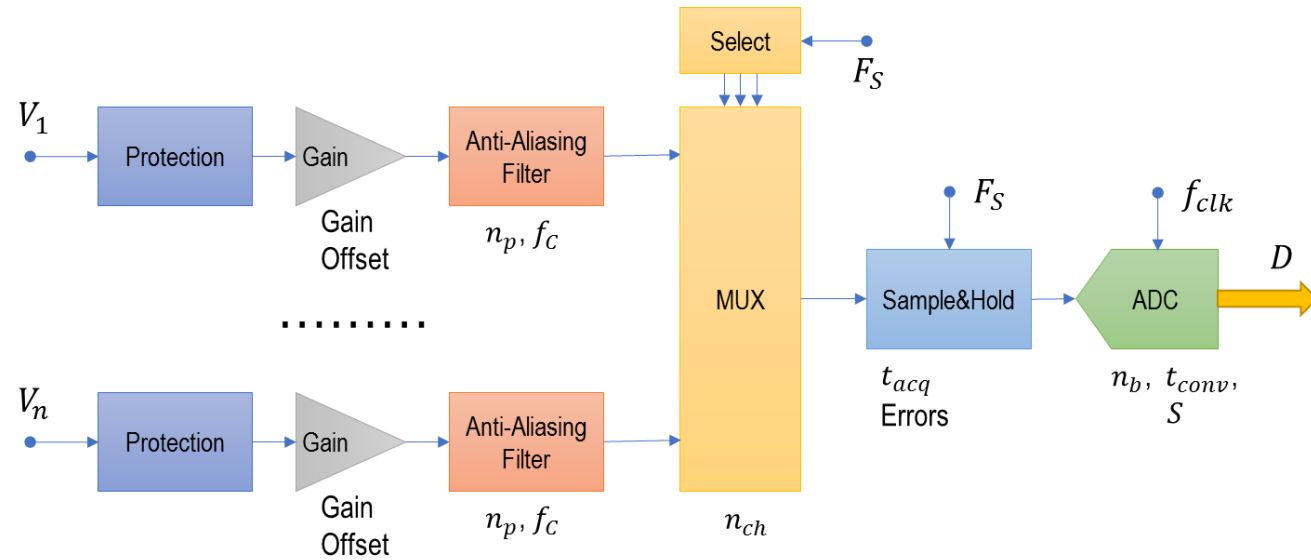
- Non uniform channel selection from the multiplexer

- Select more often channels with higher f_{Bi}
 - Example

- $f_{B1} = f_B, f_{B2} = 2f_B, f_{B3} = f_B$

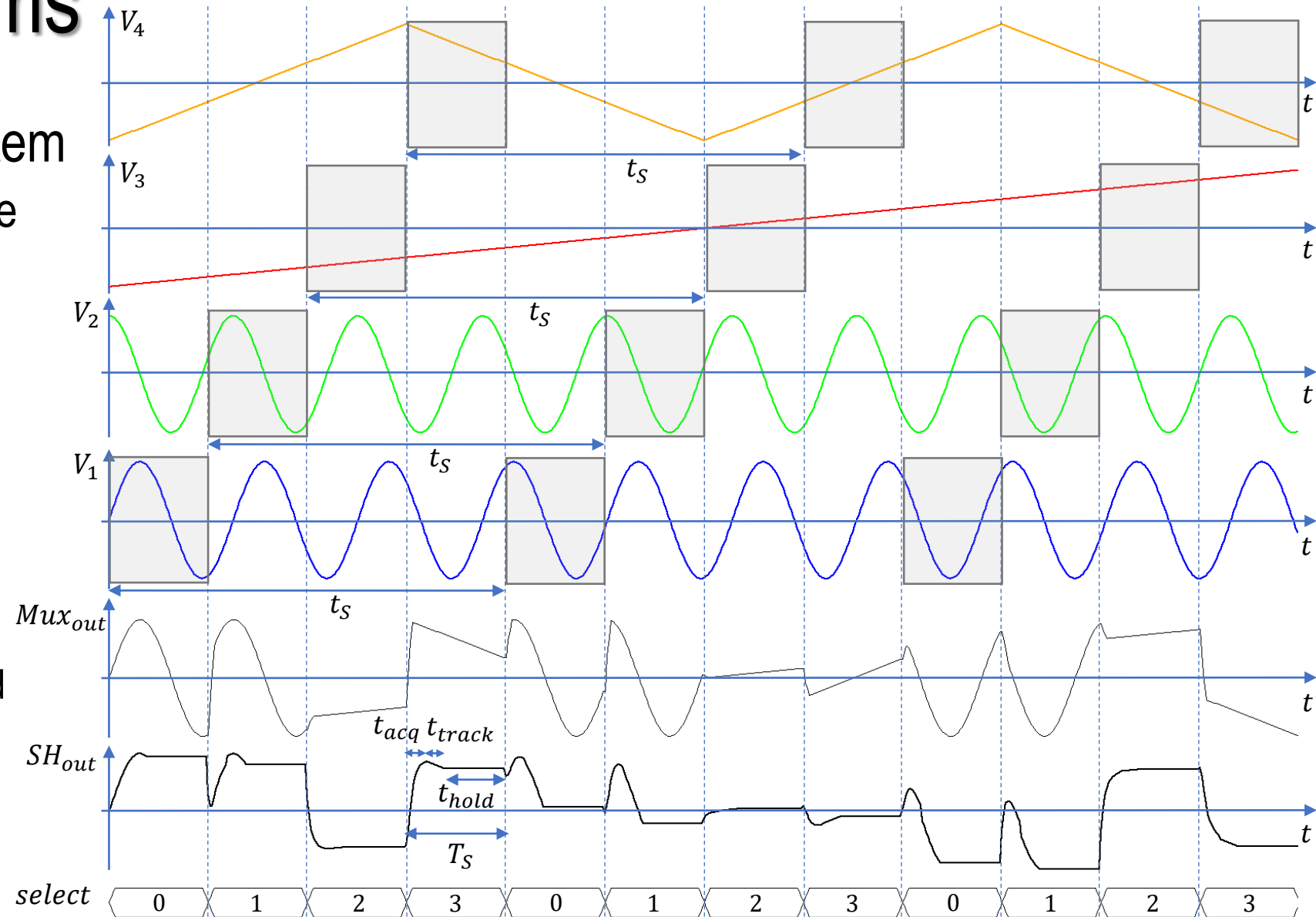
- Selection sequence at the multiplexer

- $CH_1, CH_2, CH_3, CH_2, CH_1, CH_2, CH_3, CH_2, CH_1, CH_2, CH_3, CH_2, \dots$



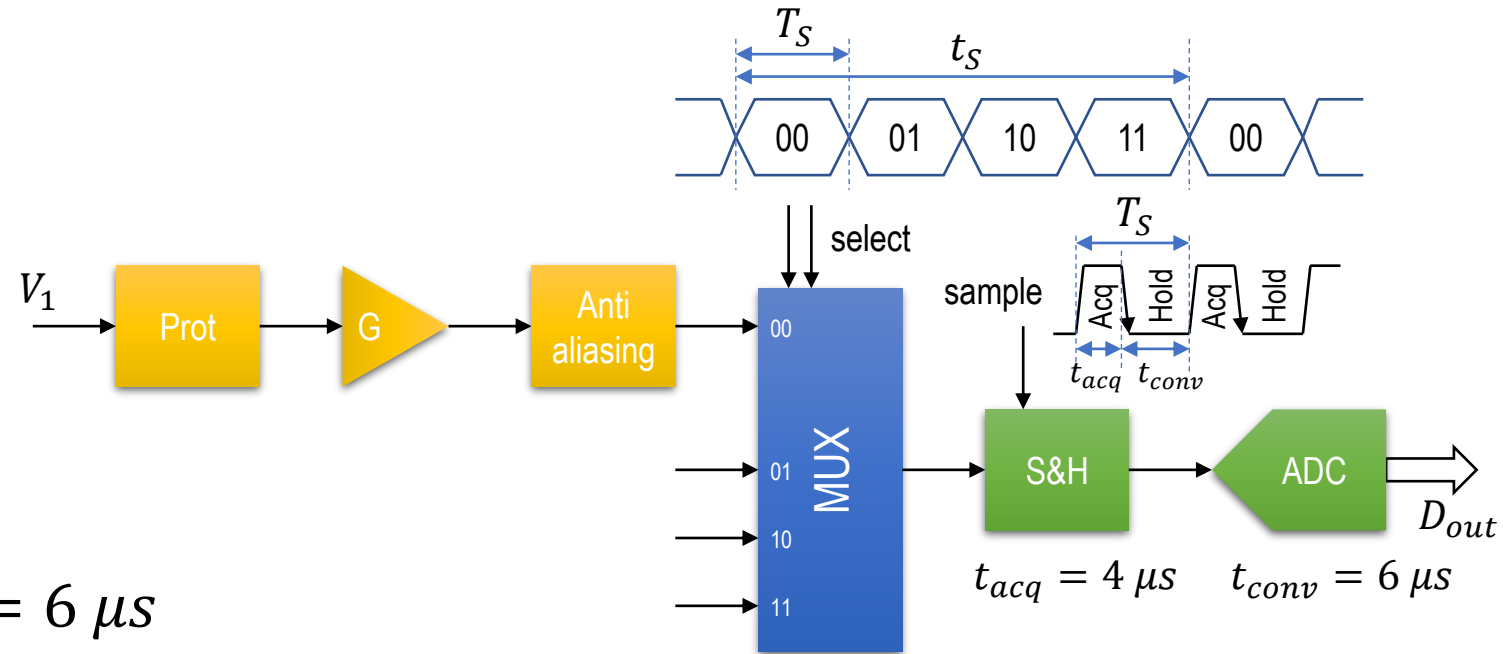
Acquisition systems

- Complete acquisition system
 - Multiple channels example
 - Multiplexer output
 - Small acquisition time
 - Then tracking
 - S&H output
 - Larger acquisition time
 - Tracking time
 - Hold time
 - Selection signal
 - Signals are undersampled in this example



Acquisition systems

- Acquisition system example
 - System with 4 input channels
 - Sample&Hold with acquisition time $t_{acq} = 4 \mu s$
 - ADC with conversion time $t_{conv} = 6 \mu s$
 - Sample at $2.5 \times$ the Nyquist frequency
 - Compute the maximum frequency f_B of the input signals



1. Make $t_{track} = 0$ so that it doesn't affect the sampling period T_s
 2. Make $t_{hold} = t_{conv}$ so that we are at the maximum possible performance

3. Then $T_s = t_{acq} + t_{conv} = 10 \mu s$
 4. Then $F_s = \frac{1}{T_s} = 100 \text{ kHz}$

5. There are 4 channels, so $t_s = 4T_s = 40 \mu s$
 6. Then $f_s = \frac{1}{t_s} = \frac{1}{4T_s} = \frac{F_s}{4} = 25 \text{ kHz}$
 7. Then $f_{Ny} = \frac{f_s}{2.5} = 10 \text{ kHz}$
 8. Then $f_B = \frac{f_{Ny}}{2} = 5 \text{ kHz}$

Acquisition systems

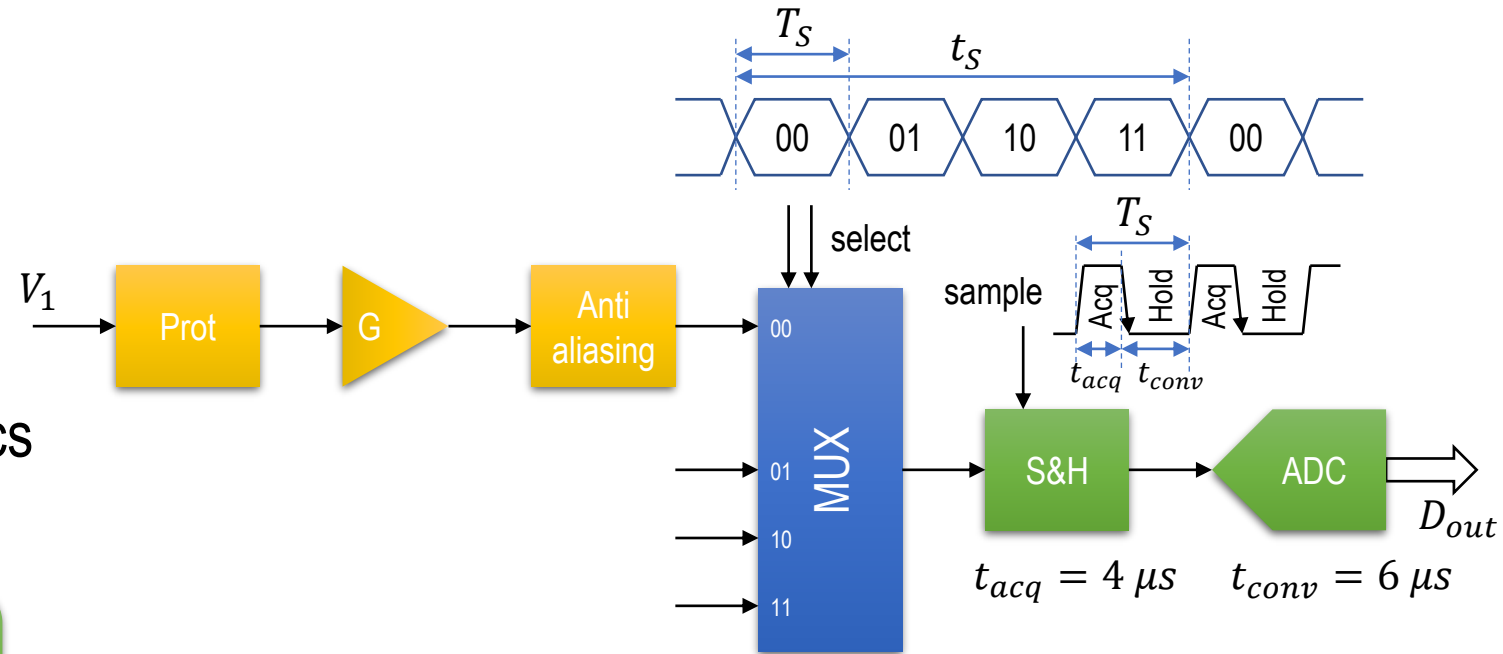
- Acquisition system example
 - Assuming quantization is on $N = 8$ bits, give the characteristics of the ADC

- For a **Flash** ADC, it is
 - $n_{comp} = 2^N - 1 = 255$
 - $t_{comp} + t_{enc} \simeq t_{comp} < 6 \mu s$

- For a **Staircase** or **Tracking** ADC, it is
 - $t_{clk} < \frac{t_{conv}}{2^N} = \frac{6 \mu s}{256} = 23.5 ns$
 - $f_{clk} = \frac{1}{t_{clk}} > 42.7 MHz$
 - $t_{comp} + t_{cnt} + t_{DAC} < 23.5 ns$

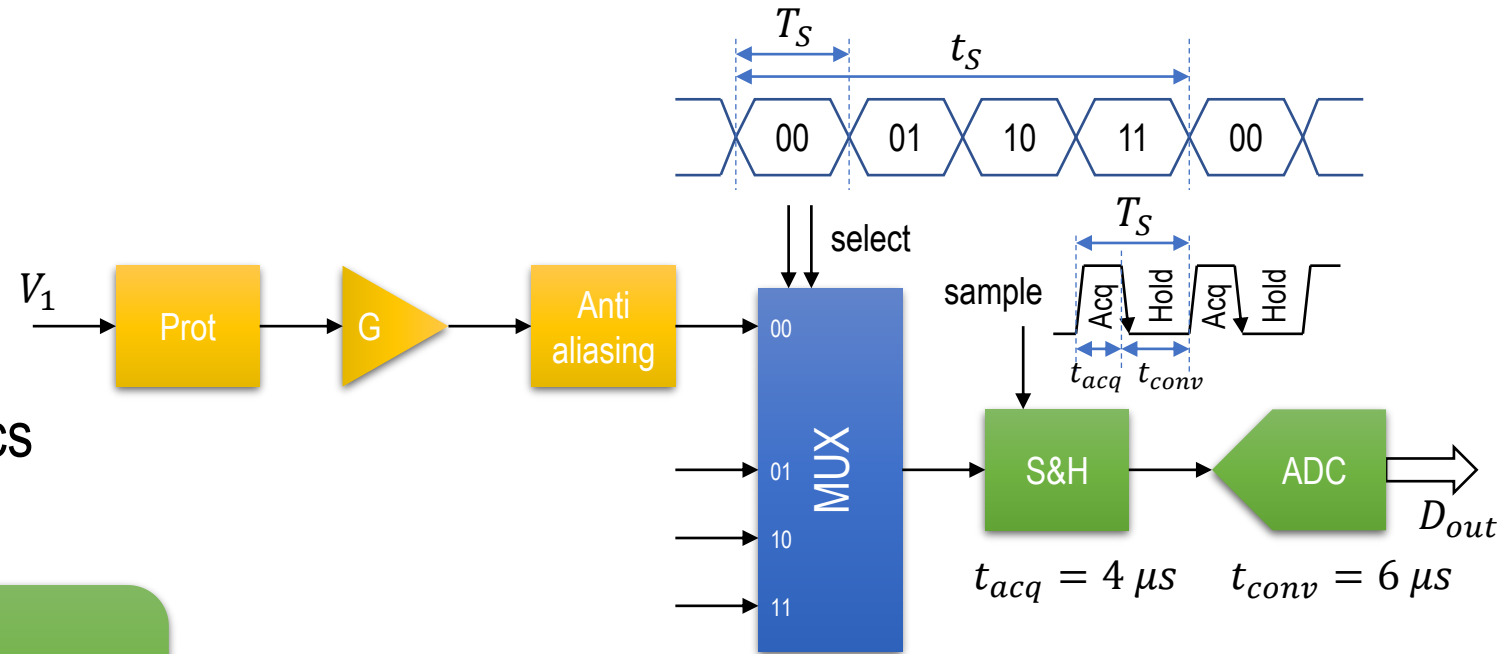
The **Tracking** ADC requires a register at the output to avoid seeing all intermediate results, as it is not generating one good sample at each clock cycle

- For a **SAR** ADC, it is
 - $t_{clk} < \frac{t_{conv}}{N} = \frac{6 \mu s}{8} = 750 ns$
 - $f_{clk} = \frac{1}{t_{clk}} > 1.33 MHz$
 - $t_{comp} + t_{cnt} + t_{DAC} < 750 ns$



Acquisition systems

- Acquisition system example
 - Assuming quantization is on $N = 8$ bits, give the characteristics of the ADC



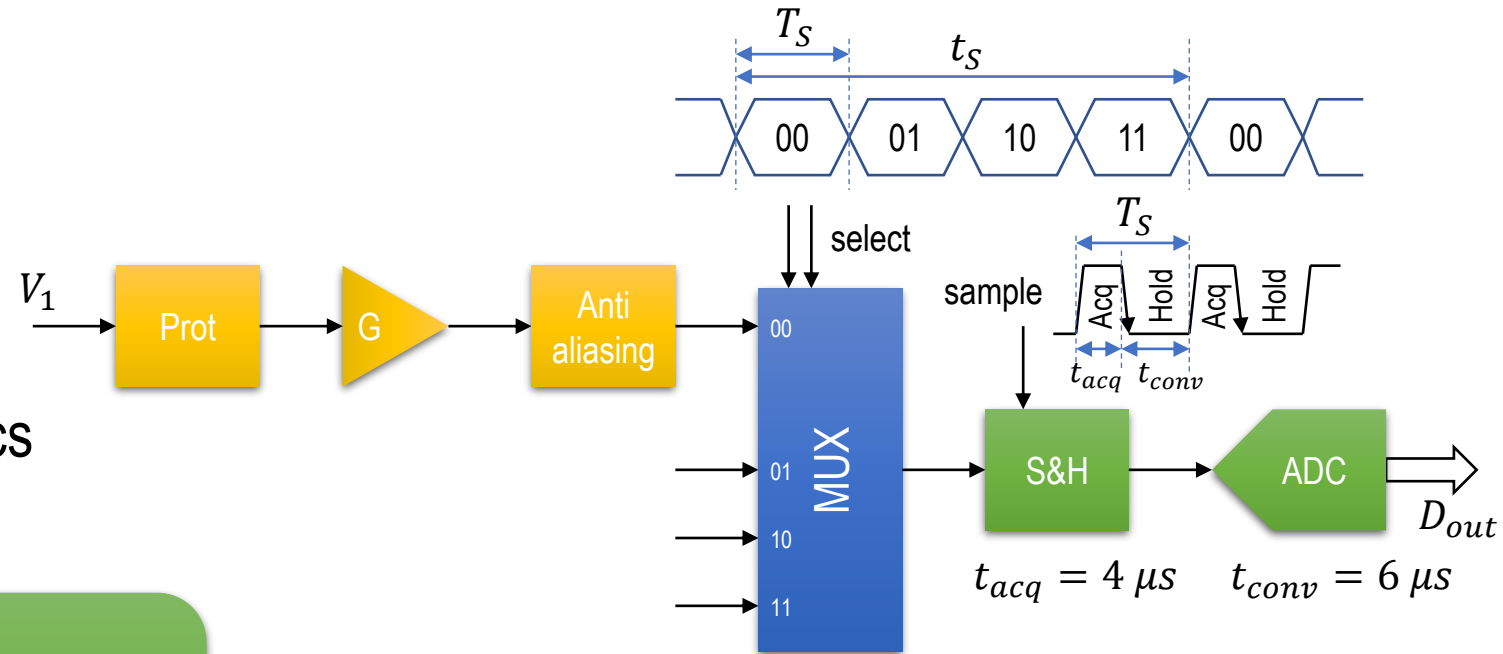
- For a **Residue** ADC with 1-bit stages, it is
 - $n_{stages} = n_{comp} = N = 8$
 - $t_{stage} \simeq t_{comp} + t_{DAC} < \frac{t_{conv}}{8} = 750 \text{ ns}$

- For a **Residue** ADC with 2-bit stages, it is
 - $n_{stages} = 4$
 - $n_{comp} = n_{stages} \cdot (2^2 - 1) = 12$
 - $t_{stage} \simeq t_{flash} + t_{DAC} < \frac{t_{conv}}{4} = 1.5 \mu s$

- For a **Residue** ADC with 4-bit stages, it is
 - $n_{stages} = 2$
 - $n_{comp} = n_{stages} \cdot (2^4 - 1) = 30$
 - $t_{stage} \simeq t_{flash} + t_{DAC} < \frac{t_{conv}}{2} = 3 \mu s$

Acquisition systems

- Acquisition system example
 - Assuming quantization is on $N = 8$ bits, give the characteristics of the ADC



- For a **Pipeline** ADC with 1-bit stages, it is
 - $n_{stages} = n_{comp} = N = 8$
 - $t_{stage} \simeq t_{comp} + t_{DAC} + t_{acq} < T_S = 10 \mu s$
 - $t_{comp} + t_{DAC} < T_S - t_{acq} = t_{conv} = 6 \mu s$

Assume that all Sample&Holds in the Pipeline converters are driven at frequency F_S and that they have an acquisition time $t_{acq} = 4 \mu s$. Then the remaining part of each stage, i.e. the Flash ADC and the DAC, should take a time no greater than t_{conv} . The first stage should be preceded by a single Sample&Hold, not two (only an internal or the external one).

- For a **Pipeline** ADC with 2-bit stages, it is
 - $n_{stages} = 4$
 - $n_{comp} = n_{stages} \cdot (2^2 - 1) = 12$
 - $t_{stage} \simeq t_{flash} + t_{DAC} + t_{acq} < T_S = 10 \mu s$
 - $t_{flash} + t_{DAC} < T_S - t_{acq} = t_{conv} = 6 \mu s$

- For a **Pipeline** ADC with 4-bit stages, it is
 - $n_{stages} = 2$
 - $n_{comp} = n_{stages} \cdot (2^4 - 1) = 30$
 - $t_{stage} \simeq t_{flash} + t_{DAC} + t_{acq} < T_S = 10 \mu s$
 - $t_{flash} + t_{DAC} < T_S - t_{acq} = t_{conv} = 6 \mu s$